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Validation Services for the geospatial products of the
Copernicus land Continental and local components
including in-situ data (lot 1)

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CLC2018 / CLCC1218 VALIDATION REPORT



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Executive Summary

This report provides the evaluation results of the CORINE Land Cover (CLC) data layers, specifically the Land Cover 2018 status and the Land Cover 2012-2018 change layers. This analysis was performed over 100% of coverage for each of the EEA39 countries. No semantic checks were envisaged for this product and documentation on previous semantic checks was not made available to the validation team.

The thematic accuracy assessment was conducted in a two-stage process:

1. An initial blind interpretation in which the validation team did not have knowledge of the product's thematic classes. However, the product polygon was provided to the validation team together with the point sample unit to consider boundary effects and geometric differences between the validation and production data (without access to the different attributes)
2. A plausibility analysis was performed by another independent team from the blind interpretation. The plausibility analysis has been carried out on all sample units in disagreement with the production data to consider the following cases:
 - Both producer and operator codes are plausible. Final validation code used is producer code
 - Error from first validation interpretation. Final validation used is producer code
 - Error from producer. Final validation code used is from first validation interpretation
 - Producer and operator are both wrong. Final Validation code used is a new code from this second interpretation.

At the end to this two-stage approach, we observe that the plausibility analysis allows a slight improvement in the results compared to the blind analysis (for each level of reporting: biogeographical or country level).

The reporting for validation was done at 3 levels: pan-European, country (or group of countries) and by biogeographical regions. A dedicated sampling design has been developed and applied using two levels of stratification according to (1) countries and group of countries greater than 90,000km² and (2) the CLC2018 or CLCC2012-2018 classes. The thematic accuracy assessment was carried out over 25695 point covering the EEA39.

The results show that the CLC2018 products meet the thematic validation requirement, i.e. 85 % in a blind and plausibility analysis. Indeed, the overall accuracies obtained for CLC2018 are **92.43 %** for the blind analysis and **92.67 %** for the plausibility analysis. The overall accuracy for the CLCCH2012-2018 data layer at CLC level 3 is **87.12%** for the blind analysis and **87.83 %** for the plausibility analysis. Those very good results are related to the fact that the 2018 dataset was the fifth update of CORINE Land Cover data, and that each time the database has undergone corrections performed by the national teams.

Regarding biogeographic regions, the lower result can be observed in Black Sea region with **84%**. The other regions have results above the recommendation threshold. For each country or biogeographical region, the overall accuracy for the CLCCH2012-2018 change layer is systematically lower than the CLC2018 overall accuracy. However, it is quite surprising to note that good results for the CLC2018 status layer does not guarantee good results for the changes CLCCH2012-2018 layer, and vice versa. But a deeper analysis shows that there is generally a link between landscape complexity, diversity of actual changes and the overall accuracy of change detection. The more various are the changes (at level 3), the weaker overall accuracy will be.

Quality checks of the validation points have been made by European experts, partners in the consortium. It should be noted that for the blind and plausibility analysis, we used high and very high-resolution satellite images as a reference:

- VHR 2018: spatial resolution <2m
- HR 2018: spatial resolution of 10m
- VHR 2012: spatial resolution of 2.5m

- HR 2012: spatial resolution between 5m and 20m

In addition, the experts also used google earth as very high-resolution ancillary data. Unlike the national photo-interpretation teams, we did not have any additional thematically source of information such as forest stand maps, peatland maps, etc. Control has thus been performed with a lack of some national source information.

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List of Abbreviations

CLC	CORINE Land Cover
CLCCH2012-2018	Corine land cover change between 2012 and 2018
EEA	European Environment Agency
EU-DEM	Digital Elevation Model over Europe
ESA	European Spatial Agency
FAO	Food and Agriculture Organization of the United Nations
FTY	Forest Type
GIO	GMES Initial Operations
GMES	Global Monitoring for Environment and Security
HRL	High Resolution Layer
IGN FI	IGN France International
JRC	Joint Research Centre
LAEA	Lambert Azimuthal Equal-Area
LUCAS	Land Use/Cover Area frame Survey
MMU	Minimum Mapping Unit
MMW	Minimum Mapping Width
PSU	Primary Sample Unit
SP	Service Provider
TCD	Tree Cover Density
UA	Urban Atlas

1. Validation Framework

The validation framework is defined by a comprehensive analysis of the product specifications to determine the criteria to be used for the validation exercise.

1.1. Products to be validated

The CORINE Land Cover (CLC) dataset was initiated for the reference year 1990 in the late 80s. Updates were then regularly produced for 2000, 2006, 2012 and the latest 2018. The CLC applies a pragmatic nomenclature of 44 land cover/land use classes with a Minimum Mapping Unit (MMU) of 25 hectares (ha) and a Minimum Mapping Width (MMW) of 100 m.

Updates are created by generating a change layer. Interpreted changes that have occurred between the two reference years considered are mapped with a 5 ha MMU. It means that the change layer has a higher level of spatial detail than the status layer. Changes are then aggregated to unchanged areas to produce the updated status layer with a 25 ha MMU.

Therefore, the products to be validated as part of the CLC2018 exercise are the following:

- CORINE Land Cover 2018 map (CLC2018) nomenclature with 44 classes
- CORINE Land Cover 2012-2018 change map (CLCCH2012-2018)

1.2. Validation Criteria

1.2.1. CLC2018

The validation exercise focuses on thematic accuracy. The CLC expected overall accuracy is greater than 85% for both the status and change layers. Former validation exercises have been undertaken for previous version of CLC; the most recent available by Büttner *et al.* (2017).

1.2.2. CLCCH2012-2018

The validation exercise should contribute to determine a suitable accuracy threshold for CLC changes between 2012 and 2018. An overall accuracy will be calculated, and another will be specified by country or group of small countries and by biogeographical region. The European countries are divided into 23 areas.

2. Validation approach

The validation approach will provide guidance on how the products will be validated by defining suitable indicators or metrics. Detailed completeness and logical consistency checks are already performed as part of the semantic checks undertaken by the QC Tool during upload of the products. Therefore, the aim of this validation exercise is not to repeat these, but to review the existing documentation and perform additional checks if deemed necessary. We recommend however to make the QC Tool reports available openly as additional quality information to the users.

Thematic accuracy will represent the bulk of the work undertaken as part of this validation exercise.

The quality assessment is performed according to INSPIRE Data Specifications. The data quality elements considered are: (i) Completeness, (ii) Logical Consistency, (iii) Positional Accuracy, (iv) Thematic Accuracy, (v) Temporal quality and (vi) Usability. Each of them forms a section in the Validation Check list.

Logical consistency checks do not consist in a duplication of Semantic checks, but are performed to identify missing information if relevant.

2.1. Completeness

Completeness should only be performed when the product is complete

Description: For non-thematic raster products (Image mosaics & EU-DEM), Completeness provides an indication for missing data or omission within the intended area. For land cover and land use products (both raster & vector), the notion of Completeness in INSPIRE provides an indication of omission and commission errors.

Indicators: the rate of excess items is used for commission errors and the rate of missing items is used to verify omission errors.

2.2. Logical consistency

Logical consistency evaluates the degree of adherence to logical rules of data structure, attribution and relationships. In INSPIRE Data Specifications; Logical Consistency comprises four sub-elements described hereafter: conceptual consistency, domain consistency, format consistency and topological consistency.

2.2.1. Conceptual consistency

Description: indicates that the data structure follows the data specifications in terms of data model and relationships.

Indicators:

- Type of feature used
- Minimum Mapping Unit (MMU)
- Coordinate Reference System
- The presence of a unique identifier for each feature
- Nomenclature used

2.2.2. Domain consistency

Description: involves the detection of attribute values that are outside the pre-defined range of values. For vector data each attribute has a pre-defined set of range of values. For raster data, the correct encoding of data is checked.

Indicator: Value domain non-conformance: number of items not in conformance with their expected value domain.

2.2.3. Format consistency

Description: includes detection of file format, file or attribute names or attribute types which do not correspond to the specifications. In addition, for raster data the pixel depth is also considered here.

Indicators:

- File format conformance
- File name conformance
- Attribute names conformance
- Attribute types conformance

2.2.4. Topological consistency

Description: topological consistency is applicable to vector data and describes the degree of correctness of the topological characteristics described in the product specification of the dataset.

Indicators:

- | | |
|-----------------------------------|--------------------------------|
| - Number of overlaps | - Number of null geometries |
| - Number of gaps | - Number of unclosed rings |
| - Number of multipart features | - Number of duplicate vertex |
| - Number of neighbouring features | - Number of pseudo nodes |
| - Number of self-intersections | - Number of non-matching nodes |

2.2.5. Additional logical consistency checks

- **Labelling or symbology:** the conformity of a layer with the symbology or style given in the product specifications is checked.
- **Map projection:** the conformity of the map projection parameters is also checked.

2.3. Positional Accuracy

Positional accuracy of CLC 2018 is directly related to the positional accuracy of the underlying VHR 2018 imagery. Visual checks were undertaken in relation to imagery used for validation and during production.

2.4. Thematic Accuracy

2.4.1. Level of reporting

The level of reporting for the validation results is at pan-European level. However, results are also provided at different levels of aggregation:

- Countries or group of countries with an area greater than 90,000km² level (Figure 1)

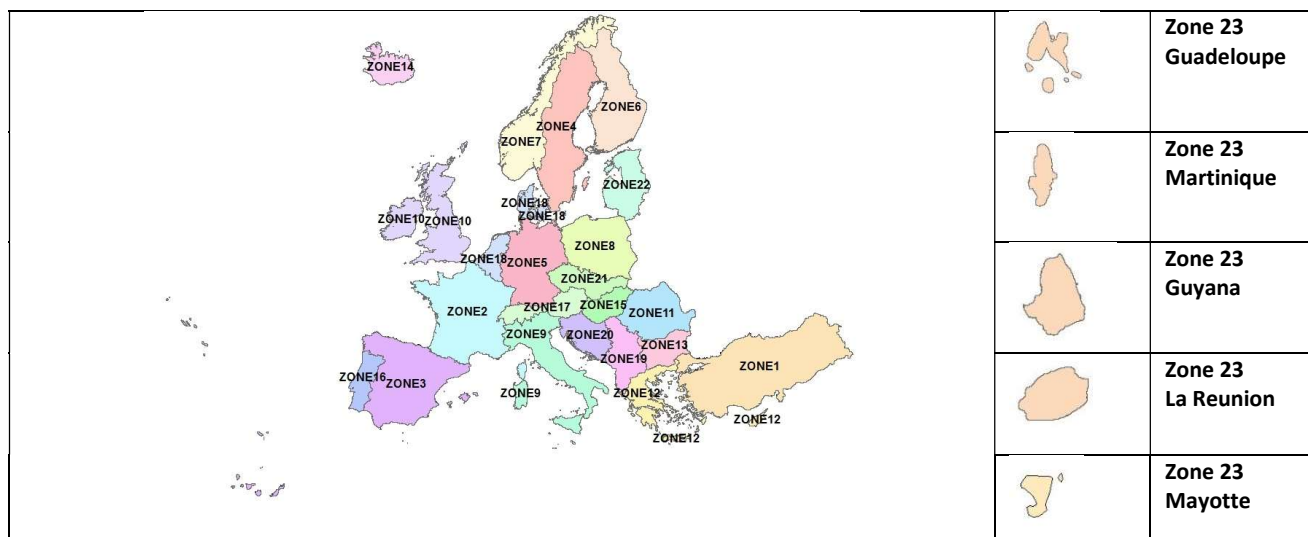


Figure 1: Stratification according to countries and group of countries greater than 90,000km² (total of 23 zones including French overseas department)

- European Biogeographical regions 2016 level (Figure 2). The French overseas departments are aggregated into one specific biogeographical region named DOM. In fact, this region corresponds to the tropical climate and includes French Guiana, Martinique, Guadeloupe, La Reunion and Mayotte.

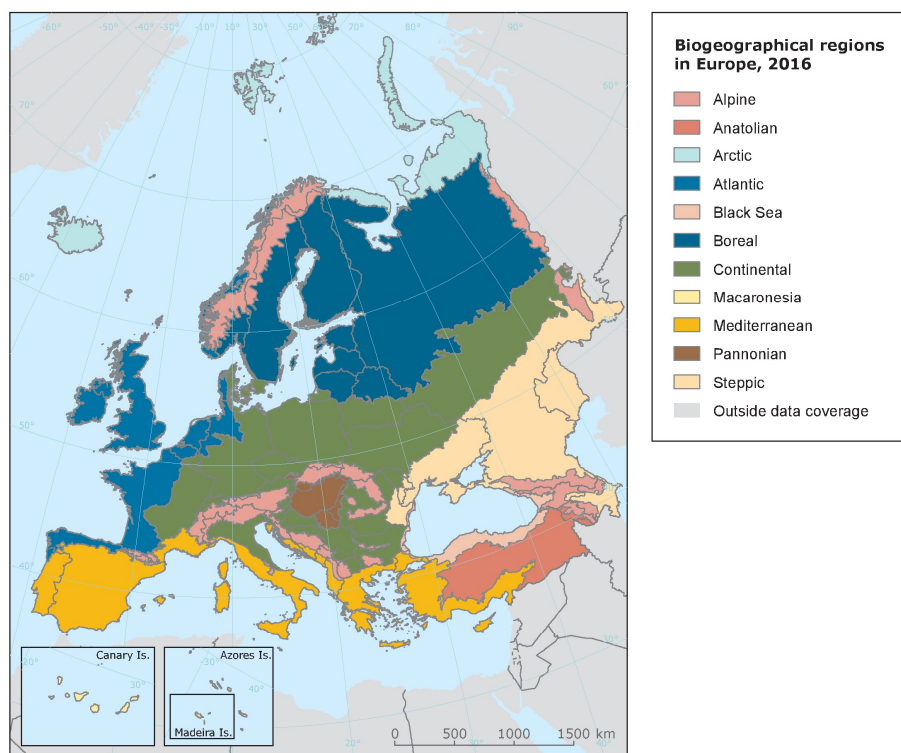


Figure 2: The European Biogeographical region 2016¹

Analysis of the results at a more disaggregated scale will contribute to assess regional differences if any and the causes of these differences.

¹ This map reflects the status of the Biogeographical Regions in Europe from 2016 onwards. Maps from European Environment agency. Link : <https://www.eea.europa.eu/data-and-maps/figures/biogeographical-regions-in-europe-2>

2.4.2. Stratification and sample design

2.4.2.1. Overview

The stratification and the sampling design primarily consist in selecting an appropriate sampling frame and sampling unit. The sampling units can either be “defined on a cartographic representation of the surveyed territory” (Gallego, 2004²), in which case it is an area frame, or on a list of the features. According to (Gallego, 2004), area frames give a better representation of the population as the spatial dimension is kept.

In an area frame, sample units can be points, lines (often referred to as transects) or areas (often referred to as segments, described by Gallego, 1995³). The first step is to define the geographical area for which the accuracy assessment is to be reported and the type of sample units. For the majority of cases, point samples will be used, but areas or segments may be used in specific cases such as when not only thematic accuracy needs to be reported, but also the geometry of mapped objects. Points are considered as the most appropriate unit for our purpose. Polygons have also the drawback of being specific to a single map. In case of changes, the sample may not be adapted anymore.

Sampling design refers to the protocol whereby the samples are selected. A probability sampling design is preferred for its objectivity. “Simple random, stratified random, clustered random and systematics designs are all examples of probability sampling designs” (Stehman et al., 1998⁴). Even though a simple random design is easy to implement, its main drawback is that some portions of the population may not be adequately sampled. Cluster sampling is often used to reduce the costs of the collection of reference data but does not resolve geographic distribution problems. A systematic approach would solve this problem, yet it is not appropriate if the map contains cyclic patterns. A stratified approach consists in allocating a pre-defined number of samples per land-cover class. As explained in (Stehman et al., 1998), stratification ensures that each class is represented.

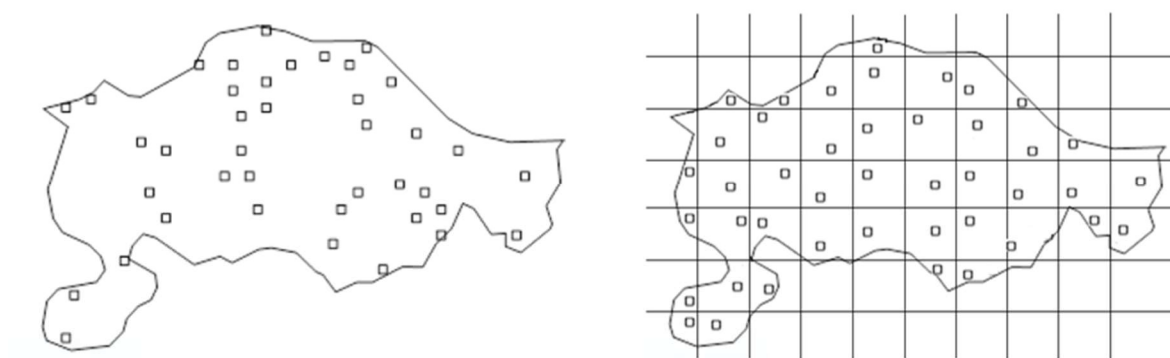


Figure 3: Simple random (left) and random systematic (right) sampling designs

A stratified systematic sampling approach based on LUCAS (Land Use/Cover Area frame statistical Survey) is used for all thematic layers adapting the number of replicates to each stratum. The choice of this approach was made for the following reasons:

- A systematic approach ensures full traceability of the results compared to a random approach in which the random selection of samples is difficult to trace

² Gallego, J. and JRC-IES, I.I., 2004. Area Frames for Land Cover Estimation: Improving the European LUCAS Survey. Presented at the Proceedings of the 3rd World Conference on Agricultural and Environmental Statistical Application, Cancun, Mexico, 2–4.

³ Gallego, F.J., 1995. Sampling frames of square segments. Office for Official Publ. of the European Communities.

⁴ Stehman, S.V. and Czaplewski, R.L., 1998. Design and analysis for thematic map accuracy assessment: fundamental principles. *Remote Sensing of Environment*, 64 (3), 331–344.

- Some of the sample units will be shared amongst different data layers fostering economies of scale for the validation of Copernicus Land pan-European and local geospatial products
- Some of the sample units will also be surveyed on the ground as part of LUCAS

LUCAS corresponds to a grid of approximately 1,100,000 points throughout the European Union. The LUCAS sampling is densified for small strata.

A set of 81 points located on an 18x18 km square constitutes a group (red points shown in Figure 4) in which every point is associated with a number comprised between 0 and 81 (the numbers do not follow each other spatially). The same pattern with the same numbers allocation is repeated all over the grid. A replicate refers to the points with the same number selected on the whole LUCAS grid.

At first, the number of samples to allocate to each stratum (or MAES4 land cover class) was calculated as a function of their area. In this manner, the sampling design is not only systematic but also stratified. A minimum number of sample units per stratum was defined to ensure that even small strata are represented in the sample.

The number of replicates to be selected for a stratum depends on its area and the number of LUCAS points intersecting the stratum.

For land cover classes covering a large proportion of the study area, 1 replicate may already exceed the defined number of samples for this class. To solve this problem, replicates are split into four sub-replicates, as illustrated by the blue numbers in Figure 4.

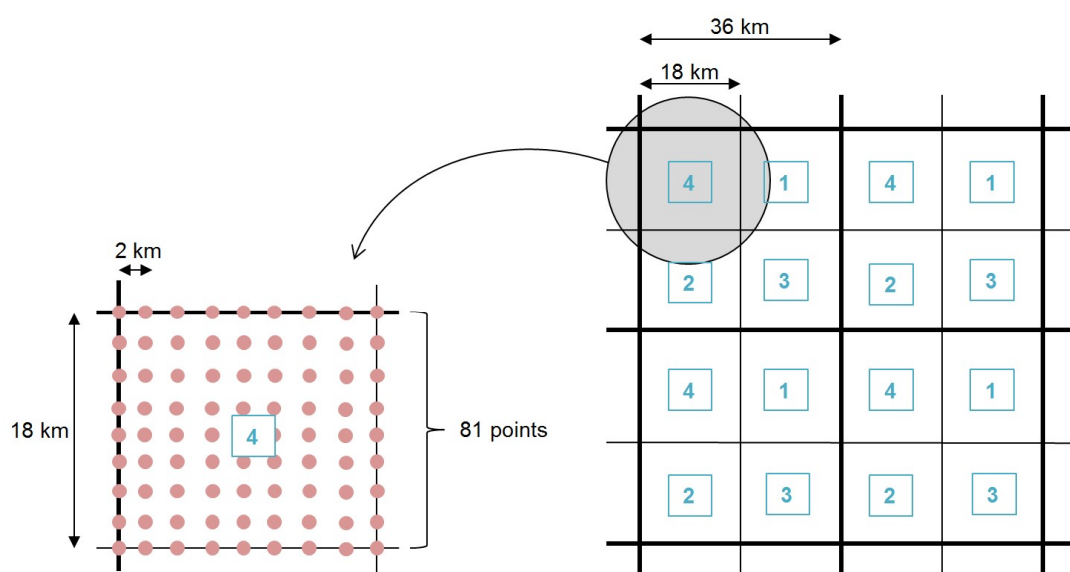


Figure 4: Replicates and sub-replicates used on LUCAS grid

The opposite problem is encountered for land cover classes covering a small proportion of the study area: even by selecting 81 replicates (the maximum number), the intersecting area between the stratum and LUCAS points is too small to reach the required number of samples. Therefore LUCAS grid was densified by creating one point every 200 m.

2.4.2.2. Stratification

For the CLC2018 status and change layers, stratification is applied at two levels:

1. Stratification according to countries and group of countries greater than 90,000km²
2. Stratification based on the CLC2012 classes and CLCCH2012-2018 classes based on the map products

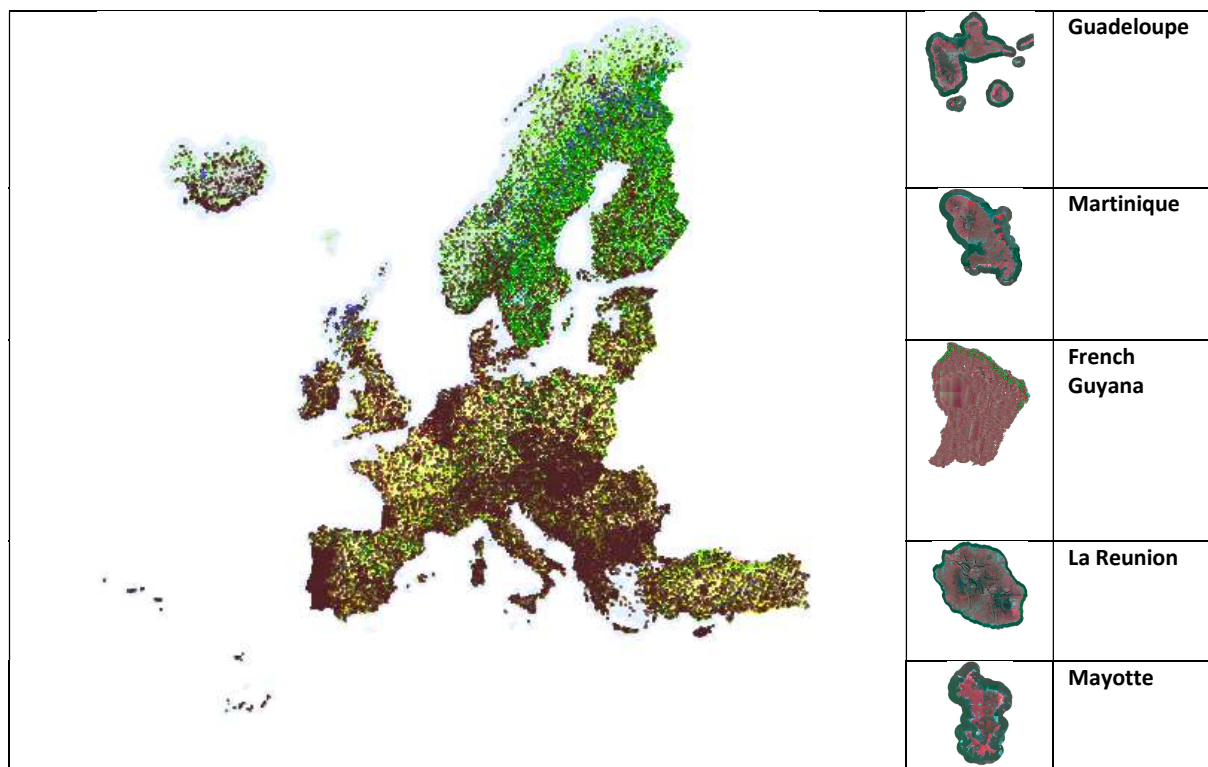


Figure 5: Distribution of the sample units according to the stratification into countries and group of countries

It should be notice that the stratification is based on the CLC 2012 map product, performed during the Second Specific Contract. This approach allows us to take advantage of the sample units already drawn and interpreted for the reference year 2012. Nevertheless, as mentioned in section 2.4.3, the sample units have been reinterpreted/updated for the reference year 2018.

The number of sample units per stratum should be such to ensure a sufficient level of precision at reporting level. The minimum number of sample units per stratum should be set at 5 if possible. Priority is given to strata which are known to be difficult to map: i.e. changes and difficult classes.

The validation exercise covers the whole study area in order to be valid.

There was a total of 25,695 sample units selected and covering the entire of EEA39 area. The overall share of sample units per main strata and per country (or aggregated countries) is presented in Table 1.

Table 1: Distribution of sample units per main strata and substrata for the CLC012 and CLCCH2006-12 products

		TR	FR	ES	SE	DE	FI	NO	PL	IT	IE+UK	RO	EL	BG	IS	HU	PT	AT+CH+LI	BE+LU+NL+DK	AL+ME+MK+RS+XK	BA+HR+SI	CZ+SK	EE+LT+LV	DOM	
	CLASS/ ZONE	Z1	Z2	Z3	Z4	Z5	Z6	Z7	Z8	Z9	Z10	Z11	Z12	Z13	Z14	Z15	Z16	Z17	Z18	Z19	Z20	Z21	Z22	Z23	Total général
111	Continuous urban fabric	30	10	20	5	9		15	9	24	19	2	14	1		2	24	20	19	2	5	10	2	1	243
112	Discontinuous urban fabric	75	139	41	53	68	66	42	68	109	90	58	36	24	20	64	23	36	95	42	30	67	53	34	1333
121	Industrial or commercial units	55	88	54	27	76	41	29	51	91	110	40	36	27	16	52	45	41	67	33	30	43	32	2	1086
122	Road and rail networks and associated land	15	24	24	4	11	2	17	59	30	25	20	32	15	1	41	24	27	27	5	27	29	19	4	482
123	Port areas	2	10	5	5	9	9	15	5	21	14	4	3	2	5	1	10	2	22	2	2	1	9	1	159
124	Airports	9	12	10	10	12	10	15	16	20	15	10	10	5	3	5	10	10	15	6	2	10	10	2	227
131	Mineral extraction sites	63	12	11	22	59	31	25	52	34	45	28	20	24	5	19	20	26	19	37	22	30	40	2	646
132	Dump sites	3	5	7	12	8	31	3	17	12	9	7	2	6		12	7	1	11	6	5	18	10	1	193
133	Construction sites	36	4	17	2	4	8	3	11	6	15	12	11	3	3	10	10	1	47	2	3	9	1		218
141	Green urban areas	4	10	10	11	12	17	19	17	17	19	14	3	10	5	19	10	19	22	10	10	19	18	1	296
142	Sport and leisure facilities	9	19	17	17	18	27	60	22	26	29	22	18	26	24	19	28	34	36	14	21	31	22	1	540
211	Non-irrigated arable land	129	95	116	73	141	97	32	61	131	91	104	34	115	6	168	52	57	90	61	36	133	103	21	1946
212	Permanently irrigated land	100		75						6		2	43				77			3	2				308
213	Rice fields	36	10	21						26	6	29	5			2	23			4				2	164
221	Vineyards	29	53	75		18				42		10	16	37		38	28	21	2	18	20	20			427
222	Fruit trees and berry plantations	52	23	80	5	18			39	34	14	21	26	33		63	23	9	22	24	21	38	17	11	573
223	Olive groves	27	9	75						23			38				75			8	22				277
231	Pastures	47	72	68	27	117	12	20	30	16	178	72	29	52	53	93	39	29	28	28	28	133	54	16	1241
241	Annual crops associated with permanent crops	1		5						20			12				18			1					57
242	Complex cultivation patterns	95	61	75	22	19		21	23	57	18	32	36	25	19	33	24	27	51	33	36	18	22	10	757
243	Land principally occupied by agriculture, with significant areas of natural vegetation	83	57	45	35	18	56	26	30	64	54	42	42	28		32	26	19	30	63	51	29	27	36	893
244	Agro-forestry areas	1		38						19							34							1	93
311	Broad-leaved forest	90	118	66	61	94	81	66	55	140	23	77	57	88	17	148	86	60	29	113	95	82	51	94	1791
312	Coniferous forest	88	101	63	240	100	170	112	95	50	104	43	44	50	9	59	63	100	73	28	51	87	50	1	1781
313	Mixed forest	44	88	37	130	47	201	53	60	23	56	64	44	52	9	73	64	72	76	27	23	67	60	2	1372
321	Natural grasslands	91	49	62	20	24	15		15	11	38	30	43	23	26	24	12	57	39	42	23	20	23	12	699
322	Moors and heathland		22	63	50	19	21	50	10	20	36	16	20	13	24		54	21	26	14	20	16	18	10	543
323	Sclerophyllous vegetation	26	29	64						72			82				23			38	33			11	378
324	Transitional woodland-shrub	174	159	135	165	62	182	107	138	97	138	107	118	98	25	181	245	76	41	127	110	123	135	35	2778
331	Beaches, dunes, sands	19	9	11	5	8	3	14	9	31	6	18	11	10	91		15	1	20	14	2		15	1	313
332	Bare rocks	32	19	17	20	9	30	22	2	22	21	16	6	19	59		6	47		19	18	5		1	390
333	Sparsely vegetated areas	99	23	31	32	7	20	51	9	31	24	7	41	21	39	5	17	23		26	23	11	6	9	555
334	Burnt areas	1	1	9	4			1		5	10		2				7			11	1				52
335	Glaciers and perpetual snow	1	10	1	10			18		18					31			47		1					137
411	Inland marshes	30	11	4	14	14	20	1	18	24	22	17	13	19	20	20	5	18	29	19	22	19	21	31	411
412	Peat bogs		5		58	20	128	52	15	1	58			1	23	10	2	4	24	1	1	10	28		441
421	Salt marshes	20	11	19	4	10	16			21	19	1	18		10		20		20	2	1			1	193
422	Salines	6	10	10						5			5	1			10			2	1				50
423	Intertidal flats		5	2		9		20		16				20			6		15		1				94
511	Water courses	31	22	10	19	20	20	19	11	20	9	19	16	5	37	11	5	6	7	12	11	10	9	10	339
512	Water bodies	159	34	40	51	59	50	54	39	30	28	30	40	24	45	43	43	23	34	30	22	26	25	1	930
521	Coastal lagoons	10	9	5	3	2			1	20	5	1	5	1	15		4		10	4	1		2	1	99
522	Estuaries	1	8	6	3	3				1	20		3		5		3			1				12	66
523	Sea and ocean	11	1	9	10	3	15	15	1	5	3	1	5	1	2		1		1	4	1		1	34	124
	TOTAL CLC2018 classes	1467	1236	1182	952	853	1018	869	734	1204	967	753	855	768	584	999	1050	757	862	755	699	933	669	370	20536
	CLCCH2012-2018	367	221	371	277	274	361	128	254	271	414	200	208	96	83	248	271	177	186	181	134	181	214	42	5159
	Total général	1834	1457	1553	1229	1127	1379	997	988	1475	1381	953	1063	864	667	1247	1321	934	1048	936	833	1114	883	412	25695

The sample units were provided to the bulk interpretation team as one shapefile along with the polygons from the original dataset in which all the information on strata or thematic classes were removed to ensure the independence of the interpretation and too to ensure that actual thematic interpretation errors are separated from thematic errors due to potential geometric shifts.

In order to ensure that unequal sampling intensities between strata are accounted for in the construction of the error matrix (section **Erreur ! Source du renvoi introuvable.2.4.4**), weights are applied to each stratum.

2.4.3. Response Design

LUCAS points are re-interpreted based on available in situ data. LUCAS thematic information is not used directly.

Response design for most data set are based on the interpretation of thematic class at the point level taking into account product specifications (MMU, MMW, class definitions, etc) based on combination of available in situ data. For CLC2018, the HR andVHR dataset can be used alongside the imagery used in production and virtual globes data (Bing or Google Earth images) as illustrated in Figure 6. These images are used in case there is a discrepancy between the independent imagery acquisition date and the imagery used in production. It is also used to confirm geometry in case of a geometric shift between the thematic data and the reference data. In detail, hereafter the reference data used during the validation are listed, covering the EEA39 area at a wall-to-wall basis:

- VHR 2018: spatial resolution <2m; spectral resolution: Red/Green/Blue/Near infraRed
- HR 2018: spatial resolution of 10m; spectral resolution: Red/Green/Blue/Near infraRed
- VHR 2012: spatial resolution of 2,5m; spectral resolution: Red/Green/Blue/Near infraRed
- HR 2012: spatial resolution between 5m and 20m; spectral resolution: Red/Green/Blue/NIR

A double-blind approach guarantees complete independence from the map products and may underestimate their accuracy for complicated and difficult classes when sometimes several LC/LU classes are possible. This is resolved by the plausibility approach for which the interpreter checks the map value to assess whether it can be considered correct or not, within the frame of accepted product specifications. However, the plausibility analysis should be combined with the double-blind approach to ensure full traceability and transparency of the validation process.

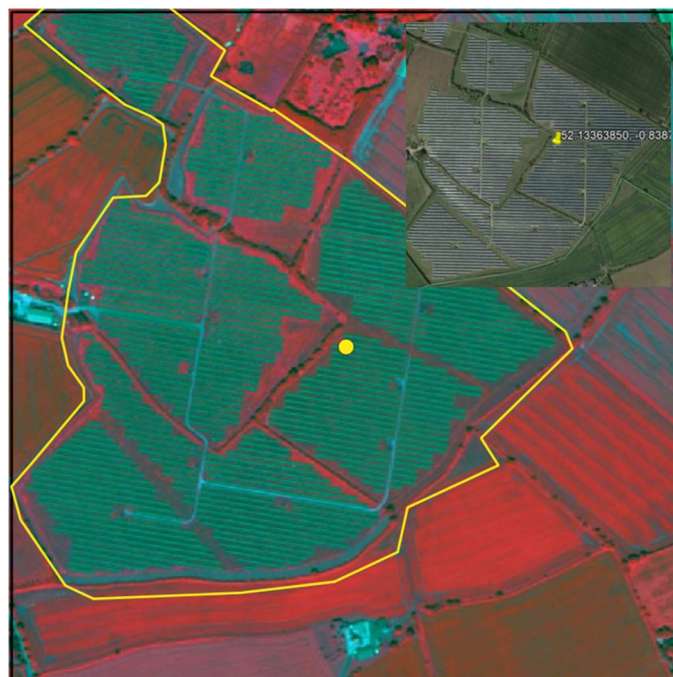


Figure 6: Example of a sample unit used for the validation of CLC2018

A blind approach was first applied. This consists in constructing the validation data set without any knowledge about the corresponding map layer information, i.e. the validation team did not have knowledge of the product's thematic classes for the selected sample units. However, the product polygon was provided as guidance to the validation team together with the sample unit to consider boundary effects and geometric differences between the validation and production data. Two parameters were considered to interpret the sample unit along with the polygon:

- the area to be checked
- the size of the polygon

The maximum area to be verified was 100ha (=1 sq.km), e.g. 4 times the size of MMU (25ha).

The expert only interpreted the polygon which contains the sample unit. For each sample unit, two thematic fields were filled:

- code2012: the land-cover class value for 2012 (one of the 44 CLC codes)
- code2018: the land-cover class value for 2018 (one of the 44 CLC codes)

Additional fields were filled if necessary:

- uncertain: drop-down menu (2012/2018/2012&2018/none) in case of uncertainty of the interpretation
- comment: free text if any comment is needed

The sample unit was first blindly interpreted by the validation expert, i.e. only the limit of the polygon is visible without knowing the CLC thematic attribute of the area; but interpretation must be made at sample unit level.

A second interpretation, carried out by a second interpreter as part of the plausibility analysis, was done only on the sample units in disagreement with product codes. The operator assigned a QC code modifying the validation code if necessary:

- Both producer and operator codes are plausible. Final validation code used is producer code
- Error from first validation interpretation. Final validation used is producer code
- Error from producer. Final validation code used is from first validation interpretation
- Producer and operator are both wrong. Final Validation code used is a new code from this second interpretation.

The Plausibility Analysis resulted in revised accuracy metrics.

2.4.4. Estimation and analyses procedures

Thematic accuracy should be presented in the form of an error matrix. Unequal sampling intensity resulting from the stratified systematic sampling approach should be accounted for by applying a weight factor (p) to each sample unit based on the ratio between the number of samples and the size of the stratum considered:

$$\hat{p}_{ij} = \left(\frac{1}{N}\right) \sum_{x \in (i,j)} \frac{1}{\pi_{uh}^*}$$

Where i and j are the columns and rows in the matrix, N is the total number of possible units (population) and π is the sampling intensity for a given stratum.

Overall accuracy and User and producer accuracy should be computed for all thematic classes and 95% confidence intervals should be calculated for each overall accuracy.

The standard error of the error rate can be calculated as follows: $\sigma_h = \sqrt{\frac{p_h(1-p_h)}{n_h}}$ where n_h is the sample size for stratum h and p_h is the expected error rate. The standard error is calculated for each stratum and an overall standard error is calculated based on the following formula:

$$\sigma = \sqrt{\sum w_h^2 \cdot \sigma_h^2}$$

In which is the proportion of the total area covered by each stratum. The 95% confidence interval is +/- 1.96.

2.5. Temporal Quality

Temporal quality is evaluated by providing an indication of the closeness of the acquired image data to the reference year, e.g. the percentage area covered outside the accepted reference period as defined in the tender/product specification i.e. 2012 +/- 1-2 year(s).

2.6. Usability

Usability relates to the appropriateness of the metadata description and accompanying documentation to describe the processes and workflows involved in the production of the data. Although it is difficult to describe usability in quantitative terms, it provides a clear evaluation based on objective criteria of any limitation in the intended use of the data.

2.7. INSPIRE compliant metadata

Presence of INSPIRE compliant metadata should be verified.

3. Validation check list

Table 2: Validation check list

PRODUCT: CLC2018						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER:				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/07/2020				REVIEW DATE:		
CONDUCTED BY: IGN FI				REVIEWED BY: Adrien Moiret	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
1	COMPLETENESS					
1.1	Commission	Rate of excess items	n/a	Please refer to the chapter 4		
1.2	Omission	Rate of missing items	n/a	Please refer to the chapter 4		
2	LOGICAL CONSISTENCY					
2.1	Format consistency	File format	X	File format is GeoPackage (.GPKG)		
2.2		File name	X	File name is 'U2018_CLC2018_V2020_20u1' that means U2018 : last modification / update of the file is from the 2018 mapping campaign CLC1218 : file contains CLC status data for reference year 2018 V2020_20u1 : file was released in 2020 as a first update of final version 20	Accepted	

PRODUCT: CLC2018						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER:				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/07/2020				REVIEW DATE:		
CONDUCTED BY: IGN FI				REVIEWED BY: Adrien Moiret	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
2.3		Attributes names	X	Attributes are - 'Name (Type)' : OBJECTID (Integer64) Code_18 (String 3) Remark (String 20) Area_Ha (Double Real) ID (String 18)	Accepted	
2.4		Attributes types	X	Attributes are - 'Name (Type)' : OBJECTID (Integer64) Code_18 (String 3) Remark (String 20) Area_Ha (Double Real) ID (String 18)	Accepted	
2.5	Conceptual consistency	Feature type	X	MultiPolygon		
2.6		MMU	X	In layer : U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1 3778 features < 25 ha (out of 2375406 features) In layer : U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1_FR_GUF 7 features < 25 ha (out of 995 features)	Conditionally accepted	

PRODUCT: CLC2018						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER:				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/07/2020				REVIEW DATE:		
CONDUCTED BY: IGN FI				REVIEWED BY: Adrien Moiret	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
2.7		Coordinate reference system	X	Layer U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1 EPSG 3035 : ETRS89-extended / LAEA Europe Layer U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1_FR_GLP EPSG 4559 : RRAF 1991 / UTM zone 20N Layer U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1_FR_GUF EPSG 2972 : RGF95 / UTM zone 22N Layer U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1_FR_MTQ EPSG 4559 : RRAF 1991 / UTM zone 20N Layer U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1_FR_MYT EPSG 4471 : RGM04 / UTM zone 38S Layer U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1_FR_REU EPSG 2975 : RGR92 / UTM zone 40S	Accepted	
2.8		Unique identifier	X	Identifiers are unique Layer U2018_CLC2018_V2020_20u1_U2018_CLC2018_V2020_20u1 ID = "EU_" + OBJECTID Other layers : ID = "FR-" + code + "-Id" . GLP : code = 971 . GUF : code = 973 . MTQ : code = 972 . MYT : code = 976 . REU : code = 974	Accepted	
2.9		Nomenclature	Conform with the criteria		Accepted	
2.10	Domaine consistency	Value domain non-conformance	Not assessed			

PRODUCT: CLC2018						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER:				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/07/2020				REVIEW DATE:		
CONDUCTED BY: IGN FI				REVIEWED BY: Adrien Moiret	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
2.11	Topological consistency	Overlaps	Not assessed			
2.12		Gaps	Not assessed			
2.13		Multipart features	Conform with the criteria	No multipart polygons	Accepted	
2.14		Neighbouring features	Not assessed			
2.15		Self-intersections	Not assessed			
2.16		Null geometry	Conform with the criteria		Accepted	
2.17		Unclosed rings	Conform with the criteria		Accepted	
2.18		Duplicate vertex	Conform with the criteria		Accepted	

PRODUCT: CLC2018						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER:				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/07/2020				REVIEW DATE:		
CONDUCTED BY: IGN FI				REVIEWED BY: Adrien Moiret	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
2.19		Pseudo nodes	This was not assessed			
2.20		Non matching nodes	Conform with the criteria		Accepted	
3	POSITIONAL ACCURACY					
3.1	Absolute or external accuracy	RMSEP	n/a			
3.2	Relative or internal accuracy	RMSEP	n/a			
4	THEMATIC ACCURACY					
4.1	Classification correctness	Overall accuracy	92.67%	Including the overall area		
4.2		Min. producer's accuracy	52.05%	Class 133 – Construction site		
4.3		Min. user's accuracy	49.97%	Class 241 – Annual crops associated with permanent crops		

PRODUCT: CLC2018						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER:				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/07/2020				REVIEW DATE:		
CONDUCTED BY: IGN FI				REVIEWED BY: Adrien Moiret	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
4.4		Kappa				
5	TEMPORAL QUALITY					
5.1	Temporal quality	Closeness of the acquired image data to the reference		Acquisition windows from 2017 to 2019 'More detailed is not assessed'		
6	USABILITY					
6.1	Usability	Usability description		N/A		
7	METADATA					
7.1	INSPIRE compliant metadata	Presence		N/A		
7.2		File format		N/A		
7.3		File name		N/A		

PRODUCT: CLC2018						
VALIDATION LEVEL: EEA39						
SERVICE PROVIDER:				SERVICE USER:	ISSUE/REVISION:	
VALIDATION DATE: 28/07/2020				REVIEW DATE:		
CONDUCTED BY: IGN FI				REVIEWED BY: Adrien Moiret	APPROVED BY:	
No.	DATA QUALITY SUB-	DATA QUALITY MEASURE	DATA QUALITY RESULT	COMMENTS BY AUDIT TEAM	DRAFT AUDIT CONCLUSION	FINAL AUDIT CONCLUSION
7.4		INSPIRE compliance		N/A		

4. Additional tables

4.1. Coordinate reference system description

Table 3: Coordinate reference system description

NAME OF COORDINATE REFERENCE SYSTEM	ETRS_1989_LAEA
WKID	EPSG:3035
PROJECTION	Lambert_Azimuthal_Equal_Area
FALSE EASTING	4321000.0
FALSE NORTHING	3210000.0
CENTRAL MERIDIAN	10.0
LATITUDE OF ORIGIN	52,0
LINEAR UNIT	(1,0)
GEOGRAPHIC COORDINATE SYSTEM	GCS_ETRS_1989
ANGULAR UNIT	Degree (0.0174532925199433)
PRIME MERIDIAN	Greenwich (0,0)
DATUM	D_ETRS_1989
SPHEROID	GRS_1980
SEMIMAJOR AXIS	6378137.0
SEMIMINOR AXIS	6356752.314140356
INVERSE FLATTENING	298.257222101

4.2. Nomenclature

Table 4: Nomenclature table

CODE	CLASS NAME	CODE	CLASS NAME
111	Continuous urban fabric	311	Broad-leaved forests
112	Discontinuous urban fabric	312	Coniferous forests
121	Industrial and commercial units	313	Mixed forests
122	Road and rail network and associated land	321	Natural grassland
123	Port areas	322	Moors and heathland
124	Airports	323	Sclerophyllous vegetation
131	Mineral extraction sites	324	Transitional woodland/shrub
132	Dump sites	331	Beaches, dunes, and sand plains
133	Construction sites	332	Bare rocks
141	Green urban areas	333	Sparsely vegetated areas
142	Sport and leisure facilities	334	Burnt area
211	non-irrigated arable land	335	Glaciers and perpetual snow
212	Permanently irrigated land	411	Inland marshes
213	Rice field	412	Peatbogs
221	Vineyard	421	Salt marshes
222	Fruit trees and berry plantation	422	Salines
223	Olive grove	423	Intertidal flats
231	Pastures	511	Water courses
241	Annual crops associated with permanent crops	512	Water bodies
242	Complex cultivation patterns	521	Coastal lagoons
243	Land principally occupied by agriculture, with significant areas of natural vegetation	522	Estuaries
244	Agro-forestry areas	523	Sea and Ocean

4.3. Thematic accuracy

4.3.1. CLC 2018 status layer

This section provides the evaluation results of the CORINE Land Cover layers, specifically the Land Cover 2018 status and the change layer for 2012 and 2018. This analysis was performed over 100% of the coverage. The thematic accuracy assessment was conducted in a two-stage process: an initial blind interpretation in which the validation team did not have knowledge of the product's thematic classes and a plausibility analysis performed on all sample units that show disagreement between the blind interpretation and the production. This chapter provides both results. Thematic evaluation is done 25,695 samples.

4.3.1.1. Results

The overall accuracies obtained in the frame of the Validation exercise are **92.43%** for the blind interpretation and **92.67%** for the plausibility analysis, so the results are above the 85% threshold expected. We note a clear improvement of the overall accuracy compared to 2012. Indeed, for the plausibility analysis, it increases from 89.66% in 2012 to 92.67% in 2018.

The confusion matrices at the European level including the blind and plausibility analysis are provided in Annex 1.

Table 5 summarizes overall accuracies obtained by zone (country or group of countries) for the blind and plausibility interpretations including the 95% confidence intervals. For the detailed analysis, for both blind and plausibility analysis, accuracy assessments and metrics (confusion matrices, producer, user and overall accuracies per zones) are provided in Annex 1, Annex 3 and Annex 6.

Table 5 Overall accuracy for blind and plausibility accuracy and confidence interval (95%) for CLC2018 per country or group of countries.

		Overall accuracy CLC2012				Overall accuracy CLC2018			
		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
		Overall accuracy	Confidence interval of 95%	Overall accuracy	Confidence interval of 95%	Overall accuracy	Confidence interval of 95%	Overall accuracy	Confidence interval of 95%
Z1	Turkey	78,61%	2,80%	85,17%	2,44%	87,38%	0,10%	87,38%	0,10%
Z2	France	86,35%	3,33%	89,33%	3,11%	94,20%	0,10%	94,52%	0,10%
Z3	Spain	85,10%	2,68%	89,42%	2,27%	91,16%	0,06%	91,36%	0,06%
Z4	Sweden	87,91%	3,92%	93,08%	3,10%	95,06%	0,13%	95,52%	0,13%
Z5	Germany	82,83%	4,16%	89,05%	3,43%	93,48%	0,06%	93,14%	0,07%
Z6	Finland	74,32%	6,11%	81,66%	5,73%	93,85%	0,09%	93,75%	0,09%
Z7	Norway	93,22%	2,66%	96,61%	1,99%	97,51%	0,04%	97,55%	0,04%
Z8	Poland	92,06%	2,37%	94,15%	2,21%	96,85%	0,03%	97,12%	0,03%
Z9	Italy - Sardinia	76,01%	4,15%	85,01%	3,53%	89,15%	0,06%	88,71%	0,06%
Z10	United Kingdom - Ireland	87,07%	3,75%	92,67%	2,78%	91,84%	0,09%	92,72%	0,09%
Z11	Roumania	78,90%	4,11%	91,21%	2,61%	88,74%	0,06%	91,63%	0,04%
Z12	Greece	80,21%	3,63%	87,28%	3,01%	87,57%	0,02%	88,86%	0,02%
Z13	Bulgaria	86,90%	3,54%	94,53%	2,03%	94,94%	0,02%	95,04%	0,02%
Z14	Island	85,13%	7,41%	93,68%	4,91%	98,93%	0,01%	98,93%	0,01%
Z15	Hungaria	86,08%	3,97%	91,28%	3,28%	96,66%	0,02%	96,71%	0,02%
Z16	Portugal	75,91%	3,68%	84,80%	3,17%	85,98%	0,02%	86,35%	0,02%
Z17	Switzerland - Austria	76,07%	4,74%	86,77%	3,72%	90,36%	0,03%	90,83%	0,03%
Z18	Belgium - Danmark - Luxembourg - Netherlands	84,75%	4,21%	88,18%	4,05%	89,96%	0,04%	90,96%	0,04%
Z19	Albania - Serbia - Macedonia - Montenegro	82,08%	3,87%	93,46%	2,18%	91,00%	0,03%	90,60%	0,03%
Z20	Bosnia Herzegovina - Croatia - Slovenia	77,38%	5,20%	89,27%	3,81%	86,79%	0,04%	88,14%	0,04%
Z21	Czech Republic - Slovakia	85,68%	3,77%	93,31%	2,40%	98,40%	0,00%	98,69%	0,00%
Z22	Estonia - Latvia - Lithuania	88,63%	3,68%	91,69%	3,22%	98,36%	0,01%	97,22%	0,01%
Z23	French overseas departments	96,41%	1,03%	97,61%	0,90%	92,23%	0,03%	92,35%	0,03%
	GLOBAL ACCURACY	83,60%	0,91%	89,66%	0,77%	92,43%	0,28%	92,67%	0,28%

	Accuracy better than 85%
	Accuracy between 80% and 85%
	Accuracy less than 80%

The overall accuracies at the biogeographical regions level illustrated in Table 6 show good to very good results except for Black Sea regions with slightly lower accuracies (like in 2012).

Annex 2 provides the producer and user accuracies for each land cover classes and the Annex 4 accuracies for each biogeographical region. Confusion matrices per biogeographical regions are presented in Annex 5.

Table 6: Overall accuracy for blind and plausibility analysis per bio-geographical region

	Overall accuracy CLC 2012		Overall accuracy CLC 2018	
	Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis
Alpine	84,29%	92,49%	93,80%	93,89%
Anatolian	77,15%	86,44%	86,49%	86,55%
Arctic	85,29%	93,75%	98,94%	98,94%
Atlantic	87,75%	91,30%	93,15%	94,00%
BlackSea	81,70%	84,45%	84,57%	84,03%
Boreal	84,35%	89,12%	95,34%	95,30%
Continental	83,91%	90,74%	93,21%	93,65%
Macaronesia	80,32%	90,33%	93,97%	96,84%
Mediterranean	80,69%	86,54%	89,46%	89,63%
Pannonian	86,87%	93,48%	96,71%	96,75%
Steppic	87,03%	96,40%	96,52%	99,13%
Oversea departments	96,41%	97,61%	92,23%	92,35%

	Accuracy better than 85%
	Accuracy between 80% and 85%
	Accuracy less than 80%

Table 7 summarizes Producer and User accuracies obtained by zone (country or group of countries), for the blind and plausibility interpretations.

Figure 7 represents the spatial distribution of user and producer accuracies below the recommendation of 85% in Europe

2018	Code CLC	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244
BLIND ANALYSIS	User accuracy	82,44%	97,20%	92,28%	92,54%	100,00%	98,53%	81,33%	93,29%	77,24%	94,81%	96,54%	94,98%	96,50%	97,67%	83,52%	85,19%	92,71%	89,45%	49,97%	86,57%	88,94%	96,63%
	Producer accuracy	95,01%	93,17%	92,94%	71,80%	96,09%	95,34%	97,05%	81,54%	69,81%	77,93%	92,22%	97,76%	91,96%	99,74%	98,69%	88,55%	95,87%	92,52%	60,77%	83,24%	83,20%	78,29%
PLAUSIBILITY ANALYSIS	User accuracy	81,07%	97,68%	93,11%	93,38%	100,00%	98,53%	81,64%	94,18%	78,97%	94,91%	96,78%	94,83%	96,56%	97,77%	83,69%	85,44%	92,31%	89,95%	49,97%	87,68%	88,66%	96,52%
	Producer accuracy	95,36%	93,89%	94,06%	72,03%	98,05%	95,34%	97,46%	80,06%	52,05%	77,98%	90,98%	97,91%	91,19%	99,74%	97,95%	89,31%	94,21%	91,55%	61,82%	84,22%	82,98%	80,32%
2018	Code CLC	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	423	511	512	521	522	523
BLIND ANALYSIS	User accuracy	94,29%	95,99%	93,15%	81,24%	92,75%	85,98%	91,71%	78,46%	92,70%	82,52%	48,73%	99,91%	96,09%	97,80%	86,40%	88,82%	86,14%	91,97%	98,96%	97,97%	92,86%	93,87%
	Producer accuracy	96,11%	96,43%	88,17%	81,08%	92,93%	85,98%	82,59%	69,47%	97,14%	94,83%	100,00%	99,80%	73,69%	99,45%	95,11%	95,45%	93,40%	96,67%	99,32%	98,52%	77,47%	90,59%
PLAUSIBILITY ANALYSIS	User accuracy	94,11%	96,15%	93,97%	82,20%	93,39%	86,89%	92,00%	78,43%	92,98%	82,61%	53,17%	99,91%	96,05%	99,56%	88,30%	90,66%	86,32%	92,26%	99,08%	98,10%	92,86%	93,96%
	Producer accuracy	97,02%	96,85%	89,14%	82,15%	92,89%	88,37%	81,97%	67,93%	96,90%	95,76%	92,29%	99,80%	74,10%	99,09%	92,70%	95,54%	96,45%	97,86%	99,31%	98,52%	77,66%	90,68%
	Accuracy better than 85%																						
	Accuracy between 80% and 85%																						
	Accuracy less than 80%																						



4.3.1.2. Main findings and discussion

The CLC2018 status product exceeds the accuracy requirements for the blind and plausibility analysis. In addition, these results are relatively homogenous across the European zones from an overall accuracy of 86.35% for Portugal to 98.93% for Island.

Concerning the bio-geographical regions, the accuracy requirement is reached for overall regions, except of Black Sea whose accuracy slightly below the threshold, i.e 85%

In general, there is a small difference between blind analysis and plausibility. This highlights the robustness of the base and the high quality of the photointerpretation despite the complexity of the nomenclature. For information 2018 dataset is the fifth update of CORINE Land Cover.

The CLC2018 product shows both low producer and user accuracies for blind and plausibility analysis for the following land cover classes: 122,133,141,241,244,331,334,411 and 522. Reasons for main sources of errors can be summarised as follows:

Main sources of both omission and commission errors (producer and user accuracy):

- **Class 111** "Continuous urban fabric": some difficulties in the visual interpretation of the urban density. Mainly the confusion concerns the class 112 "discontinuous urban fabric" and the class 122 "road and rail network and associated land".
- **Class 122** "road and rail network and associated land: omission error, there is an underestimation of roads and confusion with Urban fabric (111 and 112).
- **Class 131** "Mineral extraction sites": commission error, there is a confusion with class 132 "Dump sites", it seems logical. On the other hand, there is a confusion too, with class 313 "mixed forest", it is not consistent, maybe problem with edge of shapefiles.
- **Class 132** "Dump sites": omission error, some confusion with the class 131, it is an internal confusion. Need to have ancillary data to identify 131 from 132.
- **Class 133** "Construction sites": commission error with several classes (surely due to the polygon boundary), but especially with the class 131 "Mineral extraction sites". And omission error, some confusion with artificial classes 112, 121 due to temporal issues (progress of construction development).
- **Class 141** "Green urban areas": omission error, we have a confusion with 211 "non-irrigated arable land". It's not logical, maybe a problem of polygon edge.
- **Class 221** "Vineyards": commission error, confusion with the classes 24 "heterogeneous agricultural areas" (242 and 243) when the density of plots planted with permanent crops is not enough sufficient and not representative (Density of permanent crops must occupy more than 75 %).
- **Class 241** "annual crops associated with permanent crops": Class which remains very difficult to identify (commission and omission errors), it often confused with the other "mosaic" classes like the 242 and the 243 as well as the permanent cultures 222.
- **Class 244** "Agro-forestry areas": corresponds to agro-sylvo-pastoral systems made up of an association of forest trees (like 241), pastures or crops. Here, errors of omission with classes 211, 324,311 and 321.
- **Class 321** "Natural grassland": Commission and omission errors. Confusion with 323 "sclerophyllous vegetation", 333 "sparsely vegetation" and 324 "transitional woodland/shrub" in Mediterranean region.
- **Class 324** "transitional woodland": which characterizes both a dynamic aspect of the vegetation and a physiognomic state of the vegetation everywhere in Europe (from Boreal to Mediterranean region). Possible confusion with forest according to the forest growing 311 "broad-leaved forest", 312 "coniferous forest", 313 "mixed forest" or 321 "Natural grassland", 323 "sclerophyllous vegetation".
- **Class 411** "inland marshes": omission error, with different type land use (errors of polygon edge) or with humid grass and water.
- **Class 522** "estuaries": omission error; it is difficult to delineate estuaries and need to have bathymetry data and other marine maps. Possible confusion with 511 (or 512 sometimes)

All these confusions are represented by a map, see figure 6. The values on the map represents, Plausibility values.

- The commission errors (user accuracy) are mainly distributed in southern Europe (West and East) and concerns mainly the classes 133, 241, 331, 334
- The error of omission (producer accuracy), are distributed more uniformly over all of Europe. The land cover classes concern essentially 122, 133, 141, 241, 331, 411, 522

The stratification and sampling design proved efficient ensuring a good representation of all Land-Cover classes whilst providing precise overall thematic accuracy results at EEA39, zones and biogeographical region level.

In comparison with 2012, we see an improvement in user and producer accuracies per land cover class. Particularly for classes, 111, 221, 242 and 243. But in general, we find the same trend. Below is the table for 2012.

Table 8: Producer and user accuracy of CLC2012 given by land cover classes– Blind and plausibility analysis

2012	Code CLC	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244
BLIND ANALYSIS	User accuracy	77,34%	90,31%	89,53%	89,20%	95,43%	96,02%	84,04%	77,75%	76,25%	85,71%	88,14%	89,89%	94,25%	95,22%	67,44%	67,85%	85,41%	76,88%	31,21%	72,04%	78,49%	92,77%
	Producer accuracy	83,14%	89,14%	74,35%	66,32%	87,81%	96,46%	80,64%	80,42%	63,34%	66,32%	71,90%	93,74%	96,61%	94,07%	79,19%	76,81%	88,61%	80,16%	20,68%	62,46%	65,22%	75,11%
PLAUSIBILITY ANALYSIS	User accuracy	79,13%	94,84%	93,75%	91,87%	95,43%	98,06%	86,76%	85,97%	79,31%	88,67%	92,89%	93,20%	97,92%	98,44%	76,56%	73,98%	89,80%	84,56%	41,17%	79,79%	84,35%	93,03%
	Producer accuracy	83,71%	92,93%	83,83%	85,76%	88,36%	96,63%	97,64%	92,89%	62,44%	68,64%	76,47%	95,41%	98,09%	97,29%	93,77%	89,60%	91,76%	87,02%	27,93%	73,62%	74,69%	78,89%
2012	Code CLC	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	423	511	512	521	522	523
BLIND ANALYSIS	User accuracy	87,32%	84,66%	85,62%	68,98%	85,13%	78,01%	77,94%	86,05%	77,98%	74,41%	80,04%	98,55%	85,81%	85,37%	57,03%	84,86%	72,80%	90,24%	98,41%	96,82%	86,01%	98,94%
	Producer accuracy	94,52%	94,49%	75,28%	68,38%	83,74%	80,15%	60,70%	76,82%	92,94%	84,12%	62,64%	81,66%	79,11%	92,10%	86,53%	93,90%	88,24%	91,71%	98,54%	96,89%	63,95%	93,30%
PLAUSIBILITY ANALYSIS	User accuracy	91,84%	90,21%	94,31%	80,56%	91,16%	87,47%	88,35%	92,43%	86,22%	84,20%	78,96%	99,18%	91,35%	96,48%	58,17%	93,23%	85,49%	92,97%	98,87%	97,78%	93,80%	99,29%
	Producer accuracy	97,56%	97,64%	81,68%	81,27%	93,93%	84,69%	73,44%	86,70%	96,25%	95,45%	78,45%	86,83%	80,39%	95,77%	92,84%	95,09%	95,83%	98,55%	98,66%	98,22%	67,58%	99,82%

4.3.2. CLCCH2012-2018 change layer

4.3.2.1. Main observation at the European and bio-geographical region

The first observation is an **overestimation of changes** during the interpretation phase. This is because the satellite image used during the validation step allowed identifying a land cover change but in some case it's not clear which kind of change has occurred. Changes are identified and considered when the difference in the spectral radiometry of satellite images acquired between 2012 and 2018 corresponds to a real land cover or land use change.

For the blind analysis, the overall accuracy of CLCCH2012-2018 at the third level of nomenclature is **87.12%**. The overall accuracy of CLCCH2012-2018 at the level 2 is **89.56%**.

For the plausibility analysis, the overall accuracy of CLCCH2012-2018 at the third level is **87.83%**. The overall accuracy of CLCCH2012-2018 at the second level is **90.09%**.

The level 3 of the nomenclature permits a high number of changes combinations (44 x 43 classes = 1,936 possible changes), which makes the analysis very complex. Since the goal of the validation is to provide an overall picture about the accuracy of CLC-Change, the analyses were focused essentially at CLC level 2. Main findings are only provided where significant differences are found between the validation.

The matrix of change at the European level (CLC level 2) is provided in Annex 6 and at the biogeographical region in Annex 8.

Table 9 shows the overall accuracy of CLCCH2012-2018 for each bio-geographical region at the second and third level both for the blind and the plausibility analysis. The plausibility analysis shows a very good overall accuracy both at the second and at the third level; except for regions: Anatolian, Black Sea and Mediterranean (only for third level). At the second level, we see an upgrade of statistic Mediterranean region.

Comparing with overall accuracy of CLCCH2006-2012, we observe less good results for the three regions mentioned previously at third level; and two regions at second level: Anatolian and Black Sea.

Confusion matrix of CLCCH2012-2018 per Bio-geographical region is presented in Annex 8 of the document.

Table 9: assessment of CLCCH2012-2018 for blind and plausibility analysis per bio-geographical region at the second and third CLC level

	Overall accuracy CLCCH2006-2012 level 3		Overall accuracy CLCCH2012-2018 level 3		Overall accuracy CLCCH2006-2012 level 2		Overall accuracy CLCCH2012-2018 level 2	
	Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis
Alpine	83,47%	91,94%	89,86%	90,61%	88,55%	94,62%	92,44%	93,11%
Anatolian	75,62%	85,93%	80,43%	81,34%	79,28%	88,01%	83,72%	84,65%
Arctic	83,12%	93,38%	98,14%	98,19%	93,16%	99,47%	99,45%	99,50%
Atlantic	85,89%	89,81%	89,94%	90,82%	87,97%	91,36%	90,86%	91,53%
BlackSea	81,54%	84,31%	77,99%	78,53%	86,32%	87,89%	81,13%	81,19%
Boreal	80,21%	86,69%	91,14%	91,56%	84,78%	90,06%	91,98%	92,26%
Continental	82,98%	90,25%	87,59%	88,25%	85,31%	91,76%	89,60%	90,12%
Macaronesia	79,95%	89,95%	72,64%	89,97%	83,49%	93,23%	78,14%	90,73%
Mediterranean	79,06%	85,54%	81,37%	82,00%	85,80%	89,78%	86,66%	87,11%
Pannonian	85,29%	92,25%	91,37%	91,80%	85,83%	92,86%	92,53%	92,96%
Steppic	86,99%	96,38%	94,23%	97,02%	86,99%	96,38%	94,23%	97,02%
Oversea departments	96,38%	97,58%	86,41%	86,52%	97,58%	98,07%	86,62%	86,79%

	Accuracy better than 85%
	Accuracy between 80% and 85%
	Accuracy less than 80%

4.3.2.2. Assessment of CLCCH2012-2018 overall accuracy per country or group of countries at the second and third level

Table 10 and Table 11 shows the overall accuracy of CLCCH2012-2018 for each zone (country or group of countries) at the second and third level for the blind and plausibility analysis. The European countries have been gathered into 23 areas. Overall, the land cover changes are very good at the second level and good at the third level.

Table 10: assessment of CLCCH2012-2018 for blind and plausibility analysis and confidence interval (95%) per country or group of countries at third CLC level

		Overall accuracy CLCCH2006-2012 level 3				Overall accuracy CLCCH2012-2018 level 3			
		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
		Overall accuracy	Confidence interval of 95%	Overall accuracy	Confidence interval of 95%	Overall accuracy	Confidence interval of 95%	Overall accuracy	Confidence interval of 95%
Z1	Turkey	77,30%	2,84%	84,50%	2,47%	80,79%	0,13%	81,64%	0,13%
Z2	France	84,80%	3,36%	88,11%	3,15%	92,44%	0,10%	92,89%	0,10%
Z3	Spain	84,23%	2,69%	88,71%	2,28%	83,52%	0,09%	84,30%	0,09%
Z4	Sweden	81,42%	4,77%	89,51%	3,59%	92,58%	0,16%	92,98%	0,16%
Z5	Germany	80,59%	4,37%	88,12%	3,62%	84,41%	0,11%	85,50%	0,11%
Z6	Finland	72,92%	6,15%	80,67%	5,76%	91,08%	0,09%	91,94%	0,09%
Z7	Norway	93,07%	2,67%	96,02%	1,99%	96,53%	0,05%	96,96%	0,04%
Z8	Poland	91,52%	2,37%	93,80%	2,21%	91,82%	0,16%	91,87%	0,16%
Z9	Italy - Sardinia	75,23%	4,16%	84,74%	3,53%	83,92%	0,07%	84,17%	0,07%
Z10	United Kingdom - Ireland	85,60%	3,77%	91,67%	2,79%	87,17%	0,11%	88,09%	0,11%
Z11	Roumania	77,65%	4,30%	90,94%	2,61%	82,01%	0,09%	84,47%	0,07%
Z12	Greece	78,64%	3,67%	86,28%	3,04%	77,16%	0,03%	78,24%	0,03%
Z13	Bulgaria	86,52%	3,55%	94,36%	2,03%	93,10%	0,02%	94,04%	0,02%
Z14	Island	82,94%	7,95%	93,31%	4,91%	98,12%	0,01%	98,18%	0,01%
Z15	Hungaria	84,01%	3,98%	89,67%	3,31%	90,68%	0,04%	91,24%	0,04%
Z16	Portugal	68,26%	3,88%	80,87%	3,43%	74,06%	0,02%	75,39%	0,02%
Z17	Switzerland - Austria	75,27%	4,75%	86,39%	3,72%	83,01%	0,04%	83,27%	0,04%
Z18	Belgium - Danmark - Luxembourg - Netherlands	84,45%	4,21%	87,77%	4,05%	86,06%	0,05%	86,64%	0,05%
Z19	Albania - Serbia - Macedonia - Montenegro	80,87%	4,02%	92,61%	2,21%	84,01%	0,03%	84,19%	0,03%
Z20	Bosnia Herzegovina - Croatia - Slovenia	76,89%	5,21%	88,89%	3,82%	79,39%	0,05%	80,87%	0,05%
Z21	Czech Republic - Slovakia	84,10%	3,80%	91,82%	2,45%	93,89%	0,02%	94,77%	0,02%
Z22	Estonia - Latvia - Lithuania	86,95%	3,81%	90,17%	3,40%	87,35%	0,06%	87,41%	0,06%
Z23	French overseas departments	96,38%	1,03%	97,58%	0,90%	86,41%	0,05%	86,52%	0,05%
	GLOBAL ACCURACY	81,80%	0,95%	88,60%	0,79%	87,12%	0,39%	87,83%	0,38%

Table 11: assessment of CLCCH2012-2018 for blind and plausibility analysis per country or group of countries at the second CLC level

		Overall accuracy CLCCH2006-2012 level 2		Overall accuracy CLCCH2012-2018 level 2	
		Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis
		BLIND ANALYSIS	PLAUSIBILITY ANALYSIS	Overall accuracy	Overall accuracy
Z1	Turkey	83,16%	88,57%	84,82%	85,37%
Z2	France	88,37%	90,88%	94,18%	94,51%
Z3	Spain	89,58%	92,01%	88,81%	89,03%
Z4	Sweden	82,17%	90,02%	93,20%	93,48%
Z5	Germany	83,30%	89,96%	87,46%	88,00%
Z6	Finland	85,92%	89,87%	92,01%	92,85%
Z7	Norway	95,00%	97,59%	96,72%	97,14%
Z8	Poland	92,35%	94,81%	92,07%	92,11%
Z9	Italy - Sardinia	80,20%	87,37%	87,14%	87,28%
Z10	United Kingdom - Ireland	87,09%	92,72%	87,72%	88,62%
Z11	Roumania	79,74%	92,05%	84,42%	86,91%
Z12	Greece	85,08%	90,63%	83,07%	84,07%
Z13	Bulgaria	89,02%	95,62%	94,09%	95,13%
Z14	Island	93,09%	99,46%	99,44%	99,49%
Z15	Hungaria	84,73%	90,54%	91,50%	92,07%
Z16	Portugal	78,92%	85,63%	80,75%	81,64%
Z17	Switzerland - Austria	80,48%	88,72%	88,47%	88,72%
Z18	Belgium - Danmark - Luxembourg - Netherlands	86,81%	89,20%	86,55%	86,89%
Z19	Albania - Serbia - Macedonia - Montenegro	84,18%	94,18%	87,14%	87,25%
Z20	Bosnia Herzegovina - Croatia - Slovenia	84,83%	90,79%	85,35%	86,83%
Z21	Czech Republic - Slovakia	84,23%	92,96%	94,61%	95,48%
Z22	Estonia - Latvia - Lithuania	87,82%	90,72%	88,95%	88,37%
Z23	French overseas departments	97,58%	98,07%	86,62%	86,79%
	GLOBAL ACCURACY	85,90%	91,23%	89,56%	90,09%

At third level, we observe for Turkey, Greece, Portugal, Bosnia-Herzegovina-Croatia-Slovenia a threshold under the recommendations; and slightly below for Z3, Z9, Z11, Z17 and Z19.

Comparing with 2012, if the overall precision on a European scale is slightly improved, this is not the case for the land cover changes, indeed there is a decrease in the overall precision compared to CLCCH2006-2012.

Confusion matrices of CLCCH2012-2018 per zone/country, group of countries are presented in Annex 7.

4.3.2.1. Main findings and discussion

Table 12 below summarizes differences encountered between the validation and production interpretation at level 2 of the nomenclature for the 2012-2018 changes.

Table 12: Main flow observed at the European level (commission and omission)

CATEGORIES														
Urban	MAP-CLCC_Lvl2	11-12	12-13	12-21	13-11	13-12	13-14	13-21	13-23	13-24	13-31	13-32	13-51	14-12
	Weight of U.A	4,16477522	0,80885465	3,27435248	4,84570184	7,51635858	1,23628614	2,61360787	4,68824335	2,01734126	1,69986974	2,53861376	2,77083706	1,20586475
	User_accuracy	1,96%	10,11%	0,18%	24,41%	46,00%	14,91%	12,99%	8,84%	1,47%	0,59%	38,40%	9,74%	6,78%
	Prod_accuracy	94,14%	3,77%	100,00%	95,55%	83,72%	91,82%	78,49%	67,53%	16,77%	10,56%	46,79%	72,50%	100,00%
	Weight of U.P	0,08687836	2,16833414	0,00584491	1,23818781	4,12981981	0,20071424	0,43244224	0,61357993	0,17630353	0,09473993	2,08338952	0,37222103	0,08178622
Agricultural	MAP-CLCC_Lvl2	21-11	21-12	21-13	21-22	21-23	21-24	21-32	21-41	21-51	22-12	22-21	22-23	22-24
	Weight of U.A	5,42924878	3,19512446	3,21229213	9,10630839	96,3714615	148,339849	9,23983063	0,17659518	0,87019609	1,3802169	14,8501628	6,89794465	24,9277193
	User_accuracy	13,14%	90,91%	67,82%	41,68%	2,98%	2,39%	1,67%	37,84%	72,38%	10,52%	13,86%	1,69%	0,27%
	Prod_accuracy	61,70%	87,38%	90,34%	91,49%	81,24%	100,00%	100,00%	49,71%	82,91%	50,00%	66,04%	50,00%	29,38%
	Weight of U.P	1,15630599	3,32423308	2,41147006	4,14852027	3,53431909	3,55183383	0,15447522	0,13444286	0,7597107	0,29046586	3,11689251	0,23336706	0,22849143
		23-11	23-12	23-13	23-21	23-22	23-24	23-32	23-42	23-51	24-11	24-12	24-13	24-21
		15,0649606	8,56327472	1,34708111	96,7391592	1,0343321	26,6085153	34,5366388	0,06683206	0,20747804	10,7433773	0,68713587	2,54649281	104,507855
		0,96%	16,96%	53,91%	13,48%	33,84%	1,01%	1,82%	100,00%	63,91%	1,35%	22,49%	39,92%	0,69%
		19,76%	88,24%	10,33%	66,30%	67,55%	3,53%	28,69%	100,00%	36,15%	10,31%	39,43%	86,15%	54,01%
		0,7348311	1,64585391	7,02850958	19,6672782	0,51822586	7,61764377	2,19658779	0,06683206	0,36684017	1,40878135	0,39182739	1,1800743	1,33826762
		24-22	24-32	24-51										
		38,8297837	35,9521526	3,86967311										
		0,30%	0,57%	10,28%										
		61,70%	78,19%	68,94%										
		0,18911613	0,26340647	0,57704241										
Natural and semi-natural	MAP-CLCC_Lvl2	31-11	31-12	31-13	31-14	31-21	31-32	31-33	31-41	31-42	32-12	32-13	32-14	32-21
	Weight of U.A	0,30662576	0,45993864	1,68644168	0,32805196	14,678316	283,015899	6,99367581	0,79573285	0,01611906	0,78248312	6,0952683	4,72149688	15,6718997
	User_accuracy	75,00%	50,00%	81,82%	46,73%	1,59%	46,82%	20,01%	10,13%	100,00%	48,98%	24,24%	1,62%	6,93%
	Prod_accuracy	89,03%	7,14%	74,42%	100,00%	23,35%	84,71%	100,00%	100,00%	100,00%	56,21%	81,11%	100,00%	72,54%
	Weight of U.P	0,25830435	3,22121317	1,85405044	0,15331288	1,00127815	156,413087	1,39910376	0,0805953	0,01611906	0,68182801	1,82149554	0,07665644	1,49784166
		32-22	32-23	32-31	32-33	32-41	32-51	33-21	33-24	33-32	33-51			
		2,86960837	34,762757	242,134112	90,4668868	1,91053002	0,47879406	0,44846312	0,91150036	48,7008542	1,25811341			
		36,74%	1,46%	31,00%	3,09%	13,50%	55,56%	48,45%	7,95%	4,25%	38,06%			
Wetlands		93,96%	71,01%	82,03%	69,94%	100,00%	25,13%	100,00%	100,00%	54,93%	81,97%			
		1,12201399	0,71402924	91,5116885	4,00110824	0,25790496	1,05827966	0,2172978	0,0724326	3,76561147	0,5841123			
	MAP-CLCC_Lvl2	41-12	41-13	41-21	41-23	41-32	41-33	41-51	42-33					
	Weight of U.A	0,61537425	0,06683206	1,73341716	1,04608676	3,17577703	0,27126872	0,58688113	0,41187726					
	User_accuracy	32,58%	100,00%	24,58%	2,40%	1,79%	21,01%	11,39%	4,61%					
Water bodies	Prod_accuracy	100,00%	100,00%	90,26%	31,87%	41,90%	100,00%	69,21%	100,00%					
	Weight of U.P	0,20049618	0,06683206	0,47212257	0,07864726	0,13599288	0,05698104	0,09656579	0,01899368					
	MAP-CLCC_Lvl2	51-32	51-33	51-42										
	Weight of U.A	0,25474399	1,78021924	0,06683206										
	User_accuracy	80,87%	53,97%	100,00%										
	Prod_accuracy	71,95%	100,00%	100,00%										
	Weight of U.P	0,2863502	0,96073429	0,06683206										

	Accuracy better than 85%
	Accuracy between 80% and 85%
	Accuracy less than 80%

We will study the main changes according to the five categories of the CLC nomenclature:

CLC change classes: Urban

- **13-11 / 13-12:** When we observe the changes of urban, we acknowledge that industrial areas into the construction site are more or less clearly identified.

Generally, there is an overestimation of urbanization, with a threshold below 80%: like 21-11, 31-11, 31-12, 41-12.

CLC change classes: Agricultural

- **21-23:** Overestimation of agriculture withdrawal flow (annual cultivation in pastures). Possible confusion with temporary grass which have to be classified into 211. Confusion with 21-21 (still annual crops in 2018) or 23-23 (already grass in 2012).
- **21-24:** Overestimation of big annual cultivation plots in small plots associated to permanent crops: doubtful change?
- **23-24:** Underestimation Pastures in Heterogeneous agricultural areas.

We see that class 24 “Heterogeneous agricultural areas” is problematic in general.

CLC change classes: Natural and semi-natural

- **31-X:** We have an overestimation of the changes linked to deforestation
- **32-31:** Overestimation of forest regeneration. It was already forest in 2012 (31-31) or still transitional woodland in 2018 (32-32).

For a threshold under the recommendation, less than 80%, we see that there are more commission errors than omission errors.

CLC change classes: Wetlands and Water bodies

For wetlands we have a slight overestimation, more commission errors than omissions errors. As for the water surface, no problem with the changes, the weights are very low.

5. Conclusions and recommendations

For countries or grouping of countries the validation exercise shows anyway satisfactory overall accuracies, both for CLC2018 and for CLCCH2012-2018 changes products (excepted four areas, with a threshold very under the 85%), for the blind and the plausibility analysis. Indeed, the CLC2018 and CLCCH2012-2018 products show a global accuracy which meets the product specification in the blind and plausibility analysis, with a threshold superior of 85%. There is a slight improvement of precisions for the plausibility analysis compared to the blind analysis, because the plausibility analysis results in revised accuracy metrics based on the results of the blind analysis.

However, disparities exist from a country to another, from 75.39% of overall accuracy for Portugal to 98.18% for Island regarding the CLCCH2012-2018 product.

The CLC2018 product contains a low disparity in overall accuracy from 86.35% of overall accuracy for Portugal to 98.93% for Island. There is an obvious link between landscape complexity and weaker overall accuracy.

	Overall accuracy CLC 2018		Overall accuracy CLCCH2012-2018 level 3		Overall accuracy CLCCH2012-2018 level 2	
	Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis	Blind analysis	Plausibility analysis
Alpine	93,80%	93,89%	89,86%	90,61%	92,44%	93,11%
Anatolian	86,49%	86,55%	80,43%	81,34%	83,72%	84,65%
Arctic	98,94%	98,94%	98,14%	98,19%	99,45%	99,50%
Atlantic	93,15%	94,00%	89,94%	90,82%	90,86%	91,53%
BlackSea	84,57%	84,03%	77,99%	78,53%	81,13%	81,19%
Boreal	95,34%	95,30%	91,14%	91,56%	91,98%	92,26%
Continental	93,21%	93,65%	87,59%	88,25%	89,60%	90,12%
Macaronesia	93,97%	96,84%	72,64%	89,97%	78,14%	90,73%
Mediterranean	89,46%	89,63%	81,37%	82,00%	86,66%	87,11%
Pannonian	96,71%	96,75%	91,37%	91,80%	92,53%	92,96%
Steppic	96,52%	99,13%	94,23%	97,02%	94,23%	97,02%
Oversea departments	92,23%	92,35%	86,41%	86,52%	86,62%	86,79%

Considering the plausibility approach the CLC2018 product show that the most problematic biogeographical region is Black Sea with 84.03% overall accuracies. At the third level, the plausibility results of CLCCH2012-2018 show that problematic biogeographical regions are the ones located in the south of Europe including Black Sea, Anatolian and Mediterranean region with respectively 78.5%, 81.3% and 82% overall accuracies.

It should be noted that the results for CLC2018 are better than CLC2012, by countries, and lower for CLCCH2006-2012 product.

Better results in terms of overall accuracy might be obtained by introducing an error typology.

A typology of mistake or confusion would allow classifying the relevance of the errors without affecting the final accuracy. Indeed “minor confusion” might be filtered such as:

- The three classes 241, 244 ,411 must be improved, because confused with several types of land use
- In Mediterranean region burnt area (334) are also confused with transitional woodland (324) according to state of vegetation regeneration after fire.

Some other errors can be explained by various reasons:

- A correct identification of certain CLC classes requires the use of topical ancillary data. These were missing during our control work (Forest, peat, sclerophyllous vegetation maps...)
- The location of the validation points could sometimes be confusing inside the selected polygon (e.g. on the boundary between two land cover types bigger than 25 ha).
- The spatial resolution of satellite images might not be sufficient to identify specific classes (e.g. young vineyards and orchards)
- Generalisation of some polygons including different land cover type.
- The temporal resolution of satellite images is not enough to distinguish grasslands from arable land with vegetation.
- Different acquisition period of satellite images (1 or two years) for the same area could induce some interpretation difference (notably for land cover changes) in forest area.
- Problem of the MMU for the land cover change assessment. Indeed, in some cases the photo-interpreter could generalize the change while the check point itself falls in a part without change.

These errors and difficulties can only be overcome/resolved if:

- Very High-Resolution satellite imagery is used for the production
- Multi-temporal satellite imagery
- Auxiliary data sources are available.
- Organisation of some field control

The analysis of the **validation results at class and disaggregated zone or biogeographical level** should provide insights on where the product could be improved thematically focusing on weaker classes and geographical regions.

Another recommendation would be to use the user and producer accuracy of the change classes (or regrouped change classes at Level 2 of the nomenclature) as an indicator of the quality of the change layer rather than overall accuracy which is always likely to be dominated by unchanged areas.

A total of 25695 sample units located according to the sampling design have been interpreted by an independent expert's team. Table 13 and Table 14 shows the confusion matrix between the validation results in column and the production map in line for both blind and plausibility analysis.

[illegible]

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[illegible]

The overall accuracy obtained for CLC2018 is **92.7%**, so the plausibility results are above the 85% threshold expected.

Annex 2. Producer and user accuracy of CLC2018 per country or group of countries

Table 15: Producer, user and overall accuracy per country (from zone 1 to zone 6) – BLIND and PLAUSIBILITY analysis

CLASSES	Z1				Z2				Z3				Z4				Z5				Z6			
	Turkey				France				Spain				Sweden				Germany				Finland			
	Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
111	100,00%	51,04%	100,00%	51,04%	100,00%	100,00%	100,00%	100,00%	100,00%	88,45%	93,87%	88,45%	100,00%	100,00%	100,00%	100,00%	100,00%	97,88%	100,00%	97,88%				
112	95,84%	99,59%	95,90%	99,18%	99,56%	99,82%	99,56%	99,77%	96,77%	99,12%	95,04%	96,86%	83,72%	99,22%	83,72%	99,22%	93,28%	97,84%	93,88%	97,50%	86,85%	94,20%	87,05%	94,23%
121	91,92%	97,25%	91,35%	97,25%	98,80%	97,03%	98,79%	96,70%	97,21%	99,66%	97,22%	100,00%	94,75%	97,75%	94,75%	97,75%	87,97%	96,38%	88,32%	99,66%	75,05%	89,51%	74,60%	89,51%
122	27,93%	89,45%	27,93%	89,45%	100,00%	87,47%	100,00%	87,47%	99,15%	97,97%	99,15%	97,97%	92,72%	100,00%	92,72%	100,00%	70,65%	100,00%	72,37%	100,00%	100,00%	100,00%	95,22%	100,00%
123	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	79,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
124	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	90,00%	100,00%	90,00%	71,26%	100,00%	71,26%	100,00%	97,80%	100,00%	97,80%	100,00%	100,00%	100,00%	100,00%	100,00%
131	97,61%	89,96%	97,58%	89,04%	97,95%	100,00%	100,00%	100,00%	97,20%	98,43%	97,20%	98,43%	100,00%	92,31%	100,00%	92,31%	98,54%	96,62%	98,54%	96,62%	99,28%	17,04%	93,81%	17,04%
132	100,00%	51,12%	100,00%	51,12%	100,00%	80,80%	80,80%	80,80%	100,00%	100,00%	100,00%	100,00%	100,00%	78,90%	100,00%	78,90%	86,34%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	91,39%
133	81,15%	78,24%	78,11%	78,24%	100,00%	56,13%	100,00%	56,13%	67,88%	96,80%	80,12%	96,80%	22,30%	100,00%	22,30%	100,00%	5,05%	43,30%	5,05%	43,30%	100,00%	100,00%	79,65%	100,00%
141	87,88%	100,00%	94,12%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	79,14%	100,00%	79,14%	100,00%	99,64%	100,00%	99,64%	96,27%	100,00%	96,27%	100,00%	97,90%	70,59%	100,00%	76,47%
142	100,00%	100,00%	100,00%	100,00%	100,00%	94,54%	95,59%	94,54%	87,29%	98,67%	85,70%	98,67%	99,61%	100,00%	99,61%	100,00%	100,00%	85,54%	93,52%	85,54%	98,68%	99,34%	95,47%	99,34%
211	96,10%	86,61%	94,95%	84,01%	97,14%	97,92%	97,88%	97,89%	99,00%	96,32%	99,00%	95,07%	99,17%	85,94%	99,28%	85,94%	99,81%	98,18%	98,19%	98,20%	83,10%	96,05%	83,11%	96,05%
212	89,92%	96,21%	90,91%	96,22%	0,00%		0,00%		95,56%	97,36%	91,64%	97,37%												
213	98,56%	100,00%	98,56%	99,73%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%												
221	100,00%	47,22%	100,00%	47,22%	100,00%	96,12%	100,00%	95,99%	99,60%	82,91%	99,36%	82,35%					100,00%	94,44%	100,00%	94,44%				
222	81,05%	81,95%	81,18%	81,95%	100,00%	59,07%	100,00%	59,07%	90,17%	92,45%	92,92%	92,26%	100,00%	60,00%	100,00%	80,00%	100,00%	86,39%	93,75%	86,39%				
223	100,00%	100,00%	94,76%	100,00%	100,00%	44,44%	100,00%	44,44%	98,81%	99,20%	96,14%	99,40%												
231	76,48%	86,52%	73,47%	86,42%	95,71%	93,91%	95,69%	93,91%	77,54%	86,07%	77,50%	85,87%	52,39%	49,91%	52,39%	49,91%	97,02%	83,01%	94,79%	83,31%	22,55%	31,71%	26,33%	31,71%
241	100,00%	100,00%	100,00%	100,00%					5,65%	60,00%	5,65%	60,00%							0,00%					
242	89,63%	79,56%	84,22%	80,76%	90,56%	92,09%	90,75%	94,06%	78,20%	87,10%	79,23%	87,10%	54,49%	59,88%	56,46%	64,87%	7,07%	81,06%	8,59%	81,06%				
243	83,37%	91,65%	81,90%	88,42%	74,75%	81,63%	74,76%	81,67%	70,48%	85,04%	72,42%	87,30%	71,85%	88,79%	71,97%	89,29%	66,30%	86,77%	72,55%	86,77%	96,58%	81,85%	96,89%	81,82%
244									87,57%	97,12%	87,57%	97,12%												
311	92,96%	91,89%	95,66%	91,93%	98,95%	91,40%	98,95%	91,38%	98,90%	99,78%	98,90%	99,74%	98,08%	91,63%	98,73%	93,68%	90,19%	90,15%	93,01%	86,31%	93,11%	80,10%	92,56%	74,44%
312	91,38%	97,87%	91,38%	97,84%	97,39%	98,00%	97,97%	99,86%	98,11%	93,21%	98,11%	93,21%	99,14%	94,56%	99,14%	94,99%	96,57%	96,64%	97,54%	96,13%	96,31%	95,12%	96,30%	95,13%
313	93,82%	84,01%	94,30%	90,28%	69,17%	98,93%	70,89%	98,09%	95,50%	87,62%	96,43%	87,62%	92,23%	98,08%	93,91%	98,85%	67,55%	89,77%	69,77%	89,77%	94,15%	92,43%	94,17%	92,42%
321	74,33%	89,64%	75,97%	91,83%	95,61%	79,18%	97,60%	79,18%	89,65%	70,40%	90,54%	70,42%	100,00%	100,00%	100,00%	100,00%	75,51%	41,43%	74,22%	38,69%	100,00%	86,67%	100,00%	86,67%
322	0,00%		0,00%		91,58%	70,30%	81,03%	70,30%	93,54%	82,04%	93,70%	84,27%	99,98%	98,00%	99,98%	100,00%	93,06%	94,27%	93,06%	94,27%	86,93%	95,29%	86,96%	95,29%
323	92,41%	71,25%	99,58%	74,25%	96,12%	100,00%	96,12%	100,00%	91,95%	87,88%	91,15%	87,88%												
324	80,32%	92,26%	81,24%	92,27%	94,19%	96,71%	93,90%	96,71%	67,69%	87,82%	68,19%	90,12%	77,93%	99,41%	79,38%	99,41%	92,70%	91,68%	73,99%	97,62%	82,96%	91,25%	83,06%	91,30%
331	86,79%	81,99%	91,61%	81,99%	100,00%	88,89%	46,82%	88,89%	100,00%	19,32%	100,00%	100,00%	100,00%	60,00%	100,00%	60,00%	91,17%	80,24%	91,71%	85,95%	0,00%			0,00%
332	100,00%	77,31%	97,54%	77,31%	94,59%	94,74%	94,86%	100,00%	100,00%	100,00%	100,00%	100,00%	94,04%	95,00%	94,04%	95,00%	100,00%	68,63%	100,00%	68,63%	100,00%	49,20%	100,00%	49,20%
333	93,82%	73,61%	94,02%	74,78%	94,29%	76,24%	100,00%	76,24%	98,42%	82,92%	98,10%	68,88%	89,96%	94,03%	97,22%	94,03%	69,53%	100,00%	69,53%	100,00%	70,63%	100,00%	70,63%	100,00%
334		0,00%		0,00%		0,00%		0,00%	100,00%	55,03%	72,08%	55,03%	100,00%	50,00%	100,00%	50,00%								
335	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%								
411	92,53%	96,80%	95,51%	96,80%	98,48%	100,00%	100,00%	100,00%	4,97%	80,96%	4,97%	80,96%	100,00%	100,00%	100,00%	100,00%	72,04%	98,81%	72,04%	98,81%	99,54%	100,00%	99,70%	100,00%
412					100,00%	100,00%	100,00%	100,00%					98,55%	100,00%	98,55%	100,00%	100,00%	90,00%	100,00%	90,00%	99,88%	99,91%	97,90%	99,94%
421	94,62%	68,45%	90,43%	73,71%	100,00%	100,00%	100,00%	100,00%	97,27%	93,16%	97,28%	93,71%	100,00%	75,00%	100,00%	75,00%	100,00%	100,00%	100,00%	100,00%	79,81%	100,00%	76,01%	100,00%
422	100,00%	82,09%	100,00%	82,09%	100,00%	90,00%	100,00%	100,00%	100,00%	89,86%	100,00%	89,86%												
423					100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%					82,61%	88,89%	100,00%	88,89%				
511	100,00%	69,27%	100,00%	68,56%	100,00%	95,10%	100,00%	95,10%	100,00%	90,00%	100,00%	90,00%	100,00%	94,74%	100,00%	94,74%	90,78%	100,00%	90,78%	100,00%	100,00%	90,00%	100,00%	95,00%
512	99,26%	99,79%	99,13%	99,85%	99,90%	95,43%	99,90%	95,43%	95,69%	100,00%	95,56%	100,00%	100,00%	98,00%	100,00%	98,00%	90,37%	99,79%	90,37%	99,79%	100,00%	100,00%	100,00%	100,00%
521	90,37%	100,00%	90,37%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%				
522	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	95,71%	100,00%	95,71%	100,00%				
523	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	66,67%	100,00%	100,00%	100,00%	86,67%	100,00%	86,67%
Overall Acc	0,874		0,874		0,942		0,945		0,912		0,914		0,951		0,955		0,935		0,931		0,939		0,938	

Table 16: Producer, user and overall accuracy per country (from zone 07 to zone 12) - BLIND and PLAUSIBILITY analysis

Z7				Z8				Z9				Z10				Z11				Z12			
Norway				Poland				Italy Sardinia				United Kingdom				Roumania				Greece			
Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	86,91%	94,99%	86,64%	94,99%	67,07%	94,74%	98,64%	94,74%	96,97%	100,00%	96,97%	100,00%	100,00%	92,86%	100,00%	92,86%
99,93%	99,94%	99,93%	100,00%	95,24%	97,25%	99,31%	99,87%	88,42%	98,03%	88,39%	97,82%	91,69%	95,64%	91,91%	96,53%	92,43%	94,02%	97,51%	95,83%	93,93%	100,00%	95,78%	100,00%
97,59%	88,18%	97,54%	86,25%	99,39%	47,19%	99,39%	47,19%	99,39%	99,46%	99,22%	99,43%	93,98%	86,79%	93,63%	87,78%	94,14%	97,37%	94,14%	97,37%	51,24%	91,85%	72,14%	91,85%
100,00%	93,74%	100,00%	100,00%	96,28%	100,00%	96,28%	100,00%	97,83%	77,14%	96,76%	77,14%	97,21%	73,14%	97,59%	84,99%	94,03%	100,00%	94,03%	100,00%	94,19%	95,51%	94,19%	95,51%
47,50%	100,00%	47,50%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	93,33%	100,00%	93,33%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	95,00%	100,00%	95,00%	83,33%	100,00%	83,33%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
86,11%	76,59%	87,72%	88,30%	95,80%	90,08%	95,82%	90,53%	93,34%	54,45%	93,34%	54,45%	93,21%	98,53%	96,27%	98,53%	100,00%	50,62%	100,00%	53,59%	100,00%	93,80%	100,00%	93,80%
100,00%	100,00%	100,00%	100,00%	73,44%	94,12%	73,44%	94,12%	20,85%	100,00%	20,85%	100,00%	100,00%	95,77%	93,65%	100,00%	87,98%	100,00%	87,98%	100,00%	100,00%	100,00%	100,00%	100,00%
25,00%	79,54%	33,33%	79,54%	100,00%	65,42%	14,17%	71,12%	100,00%	75,01%	100,00%	75,01%	76,59%	84,75%	86,74%	84,75%	100,00%	2,91%	100,00%	2,91%	100,00%	84,64%	100,00%	84,64%
100,00%	94,74%	100,00%	94,74%	81,51%	94,12%	94,12%	94,12%	100,00%	94,12%	100,00%	94,12%	93,20%	89,01%	92,52%	89,01%	92,78%	92,78%	92,78%	92,78%	100,00%	100,00%	100,00%	100,00%
99,22%	100,00%	98,22%	100,00%	100,00%	95,17%	100,00%	100,00%	94,14%	100,00%	91,20%	100,00%	99,13%	100,00%	99,13%	100,00%	100,00%	78,20%	100,00%	82,77%	100,00%	75,30%	100,00%	75,30%
99,92%	89,00%	99,92%	89,02%	99,96%	98,11%	99,97%	98,11%	96,40%	91,64%	97,50%	91,61%	96,33%	95,97%	96,36%	95,90%	99,37%	93,79%	99,81%	97,94%	91,95%	84,06%	95,41%	85,27%
								84,68%	100,00%	46,18%	100,00%					100,00%	50,00%	100,00%	100,00%	94,20%	98,81%	94,23%	99,36%
								100,00%	94,50%	100,00%	94,89%					100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
								95,29%	72,24%	95,29%	72,26%					100,00%	78,20%	100,00%	78,20%	100,00%	50,00%	100,00%	56,25%
				100,00%	94,99%	100,00%	94,99%	89,08%	84,66%	89,08%	84,66%	100,00%	78,57%	100,00%	78,57%	99,88%	77,61%	99,88%	77,61%	87,16%	86,51%	87,50%	89,20%
								99,99%	90,24%	100,00%	86,69%									84,44%	75,82%	84,44%	75,82%
98,92%	95,00%	98,92%	95,00%	93,44%	88,02%	99,62%	88,02%	63,14%	75,00%	41,99%	75,00%	94,51%	94,25%	94,46%	94,24%	94,05%	83,97%	94,29%	89,71%	95,78%	53,33%	99,58%	53,33%
								83,03%	48,59%	83,03%	48,59%									100,00%	75,00%	100,00%	75,00%
83,26%	90,00%	83,26%	90,00%	70,04%	100,00%	70,04%	100,00%	82,60%	81,44%	80,69%	81,35%	89,69%	20,80%	89,69%	20,80%	57,75%	92,06%	77,09%	93,68%	86,47%	82,70%	84,69%	82,70%
93,87%	95,00%	93,87%	94,98%	99,70%	91,99%	96,39%	93,36%	69,19%	80,97%	71,31%	82,81%	61,20%	75,21%	68,00%	75,18%	73,52%	83,12%	78,24%	83,58%	77,15%	86,01%	75,54%	89,76%
								45,76%	89,47%	45,76%	89,47%											0,00%	
99,70%	97,99%	99,70%	97,99%	99,84%	99,79%	99,85%	99,94%	96,63%	97,99%	97,17%	97,08%	97,34%	100,00%	98,26%	99,96%	88,91%	94,24%	91,69%	95,62%	95,57%	97,31%	97,70%	97,31%
98,86%	99,49%	99,11%	99,40%	97,57%	99,53%	99,37%	99,47%	95,74%	100,00%	93,87%	99,92%	88,65%	92,26%	90,75%	94,12%	85,88%	93,51%	83,74%	93,51%	90,09%	97,07%	95,51%	97,07%
97,74%	99,57%	97,71%	99,57%	95,12%	99,86%	96,63%	99,86%	87,26%	99,95%	84,79%	99,90%	87,38%	77,01%	87,64%	78,81%	86,72%	91,61%	91,21%	99,96%	92,74%	80,40%	93,14%	85,36%
0,00%		0,00%		87,47%	100,00%	91,71%	100,00%	58,09%	75,98%	53,81%	75,98%	78,16%	81,50%	81,92%	81,37%	100,00%	13,92%	100,00%	13,80%	90,37%	72,50%	92,77%	72,50%
93,28%	96,00%	93,28%	96,00%	100,00%	93,70%	100,00%	100,00%	42,84%	94,40%	42,84%	94,40%	94,36%	83,29%	94,36%	83,29%	95,76%	87,50%	95,76%	87,50%	100,00%	77,94%	100,00%	90,54%
								77,47%	92,07%	82,44%	92,07%									83,83%	92,25%	86,48%	94,64%
97,98%	96,03%	97,50%	97,39%	98,82%	94,97%	92,03%	96,54%	99,65%	83,84%	99,17%	76,59%	68,55%	86,55%	70,80%	89,88%	64,20%	71,11%	62,88%	67,46%	82,90%	92,28%	80,59%	92,39%
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	95,08%	95,33%	95,08%	95,33%	90,92%	100,00%	78,96%	100,00%	4,31%	90,18%	4,31%	90,18%	84,62%	100,00%	84,62%	100,00%
95,96%	100,00%	95,96%	100,00%	100,00%	50,00%	100,00%	50,00%	95,54%	100,00%	99,42%	100,00%	80,84%	71,47%	80,50%	69,95%	90,76%	66,83%	90,76%	66,83%	100,00%	100,00%	100,00%	100,00%
97,72%	96,00%	97,73%	96,00%	100,00%	100,00%	100,00%	100,00%	99,97%	48,86%	99,97%	50,74%	69,29%	86,20%	80,53%	86,20%	89,50%	100,00%	89,50%	100,00%	85,62%	68,86%	86,30%	72,83%
100,00%	100,00%	100,00%	100,00%					100,00%	35,18%	100,00%	67,59%	100,00%	30,00%	100,00%	30,00%					100,00%	14,95%	100,00%	14,95%
100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%												
100,00%	100,00%	100,00%	100,00%	37,37%	94,44%	37,37%	94,44%	93,74%	92,66%	95,57%	92,66%	76,88%	99,02%	76,88%	99,02%	67,90%	94,12%	67,90%	94,12%	97,45%	85,06%	97,10%	74,71%
99,98%	100,00%	99,98%	100,00%	100,00%	79,49%	100,00%	79,49%			100,00%	100,00%	99,52%	93,04%	99,55%	99,02%								
								83,01%	94,91%	83,01%	94,91%	84,91%	100,00%	84,91%	100,00%		0,00%		0,00%	98,23%	100,00%	92,73%	100,00%
								75,25%	80,00%	75,25%	80,00%									100,00%	100,00%	100,00%	100,00%
100,00%	85,00%	100,00%	90,00%									90,85%	81,22%	90,85%	81,22%								
94,87%	100,00%	95,00%	100,00%	100,00%	90,06%	100,00%	90,06%	98,83%	100,00%	98,83%	100,00%	95,22%	95,22%	95,22%	95,22%	100,00%	94,74%	100,00%	94,74%	93,10%	85,00%	93,10%	85,00%
100,00%	99,97%	100,00%	99,97%	99,35%	99,68%	99,36%	99,97%	95,92%	99,48%	96,80%	98,07%	95,04%	99,87%	95,04%	99,87%	99,64%	100,00%	99,64%	100,00%	99,58%	100,00%	99,58%	100,00%
				100,00%	100,00%	100,00%	100,00%	97,38%	90,00%	97,38%	100,00%	100,00%	80,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
								0,00%		0,00%		96,33%	68,18%	100,00%	68,18%					100,00%	66,67%	100,00%	66,67%
100,00%	93,33%	100,00%	93,33%	100,00%	100,00%	100,00%	100,00%	100,00%	14,28%	100,00%	14,28%	98,11%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	94,74%	100,00%	94,74%	100,00%
0,975		0,975		0,968		0,971		0,892		0,887		0,918		0,927		0,887		0,916		0,876		0,889	

Table 17: Producer, user and overall accuracy per country (from zone 13 to zone 18) - BLIND and PLAUSIBILITY analysis

Z13				Z14				Z15				Z16				Z17				Z18				
Bulgaria				Island				Hungaria				Portugal				Swiss Austria				Belgium-Danmark- Luxembourg				
Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		
Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	
100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%	100,00%	97,99%	100,00%	78,02%	97,76%	100,00%	97,76%	100,00%	100,00%	100,00%	94,74%	100,00%	100,00%
99,95%	99,95%	99,95%	99,91%	94,36%	100,00%	94,36%	100,00%	94,03%	99,97%	94,07%	99,97%	84,90%	100,00%	80,78%	100,00%	99,35%	90,32%	99,38%	95,14%	77,24%	89,20%	77,76%	89,44%	
99,61%	94,82%	99,61%	94,82%	100,00%	87,03%	100,00%	93,51%	97,39%	99,03%	97,62%	99,03%	97,92%	98,33%	97,91%	97,87%	98,33%	94,04%	98,05%	94,04%	96,93%	96,31%	96,76%	100,00%	
95,39%	86,67%	95,39%	86,67%	100,00%	100,00%	100,00%	100,00%	100,00%	95,79%	98,01%	99,14%	98,43%	81,33%	98,43%	81,33%	94,08%	83,80%	91,58%	83,80%	32,87%	100,00%	32,87%	100,00%	
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	80,00%	100,00%	80,00%	100,00%	100,00%	100,00%	100,00%	90,12%	100,00%	90,12%	100,00%	100,00%	100,00%	100,00%	100,00%	
100,00%	79,36%	99,17%	79,36%	100,00%	100,00%	100,00%	100,00%	99,47%	100,00%	99,47%	100,00%	100,00%	94,68%	100,00%	94,68%	100,00%	87,91%	100,00%	87,91%	99,06%	94,86%	99,06%	94,86%	
96,39%	100,00%	96,39%	100,00%					98,18%	100,00%	98,18%	100,00%	66,67%	85,71%	57,03%	85,71%		0,00%		0,00%	100,00%	100,00%	100,00%	100,00%	
100,00%	100,00%	100,00%	100,00%	100,00%	50,10%	100,00%	50,10%	100,00%	67,14%	100,00%	98,95%	100,00%	91,74%	95,59%	91,74%		0,00%		0,00%	100,00%	81,93%	100,00%	81,93%	
100,00%	90,00%	100,00%	90,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	70,00%	100,00%	70,00%	100,00%	69,16%	100,00%	69,16%	26,26%	95,41%	25,94%	95,41%	
96,60%	88,15%	96,60%	88,15%	100,00%	100,00%	100,00%	100,00%	65,21%	100,00%	65,21%	100,00%	99,06%	94,01%	97,29%	94,01%	90,62%	97,54%	90,62%	97,54%	75,29%	99,64%	74,62%	99,64%	
99,57%	97,23%	99,58%	97,23%	100,00%	100,00%	100,00%	100,00%	99,00%	97,72%	99,47%	97,48%	82,86%	86,13%	83,09%	86,13%	97,94%	85,93%	97,99%	85,92%	95,96%	91,72%	98,02%	91,71%	
												98,49%	87,68%	97,26%	87,68%									
100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%	100,00%	95,82%	100,00%	95,82%									
100,00%	89,95%	99,55%	89,95%					95,90%	97,32%	96,00%	100,00%	93,59%	77,49%	83,20%	78,26%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	
100,00%	93,97%	99,77%	93,97%					90,95%	98,89%	95,34%	100,00%	93,31%	73,98%	95,57%	73,98%	100,00%	77,78%	100,00%	88,89%	100,00%	80,43%	100,00%	80,43%	
												89,96%	90,98%	90,33%	94,90%									
77,85%	87,61%	78,21%	89,66%	99,88%	99,97%	99,88%	99,97%	97,71%	95,32%	94,19%	95,32%	95,27%	83,90%	90,94%	84,34%	82,25%	79,23%	82,50%	80,60%	91,17%	90,03%	87,86%	90,03%	
												85,13%	41,14%	96,59%	41,14%			0,00%						
80,13%	72,07%	80,13%	72,07%	98,47%	94,53%	98,56%	100,00%	99,94%	84,06%	99,89%	84,12%	77,29%	90,17%	79,46%	90,17%	68,82%	93,93%	68,87%	93,93%	87,27%	84,86%	87,50%	86,75%	
90,68%	90,23%	89,55%	90,22%	0,00%				93,96%	86,90%	96,36%	81,01%	82,31%	85,50%	79,12%	85,50%	94,95%	87,67%	94,41%	87,67%	99,54%	84,05%	99,77%	83,81%	
												76,66%	97,00%	78,71%	96,54%									
97,91%	99,98%	98,60%	99,98%	100,00%	88,24%	100,00%	88,24%	92,17%	98,07%	92,07%	98,09%	95,08%	78,86%	96,12%	80,58%	97,80%	93,89%	97,82%	93,89%	96,22%	98,35%	96,22%	98,35%	
97,45%	97,41%	97,51%	97,41%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	99,90%	100,00%	99,18%	93,97%	98,54%	91,55%	87,70%	94,15%	89,14%	92,96%	87,58%	93,98%	97,43%	94,46%	
79,85%	97,96%	81,83%	97,94%	100,00%	88,89%	100,00%	88,89%	82,53%	97,61%	82,98%	97,69%	88,77%	91,33%	93,59%	91,20%	86,80%	87,56%	87,94%	90,44%	96,44%	83,08%	99,57%	92,41%	
97,91%	87,10%	92,63%	87,10%	98,49%	100,00%	98,49%	100,00%	99,59%	95,19%	100,00%	99,97%	33,71%	72,43%	33,71%	72,43%	97,30%	98,03%	97,33%	97,72%	100,00%	66,35%	100,00%	77,00%	
72,74%	100,00%	97,34%	100,00%	99,84%	99,98%	99,85%	99,98%					82,42%	85,23%	82,37%	84,91%	75,22%	100,00%	75,57%	100,00%	98,11%	92,41%	100,00%	93,69%	
												87,33%	80,52%	91,38%	80,52%									
92,91%	91,11%	92,95%	91,39%	100,00%	76,38%	100,00%	76,38%	96,00%	94,44%	96,27%	96,15%	85,42%	88,13%	84,06%	89,64%	95,13%	85,44%	93,21%	85,68%	93,31%	90,35%	88,74%	90,35%	
100,00%	80,00%	100,00%	80,00%	88,62%	98,87%	88,60%	98,65%					100,00%	86,67%	100,00%	86,67%	100,00%	100,00%	100,00%	100,00%	86,00%	100,00%	100,00%	100,00%	
100,00%	47,37%	100,00%	63,16%	98,05%	99,96%	98,05%	99,97%					100,00%	66,67%	80,00%	66,67%	96,00%	92,99%	96,00%	92,99%					
97,61%	64,14%	100,00%	62,57%	99,97%	93,36%	99,97%	93,36%	100,00%	100,00%	100,00%	100,00%	43,94%	77,53%	88,15%	76,75%	100,00%	79,89%	100,00%	79,89%					
												100,00%	57,14%	100,00%	57,14%									
				100,00%	100,00%	100,00%	100,00%									97,79%	100,00%	97,79%	100,00%					
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	98,50%	88,98%	98,50%	88,70%	21,67%	100,00%	21,67%	100,00%	94,08%	94,44%	94,08%	94,44%	60,32%	87,28%	61,86%	87,28%	
100,00%	100,00%	100,00%	100,00%	99,76%	100,00%	99,76%	100,00%	100,00%	80,00%	100,00%	80,00%	100,00%	100,00%	66,67%	100,00%	100,00%	75,00%	100,00%	75,00%	100,00%	83,84%	100,00%	88,64%	
				100,00%	100,00%	100,00%	100,00%					96,31%	100,00%	96,31%	100,00%					99,14%	100,00%	91,15%	100,00%	
100,00%	100,00%	100,00%	100,00%									93,42%	100,00%	93,42%	100,00%									
				100,00%	100,00%	100,00%	100,00%					100,00%	47,49%	100,00%	47,49%					100,00%	86,67%	100,00%	86,67%	
100,00%	60,00%	100,00%	60,00%	96,24%	99,21%	95,37%	99,21%	100,00%	99,29%	100,00%	99,29%	85,41%	100,00%	85,41%	100,00%	100,00%	100,00%	100,00%	100,00%	67,65%	80,48%	90,14%	80,48%	
100,00%	95,08%	100,00%	95,08%	99,33%	100,00%	99,52%	100,00%	100,00%	99,85%	99,85%	100,00%	99,72%	89,66%	99,72%	89,66%	100,00%	99,96%	100,00%	99,96%	98,85%	90,27%	98,91%	95,14%	
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%					80,00%	100,00%	80,00%	100,00%					100,00%	100,00%	100,00%	100,00%	
				100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%					15,31%	100,00%	15,31%	100,00%	
82,72%	100,00%	82,72%	100,00%	100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%	
0,949		0,950		0,989		0,989		0,967		0,967		0,860		0,863		0,904		0,908		0,900		0,910		

Table 18: Producer, user and overall accuracy per country (from zone 19 to zone 23 - BLIND and PLAUSIBILITY analysis)

Z19				Z20				Z21				Z22				Z23			
Albania - Serbia - Macedonia -				Bosnia Herzegovina - Croatia - Slovenia				Czech Republic - Slovakia				Estonia - Latvia- Lithuania				French overseas departments			
Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis		Blind analysis		Plausibility analysis	
Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
100,00%	100,00%	100,00%	100,00%	53,66%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
93,19%	99,54%	93,27%	99,57%	97,00%	94,53%	97,43%	94,53%	99,65%	99,94%	99,48%	99,94%	99,78%	99,89%	99,63%	100,00%	86,70%	99,83%	86,70%	99,83%
100,00%	93,57%	100,00%	97,72%	98,71%	90,56%	98,71%	90,56%	98,07%	99,57%	99,19%	99,78%	100,00%	100,00%	100,00%	99,41%	49,88%	96,94%	49,88%	96,94%
100,00%	100,00%	100,00%	100,00%	100,00%	82,51%	94,99%	82,51%	94,28%	100,00%	98,61%	100,00%	90,45%	100,00%	97,83%	100,00%	93,61%	91,62%	93,61%	91,62%
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
90,49%	82,63%	90,49%	82,63%	66,67%	100,00%	66,67%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
95,83%	92,85%	97,16%	92,85%	88,82%	87,68%	100,00%	87,68%	94,76%	83,24%	94,76%	83,24%	100,00%	96,55%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
74,95%	91,41%	76,60%	100,00%	100,00%	100,00%	100,00%	100,00%	94,55%	100,00%	94,55%	100,00%	100,00%	89,38%	90,40%	100,00%	100,00%	100,00%	100,00%	100,00%
100,00%	100,00%	100,00%	100,00%	90,35%	100,00%	90,35%	100,00%	100,00%	81,53%	100,00%	81,53%		0,00%		0,00%				
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	97,16%	100,00%	100,00%	100,00%	100,00%	100,00%
93,77%	88,28%	88,93%	94,14%	100,00%	62,00%	100,00%	62,00%	99,42%	92,92%	99,42%	92,33%	100,00%	94,54%	100,00%	94,54%	100,00%	100,00%	100,00%	100,00%
99,72%	98,67%	99,91%	97,69%	83,54%	90,99%	90,90%	90,99%	99,26%	99,90%	99,82%	99,89%	99,52%	97,95%	99,53%	96,17%	100,00%	94,55%	100,00%	94,55%
100,00%	96,93%	100,00%	96,93%	100,00%	100,00%	100,00%	100,00%												
100,00%	100,00%	100,00%	100,00%													100,00%	100,00%	100,00%	100,00%
98,56%	75,57%	98,56%	75,57%	100,00%	74,23%	100,00%	80,56%	100,00%	82,74%	100,00%	87,34%					0,00%		0,00%	
95,48%	91,49%	95,48%	91,49%	100,00%	74,17%	86,87%	80,24%	87,33%	89,32%	87,33%	89,32%	100,00%	94,92%	100,00%	100,00%	100,00%	80,19%	100,00%	100,00%
68,42%	89,28%	68,42%	89,28%	100,00%	85,26%	100,00%	90,17%												
81,77%	72,31%	87,31%	72,31%	99,89%	79,99%	99,48%	79,99%	99,75%	97,71%	98,12%	98,04%	95,01%	98,65%	90,81%	98,73%	98,57%	79,20%	98,57%	79,20%
	0,00%		0,00%																
96,04%	89,98%	96,23%	89,98%	94,47%	82,18%	94,97%	86,33%	100,00%	99,80%	100,00%	99,80%	99,95%	100,00%	100,00%	100,00%	76,28%	88,58%	79,61%	88,58%
81,60%	92,12%	80,64%	92,12%	90,35%	95,45%	88,64%	97,73%	99,82%	97,71%	100,00%	97,71%	100,00%	99,91%	100,00%	99,91%	97,25%	94,93%	99,47%	94,93%
																100,00%	100,00%	100,00%	100,00%
95,46%	96,07%	96,15%	94,44%	96,55%	90,03%	97,82%	91,94%	98,86%	99,39%	98,87%	99,43%	99,91%	94,85%	99,93%	94,85%	99,71%	83,77%	99,71%	83,77%
93,35%	95,93%	93,35%	95,93%	73,88%	91,88%	74,32%	93,91%	95,70%	98,88%	96,50%	98,85%	99,34%	99,96%	99,81%	99,51%	100,00%	100,00%	100,00%	100,00%
69,29%	93,83%	69,58%	93,83%	71,72%	85,34%	70,72%	85,34%	98,10%	95,81%	98,92%	96,66%	94,23%	99,81%	93,96%	94,83%	100,00%	100,00%	100,00%	100,00%
88,16%	85,94%	90,34%	86,53%	94,56%	72,77%	94,56%	72,77%	94,16%	91,34%	100,00%	96,63%	99,15%	99,84%	98,60%	99,84%	89,40%	100,00%	89,40%	100,00%
77,95%	80,25%	79,72%	80,25%	79,25%	85,00%	100,00%	85,00%	82,66%	100,00%	90,83%	100,00%	100,00%	94,44%	100,00%	94,44%	93,53%	90,00%	93,53%	90,00%
86,88%	46,09%	91,42%	46,09%	51,86%	84,24%	58,48%	84,24%									100,00%	67,98%	100,00%	67,98%
80,23%	85,06%	74,51%	85,45%	80,63%	80,58%	79,73%	80,65%	95,92%	93,21%	96,09%	96,23%	99,54%	95,76%	93,20%	96,01%	43,93%	87,35%	43,93%	87,35%
92,41%	40,37%	92,41%	40,37%		0,00%		0,00%					100,00%	100,00%	100,00%	100,00%	20,16%	100,00%	20,16%	100,00%
91,93%	77,40%	92,67%	85,87%	79,67%	100,00%	79,67%	100,00%	100,00%	60,00%	100,00%	60,00%					100,00%	100,00%	100,00%	100,00%
86,10%	72,14%	86,64%	77,47%	74,65%	59,52%	72,15%	52,35%	100,00%	88,03%	100,00%	88,03%	100,00%	81,32%	100,00%	81,32%	100,00%	88,89%	100,00%	88,89%
100,00%	55,41%	100,00%	55,41%	100,00%	100,00%	100,00%	100,00%												
	0,00%		0,00%																
100,00%	89,39%	100,00%	95,83%	92,89%	89,01%	92,58%	85,04%	100,00%	89,47%	100,00%	89,47%	99,52%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
100,00%	100,00%	100,00%	100,00%		0,00%		0,00%	96,06%	90,00%	96,06%	90,00%	100,00%	99,88%	100,00%	99,88%				
100,00%	100,00%	100,00%	100,00%									0,00%		0,00%		100,00%	100,00%	100,00%	100,00%
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%												
					0,00%		0,00%									0,00%		0,00%	
100,00%	97,33%	100,00%	97,33%	100,00%	99,26%	100,00%	99,26%	100,00%	90,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	77,58%	90,00%	77,58%	90,00%
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	97,95%	100,00%	97,51%	100,00%	100,00%	99,26%	100,00%	99,26%	100,00%	100,00%	100,00%	100,00%
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
																100,00%	90,12%	100,00%	90,12%
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%					81,05%	100,00%	100,00%	100,00%	88,91%	99,96%	88,91%	99,96%
0,910		0,906		0,868		0,881		0,984		0,987		0,984		0,972		0,922		0,923	

Zone 1: part of Turkey Confusion matrix CLC2018 – Blind and plausibility analysis

Table 19: Confusion matrix CLC 2018 for zone 1 (Turkey) – BLIND ANALYSIS - In column the validation and line the production

[illegible]

Overall accuracy = 87.4%

Table 20: Confusion matrix CLC 2018 for zone 1 (Turkey) – PLAUSIBILITY ANALYSIS - In column the validation and line the production

MAP2018	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	335	411	421	422	511	512	521	522	523	Total	User_accuracy							
111	5,020244	0,257551		4,558836																																							9,836631	0,510362091						
112		35,20786	0,145233						0,145233																																		35,49832	0,991817476						
121		0,126343	9,615122						0,081786																		0,049686								0,014119										9,887056	0,97249589				
122			0,049991	1,76633																						0,081786				0,076656														1,974764	0,894451196					
123					0,097174																																						0,097174	1						
124						1,388548																																					1,388548	1						
131			0,070333				6,521252									0,066832					0,221674					0,145233				0,298331														7,323655	0,890436814					
132								0,079279										0,075795																										0,155073	0,511233595					
133		0,253676	0,076656				0,076656		3,029278									0,145233			0,290466																								3,871966	0,782361704				
141										0,347548																																			0,347548	1				
142											1,392513																																		1,392513	1				
211			0,198776									414,3583	18,36093					8,187984		11,59748	20,24333				6,747776																				493,2457	0,840064661				
212		0,454743										0,189116	262,5438					0,116684		4,72017																									272,8654	0,962173373				
213														5,831013																															5,846577	0,997338095				
221												0,113616			3,715603	1,232725				1,503639	0,387597		0,387597					0,387597		0,113616															7,868654	0,472203118				
222												0,233367				47,32854				6,585077	1,972434																								57,74952	0,819548762				
223																	18,88722																												18,88722	1				
231														4,558836	0,042681			61,93856																												71,67184	0,864196576			
241																			1,630094																										1,630094	1				
242		0,023953										0,116684	3,282209	0,042681		9,602476	0,971648	3,680033		147,7897	14,33549																									183,0068	0,807563948			
243												21,33956						0,07274	4,866755	3,281815	279,8137	0,116684	0,084731																							316,4561	0,88421022			
244																		0,072433																											0,072433					
311																					2,086095		140,5403			5,387756																					152,8694	0,919349154		
312																				0,155169	0,466368		0,08986	229,9418																							235,0169	0,978405407		
313																				2,879953		0,538455	89,16329																								98,76811	0,902753844		
321		0,387597																0,051492		4,632667																											359,7861	0,918321406		
323																																															60,24974	0,742489182		
324			0,369152														0,067766				1,804096	1,804096																									334,8069	0,922713258		
331											0,021727																																				3,766704	0,819935406		
332																																															86,85724	0,773080582		
333							0,084731												4,878502																												418,6134	0,747752756		
334																																															0,081964			
335																																															0,024884	1		
411																			0,286761																												8,950887	0,967962891		
421																																															7,471317	0,737072058		
422																																															0,639106	0,82091423		
511									0,621902				0,051175											0,24684																									4,615518	0,685562269
512													0,052021																																				56,03415	0,998531896
521																																																	0,28379	1
522																																																	0,002678	1
523																																																	1,883854	1
Total	5,020244	36,71172	10,52526	6,325166	0,097174	1,388548	6,68264	0,079279	3,8782	0,369274	1,392513	436,4026	288,797	5,916376	3,715603	58,29834	19,93161	84,30023	1,630094	175,4779	341,6577	2,387148	146,9232	251,637	94,55105	434,9026	4,941475	44,92568	380,2759	3,371256	68,83922	332,9257	0,024884	9,071129	6,0898	0,524651	3,164225	56,44316	0,314032	0,002678	1,883854	3331,796								
Prod_accuracy	1	0,959036	0,913528	0,279254	1	1	0,97585	1	0,781104	0,941164	1	0,949486	0,909095	0,985572	1	0,811833	0,947601	0,734738	1	0,842213	0,818988		0,956556	0,913784	0,943017	0,759709		0,995751	0,812386	0,916114	0,975427		0,995751	0,812386	0,916114	0,975427	1	0,955132	0,904282	1	1	0,991296	0,903695	1	1	0,873785246				

Overall accuracy = 87.4%

Zone 2: France confusion matrix CLC2018 - Blind and plausibility analysis

Table 21: Confusion matrix CLC 2018 for zone2 (France) – BLIND ANALYSIS- In column the validation and line the production

MAP2018	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	335	411	412	421	422	423	511	512	521	522	523	Total	User_accuracy								
111	1,870393																																											1,870393	1						
112		100,4435																	0,183345																								100,6268	0,998177973							
121		0,166792	18,90907									0,081786						0,081786	0,021559	0,227019																								19,48801	0,970292396						
122		0,217717		2,533095																0,145233																								2,896045	0,874674093						
123					0,458603																																							0,458603	1						
124						2,259634																																						2,259634	1						
131							1,409343																																					1,409343	1						
132			0,063371					0,266771																																				0,330142	0,808048068						
133			0,145233						0,290466																																				0,517485	0,561302945					
141										1,057654									0,081786																										1,057654	1					
142											4,30981										0,248707																								4,558517	0,945441189					
211			0,021559									625,4061	0,072433						0,139605	13,05424	0,009432																								638,7033	0,979180834					
213														1,559928																															1,559928	1					
221												0,072433			47,43378				0,922251	0,922251																										49,35072	0,961156898				
222												0,37487				4,37962																														7,41424	0,590703869				
223		0,045399															0,181594		0,045399	0,090797			0,045399																							0,408587	0,444444444				
231		0,013954																335,3504	7,241837	14,46473		0,019326	0,009432																							357,0997	0,939094655				
242												4,937268						9,835883	229,2092																											248,9003	0,920887628				
243							0,029569					12,97587								63,57761	1,293925																									77,88747	0,816275187				
311																				356,9156			32,16531					1,398989											0,010495							390,4799	0,914043458				
312																					152,8691	2,891146						0,223885																			155,9841	0,980029819			
313																					0,524484	91,23449						0,466368																			92,22534	0,989256189			
321																		2,080373		4,160747		2,080373		39,56172	1,040187	1,040187																				49,96359	0,791811059				
322																		1,673261	0,83663	2,046058	0,83663				12,76247																						18,15505	0,702970822			
323																											25,79543																				25,79543	1			
324																		0,031393	0,051492	1,209428		0,637356	0,637356					75,3494																			77,91642	0,967054128			
331																													1,064883																			1,197993	0,888888889		
332																														15,87977	0,882209																	16,76198	0,947368421		
333																								1,815632				1,815632	0,907816	14,56834																		19,10742	0,762444067		
334																												0,466368																			0,466368				
335																																1,078768															1,078768	1			
411																																3,453235															3,453235	1			
412																																	0,195662														0,195662	1			
421																																		3,094958														3,094958	1		
422																																	0,053398															0,533976	0,9		
423																																			0,480579													0,348565	1		
511																																																	5,089429	0,951022966	
512																																																	4,840164	10,15941	
521																																																		2,469442	1
522																																																		0,416505	1
523																																																		0,204982	1
Total	1,870393	100,8873	19,13923	2,533095	0,458603	2,259634	1,438913	0,266771	0,290466	1,057654	4,30981	643,8483	0,072433	1,559928	47,43378	4,37962	0,181594	350,3991	253,1011	85,05595	360,7151	156,9672	131,9011	41,37736	13,93577	26,83562	79,9965	1,064883	16,78759	15,45054	1,078768	3,506632	0,195662	3,094958	0,480579	0,348565	4,840164	10,16991	2,469442	0,416505	0,204982	2392,381									
Prod_accuracy	1	0,9956	0,987974	1	1	1	0,97945	1	1	1	1	0,971356	1	1	1	1	1	0,957053	0,905603	0,74748	0,989467	0,973892	0,691689	0,95612	0,915807	0,961239	0,941909	1	0,945923	0,942901	1	0,984772	1	1	1	1	1	0,998968	1	1	1	1	1	0,942011101							

Overall accuracy = 94.2%

	MAP2018	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	335	411	412	421	422	423	511	512	521	522	523	Total	User_accuracy			
111		1,870393																																								1,870393	1				
112			100,3947																0,048761	0,183345																						100,6268	0,997693395				
121			0,166792	18,84569					0,063371				0,103345						0,081786		0,227019																							19,48801	0,96704058		
122			0,217717		2,533095																0,145233																							2,896045	0,874674093		
123						0,458603																																					0,458603	1			
124							2,259634																																				2,259634	1			
131								1,409343																																			1,409343				
132				0,063371					0,266771																																			0,330142	0,808048068		
133				0,145233						0,290466										0,081786																								0,517485	0,561302945		
141											1,057654																																	1,057654	1		
142												4,30981									0,248707																								4,558517	0,945441189	
211				0,021559								0,198635	625,2074	0,072433						0,139605	13,05424	0,009432																							638,7033	0,978869836	
213															1,559928																														1,559928	1	
221													0,072433			47,3706										0,063181																				49,35072	0,959876652
222													0,37487				4,37962																													7,41424	0,590703869
223			0,045399															0,181594																												0,408587	0,444444444
231			0,013954																																												

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Zone 3: Spain - confusion matrix CLC2018 - Blind and plausibility analysis

Table 23: Confusion matrix CLC 2018 for zone3 (SPAIN) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	335	411	421	422	423	511	512	521	522	523	Total	User_accuracy				
111	7,855715	0,513086							0,513086																																				8,881886	0,884464735			
112		22,49837	0,032214						0,074753			0,092366																																	22,6977	0,991217876			
121		0,040454	11,75535																																											11,7958	0,996570455		
122				3,443094																									0,07142																	3,514514	0,979678478		
123					0,188339																																								0,188339	1			
124						0,745898	0,082878																																						0,828776	0,9			
131							2,87419						0,045899																																2,920089	0,984281538			
132								0,47707																																					0,47707	1			
133			0,07142						2,161792																																				2,233212	0,968019038			
141		0,149507								0,567371																																			0,716878	0,791447255			
142		0,047558									3,522																																		3,569558	0,986676918			
211			0,102343									413,8606				0,116684	0,035257	2,844649		8,456044	4,202845	0,072433																								429,6909	0,96315897		
212												0,398166	115,4343																	0,072433	0,072433															118,5607	0,973630912		
213														5,874414																															5,874414	1			
221												1,473505			36,7245	1,249577	1,164513			3,682912																										44,295	0,829088894		
222																45,03837																													48,71665	0,924496445			
223															0,116684	0,683834	99,6078		2,416661	1,261619																									100,4083	0,992027383			
231			0,131527	0,029569					0,37894		0,512893		0,063445					56,64139		0,029569	2,331343						2,597083				3,026588									0,065994					65,80834	0,860702295			
241																			0,361181	0,120394																									0,601968	0,6			
242												2,080787			0,030182	0,155651		2,583306	1,960393	72,78443	0,07142				1,818258				2,080787																83,56521	0,870989576			
243																2,705185		4,185774	1,496855		56,18534									1,496855																66,07001	0,85039098		
244																		2,608935				87,90506																								90,514	0,971176443		
311																							206,0243	0,15775	0,072576																					206,473	0,997826771		
312																																															197,0644	0,932087348	
313																																															52,70335	0,87622983	
321																		4,188395	0,156258	4,155542																											131,0533	0,703958326	
322																																															93,17964	0,820412812	
323									0,056058				5,252902																																			194,3936	0,878830851
324												0,051492																																			119,3582	0,878191863	
331																																															6,689087	0,193235079	
332																																															9,73584	1	
333																																															37,41368	0,829228563	
334																																															2,188439	0,550260421	
335																																															0,009491	1	
411												0,066832																																			0,351063	0,80962944	
421																																															2,964596	0,931625328	
422																																															0,765618	0,89858796	
423																																															0,026195	1	
511																																															1,796298	1	
512																																															11,8104	1	
521																																															0,348754	1	
522																																															0,378164	1	
523																																															0,473893	1	
Total	7,855715	23,24897	12,09285	3,472663	0,188339	0,745898	2,957067	0,47707	3,184629	0,567371	4,034893	418,0238	120,7966	5,874414	36,87136	49,9493	100,8076	73,05245	6,391348	93,07382	79,71316	100,3816	208,3201	187,2139	48,35781	102,9031	81,7289	185,797	154,8407	1,292566																			

Overall accuracy = 91.2%

CIC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	335	411	421	422	423	511	512	521	522	523	Total	User accuracy				
111	7,855715	1,026171																																									8,881886	0,884464					
112	0,513086	21,98528	0,032214						0,074753			0,092366																																22,6977	0,96861				
121			11,7958																																										11,7958				
122				3,443094																																									3,514514	0,979678			
123					0,188339																							0,07142																		0,188339			
124						0,745898	0,082878 2,87419																																							0,828776			
131														0,045899																																2,920089	0,984281		
132								0,47707																																						0,47707			
133			0,07142						2,161792																																						2,233212	0,968019	
141		0,074753								0,567371	0,074753 3,522																																				0,716878	0,791447	
142		0,047558																																														3,569558	0,986676
211			0,102343									408,4934	5,166612		0,08937	0,116684	0,090459	2,844649		8,456044	4,202845	0,072433					0,056058																					429,6909	0,950686
212												0,281482	115,4429								2,583306								0,072433	0,164799																	118,5607	0,973702	
213														5,874414																																		5,874414	
221												1,581819																																				44,295	0,823524
222															36,47803	1,340426 44,94752	1,211816 1,20833		2																														

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Table 25: Confusion matrix CLC 2018 for zone4 (Sweden – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy = 95.05%

Table 26: Confusion matrix CLC 2018 for zone4 (Sweden – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	335	411	412	421	511	512	521	522	523	Total	User_accuracy	
111	0,156321																																			0,156321	1	
112		18,80077	0,14188															0,006032																		18,94869	0,992194061	
121			3,327491						0,076656																											3,404147	0,977481456	
122				0,37637																																0,37637	1	
123					0,115086																															0,115086	1	
124						0,624809																														0,624809	1	
131		0,062603					0,751124																													0,813727	0,923066408	
132								0,484602	0,076656																											0,561258	0,863420365	
133									0,043997																											0,043997	1	
141			0,004362							1,202119																											1,20648	0,996384773
142											2,385068																									2,385068	1	
211		2,537884		0,029569								108,9604		5,066714	2,52619	5,06225		2,52619					0,072433													126,7817	0,859433704	
222													0,060507					0,015127																		0,075634	0,8	
231		0,588396												5,619426		1,680184	0,560061	1,685558	0,560061				0,565435												11,25912	0,499099896		
242						0,25202								0,00448	3,27626	1,00808		0,257394																		5,050254	0,64873169	
243			0,00964									0,004935	0,789355				19,89726			1,57871				0,004935											22,28484	0,892861012		
311																	80,63901		3,650562					0,01875											86,07559	0,936839383		
312			0,028335							0,004362	0,004935							1112,581	1,932424				56,74792						1,767275							1171,299	0,949869339	
313		0,466368															0,466368	0,453183	119,0714																	120,4573	0,988494529	
321																				8,125175																8,125175	1	
322																					116,5208															116,5208	1	
324																		1,32855																		226,491	0,994134205	
331															0,036405			0,036405						0,109214												0,182024	0,6	
332																																					16,08678	0,95
333																					0,026746	0,789355				15,28244	0,804339									29,88476	0,940268663	
334																										0,968955	28,0997									0,046862	0,5	
335																											0,023431										0,5	
411																												1,126684									1,126684	1
412																													2,247814								2,247814	1
421																	0,011099																			120,1329	1	
511																		0,24964													0,033298	4,493521					0,044397	0,75
512																		3,055519																			4,743161	0,947368421
521																																					152,9468	0,980022341
522																																					0,068242	1
523																																					0,080239	1
Total	0,156321	22,45603	3,511708	0,405939	0,115086	0,876829	0,751124	0,484602	0,19731	1,202119	2,394364	109,7547	0,060507	10,72703	5,80245	27,64778	81,67653	1122,194	126,7932	8,125175	116,5476	283,6368	0,109214	16,2514	28,90404	0,023431	1,126684	2,247814	121,9001	0,033298	4,493521	149,8913	0,068242	0,080239	1,613215	1,613215	2252,26	
Prod_accuracy	1	0,837226	0,947542	0,927158	1	0,712578	1	1	0,222985	1	0,996117	0,992763	1	0,523857	0,564634	0,71967	0,987297	0,991433	0,9391	1	0,999771	0,793841	1	0,940377	0,972172	1	1	1	0,985502	1	1	1	1	1	1	1	0,955210882	

Overall accuracy = 95.5%

Zone 5: Germany confusion matrix CLC2018 - Blind and plausibility analysis

Table 27: Confusion matrix CLC 2018 for zone5 (Germany) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	412	421	423	511	512	521	522	523	Total	User_accuracy				
111	2,459355		0,053258																																		2,512613	0,978803674			
112		98,61332							2,086548								0,087592																				100,7875	0,978428468			
121		0,69684	22,91661		0,081786							0,081786																									23,77702	0,963813282			
122				0,878929																																	0,878929	1			
123					0,30771																																0,30771	1			
124						1,804427																															1,804427	1			
131							4,150031					0,145233																									4,295264	0,966187661			
132								0,493755																													0,493755	1			
133		0,145233							0,110891																												0,256124	0,432957997			
141										3,920323																											3,920323	1			
142										0,151826	4,961307					0,343534																					5,800202	0,855367991			
211		2,086548					0,035026					586,5947			0,132154	5,560379	0,151826							2,933972												0,343534		597,4946	0,981757316		
221													4,437227			0,261013																					4,69824	0,944444444			
222												0,115916		4,792758		0,319517	0,319517																				5,547708	0,863916788			
231		4,173096	0,145514	0,365176								0,562609			218,7942	22,26359	0,319517	10,88893			4,995687	0,117741		0,124814							0,315513	0,513626					263,58	0,830086469			
242															0,18705	2,186434	0,323702																				2,697185	0,810635307			
243												0,18705				3,028208	0,18705							0,087592													3,4899	0,86770637			
311															0,082343		147,8628	6,612704	9,290291					0,17406													164,0222	0,901480394			
312						0,040656											0,337065	4,995687	234,2193	1,253808				1,064004										0,459909				242,3704	0,966369125		
313			2,933972													0,204584		0,019204	38,45204	1,226244																	42,83604	0,897656185			
321																4,774571						0,204584	0,040678					0,337065						0,048163			11,52365	0,414327854			
322																6,118593						3,368192															3,572776	0,942738014			
324																																						20,66013	0,916823306		
331																				1,692612																		0,332317	0,802373003		
332																																						0,408124	0,686280068		
333																																						0,292229	1		
411																			0,020309																			1,701465	0,988064032		
412																																						3,155037	0,9		
421																																						0,887839	1		
423																																						0,400499	0,888888889		
511																																						3,105893	1		
512							0,0266																															12,84262	0,997928796		
521																																							0,055574	1	
522																																							0,99368	0,99368	1
523																																							0,111906	0,167858	0,666666667
Total	2,459355	105,715	26,04936	1,244105	0,389496	1,845084	4,211657	0,493755	2,197439	4,072149	4,961307	587,6873	4,437227	4,792758	225,5189	30,93447	4,567427	163,9536	242,5449	56,9258	6,323139	3,619457	20,43284	0,292472	0,280087	0,420265	2,333725	2,839534	0,887839	0,430946	3,421406	14,18125	0,055574	1,03818	0,111906	1531,67					
Prod_accuracy	1	0,932822	0,879738	0,706475	0,79002	0,977965	0,985368	1	0,050464	0,962716	1	0,998141	1	1	0,970181	0,07068	0,663001	0,901857	0,965674	0,675477	0,755095	0,930579	0,927022	0,911686	1	0,695343	0,720375	1	1	0,826089	0,907783	0,90373	1	0,957137	1		0,934767591				

Overall accuracy = 93.4%

Table 28: Confusion matrix CLC 2018 for zone5 (Germany) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	241	242	243	311	312	313	321	322	324	331	332	333	411	412	421	423	511	512	521	522	523	Total	User_accuracy			
111	2,459355		0,053258																																		2,512613	0,978803674			
112		98,26978							2,086548		0,343534						0,087592																				100,7875	0,975019965			
121			23,69524									0,081786																									23,77702	0,996560284			
122				0,878929																																		0,878929	1		
123					0,30771																																	0,30771	1		
124						1,804427																																1,804427	1		
131							4,150031					0,145233																										4,295264	0,966187661		
132								0,493755																														0,493755	1		
133		0,145233							0,110891																													0,256124	0,432957997		
141										3,920323																												3,920323	1		
142										0,151826	4,961307						0,343534																					5,800202	0,855367991		
211		2,086548					0,035026					586,7269			0,115916		5,444464	0,151826			2,933972													0,343534				597,4946	0,981978496		
221													4,437227				0,261013																					4,69824	0,944444444		
222																0,319517																						5,547708	0,863916788		
231		4,173096	0,145514	0,335607								10,25992		0,319517	219,5999		16,81882	0,333443	5,444464		4,995687	0,117741		0,207157							0,315513	0,513626						263,58	0,833143294		
242															0,18705		2,186434	0,323702																				2,697185	0,810635307		
243												0,18705					3,028208	0,18705							0,087592													3,4899	0,86770637		
311															5,444464			141,5633	5,38646	6,265608					5,362284													164,0222	0,863074501		
312					0,040656												0,337065	4,995687	232,993	2,466126					1,077929									0,459909					242,3704	0,961309745	
313			2,933972												0,204584				0,019204	38,45204	1,226244																	42,83604	0,897656185		
321															6,118593							4,459057	0,204584	0,356191				0,337065					0,048163					11,52365	0,386948239		
322																						0,204584	3,368192															3,572776	0,942738014		
324																				0,466368					20,16793	0,025829												20,66013	0,976176464		
331																										0,285636		0,046681										0,332317	0,859528258		
332																										0,046681		0,280087	0,081355									0,408124	0,686280068		
333																												0,292229											0,292229	1	
411																				0,020309																		1,701465	0,988064032		
412																													1,681156										1,701465	0,988064032	
421																												0,315504	2,839534		0,887839								3,155037	0,9	
423																																							0,887839	1	
511																																							0,356	0,400499	0,888888889
512							0,0266																																3,105893	1	
521																																							12,84262	0,997928796	
522																																									

Overall accuracy = 93.1%

Zone 6: Finland confusion matrix CLC2018 - Blind and plausibility analysis

Table 29: Confusion matrix CLC 2018 for zone6 (Finland) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	112	121	122	123	124	131	132	133	141	142	211	231	243	311	312	313	321	322	324	332	333	411	412	421	511	512	523	Total	User_accuracy
112	11,8641	0,724283							0,003803	0,002146																		12,59434	0,942018977
121	0,276292	2,356324																										2,632616	0,895050365
122			0,095002																									0,095002	1
123				0,198427																								0,198427	1
124					0,371019																							0,371019	1
131		0,001689				0,588195									0,001689	2,860336												3,451908	0,170396947
132							0,555807																					0,555807	1
133								0,232525																				0,232525	1
141	0,073775								0,17706																			0,250835	0,705882353
142						0,004282				0,64095																		0,645232	0,993363228
211											70,0682	0,242364				2,467198			0,072433				0,099922					72,95012	0,960494696
231												0,070577	0,013631			0,086063								0,052328				0,222598	0,31705847
243											2,306558		36,54621		1,153279	4,636642						0,002423	0,002423					44,64753	0,818549324
311	0,58327													25,38077	0,590347	4,549255			0,58327									31,68691	0,800985933
312	0,138146										11,92892		1,153279		589,6734	1,523776			15,53305									619,9506	0,951161973
313	0,724283	0,057272							0,004265					0,007077	8,557524	269,2984		2,860336	9,83759			0,004265						291,351	0,924309137
321																	0,584739				0,08996							0,674699	0,866666667
322														1,474046				29,80006										31,27411	0,952866873
324										0,002146					11,94674	0,420263			128,9522									141,3214	0,912475048
331																		0,01712			0,034241							0,051361	
332														0,321259	0,316534			1,60469	0,392556	3,165345	0,633069						6,433452	0,49201338	
333																					1,820689							1,820689	1
411																						1,439459						1,439459	1
412											0,016119								0,064476				87,32012					87,40071	0,999077864
421																								0,206786				0,206786	1
511													0,128402			0,128402									2,311238			2,568042	0,9
512																										132,1183		132,1183	1
523														0,074301		0,074301											0,965907	1,114508	0,866666667
Total	13,65987	3,139567	0,095002	0,198427	0,371019	0,592477	0,555807	0,232525	0,180863	0,649508	84,3198	0,312941	37,84152	27,25745	612,2395	286,0447	0,584739	34,28221	155,4356	3,165345	2,577958	1,446147	87,42246	0,259113	2,311238	132,1183	0,965907	1488,26	
Prod_accuracy	0,868537	0,750525	1	1	1	0,992772	1	1	0,978971	0,986824	0,830982	0,225527	0,96577	0,93115	0,963142	0,941456	1	0,869257	0,829618	1	0,706252	0,995375	0,998829	0,798051	1	1	1		0,938546964

Overall accuracy = 93.8%

Table 30: Confusion matrix CLC 2018 for zone6 (Finland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	112	121	122	123	124	131	132	133	141	142	211	231	243	311	312	313	321	322	324	332	333	411	412	421	511	512	523	Total	User_accuracy
112	11,86791	0,724283								0,002146																		12,59434	0,942320969
121	0,276292	2,356324																										2,632616	0,895050365
122			0,095002																									0,095002	1
123				0,198427																								0,198427	1
124					0,371019																							0,371019	1
131		0,001689				0,588195									0,001689	2,860336												3,451908	0,170396947
132						0,025953	0,507973			0,021881																		0,555807	0,913937635
133								0,232525																				0,232525	1
141	0,044265	0,014755							0,191815																			0,250835	0,764705882
142						0,004282				0,64095																		0,645232	0,993363228
211											70,0682	0,194997			0,072433	2,467198			0,072433				0,074856					72,95012	0,960494696
231												0,070577	0,013631			0,086063								0,052328				0,222598	0,31705847
243			0,002383								2,306558	0,002423	36,5309		1,153279	4,636642							0,002423	0,012924				44,64753	0,81820648
311	0,58327													23,58759	0,590347	4,549255				0,58327			1,793178					31,68691	0,744395435
312	0,138146					0,008565		0,055125			11,93134		1,153279		589,7565	1,52941			15,37828									619,9506	0,95129589
313	0,724283	0,061554	0,002383							0,004265			0,007077	0,007077	8,557524	269,2771		2,860336	9,83759			0,004265	0,007631					291,351	0,924235776
321																	0,584739				0,08996							0,674699	0,866666667
322														1,474046				29,80006										31,27411	0,952866873
324								0,004282		0,002146				0,013631	11,94674	0,322171			129,0324									141,3214	0,913042402
331																			0,01712		0,034241							0,051361	
332														0,325983	0,316534	0,013631		1,591059	0,387831	3,165345	0,633069							6,433452	0,49201338
333																					1,820689							1,820689	1
411																						1,439459						1,439459	1
412																			0,048357				87,35235					87,40071	0,999446719
421																								0,206786				0,206786	1
511																0,128402										2,43964		2,568042	0,95
512																											132,1183	132,1183	1
523														0,074301		0,074301											0,965907	1,114508	0,866666667
Total	13,63416	3,158604	0,099768	0,198427	0,371019	0,626994	0,507973	0,291932	0,191815	0,671389	84,3061	0,267997	37,70489	25,48263	612,395	285,9445	0,584739	34,26858	155,3402	3,165345	2,577958	1,443724	89,23044	0,272037	2,43964	132,1183	0,965907	1488,26	
Prod_accuracy	0,870454	0,746002	0,952229	1	1	0,938118	1	0,796503	1	0,954663	0,831117	0,263349	0,968864	0,925634	0,963033	0,941711	1	0,869603	0,830644	1	0,706252	0,997046	0,978952	0,760137	1	1	1		0,937515362

Overall accuracy = 93.7%

Zone 7: Norway confusion matrix CLC2018- Blind and plausibility analysis

Table 31: Confusion matrix CLC 2018 for zone 7 (Norway) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	231	242	243	311	312	313	321	322	324	331	332	333	334	335	411	412	423	511	512	523	Total	User_accuracy		
111	0,082994																																0,0829944	1		
112		7,961314															0,002473																7,96626086	0,999379066		
121			0,903209		0,044367				0,076656																								0,002473		1,02423303	0,881839458
122				0,140702													0,004695				0,004695													0,15009288	0,937434607	
123					0,044536																													0,04453605	1	
124						0,300792																												0,30079185	1	
131							0,501689		0,153313																									0,65500158	0,765935099	
132								0,019873																										0,01987341	1	
133									0,076656							0,014215													0,0055					0,09637183	0,795423725	
141										0,250418		0,013912																						0,2643299	0,947368421	
142											2,382662																							2,38266231	1	
211			0,02232				0,075439					21,84986	0,0055	1,21327	1,217194	0,072433	0,0055				0,072433							0,016246					24,5501966	0,890007585		
231													1,005403		0,052916																			1,0583192	0,95	
242														6,036309	0,670426																			6,70673526	0,900036877	
243		0,003007															2,0353																	40,84158152	0,949957685	
311																180,9361	0,014925	0,004777			0,002881			3,691336									184,650067	0,979886717		
312		0,002473									0,004621	0,003007					251,9656	0,469249			0,811878													253,2567892	0,994901504	
313																		21,15429												0,091662				21,24594984	0,995685682	
322																				190,4816			3,968367	3,968367										198,418333	0,96	
324							0,0055								0,466368	0,466368	0,845221	0,005763			43,25722												45,04644079	0,960280545		
331																						0,066875												0,06687478	1	
332																								94,19374										94,19373976	1	
333																				13,71235					329,1163									342,8286899	0,960002316	
334																										0,032527									0,03252707	1
335																										11,06402									11,06401644	1
411																											0,012457								0,01245731	1
412																													89,91967						89,91966839	1
423					0,004859						0,004859													0,004859										0,097172	0,85	
511																																			1,741574	1
512											0,009243				0,004621	0,004621																			56,32289752	0,999671794
523															0,122727																				1,718181	1,8409083
Total	0,082994	7,966794	0,925529	0,140702	0,093762	0,300792	0,582628	0,019873	0,306626	0,250418	2,401385	21,86678	1,016404	7,249579	41,33203	181,4892	254,8634	21,64422	0,004777	204,194	44,14911	0,066875	98,16211	336,7809	0,032527	11,06402	0,012457	89,94142	0,082596	1,835709	56,30441	1,718181	1386,882086			
Prod_accuracy	1	0,999312	0,975884	1	0,47499	1	0,861079	1	0,25	1	0,992203	0,999226	0,989177	0,832643	0,938685	0,996953	0,98863	0,977364		0,932846	0,979798	1	0,959573	0,977242	1	1	1	0,999758	1	0,94872	1	1		0,975137979		

Overall accuracy = 97.5%

Table 32: Confusion matrix CLC 2018 for zone 7 (Norway) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	231	242	243	311	312	313	321	322	324	331	332	333	334	335	411	412	423	511	512	523	Total	User_accuracy	
111	0,082994																																0,082994	1	
112		7,966261																															7,966261	1	
121			0,883363		0,044367				0,076656		0,019846																						1,024233	0,862462666	
122				0,150093																													0,150093	1	
123					0,044536																												0,044536	1	
124						0,300792																											0,300792	1	
131							0,578345		0,076656																								0,655002	0,882967549	
132								0,019873																									0,019873	1	
133									0,076656							0,014215												0,0055					0,096372	0,795423725	
141										0,250418		0,013912																					0,26433	0,947368421	
142											2,382662																						2,382662	1	
211			0,02232				0,075439					21,85536	0,0055	1,21327	1,217194	0,072433					0,072433							0,016246				24,5502	0,890231621		
231													1,005403		0,052916																		1,058319	0,95	
242														6,036309	0,670426																		6,706735	0,900036877	
243		0,003007											0,0055		38,79227		2,0353	0,0055															40,84158	0,949823015	
311																180,9322		0,014925	0,004777		0,006806			3,691336									184,6501	0,979865464	
312		0,002473								0,009243	0,003007						251,7267	0,469249			1,046142												253,2568	0,993958249	
313																	21,15429												0,091662				21,24595	0,995685682	
322																			190,4816			3,968367	3,968367										198,4183	0,96	
324							0,0055								0,466368	0,466368	0,23034	0,005763			43,8721												45,04644	0,97393049	
331																						0,066875											0,066875	1	
332																							94,19374										94,19374	1	
333																				13,71235				329,1163									342,8287	0,960002316	
334																									0,032527								0,032527	1	
335																										11,06402							11,06402	1	
411																											0,012457						0,012457	1	
412																													89,91967				89,91967	1	
423					0,004859					0,004859																				0,087455				0,097172	0,9
511																															1,741574			1,741574	1
512											0,009243				0,004621		0,004621															56,30441	56,3229	0,999671794	
523															0,122727																		1,718181	1,840908	0,933333333
Total	0,082994	7,971741	0,905682	0,150093	0,093762	0,300792	0,659285	0,019873	0,229969	0,250418	2,425853	21,87228	1,016404	7,249579	41,32653	181,4852	253,9969	21,64972	0,004777	204,194	44,99748	0,066875	98,16211	336,776	0,032527	11,06402	0,012457	89,94142	0,087455	1,833236	56,30441	1,718181	1386,882		
Prod_accuracy	1	0,999313	0,975356	1	0,47499	1	0,877231	1	0,333333	1	0,982196	0,999226	0,989177	0,832643	0,938677	0,996953	0,991062	0,977116		0,932846	0,97499	1	0,959573	0,977256	1	1	1	0,999758	1	0,95	1	1	0,975461061		

Overall accuracy = 97.5%

Zone 8: Poland- Confusion matrix CLC2018 - Blind and plausibility analysis

Table 33: Confusion matrix CLC 2018 for zone 8 (Poland) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	411	412	511	512	521	523	Total	User_accuracy	
111	0,299801																															0,299801	1	
112		64,55899	0,033311									0,024106							1,769362													66,38577	0,97248237	
121		0,315987	5,455285											5,78999																		11,56126	0,471858971	
122				2,533649																												2,533649	1	
123					0,108988																											0,108988	1	
124						0,888701																										0,888701	1	
131							3,312948	0,145233				0,03345								0,076656									0,109688			3,677975	0,900753128	
132								0,401679														0,025105										0,426784	0,941176471	
133		0,221889		0,097828			0,145233		0,879478																								1,344428	0,654164818
141										0,575823						0,035989																	0,611811	0,941176471
142										0,13066	2,574207																						2,704867	0,951694601
211		2,674756										546,1846		0,275108	1,769362	0,008096											5,78999					556,7019	0,98110782	
222													6,670639		0,351542																		7,022181	0,949938299
231												0,029427		87,31078	11,57998			0,257317				0,015382										99,19288	0,880212113	
242															32,03069																		32,03069	1
243														0,035988		74,98726		3,793908	2,674756			0,029148										81,52106	0,919851387	
311																0,035988	63,59583	0,072245				0,027531										63,7316	0,997869763	
312												0,084338				0,008443	238,6977	0,932736				0,105624										239,8288	0,99528355	
313																0,06858	0,008443	0,06858	105,7815													105,9271	0,998625437	
321																			1,081184														1,081184	1
322																			0,021106	0,313815												0,334921	0,936981823	
324																0,069488	0,059935	1,748438	0,050965	0,035988		37,12122										39,08604	0,949731058	
331																							0,176829										0,176829	1
332																				0,021106				0,021106								0,042212	0,5	
333																										0,348594						0,348594	1	
411																											3,644492						3,858874	0,944444444
412														0,025657			0,025657					0,025657					0,298344					0,375314	0,794917465	
511																											0,317182		2,874285				3,191467	0,900615632
512		0,016725										0,03315				0,004314																	16,90895	0,996795258
521																																0,022981	1	
523																																0,046602	0,046602	1
Total	0,299801	67,78835	5,488596	2,631477	0,108988	0,888701	3,458181	0,546912	0,879478	0,706482	2,574207	546,389	6,670639	93,43752	45,73157	75,20971	63,69831	244,6381	111,2093	1,236041	0,313815	37,56405	0,176829	0,021106	0,348594	9,751664	0,298344	2,874285	16,96445	0,022981	0,046602	1341,974		
Prod_accuracy	1	0,952361	0,993931	0,962824	1	1	0,958003	0,734449	1	0,815056	1	0,999626	1	0,93443	0,700406	0,997042	0,998391	0,975717	0,951193	0,874716	1	0,988211	1	1	1	0,37373	1	1	1	0,993534	1	1	0,96846325	

Overall accuracy = 96.8%

Table 34: Confusion matrix CLC 2018 for zone 8 (Poland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	411	412	511	512	521	523	Total	User_accuracy		
111	0,299801																															0,299801	1		
112		66,3004	0,033311							0,035989		0,016071																					66,38577	0,99871402	
121		0,315987	5,455285						5,78999																								11,56126	0,471858971	
122				2,533649																													2,533649	1	
123					0,108988																												0,108988	1	
124						0,888701																											0,888701	1	
131							3,329673	0,145233				0,016725								0,076656									0,109688				3,677975	0,905300524	
132								0,401679														0,025105											0,426784	0,941176471	
133		0,145233		0,097828			0,145233		0,956134																								1,344428	0,7111827	
141										0,575823						0,035989																	0,611811	0,941176471	
142											2,704867																						2,704867	1	
211												546,1565		0,275108	1,769362	2,682852				0,0281								5,78999					556,7019	0,981057344	
222													6,670639		0,351542																		7,022181	0,949938299	
231												0,029427		87,31078	11,57998				0,257317				0,015382										99,19288	0,880212113	
242															32,03069																		32,03069	1	
243															0,035988																		81,52106	0,933579772	
311																76,10641				2,674756													63,7316	0,999435323	
312												0,084338				0,035988	63,69561																63,7316	0,999435323	
313																0,017997	0,008443	238,5686	0,932736				0,216735										239,8288	0,994745217	
321																	0,008443	0,06858	105,7815				0,06858										105,9271	0,998625437	
322																				1,081184														1,081184	1
324																					0,334921													0,334921	1
331																0,069488	0,051492	1,179087	0,050965														39,08604	0,965434414	
332																								0,176829										0,176829	1
333																					0,021106				0,021106									0,042212	0,5
411																										0,348594								0,348594	1
412														0,025657				0,025657									3,644492						3,858874	0,944444444	
511																											0,298344							0,375314	0,794917465
512																											0,317182		2,874285					3,191467	0,900615632
521																0,004314																		16,90895	0,999744865
523																															0,022981			0,022981	1
Total	0,299801	66,76162	5,488596	2,631477	0,108988	0,888701	3,474906	0,546912	6,746124	0,611811	2,704867	546,303	6,670639	87,64753	45,73157	78,95304	63,78964	240,0735	109,4681	1,178947	0,334921	41,00475	0,176829	0,021106	0,348594	9,751664	0,298344	2,874285	17,01433	0,022981	0,046602	1341,974			
Prod_accuracy	1	0,993092	0,993931	0,962824	1	1	0,958205	0,734449	0,141731	0,941176	1	0,999732	1	0,996158	0,700406	0,963945	0,998526	0,993731	0,966323	0,917076	1	0,920259	1	1	1	0,37373	1	1	0,993553	1	1		0,971229274		

Overall accuracy = 97.1%

Zone 9: Italy confusion matrix CLC2018 - Blind and plausibility analysis

Table 35: Confusion matrix CLC 2018 for zone 9 (Italy) – BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	335	411	412	421	422	511	512	521	523	Total	User_accuracy				
111	5,740912	0,302919																																										6,043831	0,949879629			
112	0,864702	43,85903	0,003883				0,003986													0,003883								0,006157																44,74164	0,980273159			
121		0,020266	13,54394								0,006098	0,024357								0,023127																								13,61779	0,994576992			
122			0,063897	0,559647								0,031948								0,031948	0,006098					0,031948																		0,725488	0,771408012			
123					0,434197																																						0,434197	1				
124						0,925298																																						0,973998	0,95			
131			0,006098				1,577003	0,878657			0,107596																																0,215191			2,896126	0,544521385	
132								0,232444																			0,003986					0,107596													0,232444	1		
133							0,006157				0,073962												0,01849																						0,098609	0,750051212		
141										0,384578		0,024036																																	0,408614	0,941176471		
142											2,252054																																		2,252054	1		
211		0,864702						0,003986			0,026423	311,1501						0,016621		0,003986	20,45229	6,810026	0,026083		0,020767			0,026083	0,072433											0,066832					339,5403	0,916386266		
212													1,843295																																1,843295	1		
213												0,622264	0,076016	11,99891																																12,69719	0,945005112	
221												3,059175	0,233367		18,30474	0,005432	0,005432			1,300381	1,214884					1,214884																			25,3383	0,722414034		
222																15,08186				1,844291	0,888657																									17,81481	0,846591283	
223																	46,90911				2,537674																								51,98446	0,902367963		
231		0,904037													0,904037			10,84844		0,904037	0,904037								2,537674																14,46458	0,75		
241							0,077204					0,429929	0,024036					0,453965	4,299291	3,134078		0,429929																								8,848433	0,485881631	
242		1,870373	0,008583	0,00628																64,93872	3,688582		1,844291																							79,734	0,814442109	
243																				5,39565	77,01772		3,603257			3,642841			0,026083																	95,12026	0,809687781	
244																				0,718613	6,108211																									6,826824	0,894736842	
311												0,102034									0,033558		246,265	2,322866	0,019926		2,322866		0,033558	0,218673																	251,3184	0,979892119
312																																															52,55788	1
313																																															33,55683	0,999504683
321		0,904037																																													27,9954	0,759837596
322																																															7,066649	0,944003581
323																					2,865226																										47,20457	0,920730776
324																																															58,22954	0,838353216
331		0,878657										0,878657	0,878657						0,878657	0,878657	0,980691		3,102338					1,757313																			48,81692	0,838353216
332																				0,155827																											4,436731	0,953301214
333								0,031393																																							18,50946	1
334																																															40,20867	0,488555128
335																																															1,438955	0,351796443
411																																															1,482746	1
412																																															0,910486	0,926597339
421																																															0,017291	
422																																								</								

Overall accuracy = 89.15%

[illegible]

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Zone 10: United Kingdom/Ireland confusion matrix CLC2018 - Blind and plausibility analysis

Table 37: Confusion matrix CLC 2018 for zone 10 (United Kingdom & Ireland) – BLIND ANALYSIS- In column the validation and line the production

MAP2018	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	423	511	512	521	522	523	Total	User_accuracy		
111	1,344513									0,074695																										1,419208	0,947368421		
112	0,641663	68,78994	0,491392											0,005989		1,99625																				71,92523	0,956408989		
121		0,081786	11,33843		0,029153	0,323006	0,076656		0,145233	0,059242				0,087775						0,562875		0,117457						0,243022								13,06464	0,867871957		
122		0,038733		0,896425					0,145233					0,145233																						1,225623	0,731403028		
123					0,408139																															0,408139	1		
124						1,615029																														1,615029	1		
131							2,240799							0,007689														0,025742								2,27423	0,985300315		
132			0,021706					0,49174																												0,513445	0,957725622		
133		0,145233					0,025742		0,95024																											1,121214	0,847509345		
141			0,14599							2,365935	0,14599																									2,657915	0,890146753		
142										16,61384																										16,61384	1		
211		5,915841	0,037348									289,8295		6,199774						0,010352																301,9928	0,959723165		
222		0,053307											0,293189						0,026654																	0,373149	0,785714286		
231			0,029569	0,025742			0,029569			0,038733		10,31523		408,0278		3,63851	0,052397	0,045035		0,029569	0,021706	10,61463					0,029569									432,8981	0,942549385		
242												0,194081		9,15903	2,558239					0,194081								0,194081								12,29951	0,207995198		
243												0,517228		3,79408		16,24032			0,007361		0,517228	0,517228														21,59345	0,75209491		
311																	22,52348																			22,52348	1		
312																		72,94358	0,398055		1,471516	4,24922														79,06237	0,922608037		
313														0,228048	0,293931	0,587862	0,488012	0,881793	12,21749			1,168204														15,86534	0,770074293		
321							0,031393							3,851154		0,517228				47,5269		0,062786			3,113921		3,113921		0,100618						58,31792	0,814962198			
322																0,653451				9,673848	74,83342	0,10497														89,849	0,83287982		
324												0,102983		0,102983				3,212646	1,358394	0,010352	1,119819	38,17707					0,027542									44,11179	0,865461808		
331																							0,603709													0,603709	1		
332																	0,074638							1,220195	0,412527											1,70736	0,714667528		
333																					1,338901	0,65658			0,28923	14,27577										16,56048	0,862038447		
334																									0,056981		0,132956									0,189937	0,3		
411													0,07136								0,021706						10,45548					0,010352				10,5589	0,990205594		
412																2,79926		0,653451		2,79926									130,4881				0,708687				140,248	0,93040956	
421																																					1,583208	1	
423														0,060261										0,060261													0,962747	0,812222111	
511	0,018477																															0,368056					0,386533	0,95219885	
512																																						0,018477	0,998660211
521																																	0,05147	0,012867				13,77229	0,998660211
522																																					0,064337	0,8	
523																																					0,958938	0,958938	1
Total	2,004652	75,02484	12,06444	0,922166	0,437291	1,938035	2,404159	0,49174	1,240705	2,538605	16,75983	300,8561	0,293189	431,7412	2,85217	26,53587	23,13852	82,28621	13,9813	60,80724	79,30259	55,68985	0,663969	1,509424	20,60148	0,056981	13,59897	131,1114	1,864608	0,860702	0,386533	14,49133	0,05147	0,350279	0,977415	1379,835			
Prod_accuracy	0,670696	0,916896	0,939823	0,972086	0,933333	0,833333	0,932051	1	0,765887	0,931982	0,991289	0,963349	1	0,945075	0,896945	0,612014	0,973419	0,886462	0,873845	0,781599	0,943644	0,68553	0,909242	0,808384	0,692949	1	0,768843	0,995246	0,849083	0,908519	0,952199	0,950381	1	0,963265	0,981096		0,918355738		

Overall accuracy = 91.8%

Table 38: Confusion matrix CLC 2018 for zone 10 (United Kingdom & Ireland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

MAP2018	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	423	511	512	521	522	523	Total	User_accuracy		
111	1,344513									0,074695																										1,419208	0,947368421		
112		69,42619	0,491392					0,005415						0,005989		1,99625																			71,92523	0,965254947			
121			11,46772		0,029153	0,323006				0,145233	0,059242			0,116928						0,562875		0,117457						0,243022								13,06464	0,877768129		
122			0,038733	1,041658										0,145233																						1,225623	0,849900216		
123					0,408139																															0,408139	1		
124						1,615029																														1,615029	1		
131							2,240799							0,007689															0,025742							2,27423	0,985300315		
132								0,513445																												0,513445	1		
133			0,145233					0,025742		0,95024																											1,121214	0,847509345	
141				0,14599							2,365935	0,14599																									2,657915	0,890146753	
142										16,61384																										16,61384	1		
211		5,915841	0,074696									289,6066		6,38532						0,010352																301,9928	0,958985091		
222		0,053307											0,293189					0,026654																		0,373149	0,785714286		
231			0,029569	0,025742			0,029569	0,021706		0,038733		10,23374		407,9479		3,63851	0,052397	0,045035		0,029569	0,021706	10,75436					0,029569									432,8981	0,94236473		
242												0,194081		9,15903	2,558239				0,194081									0,194081								12,29951	0,207995198		
243												0,517228		3,79408		16,23296		0,007361	0,007361	0,517228	0,517228															21,59345	0,751754016		
311																	22,51493																			22,52348	0,999620487		
312																		74,4151	0,398055		1,471516	2,777704														79,06237	0,941220133		
313														0,228048	0,293931	0,781943	0,293931	0,587862	12,5038			1,175827														15,86534	0,78812042		
321						0,031393								3,851154		0,517228				47,45455	0,135139			3,113921	3,113921			0,100618								58,31792	0,813721521		
322							0,007689									0,517228				4,52305	0,10497									0,060261						89,849	0,83287982		
324														0,102983		0,051492	0,051492	1,74113	1,358394	0,010352	1,119819	39,64859					0,027542									44,11179	0,89882061		
331																							0,603709														0,603709	1	
332																					0,074638	0,100618	1,194215	0,337888												1,70736	0,699451252		
333																					1,338901	0,65658		0,28923	14,27577												16,56048	0,862038447	
334																								0,056981			0,132956										0,189937	0,3	
411														0,07136							0,021706						10,45548					0,010352					10,5589	0,990205594	
412																		0,653451			0,008548							138,8773			0,708687						140,248	0,990226694	
421																																					1,583208	1	
423														0,060261									0,060261							0,781964							0,962747	0,812222111	
511	0,018477																														0,368056							0,386533	0,95219885
512																																13,77229						13,79077	0,998660211
521																																	0,064337					0,064337	1
522										0,018477																			0,120522		0,018477					0,337412	0,494887	0,68179565	
523																																				0,958938	0,958938	1	
Total	1,36299	75,54057	12,2481	1,067399	0,437291	1,938035	2,327503	0,548255	1,095473	2,557082	16,75983	300,5517	0,293189	431,876	2,85217	23,87184	22,91275	81,99964	14,26761	57,92794	79,30259	56,00129	0,764588	1,483445	17,72758	0,056981	13,59897	139,5006	1,864608	0,860702	0,386533	14,49133	0,064337	0,337412	0,958938	1379,835			
Prod_accuracy	0,986444	0,919058	0,936286	0,975884	0,933333	0,833333	0,962748	0,936509	0,867424	0,925248	0,991289	0,963583	1	0,944595	0,896945	0,680005	0,982638	0,907505	0,876377	0,8192	0,943644	0,707994	0,789587	0,805028	0,805286	1	0,768843	0,995532	0,849083	0,908519	0,952199	0,950381	1	1	1		0,92715885		

Overall accuracy = 92.7%

Zone 11: Romania confusion matrix CLC2018 - Blind and plausibility analysis

Table 39: Confusion matrix CLC 2018 for zone 11 (Romania) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	511	512	521	523	Total	User_accuracy	
111	0,386562																																	0,386562	1	
112		41,76859	0,005322														0,924659	0,924659		0,803698														44,42693	0,94016379	
121		0,005322	5,578714														0,145233																	5,729269	0,973721721	
122				0,465197																														0,465197	1	
123					0,107208																													0,107208	1	
124						0,292919																												0,292919	1	
131			0,062472				1,06647										0,024586		0,803698	0,062472				0,062472							0,024586		2,106756	0,506214371		
132								0,107975																										0,107975	1	
133		0,165313	0,267344	0,029516				0,014758	0,014758											0,014758														0,506447	0,029140386	
141			0,012072							0,155242											0,014758													0,167314	0,927848115	
142	0,012072	0,021431								0,012072	0,206849											0,012072												0,264496	0,782049647	
211		0,924659										331,0659					0,92974	17,1233	2,173926	0,768476													352,986	0,937900922		
212		0,012051											0,012051																					0,024101	0,5	
213														1,319254																				1,319254	1	
221												0,662866			4,756745		0,662866																	6,082477	0,78204079	
222																7,991897				2,305429														10,29733	0,776113787	
231		2,173926										0,662866					104,494	2,173926	2,173926	9,474285	1,09397				2,20201									124,4489	0,839653823	
242		0,116684																37,64959	3,132132															40,8984	0,92056376	
243																	3,227085	25,22921		0,803698		1,09397											30,35396	0,831166933		
311												0,768476						7,327019		200,6066	1,104074	3,066495											212,8726	0,942378388		
312																				4,147797	36,3611	2,487698		0,037611									38,88641	0,935059349		
313																				4,147797		45,3776		0,008781									49,53418	0,91608667		
321																			1,248459	1,250025		4,314192		9,205578	9,987669			4,993835					30,99976	0,139168569		
322																	0,312111							2,184779									2,496891	0,875		
324													0,009359					0,803698	5,346831	2,187939	0,299758			21,28602									29,93361	0,711107792		
331																				0,020848					0,449337	0,028084							0,498269	0,901796119		
332																				0,034234	0,017117			0,034234	0,034234		0,275873	0,017117					0,412807	0,668285445		
333																											0,145939							0,145939	1	
411														0,697515															11,16023					11,85775	0,941176471	
421																													0,281568						0,281568	
511																									0,38344					6,901925				7,285365	0,947368421	
512																																	6,877523		6,877523	1
521																																	2,83367		2,83367	1
523																																		0,089409	0,089409	1
Total	0,398634	45,18797	5,925923	0,494713	0,107208	0,292919	1,06647	0,122734	0,014758	0,167314	0,206849	333,1601	0,012051	1,319254	4,756745	8,001255	111,1057	65,19849	34,31659	225,6375	42,33841	52,32553	4,314192	2,281485	33,15767	10,43701	0,303957	0,163055	16,43564	6,901925	6,90211	2,83367	0,089409	1015,977		
Prod_accuracy	0,969716	0,92433	0,941408	0,940337	1	1	1	0,879755	1	0,927848	1	0,993714	1	1	1	0,99883	0,940492	0,577461	0,73519	0,889066	0,858821	0,867217	1	0,957613	0,641964	0,043052	0,907607	0,895025	0,679027	1	0,996438	1	1		0,887355764	

Overall accuracy = 88.7%

Table 40: Confusion matrix CLC 2018 for zone 11 (Romania) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	511	512	521	523	Total	User_accuracy	
111	0,386562																																	0,386562	1	
112		42,57228	0,005322														0,924659	0,924659																44,42693	0,958254118	
121		0,005322	5,578714														0,145233																	5,729269	0,973721721	
122				0,465197																														0,465197	1	
123					0,107208																													0,107208	1	
124						0,292919																												0,292919	1	
131			0,062472				1,128942										0,024586		0,803698					0,062472								0,024586		2,106756	0,535867441	
132								0,107975																										0,107975	1	
133		0,02008	0,267344	0,029516				0,014758	0,014758								0,145233			0,014758														0,506447	0,029140386	
141			0,012072							0,155242																								0,167314	0,927848115	
142	0,012072	0,021431								0,012072	0,218921																							0,264496	0,827691326	
211		0,924659										345,7199					0,92974	2,469266	2,173926	0,768476													352,986	0,979415422		
212													0,024101																					0,024101	1	
213														1,319254																				1,319254	1	
221												0,662866			4,756745		0,662866																	6,082477	0,78204079	
222																7,991897				2,305429														10,29733	0,776113787	
231																	111,6434		0,803698	8,705809	1,09397				2,20201									124,4489	0,897102478	
242		0,116684																38,31245	2,469266															40,8984	0,936771384	
243																		0,662866	25,37004															30,35396	0,835806582	
311																		7,327019		203,549	1,104074	0,892568												212,8726	0,956200751	
312																					36,3611	2,487698			0,037611									38,88641	0,935059349	
313																						49,51662			0,017562									49,53418	0,999645456	
321																				1,248459	1,250025		4,276581		9,243189	9,987669			4,993835					30,99976	0,13795531	
322																					0,312111			2,184779										2,496891	0,875	
324													0,009359					0,803698	5,346831	3,281909	0,299758				20,19205									29,93361	0,674561252	
331																				0,020848						0,449337	0,028084							0,498269	0,901796119	
332																				0,034234	0,017117			0,034234	0,034234		0,275873	0,017117						0,412807	0,668285445	
333																												0,145939							0,145939	1
411																	0,697515																	11,16023	0,941176471	
421																																			0,281568	
511																									0,38344									7,285365	0,947368421	
512																																			6,877523	1
521																																			2,83367	1
523																																			0,089409	0,089409
Total	0,398634	43,66046	5,925923	0,494713	0,107208	0,292919	1,128942	0,122734	0,014758	0,167314	0,218921	346,3828	0,024101	1,319254	4,756745	8,001255	118,4004	49,69626	32,42432	221,9938	43,42031	54,29062	4,276581	2,281485	32,1101	10,43701	0,303957	0,163055	16,43564	6,901925	6,90211	2,83367	0,089409	1015,977		
Prod_accuracy	0,969716	0,975076	0,941408	0,940337	1	1	1	0,879755	1	0,927848	1	0,998086	1	1	1	0,99883	0,942932	0,770932	0,782438	0,916913	0,837422	0,912066	1	0,957613	0,628838	0,043052	0,907607	0,895025	0,679027	1	0,996438	1	1		0,916339964	

Overall accuracy = 91.6%

Zone 12: Greece confusion matrix CLC2018 - Blind and plausibility analysis

Table 41: Confusion matrix CLC 2018 for zone 12 (Greece) –BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	311	312	313	321	322	323	324	331	332	333	334	411	421	422	511	512	521	522	523	Total	User_accuracy
111	0,939956	0,072304																																							1,01226	0,928571429
112		11,86502																																							11,86502	1
121		0,138024	3,236431																							0,149313															3,523767	0,918457523
122				1,242854																0,058488																				1,301342	0,955055689	
123					0,034025																																			0,034025	1	
124						0,637332																																		0,637332	1	
131			0,096214				1,455349																																	1,551563	0,937988818	
132								0,100919																																0,100919	1	
133		0,020284		0,076656					0,534347																															0,631287	0,84644034	
141										0,058869																														0,058869	1	
142		0,120975									0,810866																													0,0863	1,076886	0,752973271
211			1,829517									47,14874									1,869909	3,410494					0,058744													56,08818	0,84061815	
212													36,01956								0,405403	0,009574					1,829517					0,019623								36,45416	0,988078195	
213														2,699039																										2,699039	1	
221															1,621611		0,405403				0,810805	0,405403																		3,243221	0,5	
222		0,244333										0,244333				7,831483					0,488666																				9,053149	0,865056246
223																	26,68225				1,034989	2,669558					4,804462													35,19126	0,758206733	
231		0,171001																4,289621			0,171001					1,466685		1,7746	0,171001												8,043908	0,533275693
241												0,193865							0,872391	0,096932																				1,163188	0,75	
242			1,153507									1,842111	1,139989				1,153507	1,034989	0,018203		30,77457	0,096932																		37,21381	0,826966431	
243												1,829517									41,92074																				48,73892	0,86010796
311																								54,70669			1,580977														56,21957	0,973089783
312																																									43,00344	0,970721327
313													0,765979																												25,13671	0,803965629
321												0,018994																													43,47557	0,725046434
322																																									3,270262	0,779362778
323																																									110,0276	0,922479418
324																																									46,03686	0,922757984
331																																									0,994367	1
332																																									0,920855	1
333																																									12,33696	0,688623128
334																																									0,548326	0,149469126
411													0,065762																												0,880282	0,850589887
421																																									1,509275	1
422																																									0,120125	1
511																																									0,887492	0,84999737
512																																									4,295982	1
521																																									0,361479	1
522																																									0,054417	1
523																																									1,553804	1
Total	0,939956	12,63194	6,315669	1,31951	0,034025	0,637332	1,455349	0,100919	0,534347	0,058869	0,810866	51,27756	38,23562	2,699039	1,621611	8,984989	31,59926	4,478825	0,872391	35,59125	54,33694	57,24016	46,3368	21,79003	34,87945	2,54872	121,0762	51,24298	1,175161	0,920855	9,922313	0,081958	0,768382	1,536484	0,120125	0,953253	4,314185	0,361479	0,054417	1,640105	611,4993	
Prod_accuracy	1	0,939287	0,512445	0,941905	1	1	1	1	1	1	1	0,919481	0,942042	1	1	0,871618	0,844395	0,957756	1	0,864667	0,771496	0,95574	0,90089	0,927445	0,903736	1	0,8383	0,829009	0,846154	1	0,856203	1	0,974462	0,982292	1	0,931014	0,995781	1	1	0,947381		0,875732304

Overall accuracy = 87.6%

Table 42: Confusion matrix CLC 2018 for zone 12 (Greece) –PLAUSIBILITY ANALYSIS - In column the validation and line the production

MAP2018	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	411	421	422	511	512	521	522	523	Total	User_accuracy					
111	0,939956	0,072304																																									1,01226	0,928571429				
112		11,86502																																									11,86502	1				
121		0,138024	3,236431																																								3,523767	0,918457523				
122				1,242854																0,058488							0,149313																	1,301342	0,955055689			
123					0,034025																																							0,034025	1			
124						0,637332																																						0,637332	1			
131			0,096214				1,455349																																					1,551563	0,937988818			
132								0,100919																																				0,100919	1			
133		0,020284		0,076656					0,534347																																			0,631287	0,84644034			
141										0,058869																																		0,058869	1			
142		0,120975									0,810866																																	0,0863	1,076886	0,752973271		
211												47,82476																																	56,08818	0,852670792		
212													36,22226																																36,45416	0,99363864		
213														2,699039																															2,699039	1		
221															1,824312		0,405403																												3,243221	0,5625		
222												0,244333	0,244333			8,075816	0,488666																												9,053149	0,892044997		
223																26,68225																													35,19126	0,758206733		
231																		4,289621																											8,043908	0,533275693		
241												0,193865							0,872391	0,096932																									1,163188	0,75		
242			1,153507									1,842111	1,139989			1,153507	1,034989	0,018203																											37,21381	0,826966431		
243																	0,060341																												48,73892	0,897645053		
311																	0,019623																												56,21957	0,973089783		
312																							54,70669																						43,00344	0,970721327		
313													0,765979											1,247028	41,74436																					25,13671	0,853575442	
321												0,018994												0,039422		21,45608																					43,47557	0,725046434
322																																															3,270262	0,905441191
323																	2,630679																												110,0276	0,946388675		
324																																															0,309232	2,96103
331																																															104,1289	0,946388675
332																																															0,309232	2,96103
333																																															0,309232	2,96103
334																																															0,309232	2,96103
411																																															0,309232	2,96103
421																																															0,309232	2,96103
422														</																																		

Overall accuracy = 88.8%

Zone 13: Bulgaria confusion matrix CLC2018- Blind and plausibility analysis

Table 43: Confusion matrix CLC 2018 for zone 13 (Bulgaria) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	412	422	511	512	521	523	Total	User_accuracy				
111	0,034319																																			0,034319	1			
112		16,53504		0,007629																																16,54267	0,999538845			
121			3,094942									0,007629									0,161459															3,26403	0,94819664			
122			0,012134	0,157746								0,012134																								0,182014	0,866666667			
123					0,032255																															0,032255	1			
124						0,241416																														0,241416	1			
131							0,913215	0,005304										0,077412			0,077412			0,077412												1,150754	0,793579276			
132								0,141614																												0,141614	1			
133									0,038081																											0,038081	1			
141										0,160007	0,017779																									0,177786	0,9			
142		0,007629									0,504572								0,027448					0,032752												0,5724	0,881502631			
211												159,0416							0,862169	0,007087				0,007593												163,5685	0,972324435			
213													0,817024																							0,817024	1			
221														5,128373			0,282611	0,282611						0,007629												5,701224	0,899521283			
222															1,936908			0,007629						0,116684												2,06122	0,939689949			
231												0,007087					14,43771	0,678467					0,339233	1,0177												16,4802	0,876064169			
242												0,574736					0,574736	10,58159	1,226883																		14,68215	0,720711145		
243												0,017006						41,22999																			45,69261	0,902333954		
311																			97,40642						0,010312												97,42189	0,999841226		
312																				21,88108	0,582234																22,46331	0,974080664		
313																				0,558045	26,80161																27,35966	0,979603375		
321																		0,855082	0,855082			14,69447		0,465388													16,87002	0,871040384		
322																							0,976416														0,976416	1		
324												0,058579						0,619652	0,072286	0,007859	1,96562			27,9064													30,6304	0,911068854		
331												0,009024														0,07219											0,009024	0,090238	0,8	
332																					0,026647	0,053293	0,026647	0,133233		0,239819	0,026647										0,506285	0,473684211		
333																		0,086786				0,260359		0,260359				1,086657									1,694162	0,641412869		
411																												0,363618										0,363618	1	
412																													0,055519									0,055519	1	
422																														0,033373								0,033373	1	
511																		0,264063	0,264063																		1,320314	0,6		
512																		0,137858																				2,801134	0,950784822	
521																																						2,663276		
523																																						0,030999	0,030999	1
Total	0,034319	16,54267	3,107076	0,165374	0,032255	0,241416	0,913215	0,146918	0,038081	0,160007	0,522351	159,7278	0,817024	5,128373	1,936908	18,54577	13,20594	45,46644	99,48747	22,45407	33,56491	15,00812	1,342296	30,03546	0,07219	0,239819	1,113304	0,363618	0,055519	0,033373	0,792188	2,663276	0,030999	0,052227	474,0408					
Prod_accuracy	1	0,999539	0,996095	0,95387	1	1	1	0,963898	1	1	0,965964	0,995704	1	1	1	0,778491	0,801275	0,906823	0,979082	0,974482	0,798501	0,979101	0,727422	0,929115	1	1	0,976065	1	1	1	1	1	1	0,827221		0,949440835				

Overall accuracy = 94.9%

Table 44: Confusion matrix CLC 2018 for zone 13 (Bulgaria) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

MAP2018	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	412	422	511	512	521	523	Total	User_accuracy	
111	0,034319																																		0,034319	1	
112		16,52741		0,007629																				0,007629											16,54267	0,999077689	
121			3,094942									0,007629							0,161459																3,26403	0,94819664	
122			0,012134	0,157746								0,012134																							0,182014	0,866666667	
123					0,032255																														0,032255	1	
124						0,241416																													0,241416	1	
131							0,913215	0,005304										0,077412			0,077412			0,077412											1,150754	0,793579276	
132								0,141614																											0,141614	1	
133									0,038081																										0,038081	1	
141										0,160007	0,017779																								0,177786	0,9	
142		0,007629									0,504572								0,027448					0,032752											0,5724	0,881502631	
211												159,0325		0,011702	0,004491	3,533318	0,116684		0,007087			0,855082		0,007593											163,5685	0,972268767	
213													0,817024																						0,817024	1	
221							0,007629							5,128373			0,282611	0,282611																	5,701224	0,899521283	
222															1,936908			0,007629						0,116684											2,06122	0,939689949	
231												0,007087				14,77695			0,678467					1,0177											16,4802	0,896648469	
242												0,574736				0,574736	10,58159	1,801619				1,149471													14,68215	0,720711145	
243												0,005304		0,011702			2,225051	41,2224				2,22056		0,007593											45,69261	0,902167786	
311																			97,39856					0,005156		0,018171									97,42189	0,999760556	
312																				21,88108	0,582234														22,46331	0,974080664	
313																				0,558045	26,79646			0,005156											27,35966	0,979414922	
321																		0,855082	0,855082			14,69447		0,465388											16,87002	0,871040384	
322																							0,976416												0,976416	1	
324												0,058579						0,619652	0,072286		1,888208			27,99167											30,6304	0,913852719	
331																0,009024										0,07219									0,009024	0,090238	0,8
332																						0,026647	0,026647	0,026647	0,106586			0,319759							0,506285	0,631578947	
333																		0,086786					0,287006		0,260359			1,060011							1,694162	0,625684398	
411																												0,363618								0,363618	1
412																													0,055519							0,055519	1
422																														0,033373						0,033373	1
511																	0,264063	0,264063													0,792188				1,320314	0,6	
512																		0,137858																	2,801134	0,950784822	
521																																	0,030999			0,030999	1
523																																	0,043203			0,043203	1
Total	0,034319	16,53504	3,107076	0,165374	0,032255	0,241416	0,920843	0,146918	0,038081	0,160007	0,522351	159,698	0,817024	5,151776	1,941398	18,89402	13,20594	46,03358	98,78599	22,43912	32,74615	15,86321	1,003063	30,11469	0,07219	0,319759	1,060011	0,363618	0,055519	0,033373	0,792188	2,663276	0,030999	0,052227	474,0408		
Prod_accuracy	1	0,999539	0,996095	0,95387	1	1	0,991715	0,963898	1	1	0,965964	0,995833	1	0,995457	0,997687	0,782096	0,801275	0,895485	0,985955	0,975131	0,818309	0,926324	0,973435	0,929502	1	1	1	1	1	1	1	1	1	0,827221	0,950369986		

Overall accuracy = 95.03%

Zone 14: Island confusion matrix CLC2018 - Blind and plausibility analysis

Table 45: Confusion matrix CLC 2018 for zone 14 (Island) –BLIND ANALYSIS - In column the validation and line the production

CLC Classes	112	121	122	123	124	131	133	141	142	211	231	242	243	311	312	313	321	322	324	331	332	333	335	411	412	421	423	511	512	521	522	523	Total	User_accuracy
112	0,434877																															0,434877	1	
121		0,246524																0,01837				0,01837										0,283265	0,870295642	
122			0,012019																													0,012019	1	
123				0,029903																												0,029903	1	
124					0,118513																											0,118513	1	
131						0,049736																										0,049736	1	
133	0,025971						0,02608																									0,052051	0,501044546	
141								0,070456																								0,070456	1	
142									0,655897																							0,655897	1	
211										0,098413																						0,098413	1	
231											10,74798	0,002925																				10,7509	0,999727923	
242												0,188863	0,010938																			0,199801	0,945257773	
311														1,010877				0,067392							0,067392							1,14566	0,882352941	
312															0,116659																	0,116659	1	
313																0,169149		0,021144														0,190292	0,888888889	
321																	13,51783															13,51783	1	
322																		149,5998										0,034721				149,6345	0,999767962	
324											0,01304						0,206876	0,1313	1,135995													1,487211	0,763842388	
331																				14,96041	0,075577							0,095987				15,13198	0,988662135	
332																					102,8136								0,045036			102,8586	0,999562158	
333																				1,894309	1,969886	54,37007										58,23427	0,933643963	
335																							44,76702									44,76702	1	
411																								1,838871								1,838871	1	
412																									28,49133							28,49133	1	
421																										0,109774						0,109774	1	
423																											0,358474					0,358474	1	
511																				0,0266								0,348642				3,375241	0,99211918	
512																																6,637691	1	
521																															0,622728	1		
522																															0,119445	1		
523																															0,132759	1		
Total	0,460849	0,246524	0,012019	0,029903	0,118513	0,049736	0,02608	0,070456	0,655897	0,098413	10,76102	0,191788	0,010938	1,010877	0,116659	0,169149	13,72471	149,838	1,135995	16,88132	104,8591	54,38844	44,76702	1,838871	28,55872	0,109774	0,358474	3,47935	6,682727	0,622728	0,119445	0,132759	441,5262	
Prod_accuracy	0,943645	1	1	1	1	1	1	1	1	1	0,998788	0,984748		1	1	1	0,984927	0,99841	1	0,886211	0,980493	0,999662	1	1	0,99764	1	1	0,962433	0,993261	1	1	1	0,989296595	

Overall accuracy = 98.9%

Table 46: Confusion matrix CLC 2018 for zone 14 (Island) –PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classes	112	121	122	123	124	131	133	141	142	211	231	242	311	312	313	321	322	324	331	332	333	335	411	412	421	423	511	512	521	522	523	Total	User_accuracy
112	0,434877																														0,434877	1	
121		0,264894																			0,01837										0,283265	0,935147821	
122			0,012019																												0,012019	1	
123				0,029903																											0,029903	1	
124					0,118513																										0,118513	1	
131						0,049736																									0,049736	1	
133	0,025971						0,02608																								0,052051	0,501044546	
141								0,070456																							0,070456	1	
142									0,655897																						0,655897	1	
211										0,098413																					0,098413	1	
231											10,74798	0,002925																			10,7509	0,999727923	
242												0,199801																			0,199801	1	
311													1,010877				0,067392							0,067392							1,14566	0,882352941	
312														0,116659																	0,116659	1	
313															0,169149			0,021144													0,190292	0,888888889	
321																13,51783															13,51783	1	
322																	149,5998										0,034721				149,6345	0,999767962	
324											0,01304						0,206876	0,1313	1,135995												1,487211	0,763842388	
331																			14,92842	0,075577							0,127983				15,13198	0,986547684	
332																				102,8266							0,031996				102,8586	0,999688934	
333																			1,894309	1,969886	54,37007										58,23427	0,933643963	
335																						44,76702									44,76702	1	
411																							1,838871								1,838871	1	
412																								28,49133							28,49133	1	
421																									0,109774						0,109774	1	
423																										0,358474					0,358474	1	
511																			0,0266								3,348642				3,375241	0,99211918	
512																												6,637691				6,637691	1
521																													0,622728			0,622728	1
522																														0,119445		0,119445	1
523																															0,132759	0,132759	1
Total	0,460849	0,264894	0,012019	0,029903	0,118513	0,049736	0,02608	0,070456	0,655897	0,098413	10,76102	0,202726	1,010877	0,116659	0,169149	13,72471	149,8196	1,135995	16,84933	104,8721	54,38844	44,76702	1,838871	28,55872	0,109774	0,358474	3,511346	6,669687	0,622728	0,119445	0,132759	441,5262	
Prod_accuracy	0,943645	1	1	1	1	1	1	1	1	1	0,998788	0,985571	1	1	1	0,984927	0,998533	1	0,885995	0,980496	0,999662	1	1	0,99764	1	1	0,953663	0,995203	1	1	1	0,989320041	

Overall accuracy = 98.9%

Zone 15: Hungary confusion matrix CLC2018 - Blind and plausibility analysis

Table 47: Confusion matrix CLC 2018 for zone 15 (Hungary) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	324	333	411	412	511	512	Total	User_accuracy
111	0,073506																												0,073506	1
112		19,52914	0,005278																										19,53442	0,999729819
121		0,008833	3,470767															0,025123											3,504723	0,990311262
122				0,575703								0,005194						0,020103											0,601	0,957907867
123					0,01206																								0,01206	1
124			0,059096			0,236385																							0,295481	0,8
131							0,544362																						0,544362	1
132								0,280381																					0,280381	1
133								0,005194	0,332392						0,145233								0,012282						0,495101	0,671361146
141										0,213254																			0,213254	1
142											1,271268																		1,271268	1
211			0,011747				0,002903					194,5543		0,017429	0,141808	0,124422	0,006805	0,10454	4,129678										199,0936	0,977200005
213													0,348184																0,348184	1
221												0,116684		4,232062															4,348746	0,973168462
222														3,401843				0,038326											3,440169	0,988859241
231												0,954273				31,41776			0,580672				0,007564						32,96027	0,953200938
242		1,230959									0,678199	0,009587		0,163657			0,006805	12,29502	0,242382										14,62661	0,840592768
243			0,002903													0,605428		8,664183						0,697217					9,969731	0,869048821
311																			68,11169		1,247466				0,096176				69,45533	0,980654583
312																				3,94405									3,94405	1
313																			0,140482		6,112618		0,009319						6,262419	0,976079339
321												0,48205							0,003266			9,611126							10,09644	0,95193197
324												0,404546			0,051492			0,126816	0,773302		0,046379		23,81801						25,22055	0,944389206
333																								0,114331					0,114331	1
411																			0,161695				0,161695		2,611346				2,934736	0,889806111
412																						0,039771			0,039771	0,318169			0,397711	0,8
511			0,014061																								1,955482		1,969543	0,992860752
512																							0,008308					5,66416	5,672468	0,998535465
Total	0,073506	20,76893	3,563852	0,575703	0,01206	0,236385	0,547266	0,285575	0,332392	0,213254	1,949467	196,5266	0,348184	4,413148	3,740376	32,15441	12,30183	9,221473	73,90078	3,94405	7,406463	9,650897	24,81058	0,114331	2,651118	0,318169	1,955482	5,66416	417,6804	
Prod_accuracy	1	0,940305	0,973881	1	1	1	0,994695	0,981812	1	1	0,652111	0,989964	1	0,958967	0,909492	0,97709	0,999447	0,939566	0,921664	1	0,825309	0,995879	0,959994	1	0,984998	1	1	1		0,96656075

Overall accuracy = 96.6%

Table 48: Confusion matrix CLC 2018 for zone 15 (Hungary) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	213	221	222	231	242	243	311	312	313	321	324	333	411	412	511	512	Total	User_accuracy
111	0,073506																												0,073506	1
112		19,52914	0,005278																										19,53442	0,999729819
121			3,470767	0,008833														0,025123											3,504723	0,990311262
122				0,595806								0,005194																	0,601	0,99135749
123					0,01206																								0,01206	1
124			0,059096			0,236385																							0,295481	0,8
131							0,544362																						0,544362	1
132								0,280381																					0,280381	1
133								0,005194	0,489907																				0,495101	0,989508917
141										0,213254																			0,213254	1
142											1,271268																		1,271268	1
211			0,00344				0,002903					194,0741		0,017429	0,116684	0,705094	0,006805	0,037417	4,129678										199,0936	0,974788494
213													0,348184																0,348184	1
221														4,348746															4,348746	1
222															3,440169														3,440169	1
231												0,954273				31,41776			0,580672					0,007564					32,96027	0,953200938
242		1,230959									0,678199			0,163657		0,006805	12,30461	0,242382											14,62661	0,84124821
243			0,002903													1,1861	0,006805	8,076707											9,969731	0,810122808
311																			68,1263		1,255268				0,697217				69,45533	0,980864874
312																				3,94405									3,94405	1
313																			0,140482	0,003901	6,118036								6,262419	0,976944596
321																			0,003266			10,09318							10,09644	0,999676553
324				0,003266							0,067123			0,051492					0,849954					24,24871					25,22055	0,961466561
333																								0,114331					0,114331	1
411																			0,161695						0,161695				2,603039	0,886975353
412																0,039771									0,039771	0,318169			0,397711	0,8
511			0,014061																								1,955482		1,969543	0,992860752
512																												5,672468	5,672468	1
Total	0,073506	20,7601	3,555545	0,607905	0,01206	0,236385	0,547266	0,285575	0,489907	0,213254	1,949467	195,1007	0,348184	4,529832	3,608344	33,35553	12,31822	8,381628	73,99204	3,94795	7,373304	10,09318	25,18896	0,114331	2,64281	0,318169	1,955482	5,680775	417,6804	
Prod_accuracy	1	0,940706	0,976156	0,980097	1	1	0,994695	0,981812	1	1	0,652111	0,994738	1	0,960024	0,953393	0,941906	0,998895	0,96362	0,920725	0,999012	0,829755	1	0,962672	1	0,984951	1	1	0,998538		0,967057296

Overall accuracy = 96.7%

Zone 16: Portugal confusion matrix CLC2018 - Blind and plausibility analysis

Table 49: Confusion matrix CLC 2018 for zone 16 (Portugal) – BLIND ANALYSIS - In column the validation and line the production

CL Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User_accuracy					
111	2,578688	0,052946																																										2,631634	0,979880965					
112		8,557148																																										8,557148	1					
121			2,069475					0,006391				0,009725														0,018975																			2,104566	0,983326573				
122				0,451073																	0,021768				0,081786																				0,554627	0,813291056				
123					0,07552																																								0,07552	1				
124						0,267615																																							0,267615	1				
131			0,031642				0,56319																																						0,594832	0,946805844				
132								0,038345																																					0,044736	0,857142857				
133									0,210838																		0,018975																				0,229813	0,917433013		
141		0,01225	0,01225							0,085752											0,01225																									0,122503	0,7			
142		0,042703									0,670109																																			0,712812	0,940092457			
211												24,944							0,116684		1,673713	1,484981				0,74249																				28,96186	0,861270388			
212												0,116684	11,24151								0,566198	0,566198								0,072433					0,257828											12,82085	0,876814719			
213														2,213118									0,096626																								2,309744	0,958165893		
221													0,055939		7,206947	0,072433				0,946605	0,946605		0,072433																							9,30096	0,774860518			
222													0,116684			2,856661				0,644498	0,171196								0,072433																	3,861472	0,739785592			
223												0,055939			0,350051	0,132411	15,71302									0,072433			0,026326	0,784083																	17,27054	0,909816585		
231												0,072433						13,92558		0,84792	0,171196																										16,598	0,838991251		
241		1,387141																		4,854992	2,774281	0,009725				1,387141				0,69357																	11,80042	0,411425378		
242																					24,84384																										27,55307	0,901672243		
243																		1,732416												0,778452																34,08773	0,854994038			
244															0,143855																																31,39843	0,969989432		
311																					30,45614																										56,66088	0,788554964		
312																					6,219006	44,68022				0,031701																						25,81134	0,939688363	
313																					1,027866					24,25462																						24,11261	0,913287402	
321													0,031701																																			2,171922	0,724304352	
322													0,051492																																			23,64008	0,852251178	
323																																																	11,28944	0,805187876
324																																																	79,62725	0,88127885
331		0,027321																			0,466368	1,432729	0,09035	2,237444		0,077693			0,460775	70,17381		1,407237																0,409818	0,866666667	
332																																																	0,114655	0,666666667
333																																																	2,467499	0,775264595
334																																																	3,264575	0,571428571
411																																																	0,071346	1
412																																																	0,151048	1
421																																																	0,713832	1
422																																																		

Overall accuracy = 86%

Table 50: Confusion matrix CLC 2018 for zone 16 (Portugal) – PLAUSIBILITY ANALYSIS - In column the validation and line the production

[illegible]

Overall accuracy = 86.3%

Zone 17: Austria/Switzerland confusion matrix CLC2018 - Blind and plausibility analysis

Table 51: Confusion matrix CLC 2018 for zone 17 (Austria/Switzerland) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	141	142	211	221	222	231	242	243	311	312	313	321	322	324	331	332	333	335	411	412	511	512	Total	User_accuracy
111	0,596104																													0,596104	1
112		24,15739	0,004154							1,288562					0,006405			1,288562												26,74507	0,903246338
121		0,131867	2,261225							0,011336																				2,404428	0,940441723
122	0,013678	0,013678	0,034321	0,319063																										0,380739	0,838010149
123					0,017963																									0,017963	1
124						0,254938																								0,254938	1
131							0,431726						0,025061	0,009259	0,025061															0,491108	0,879086192
132													0,019717																	0,019717	
133				0,006405																										0,006405	
141								0,208918	0,067053	0,013057						0,013057														0,302085	0,691585298
142						0,027937			1,295331										0,004693											1,327961	0,975428849
211		0,013678		0,013678					0,067053	63,65674			3,461954	6,814179							0,056513									74,0838	0,859253256
221											3,488172																			3,488172	1
222										0,024128		0,168899		0,024128																0,217156	0,777777778
231													35,40622	3,405336			5,193318	0,61275			0,072433								44,69005	0,792261699	
242														22,63336	0,61275		0,003508	0,847595											24,09721	0,939252183	
243													1,702668		12,10271															13,80538	0,87666634
311																22,36016		1,456271												23,81643	0,938854355
312													2,430662				100,0845	3,683196			0,099306								106,2976	0,941549361	
313																	8,593538	60,62548	0,019162										69,23818	0,875607639	
321																	0,110157	0,009543	49,53073	0,767059	0,060009					0,050213			50,52771	0,980268642	
322																				12,95573										12,95573	1
324																0,003957	0,072168	1,320713			8,195076								9,591913	0,854373438	
331																						0,01693								0,01693	1
332																0,485424			1,348392				26,10769		0,132819				28,07433	0,92994899	
333																			0,003508	3,500277	0,131362		1,088777	18,76883					23,49276	0,798919959	
335																									5,879403					5,879403	1
411																	0,050213									0,853623				0,903836	0,944444444
412																	0,014485										0,043455			0,05794	0,75
511																												1,099601		1,099601	1
512																										0,003508			7,882281	7,885789	0,99955519
Total	0,609782	24,31661	2,299699	0,339146	0,017963	0,282874	0,431726	0,208918	1,429437	64,99383	3,488172	0,168899	43,04628	32,88626	12,74693	22,86259	114,1218	69,84411	50,90649	17,22307	8,614697	0,01693	27,19647	18,76883	6,012222	0,907343	0,043455	1,099601	7,882281	532,7665	
Prod_accuracy	0,97757	0,993452	0,98327	0,940784	1	0,90124	1	1	0,906183	0,979428	1	1	0,822515	0,688231	0,949461	0,978024	0,876997	0,868011	0,972975	0,752231	0,95129	1	0,959966	1	0,977908	0,940793	1	1	1		0,90358956

Overall accuracy = 90.3%

Table 52: Confusion matrix CLC 2018 for zone 17 (Austria / Switzerland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	141	142	211	221	222	231	241	242	243	311	312	313	321	322	324	331	332	333	335	411	412	511	512	Total	User_accuracy	
111	0,596104																														0,596104	1	
112		25,44595	0,004154							1,288562						0,006405																26,74507	0,951425771
121		0,131867	2,261225	0,009259						0,002077																						2,404428	0,940441723
122	0,013678	0,013678	0,034321	0,319063																												0,380739	0,838010149
123					0,017963																											0,017963	1
124						0,254938																										0,254938	1
131							0,431726						0,025061		0,009259	0,025061																0,491108	0,879086192
132												0,019717																				0,019717	
133				0,006405																												0,006405	
141								0,208918	0,067053	0,013057							0,013057															0,302085	0,691585298
142						0,027937			1,295331												0,004693											1,327961	0,975428849
211		0,013678	0,006405	0,013678					0,067053	63,65034			3,461954		6,814179							0,056513										74,0838	0,859166797
221											3,488172																					3,488172	1
222												0,193028		0,024128																		0,217156	0,888888889
231													36,01897		3,405336	0,072433	5,114961					0,078357										44,69005	0,805972805
242															22,63336	0,61275	0,003508	0,847595														24,09721	0,939252183
243													1,702668			12,10271																13,80538	0,87666634
311																	22,36016		1,456271													23,81643	0,938854355
312													2,430662				98,811	4,950904				0,105061										106,2976	0,929569197
313																	6,614456	62,61796				0,005755										69,23818	0,904384957
321																	0,183426	0,009543	49,37402	0,688702	0,221809						0,050213					50,52771	0,977167112
322																					12,95573											12,95573	1
324																		0,053005	1,320713			8,218195										9,591913	0,856783703
331																							0,01693									0,01693	1
332																0,485424				1,348392				26,10769		0,132819						28,07433	0,92994899
333																				0,003508	3,500277	0,131362		1,088777	18,76883						23,49276	0,798919959	
335																										5,879403						5,879403	1
411																		0,050213										0,853623				0,903836	0,944444444
412																		0,014485											0,043455			0,05794	0,75
511																														1,099601		1,099601	1
512																																7,885789	0,99955519
Total	0,609782	25,60517	2,306104	0,348405	0,017963	0,282874	0,431726	0,208918	1,429437	64,95404	3,488172	0,193028	43,65903	0,024128	32,86213	12,81936	22,85864	110,8451	71,20299	50,73061	17,14471	8,81705	0,01693	27,19647	18,76883	6,012222	0,907343	0,043455	1,099601	7,882281	532,7665		
Prod_accuracy	0,97757	0,993782	0,980539	0,915781	1	0,90124	1	1	0,906183	0,979929	1	1	0,825006		0,688737	0,944096	0,978193	0,891434	0,879429	0,973259	0,755669	0,93208	1	0,959966	1	0,977908	0,940793	1	1	1		0,908290434	

Overall accuracy = 90.8%

Zone 18: Denmark/Netherland/Belgium /Luxembourg confusion matrix CLC2018 - Blind and plausibility analysis

Table 53: Confusion matrix CLC 2018 for zone 18 (Denmark, Netherland, Belgium, Luxembourg) – BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	331	411	412	421	423	511	512	521	522	523	Total	User_accuracy	
111	0,460203	0,025567																																0,48577	0,947368421	
112		40,44692	0,072887	2,256526							0,072887	2,268052								0,228801													45,34607	0,891960803		
121		0,290675	7,587624																															7,878298	0,963104395	
122				1,125164																														1,125164	1	
123					1,064262																													1,064262	1	
124						0,847207																												0,847207	1	
131							0,67125																											0,707603	0,948625103	
132								0,134559																							0,036353			0,134559	1	
133			0,010044						1,103542	0,145233								0,088183																1,347001	0,819258126	
141										1,321766	0,063648																							1,385414	0,954058395	
142											9,963599																				0,036353			9,999952	0,99636468	
211		7,1374	0,01169							3,565778		158,918				3,565778			0,025587	0,012794			0,025587											173,2626	0,917208828	
221													0,064006																					0,064006	1	
222												0,122249		0,670077			0,04075																	0,833076	0,80434086	
231										3,086813					55,72019	3,086813																		61,89382	0,900254573	
242		1,020937	0,029569									4,083748				3,062811	45,94725																	54,14431	0,848607093	
243		1,572047	0,072887								0,046454	0,006595				1,176669	23,67205		0,025587						1,572047				0,020732					28,16507	0,840475498	
311							0,006382											17,36008		0,056927			0,228801											17,65219	0,983451888	
312		0,822427																	19,67161	0,436801														20,93083	0,939838603	
313		0,874879																0,437439	2,739337	19,92771			0,006595											23,98596	0,830807257	
321			0,024657									0,207224				0,397682	0,051492	0,056927				2,296202			0,155732	0,219165					0,051492				3,460572	0,663532412
322			0,018275	0,010044																			4,081829	0,078353										4,417303	0,92405474	
324		0,031393		0,031393																			4,954814											5,483968	0,903508995	
331																								0,95676										0,95676	1	
411		0,143608														0,066832																			3,119378	0,872762829
412																	0,012794	0,156941								1,371546								1,635871	0,838419582	
421																																			1,624186	1
423																											1,624186								0,211596	0,866666667
511																											0,014106	0,183383	0,014106						2,269656	0,804809166
512																													0,673567	12,4993			0,673567		13,84643	0,902708985
521																																0,167662			0,167662	1
522																																	0,201809		0,201809	1
523																																		0,340171	0,340171	1
Total	0,460203	52,36585	7,827632	3,423126	1,064262	0,847207	0,677632	0,134559	1,103542	5,032776	13,2334	165,6059	0,064006	0,670077	61,11935	52,65133	23,78252	18,04264	22,46212	20,66303	2,296202	4,1603	5,31027	1,112492	4,513689	1,371546	1,638293	0,183383	2,700079	12,64492	0,167662	1,318392	0,340171	488,9885		
Prod_accuracy	1	0,772391	0,969338	0,328695	1	1	0,990582	1	1	0,262632	0,752913	0,959616	1	1	0,911662	0,87267	0,995355	0,962169	0,875768	0,964414	1	0,981138	0,933063	0,860015	0,60316	1	0,99139	1	0,676513	0,988484	1	0,153072	1		0,899619929	

Overall accuracy = 89.96%

Table 54: Confusion matrix CLC 2018 for zone 18 (Denmark, Netherland, Belgium, Luxembourg) –PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	331	411	412	421	423	511	512	521	522	523	Total	User_accuracy	
111	0,48577																																	0,48577	1	
112		40,55638	0,072887	2,256526							0,192227	0,011526			2,256526																			45,34607	0,894374703	
121			7,878298																															7,878298	1	
122				1,125164																														1,125164	1	
123					1,064262																													1,064262	1	
124						0,847207																												0,847207	1	
131							0,67125																											0,707603	0,948625103	
132								0,134559																										0,134559	1	
133			0,010044						1,103542	0,145233								0,088183																	1,347001	0,819258126
141										1,321766	0,063648																								1,385414	0,954058395
142											9,963599																							9,999952	0,99636468	
211		7,131555	0,02338									158,8914			0,020732	3,565778			0,025587	0,012794			0,025587											173,2626	0,917055435	
221													0,064006																					0,064006	1	
222												0,122249		0,670077			0,04075																	0,833076	0,80434086	
231											3,086813				55,72019	3,086813																		61,89382	0,900254573	
242		1,020937	0,029569									3,062811			3,062811	46,96818																		54,14431	0,867462941	
243		1,572047	0,072887							0,063648	0,046454	0,006595			1,176669	0,003767	23,60463		0,025587							1,572047			0,020732					28,16507	0,838081932	
311							0,006382											17,36008		0,056927			0,228801											17,65219	0,983451888	
312		0,822427																	19,77179	0,025587			0,311026											20,93083	0,944625199	
313		0,874879	0,012233															0,437439	0,469622	22,1658			0,025983											23,98596	0,924115785	
321			0,024657												0,418414	0,051492					2,664593				0,106316		0,143608			0,051492				3,460572	0,769986255	
322			0,018275	0,010044											0,228801							4,138756	0,021426											4,417303	0,936942005	
324		0,031393		0,031393								0,466368			0,466368								4,954814											5,483968	0,903508995	
331																								0,95676										0,95676	1	
411		0,143608													0,066832											2,722477			0,165034	0,021426					3,119378	0,872762829
412																	0,012794	0,156941					0,016119			1,450017								1,635871	0,886388308	
421																											1,624186								1,624186	1
423																											0,014106	0,183383	0,014106						0,211596	0,866666667
511																												1,82664			0,443016				2,269656	0,804809166
512																													13,17286			0,673567			13,84643	0,951354493
521																															0,167662				0,167662	1
522																																0,201809			0,201809	1
523																																	0,340171	0,340171		1
Total	0,48577	52,15322	8,142229	3,423126	1,064262	0,847207	0,677632	0,134559	1,103542	5,096424	13,35274	162,0946	0,064006	0,670077	63,41734	53,67603	23,65818	18,04264	20,29259	22,26111	2,664593	4,138756	5,583758	0,95676	4,400841	1,450017	1,781901	0,183383	2,026513	13,31849	0,167662	1,318392	0,340171	488,9885		
Prod_accuracy	1	0,777639	0,967585	0,328695	1	1	0,990582	1	1	0,259352	0,746184	0,980239	1	1	0,878627	0,875031	0,997737	0,962169	0,974336	0,995719	1	1	0,887362	1	0,618627	1	0,911491	1	0,901371	0,989066	1	0,153072	1	0,909575755		

Overall accuracy = 90.95%

Zone 19: Albania/Serbia/Macedonia confusion matrix CLC2018 - Blind and plausibility analysis

Table 55: Confusion matrix CLC 2018 for zone 19 (Albania/Serbia/Macedonia) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	511	512	521	523	Total	User_accuracy	
111	0,048159																																								0,048159	1
112		17,35936																	0,074402								0,005614														17,43937	0,995411747
121			1,877352																		0,005614		0,083252				0,005614														2,006321	0,93571854
122				0,153003																							0,040103														0,153003	1
123					0,013351																																			0,013351	1	
124						0,14539						0,015278	0,015278																											0,175946	0,826331577	
131							0,991042	0,038139													0,038139																			1,06732	0,928532602	
132							0,014101	0,150092																																0,164193	0,91412088	
133									0,046237																															0,046237	1	
141										0,176481																														0,176481	1	
142		0,015278				0,015278					0,230087																													0,260644	0,882765923	
211		1,253714					0,008486					112,9997						0,021324	0,212267	0,025066																				114,5206	0,986719789	
212													2,296311						0,072782																					2,369093	0,969278397	
213														0,349847																										0,349847	1	
221												0,161716								0,11519				0,11519																1,936089	0,755717204	
222															1,463136	0,080858				0,159037																				1,868723	0,914895468	
223																1,709686				0,318074																				2,968375	0,892845918	
231								0,012032										12,33054		0,159037	0,021324		1,098448	2,196896			1,234902													17,05317	0,723063968	
241																			0,00535																					0,00535		
242												0,021324							63,98107	7,100101																				71,10249	0,899842818	
243															0,021324			0,198442	2,264536	64,54594	2,897191	0,129136				0,008486													70,06506	0,921228723		
311																	1,223208			0,466368	177,4677		1,517632				4,053612													184,7286	0,960694654	
312																					14,06754						0,596834													14,66437	0,959300376	
313																					0,872384	13,25532																		14,1277	0,938250107	
321																		1,072941		1,072941	0,212267						2,66981	0,058661												36,17644	0,859394131	
322																					0,068054	0,148038	1,709342				0,204509														2,129943	0,802529673
323																				1,706739	2,255927		0,136454	0,136454	8,192802		4,27596			1,072941										17,77728	0,460858061	
324							0,012032											1,456733	0,011896	2,96228	3,007677		3,106748	0,148092			60,93417												71,63962	0,850565136		
331																											1,269365	0,859211												2,128576	0,403655463	
332																					0,097577	0,048789					0,097577	1,336403	0,146366										1,726711	0,77395872		
333																					1,235306	0,298417	1,223208				0,159967			7,551536										10,46843	0,721362504	
334																			0,466368				0,084921				0,343193				1,111666									2,006147	0,554129702	
335																																									0,058661	
411							0,008486					0,116468															0,066832		0,058661				1,615054								1,80684	0,893855135
412																																	0,001524								0,001524	1
421																																		0,094313							0,094313	1
422																																				0,086378					0,086378	1
511																								0,012032				0,070607												3,090012	0,97325601	
512																																									6,486278	1
521																																									0,220983	1
523																																									0,277344	1
Total	0,048159	18,62835	1,877352	0,153003	0,013351	0,160668	1,034147	0,200263	0,046237	0,176481	0,245366	113,3145	2,296311	0,349847	1,484459	1,790544	3,87351	15,07997	66,61695	79,10249	185,9059	15,06906	19,12945	35,26433	2,193002	9,430111	75,94683	0,929819	1,453725	8,770843	1,111666	1,615054	0,001524	0,094313	0,086378	3,007373	6,486278	0,220983	0,277344	673,4859		
Prod_accuracy	1	0,931878	1	1	1	0,904909	0,958318	0,749474	1	1	0,937733	0,997222	1	1	0,985635	0,954842	0,684212	0,817676	0,960432	0,815979	0,954611	0,933538	0,692927	0,881622	0,779453	0,868792	0,802327	0,924063	0,919296	0,860982	1	1	1	1	1	1	1	1	1	1	0,91001439	

Overall accuracy = 91%

Table 56: Confusion matrix CLC 2018 for zone 19 (Albania/Serbia/Macedonia) – PLAUSIBILITY ANALYSIS In column the validation and line the production

UC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	242	243	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	511	512	521	523	Total	User_accuracy				
111	0,048159																																								0,048159	1			
112		17,36497																	0,074402																							17,43937	0,995733684		
121			1,960604																									0,040103														2,006321	0,977213328		
122				0,153003																			0,005614																			0,153003	1		
123					0,013351																																					0,013351	1		
124						0,14539																																				0,175946	0,826331577		
131							0,991042	0,038139														0,038139																				1,06732	0,928532602		
132								0,164193																																		0,164193	1		
133									0,046237																																	0,046237	1		
141										0,176481																																0,176481	1		
142						0,015278					0,245366																															0,260644	0,941382962		
211		1,253714					0,008486					111,8717							0,021324		1,153029							0,212267														114,5206	0,97687035		
212													2,296311							0,072782																						2,369093	0,969278397		
213														0,349847																												0,349847	1		
221												0,080858			1,463136	0,080858		0,11519	0,080858	0,11519																						1,936089	0,755717204		
222																1,709686			0,159037	0,159037																						1,868723	0,914895468		
223																	2,650302		0,318074	0,318074																						2,968375	0,892845918		
231								0,012032										12,33054	0,159037	0,159037	0,021324		1,098448	2,196896				1,234902														17,05317	0,723063968		
241																			0,00535																							0,00535			
242											0,021324								7,100101																							71,10249	0,899842818		
243															0,021324			0,198442	2,264536	64,54594	1,448595	0,129136				0,008486	1,448595															70,06506	0,921228723		
311																	1,223208		0,466368	174,4604		1,517632																					184,7286	0,944414987	
312																					14,06754							0,596834															14,66437	0,959300376	
313																					0,872384		13,25532																				14,1277	0,938250107	
321																			1,072941	0,212267				31,30181		0,76044	2,770321		0,058661														36,17644	0,865254019	
322																							0,068054	0,068054	1,709342		0,204509																2,129943	0,802529673	
323																					0,136454	0,136454			8,192802				0,079983														17,77728	0,460858061	
324							0,012032											1,456733	0,011896	2,833145	3,001729		3,106748																				71,63962	0,854517914	
331																																											2,128576	0,403655463	
332																								0,097577		0,048789																	1,726711	0,8587242	
333																								0,837044	0,298417																		10,46843	0,774687494	
334																		0,466368									0,428114				1,11666												2,006147	0,554129702	
335																																											0,058661		
411						0,008486																						0,066832		0,058661														1,80684	0,958314851
412																																											0,001524	1	
421																																											0,094313	1	
422																																												0,086378	1
511																								0,012032				0,070607																3,090012	0,97325601
512																																												1	
521																																												6,486278	0,220983
523																																												0,220983	1
Total	0,048159	18,61869	1,960604	0,153003	0,013351	0,160668	1,020046	0,214364	0,046237	0,176481	0,275922	111,9739	2,296311	0,349847	1,484459	1,790544	3,87351	14,12222	66,48554	80,03769	181,4384	15,06906	19,05182	34,64987	2,144214	8,961728	82,1617	0,929819	1,60009	9,360267	1,111666	1,731522	0,001524	0,094313	0,086378	3,007373	6,486278	0,220983	0,277344	673,4859	0,906004197				
Prod_accuracy	1	0,932664	1	1	1	0,904909	0,971565	0,765953	1	1	0,889257	0,999087	1	1	0,985635	0,954842	0,684212	0,87313	0,962331	0,806444	0,961541	0,933538	0,695751	0,903375	0,797188	0,914199	0,745084	0,924063	0,926678	0,866403	1	1	1	1	1	1	1	1	1	1	1				

Overall accuracy = 90.6%

Zone 20: Bosnia and Herzegovina/Croatia confusion matrix CLC2018 - Blind and plausibility analysis

Table 57: Confusion matrix CLC 2018 for zone 20 (Bosnia Herzegovina/Croatia) – BLIND ANALYSIS -In column the validation and line the production

LC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	223	231	242	243	311	312	313	321	322	323	324	332	333	334	411	421	422	511	512	521	523	Total	User_accuracy	
111	0,040137																																				0,040137	1	
112	0,034655	10,21952	0,007087									0,007087								0,010009		0,532022																10,81038	0,945343182
121		0,113272	1,086053																																			1,199325	0,905553694
122				0,657515		0,064204	0,030534		0,010009																0,034655													0,796918	0,825072235
123					0,02249																																	0,02249	1
124						0,128409																																0,128409	1
131		0,030534					0,484994												0,007087								0,030534											0,55315	0,876785578
132								0,114379																														0,114379	1
133									0,093752																													0,093752	1
141										0,096999																												0,096999	1
142		0,078769									0,285701	0,019692									0,076656																	0,460819	0,619985344
211												42,96536						0,027815		4,217524										0,009272								47,21998	0,909898063
212													0,306375																									0,306375	1
221														1,199111				0,409297								0,007086												1,615494	0,742256473
222															0,43915			0,045179	0,035907			0,035907				0,035907											0,59205	0,741745107	
223		0,093011														0,806951			0,046505																			0,946467	0,852592446
231																	25,04824		0,013503							1,559402	1,576194											31,31614	0,799850731
242			0,007087									8,438373						58,38501		0,030534																		71,04799	0,821768582
243																		0,035907	45,56521	0,030534																	47,73635	0,954517974	
311																			146,4475						0,430616	6,304199	3,154311										162,6677	0,900286724	
312																																						26,45138	0,918790884
313																		2,45681																			50,27951	0,853410857	
321																																						22,36524	0,727674834
322																																						1,935186	0,85
323																		0,466368	0,430616																		17,63414	0,842444614	
324							0,030534												0,102983	4,732596	2,163621		0,935414				33,0409										41,00605	0,805756725	
331																			0,017564			0,017564															0,035127		
332																																						1,062698	1
333																																						6,010655	0,595167364
334																																						0,23182	1
411																				0,111924																		0,906945	0,890148593
412																																						0,029473	
421																																						0,030706	
422																																						0,04168	1
423																																						0,000798	
511																																						1,825995	0,992605304
512																																						2,223559	1
521																																						0,001875	1
523																																						0,18065	1
Total	0,074792	10,5351	1,100227	0,657515	0,02249	0,192613	0,546063	0,114379	0,103761	0,096999	0,285701	51,43052	0,306375	1,199111	0,43915	0,806951	25,07605	61,80566	50,42981	151,6805	32,89606	59,82532	17,21004	2,075525	28,64815	40,97655	1,333947	4,79205	0,23182	0,976396	0,000798	0,04168	1,812493	2,223559	0,001875	0,18065	550,1307		
Prod_accuracy	0,536644	0,970044	0,987117	1	1	0,666667	0,888165	1	0,90354	1	1	0,835406	1	1	1	1	0,998891	0,944655	0,903537	0,9655	0,73879	0,717239	0,945647	0,792526	0,51856	0,806337	0,796657	0,746517	1	0,92887		1	1	1	1	1	1	0,867903857	

Overall accuracy = 86.8%

Table 58: Confusion matrix CLC 2018 for zone 20 (Bosnia Herzegovina/Croatia) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	223	231	242	243	311	312	313	321	322	323	324	332	333	334	411	421	422	511	512	521	523	Total	User_accuracy	
111	0,040137																																				0,040137	1	
112		10,21952	0,007087	0,034655								0,007087								0,010009		0,532022															10,81038	0,945343182	
121		0,113272	1,086053																																		1,199325	0,905553694	
122				0,657515		0,064204			0,010009									0,030534							0,034655												0,796918	0,825072235	
123					0,02249																																0,02249	1	
124						0,128409																															0,128409	1	
131		0,030534					0,484994											0,007087								0,030534											0,55315	0,876785578	
132								0,114379																													0,114379	1	
133									0,093752																												0,093752	1	
141									0,096999																												0,096999	1	
142		0,078769									0,285701	0,019692								0,076656																	0,460819	0,619985344	
211												42,96536					0,027815		4,217524											0,009272							47,21998	0,909898063	
212													0,306375																								0,306375	1	
221														1,301436			0,102324	0,204649							0,007086												1,615494	0,805595794	
222															0,475057			0,081086						0,035907													0,59205	0,802393698	
223		0,046505														0,853456		0,046505																			0,946467	0,901728297	
231												0,013503					25,04824									1,559402	1,576194										31,31614	0,799850731	
242			0,007087									4,220849						61,3343	1,161763			4,217524				0,034655											71,04799	0,863279845	
243																		46,65347				0,030534				1,052352										47,73635	0,97731526		
311																			149,5515			6,331				3,166681	3,618512										162,6677	0,91936816	
312																					24,84031	1,61107															26,45138	0,939093163	
313																					4,91362	42,90908															50,27951	0,853410857	
321																						1,214705		16,27462		3,651199	0,010009		1,214705								22,36524	0,727674834	
322																						0,290278			1,644908												1,935186	0,85	
323																		0,466368	0,430616						14,85579	1,881369										17,63414	0,842444614		
324																		0,102983	2,861768	2,163621	1,870828	0,935414				33,07143										41,00605	0,806501358		
331																		0,017564				0,017564															0,035127		
332																											1,062698											1,062698	1
333																				0,271249					1,077062	1,244365	0,271249	3,146729									6,010655	0,52352518	
334																															0,23182							0,23182	1
411											0,040508								0,111924																			1,01887	0,850390768
412																																						0,029473	
421																																						0,030706	
422																																						0,04168	1
423																																						0,000798	
511																																						0,000798	
512																																						1,812493	0,992605304
521																																						2,223559	1
523																																						0,001875	1
Total	0,040137	10,4886	1,100227	0,692171	0,02249	0,192613	0,484994	0,114379	0,103761	0,096999	0,285701	47,267	0,306375	1,301436	0,546872	0,853456	25,17838	64,58083	52,63042	152,8831	33,42254	60,67433	17,21004	1,644908	25,40422	41,48058	1,333947	4,361434	0,23182	0,935888	0,000798	0,04168	1,812493	2,223559	0,001875	0,18065	550,1307		
Prod_accuracy	1	0,974345	0,987117	0,949932	1	0,666667	1	1	0,90354	1	1	0,908993	1	1	0,868682	1	0,994831	0,949729	0,886435	0,978208	0,74322	0,707203	0,945647	1	0,584776	0,797275	0,796657	0,72149	1	0,925792		1	1	1	1	1	1	0,881396386	

Overall accuracy = 88.13%

Zone 21: Czech Republic/Slovakia confusion matrix CLC2018 - Blind and plausibility analysis

Table 59: Confusion matrix CLC 2018 for zone 21 (Czech Republic/Slovakia) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	332	333	411	412	511	512	Total	User_accuracy	
111	0,098557																													0,098557	1	
112		26,38881	0,007676									0,007676																		26,40416	0,999418576	
121		0,008227	4,016369	0,008939																										4,033536	0,995743985	
122					0,634909																									0,634909	1	
123						0,013726																								0,013726	1	
124							0,30641																							0,30641	1	
131				0,046244				0,836331	0,020718										0,046244	0,008939										0,046244	0,832401616	
132									0,359594																					0,359594	1	
133				0,025319						0,147004			0,00798																		0,180303	0,815317475
141											0,317046																				0,317046	1
142												1,275266	0,097228																		1,372493	0,929159814
211			0,00798					0,046244				0,007388	192,1328			0,00754								0,116684							192,3186	0,999033711
221													0,086156	1,550806	0,237325																1,874287	0,827411065
222			0,060174										0,13543		1,63626																1,831864	0,893221349
231		0,016431		0,029569								0,210316			53,11679		0,016687		0,895307	0,077202										54,36231	0,977088685	
242															0,00798	3,890897														3,898876	0,997953318	
243												0,887893					45,40661	0,094938		0,059079			0,024571							46,47309	0,977051632	
311																		58,77608		0,338621				0,024571						59,13927	0,993858697	
312															0,11453				84,21198					0,840695						85,16721	0,988784123	
313															0,005189				2,015601	46,46092				0,011718						48,49343	0,958086944	
321																					1,803369	0,111923	0,059099							1,97439	0,913380071	
322																						1,029097								1,029097	1	
324																		0,513476	0,862552	0,426471	0,111923		26,40241				0,007414		28,32424	0,93214872		
332																						0,103951		0,155926						0,259877	0,6	
333																		0,022006					0,004921		0,198055					0,224982	0,88031586	
411																							0,020897			0,35525			0,020897	0,397044	0,894736842	
412																							0,020079				0,180715			0,200795	0,9	
511																	0,064501											0,580506		0,645007	0,9	
512																														3,214614	3,214614	1
Total	0,098557	26,48162	4,095608	0,673418	0,013726	0,30641	0,882575	0,380312	0,147004	0,317046	1,282653	193,5654	1,550806	1,873585	53,25203	3,890897	45,4878	59,45274	87,99438	47,36229	1,915292	1,244971	27,52564	0,155926	0,198055	0,35525	0,188129	0,580506	3,281755	564,5544		
Prod_accuracy	1	0,996495	0,980653	0,942816	1	1	0,947603	0,945524	1	1	0,99424	0,992598	1	0,873331	0,99746	1	0,998215	0,988618	0,957015	0,980969	0,941564	0,826603	0,959193	1	1	1	0,96059	1	0,979541		0,983956617	

Overall accuracy = 98.4%

Table 60: Confusion matrix CLC 2018 for zone 21 (Czech Republic/Slovakia) –PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	221	222	231	242	243	311	312	313	321	322	324	332	333	411	412	511	512	Total	User_accuracy	
111	0,098557																													0,098557	1	
112		26,38881	0,007676									0,007676																		26,40416	0,999418576	
121			4,024596	0,008939																										4,033536	0,997783714	
122				0,634909																										0,634909	1	
123					0,013726																									0,013726	1	
124						0,30641																								0,30641	1	
131		0,046244					0,836331	0,020718										0,046244	0,008939											0,046244	1,00472	0,832401616
132								0,359594																						0,359594	1	
133			0,025319						0,147004			0,00798																		0,180303	0,815317475	
141										0,317046																				0,317046	1	
142		0,00798									1,267286	0,097228																		1,372493	0,923345754	
211		0,023883					0,046244				0,007388	192,102			0,00754								0,116684						0,014828	192,3186	0,998873916	
221													1,636962	0,237325																1,874287	0,873378346	
222		0,060174										0,13543		1,63626																1,831864	0,893221349	
231												0,093633			53,29616			0,007414	0,887893	0,077202									54,36231	0,980388222		
242															0,00798	3,890897														3,898876	0,997953318	
243															0,887893		45,40661	0,08367		0,059079			0,035839						46,47309	0,977051632		
311																		58,80015		0,314548			0,024571						59,13927	0,994265748		
312															0,11453				84,18718	0,059099			0,806393						85,16721	0,988492972		
313															0,005189				1,604926	46,87159			0,011718						48,49343	0,966555612		
321																					1,907877		0,066513						1,97439	0,966312237		
322																						1,029097							1,029097	1		
324																		0,513476	0,548003				27,25535				0,007414		28,32424	0,962262243		
332																						0,103951		0,155926					0,259877	0,6		
333																		0,022006					0,004921		0,198055				0,224982	0,88031586		
411																							0,020897			0,35525			0,397044	0,894736842		
412																							0,020079				0,180715		0,200795	0,9		
511																												0,645007		0,645007	1	
512																													3,214614	3,214614	1	
Total	0,098557	26,52709	4,057591	0,643849	0,013726	0,30641	0,882575	0,380312	0,147004	0,317046	1,274674	192,444	1,636962	1,873585	54,3193	3,890897	45,40661	59,47296	87,23695	47,38152	1,907877	1,133048	28,36296	0,155926	0,198055	0,35525	0,188129	0,645007	3,296584	564,5544		
Prod_accuracy	1	0,994787	0,991868	0,986116	1	1	0,947603	0,945524	1	1	0,994204	0,998223	1	0,873331	0,981164	1	1	0,988687	0,96504	0,989238	1	0,908256	0,960949	1	1	1	0,96059	1	0,975135		0,986909291	

Overall accuracy = 98.7%

Zone 22: Estonia/Latvia/Lithuania confusion matrix CLC2018 - Blind and plausibility analysis

Table 61: Confusion matrix CLC 2018 for zone 22 (Estonia/Latvia/Lithuania) – BLIND ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	141	142	211	222	231	242	243	311	312	313	321	322	324	331	333	411	412	421	511	512	521	523	Total	User_accuracy	
111	0,055973																														0,055973	1	
112		11,88188												0,012799																	11,89468	0,998923959	
121			3,176702																												3,176702	1	
122				0,354121																											0,354121	1	
123					0,133922																										0,133922	1	
124						0,328481																									0,328481	1	
131							0,945545										0,033781														0,979326	0,965505992	
132								0,157174										0,018667													0,17584	0,893842765	
133														0,018196																	0,018196		
141									1,20418																						1,20418	1	
142										0,60936								0,035208													0,644568	0,94537701	
211											173,0576		3,494709				0,133594														176,6859	0,979464671	
222												0,592611	0,031729																		0,62434	0,949180691	
231				0,029569							0,779489		67,47022								0,116684										68,39596	0,986464965	
242														57,58115																	57,58115	1	
243		0,008039											0,015937		41,47102						0,012049										41,50705	0,999132049	
311																62,74379		3,329148			0,079575										66,15252	0,948471764	
312		0,018196															92,2725	0,021175													92,31187	0,9995735	
313																0,013735		115,0503			0,208406										115,2724	0,998072903	
321				0,00784															4,92338												4,931219	0,998410166	
322																				0,472359				0,027786							0,500145	0,944444444	
324										0,063541							0,252129	3,678983			90,12174										94,11639	0,957556248	
331																						0,311894									0,311894	1	
333																0,042156			0,042156				0,550679							0,042156	0,677147	0,813235162	
411																								5,709729							5,709729	1	
412																									12,23667	0,014214					12,25088	0,998839797	
511																															1,428387	1	
512																	0,15871														21,50677	0,992620478	
521																															0,016718	0,016718	1
523																															0,180254	0,180254	1
Total	0,055973	11,90812	3,176702	0,39153	0,133922	0,328481	0,945545	0,157174	1,20418	0,60936	173,9006	0,592611	71,01259	57,61215	41,47102	62,79969	92,88593	122,0983	4,965535	0,472359	90,53845	0,311894	0,550679	5,737514	12,23667	0,014214	1,428387	21,34806	0,016718	0,22241	779,1267		
Prod_accuracy	1	0,997797	1	0,904454	1	1	1	1	1	1	0,995152	1	0,950116	0,999462	1	0,99911	0,993396	0,942276	0,99151	1	0,995397	1	1	0,995157	1		1	1	1	0,810459		0,983647906	

Overall accuracy = 98.4%

Table 62: Confusion matrix CLC 2018 for zone 22 (Estonia/Latvia/Lithuania) – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	141	142	211	222	231	242	243	311	312	313	321	322	324	331	333	411	412	421	511	512	521	523	Total	User_accuracy
111	0,055973																														0,055973	1
112		11,89468																													11,89468	1
121			3,158036					0,018667																							3,176702	0,994123869
122				0,354121																											0,354121	1
123					0,133922																										0,133922	1
124						0,328481																									0,328481	1
131							0,979326																								0,979326	1
132								0,17584																							0,17584	1
133		0,018196																													0,018196	
141									1,20418																						1,20418	1
142									0,035208	0,60936																					0,644568	0,94537701
211											169,9182		6,76769																		176,6859	0,961696488
222												0,62434																			0,62434	1
231											0,794174		67,52583								0,075956									68,39596	0,987278038	
242														57,58115																	57,58115	1
243		0,008039											0,015937		41,47102						0,012049										41,50705	0,999132049
311																62,74379		3,329148			0,079575										66,15252	0,948471764
312		0,018196															91,85569	0,021175			0,416811										92,31187	0,995058248
313																		109,3099			5,96256										115,2724	0,94827419
321				0,00784															4,92338												4,931219	0,998410166
322																			0,027786	0,472359										0,500145	0,944444444	
324										0,012049			0,051492				0,015937	3,678983			90,35793										94,11639	0,960065816
331																						0,311894									0,311894	1
333																0,042156			0,042156		0,042156		0,550679								0,677147	0,813235162
411																								5,709729							5,709729	1
412																									12,23667	0,014214					12,25088	0,998839797
511																											1,428387				1,428387	1
512																0,15871															21,50677	0,992620478
521																													0,016718		0,016718	1
523																														0,180254	0,180254	1
Total	0,055973	11,93911	3,158036	0,36196	0,133922	0,328481	0,979326	0,194507	1,239388	0,60936	170,7244	0,62434	74,36095	57,58115	41,47102	62,78595	92,03034	116,3392	4,993321	0,472359	96,94704	0,311894	0,550679	5,709729	12,23667	0,014214	1,428387	21,34806	0,016718	0,180254	779,1267	
Prod_accuracy	1	0,996278	1	0,978341	1	1	1	0,904031	0,971592	1	0,995278	1	0,908082	1	1	0,999329	0,998102	0,939579	0,985993	1	0,932034	1	1	1	1	1	1	1	1		0,97219084	

Overall accuracy = 97.2%

Zone 23: Overseas departments confusion matrix CLC2018 - Blind and plausibility analysis

Table 63: Confusion matrix CLC 2018 for zone 23 (Guyana/Guadeloupe/La reunion/Martinique/Mayotte) – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	141	142	211	213	221	222	231	242	243	244	311	312	313	321	322	323	324	331	332	333	411	421	423	511	512	521	522	523	Total	User_accuracy				
111	0,01789																																				0,01789	1				
112		2,916862															0,005092																				2,921954	0,998257283				
121			0,161577												0,005092			0,006825																			0,166669	0,969447525				
122				0,074629																																		0,081454	0,916208237			
123					0,023021																																	0,023021	1			
124						0,064595																																0,064595	1			
131							0,044961																															0,044961	1			
132								0,003158																														0,003158	1			
141									0,019232																													0,019232	1			
142										0,031181																												0,031181	1			
211											2,38941																											2,527055	0,94553126			
213												0,277662																										0,277662	1			
222														0,577819			0,071374	0,071374																				0,720568	0,801894381			
231			0,110781													0,899972							0,004496				0,117606							0,003449					1,136305	0,792016677		
242			0,005092	0,16235													1,298797																						1,466239	0,885801843		
243																0,003449	0,16235	3,107423																					3,273221	0,94934696		
244																			0,092179																				0,092179	1		
311			0,165748																	40,52414						1,330481													48,37311	0,837741095		
312																				0,093109																			0,093109	1		
313																					0,23056																		0,23056	1		
321																						1,006519																	1,006519	1		
322																							0,679496																0,754996	0,9		
323																									0,750142															1,103425	0,679830082	
324																0,004496	0,170244	0,004496																						1,450848	0,873474357	
331																																								0,011514	1	
332																																								0,311411	1	
333																																								0,376002	1	
411																																									4,676151	1
421																																									0,031668	1
511																																									0,455841	0,9
512																																									0,410257	1
521																																									0,018621	1
522																																									0,007951	1
523																																									1,164754	0,901171947
Total	0,01789	3,364232	0,323926	0,079721	0,023021	0,064595	0,044961	0,003158	0,019232	0,031181	2,38941	0,277662	0,132553	0,577819	0,913009	1,702765	3,19521	0,092179	40,64226	0,093109	0,23056	1,125902	0,726496	0,750142	2,884633	0,057098	0,311411	0,376002	4,676151	0,031668	0,003055	0,528816	0,018621	0,007951	1,049643	56,2315	122,9975	0,922307277				
Prod_accuracy	1	0,867022	0,498807	0,936125	1	1	1	1	1	1	1	1	1	1	0,985721	0,762758	0,972525	1	0,997094	1	1	0,893967	0,935306	1	0,43932	0,201649	1	1	1	1	1	1	1	1	1	0,889123	0,922307277					

Overall accuracy = 92.2%

Table 64: Confusion matrix CLC 2018 for zone 23 (Guyana/Guadeloupe/La reunion/Martinique/Mayotte) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	141	142	211	213	221	222	231	242	243	244	311	312	313	321	322	323	324	331	332	333	411	421	423	511	512	521	522	523	Total	User_accuracy			
111	0,01789																																					0,01789	1		
112		2,916862															0,005092																					2,921954	0,998257283		
121			0,161577												0,005092																							0,166669	0,969447525		
122				0,074629													0,006825																					0,081454	0,916208237		
123					0,023021																																	0,023021	1		
124						0,064595																																0,064595	1		
131							0,044961																															0,044961	1		
132								0,003158																														0,003158	1		
141									0,019232																													0,019232	1		
142										0,031181																												0,031181	1		
211											2,38941																											2,527055	0,94553126		
213												0,277662																										0,277662	1		
222														0,720568																								0,720568	1		
231		0,110781														0,899972						0,004496				0,117606							0,003449					1,136305	0,792016677		
242		0,005092	0,16235														1,298797																					1,466239	0,885801843		
243																0,003449	0,16235	3,107423																				3,273221	0,94934696		
244																			0,092179																			0,092179	1		
311		0,165748																		40,52414			0,114887			1,330481						0,003055					6,234796	48,37311	0,837741095		
312																					0,093109																	0,093109	1		
313																						0,23056																0,23056	1		
321																						1,006519																1,006519	1		
322																							0,679496		0,0755													0,754996	0,9		
323		0,165748																	0,093768					0,750142	0,093768													1,103425	0,679830082		
324															0,004496	0,170244	0,004496		0,004333						1,267278													1,450848	0,873474357		
331																										0,011514													0,011514	1	
332																											0,311411												0,311411	1	
333																								0,047				0,376002											0,423002	0,888888889	
411																													4,676151										4,676151	1	
421																														0,031668									0,031668	1	
511																										0,045584							0,410257						0,455841	0,9	
512																																		0,018621					0,018621	1	
521																																			0,007951					0,007951	1
522																																							1,164754	0,901171947	
523																																							49,9967	50,01673	0,999599669
Total	0,01789	3,364232	0,323926	0,079721	0,023021	0,064595	0,044961	0,003158	0,019232	0,031181	2,38941	0,277662	0,132553	0,720568	0,913009	1,631391	3,123836	0,092179	40,64226	0,093109	0,23056	1,125902	0,726496	0,750142	2,884633	0,057098	0,311411	0,376002	4,676151	0,031668	0,003055	0,528816	0,018621	0,007951	1,049643	56,2315	122,9975				
Prod_accuracy	1	0,867022	0,498807	0,936125	1	1	1	1	1	1	1	1	1	1	0,985721	0,796129	0,994746	1	0,997094	1	1	0,893967	0,935306	1	0,43932	0,201649	1	1	1	1	0,775802	1	1	1	0,889123		0,923467858				

Overall accuracy = 92.3%

Annex 4. Producer and user accuracy per bio-geographical region – Blind and plausibility analysis

Table 65: Producer, user and overall accuracy for Blind and plausibility analysis per bio-geographical region (6 regions from ALP to BOR region)

	ALP				ANA				ARC				ATL				BLS				BOR			
	Alpine				Anatolian				Arctic				Atlantic				BlackSea				Boreal			
	Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis	
	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
111	89,96%	100,00%	96,94%	100,00%	100,00%	69,20%	100,00%	69,20%					82,81%	95,40%	99,43%	96,17%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
112	89,84%	92,70%	95,66%	95,66%	93,41%	99,23%	93,41%	99,23%	94,36%	100,00%	94,36%	100,00%	91,60%	95,03%	91,69%	95,38%	99,71%	100,00%	99,71%	100,00%	88,84%	98,04%	88,87%	98,09%
121	99,47%	97,11%	99,47%	96,79%	82,40%	94,04%	80,95%	94,04%	100,00%	87,03%	100,00%	93,51%	96,89%	93,54%	96,72%	94,10%	95,43%	100,00%	95,43%	100,00%	90,17%	95,28%	89,98%	95,07%
122	100,00%	86,76%	93,77%	86,76%	100,00%	43,19%	100,00%	43,19%	100,00%	100,00%	100,00%	100,00%	60,30%	87,99%	61,15%	91,18%	98,48%	98,75%	98,48%	98,75%	93,44%	99,03%	95,80%	100,00%
123	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	92,28%	100,00%	96,07%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
124	71,93%	100,00%	71,93%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	92,59%	100,00%	92,59%	100,00%	100,00%	100,00%	100,00%	100,00%	83,55%	100,00%	83,55%	100,00%
131	92,80%	85,44%	94,39%	85,44%	96,44%	100,00%	96,44%	100,00%	100,00%	100,00%	100,00%	100,00%	96,72%	98,65%	98,15%	98,65%	100,00%	88,26%	100,00%	78,12%	96,64%	43,18%	95,50%	45,20%
132	95,61%	82,42%	96,35%	100,00%	100,00%	51,12%	100,00%	51,12%					100,00%	97,56%	96,24%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	91,58%	98,18%	89,01%
133	100,00%	100,00%	100,00%	100,00%	100,00%	80,68%	100,00%	80,68%	100,00%	50,10%	100,00%	50,10%	47,02%	82,50%	48,59%	82,50%	100,00%	95,58%	100,00%	95,58%	41,91%	92,11%	43,03%	92,11%
141	100,00%	100,00%	100,00%	100,00%	80,00%	100,00%	80,00%	100,00%	100,00%	100,00%	100,00%	100,00%	57,76%	93,72%	57,64%	93,72%	71,11%	100,00%	71,11%	100,00%	99,83%	95,94%	98,42%	96,59%
142	99,70%	98,05%	99,56%	98,43%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	88,63%	97,67%	87,63%	97,67%	100,00%	85,01%	100,00%	85,01%	99,46%	99,05%	98,93%	99,05%
211	98,08%	88,82%	99,81%	88,84%	90,77%	92,21%	91,49%	88,66%	100,00%	100,00%	100,00%	100,00%	96,37%	97,04%	97,18%	97,01%	99,73%	83,91%	97,25%	83,91%	95,43%	93,81%	95,41%	92,92%
212	100,00%	100,00%	100,00%	100,00%	87,87%	96,94%	89,25%	96,96%					95,52%	89,10%	95,52%	89,10%	100,00%	50,00%	100,00%	50,00%				
213					96,56%	100,00%	96,56%	100,00%									100,00%	100,00%	100,00%	100,00%				
221	100,00%	90,12%	100,00%	90,12%	100,00%	9,77%	100,00%	9,77%					100,00%	100,00%	100,00%	100,00%	100,00%	30,75%	100,00%	30,75%				
222	94,56%	84,17%	94,56%	84,17%	97,18%	91,37%	95,95%	91,37%					100,00%	73,31%	100,00%	73,31%	62,62%	43,49%	62,13%	43,49%	100,00%	90,95%	100,00%	97,79%
223					100,00%	100,00%	3,00%	100,00%																
231	93,05%	84,78%	92,49%	86,30%	78,69%	84,39%	76,69%	84,27%	99,88%	99,97%	99,88%	99,97%	94,99%	91,98%	94,58%	92,76%	36,84%	93,37%	38,45%	100,00%	88,98%	92,78%	85,53%	92,85%
241					100,00%	100,00%	100,00%	100,00%						0,00%		0,00%								
242	89,79%	84,90%	89,79%	84,90%	98,01%	79,87%	90,59%	83,18%	98,47%	94,53%	98,56%	100,00%	94,56%	90,15%	95,25%	92,73%	46,13%	53,32%	44,04%	53,32%	99,95%	95,89%	100,00%	96,28%
243	89,06%	86,51%	88,38%	90,18%	86,31%	83,69%	85,79%	81,63%	0,00%				71,74%	77,53%	73,07%	77,52%	67,27%	99,87%	64,05%	87,49%	91,35%	90,62%	91,45%	90,60%
244					0,00%		0,00%										0,00%		0,00%					
311	97,23%	96,18%	97,72%	95,20%	96,20%	89,72%	96,20%	89,72%	100,00%	88,24%	100,00%	88,24%	96,49%	97,36%	98,20%	97,32%	85,11%	90,62%	92,10%	90,62%	99,08%	90,10%	99,49%	90,10%
312	92,02%	96,93%	92,09%	96,40%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	94,32%	95,98%	95,39%	97,69%	77,33%	100,00%	77,33%	100,00%	98,27%	95,31%	98,30%	95,58%
313	87,86%	94,03%	87,95%	93,45%	100,00%	91,26%	100,00%	91,26%	100,00%	88,89%	100,00%	88,89%	81,26%	90,17%	87,11%	91,18%	83,61%	84,62%	85,02%	94,14%	94,00%	95,23%	94,05%	94,28%
321	89,95%	93,40%	90,50%	93,35%	71,85%	93,85%	73,19%	95,14%	98,49%	100,00%	98,49%	100,00%	80,14%	75,36%	85,50%	75,72%	82,98%	100,00%	82,98%	100,00%	99,22%	99,85%	98,70%	99,85%
322	93,54%	94,74%	93,13%	95,41%	0,00%		0,00%		99,84%	99,98%	99,85%	99,98%	93,65%	87,05%	93,73%	88,16%	0,00%				72,93%	99,71%	73,01%	99,71%
323	100,00%	33,50%	100,00%	33,50%	98,76%	55,52%	98,76%	55,52%																
324	90,22%	90,39%	88,52%	90,42%	80,54%	91,04%	80,61%	91,04%	100,00%	76,38%	100,00%	76,38%	81,70%	89,97%	82,22%	92,54%	74,77%	78,95%	74,77%	78,95%	83,80%	96,14%	83,80%	96,21%
331	69,51%	78,92%	25,88%	78,92%	61,86%	66,40%	72,95%	66,40%	88,62%	98,87%	88,60%	98,86%	90,54%	97,81%	92,84%	98,71%	100,00%	98,21%	100,00%	98,21%	100,00%	90,27%	100,00%	90,27%
332	95,37%	96,45%	95,81%	97,06%	100,00%	74,86%	96,64%	74,86%	98,13%	99,96%	98,13%	99,97%	95,72%	92,74%	95,43%	92,37%	100,00%	100,00%	100,00%	100,00%	100,00%	55,13%	100,00%	55,13%
333	96,40%	91,09%	97,32%	91,29%	95,07%	73,97%	95,29%	75,37%	99,97%	93,36%	99,97%	93,36%	92,87%	90,13%	95,88%	90,11%	59,16%	66,70%	59,16%	66,70%	96,09%	89,89%	96,09%	89,89%
334	100,00%	45,76%	100,00%	45,76%									100,00%	56,83%	100,00%	56,83%					100,00%	50,00%	100,00%	50,00%
335	99,36%	99,72%	99,36%	99,72%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%												
411	72,90%	100,00%	72,90%	100,00%	94,82%	96,29%	94,82%	96,29%	100,00%	100,00%	100,00%	100,00%	76,17%	96,96%	76,78%	96,96%	94,67%	100,00%	94,67%	100,00%	99,62%	100,00%	99,95%	100,00%
412	98,29%	99,78%	96,61%	99,78%					99,76%	100,00%	99,76%	100,00%	99,58%	93,95%	99,60%	99,15%					99,94%	99,95%	99,95%	99,97%
421					100,00%	62,50%	100,00%	68,75%	100,00%	100,00%	100,00%	100,00%	93,67%	99,63%	90,91%	100,00%		0,00%		0,00%	78,30%	100,00%	75,13%	100,00%
422					100,00%	100,00%	100,00%	100,00%					100,00%	80,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%				
423	100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%	94,64%	86,56%	95,65%	86,80%					100,00%	100,00%	100,00%	100,00%
511	95,18%	96,56%	95,34%	100,00%	100,00%	77,10%	100,00%	77,10%	96,24%	99,21%	95,37%	99,21%	84,08%	93,25%	92,27%	93,25%	100,00%	83,37%	100,00%	81,13%	99,97%	94,37%	100,00%	95,80%
512	100,00%	99,98%	100,00%	99,98%	99,87%	100,00%	99,87%	100,00%	99,33%	100,00%	99,52%	100,00%	96,69%	96,64%	96,74%	98,30%	86,22%	95,55%	85,13%	97,28%	100,00%	98,83%	100,00%	98,83%
521					100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	86,85%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
522									100,00%	100,00%	100,00%	100,00%	64,82%	93,21%	65,07%	93,21%					100,00%	100,00%	100,00%	100,00%
523	100,00%	100,00%	100,00%	100,00%					100,00%	100,00%	100,00%	100,00%	99,21%	94,95%	100,00%	94,95%	97,39%	100,00%	97,39%	100,00%	98,45%	94,75%	100,00%	94,75%
Global Acc	0,938		0,939		0,865		0,866		0,989		0,989		0,931		0,940		0,846		0,840		0,953		0,953	

Table 66: Producer, user and overall accuracy for Blind and plausibility analysis per bio-geographical region (6 regions from CON to STE region)

CON				DOM				MAC				MED				PAN				STE			
Continental				Oversea department				Macaronesia				Mediterranean				Pannonian				Steppic			
Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis		Blind Analysis		Plausibility analysis	
Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User	Producer	User
83,47%	98,81%	83,47%	98,81%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	77,81%	97,45%	75,85%	100,00%	100,00%	100,00%	100,00%				
95,90%	98,02%	96,32%	98,79%	86,70%	99,83%	86,70%	99,83%	100,00%	100,00%	100,00%	100,00%	93,04%	99,61%	92,42%	98,79%	95,84%	99,98%	95,91%	99,98%	68,07%	100,00%	100,00%	100,00%
93,27%	86,64%	93,44%	88,38%	49,88%	96,94%	49,88%	96,94%	100,00%	100,00%	100,00%	100,00%	89,67%	97,96%	94,32%	98,04%	97,92%	99,23%	98,10%	99,23%	95,53%	100,00%	95,53%	100,00%
95,76%	98,04%	96,80%	98,04%	93,61%	91,62%	93,61%	91,62%					57,50%	91,99%	57,47%	91,99%	100,00%	97,04%	98,60%	99,39%	100,00%	100,00%	100,00%	100,00%
100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
99,01%	99,25%	99,01%	99,25%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	99,54%	96,15%	99,54%	96,15%	100,00%	88,74%	100,00%	88,74%	100,00%	100,00%	100,00%	100,00%
97,59%	84,25%	98,04%	84,83%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	96,83%	83,31%	97,13%	83,31%	99,50%	93,85%	99,50%	93,85%		0,00%		0,00%
88,94%	93,55%	85,86%	93,55%	100,00%	100,00%	100,00%	100,00%					49,98%	99,29%	49,72%	99,29%	88,76%	100,00%	88,76%	100,00%	100,00%	100,00%	100,00%	100,00%
100,00%	55,81%	21,90%	58,58%					100,00%	100,00%	100,00%	100,00%	68,85%	84,23%	73,21%	84,23%	100,00%	68,29%	100,00%	98,47%				
95,20%	96,84%	95,71%	96,84%	100,00%	100,00%	100,00%	100,00%		0,00%		0,00%	97,90%	87,51%	100,00%	87,51%	100,00%	100,00%	100,00%	100,00%				
97,80%	95,36%	95,66%	96,12%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	98,93%	100,00%	92,31%	92,56%	90,30%	92,56%	67,21%	94,97%	67,44%	96,01%	100,00%	100,00%	100,00%	100,00%
99,09%	96,39%	99,02%	97,08%	100,00%	94,55%	100,00%	94,55%	100,00%	100,00%		0,00%	97,01%	92,20%	97,51%	90,86%	99,31%	98,60%	99,62%	98,45%	100,00%	99,89%	100,00%	99,89%
25,92%	90,64%	5,65%	100,00%							0,00%		95,16%	97,92%	94,16%	98,01%	100,00%	100,00%	100,00%	100,00%				
100,00%	95,15%	100,00%	95,50%	100,00%	100,00%	100,00%	100,00%					99,74%	99,40%	99,74%	99,31%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
96,01%	80,10%	95,92%	80,10%	0,00%		0,00%						99,25%	82,35%	98,04%	82,42%	97,13%	95,07%	97,22%	98,21%	100,00%	80,00%	100,00%	80,00%
92,49%	84,68%	91,04%	84,77%	100,00%	80,19%	100,00%	100,00%					87,44%	88,95%	89,04%	89,12%	92,44%	96,31%	96,13%	97,20%	100,00%	100,00%	100,00%	100,00%
												95,87%	92,71%	94,62%	92,31%								
94,74%	88,49%	94,90%	88,06%	98,57%	79,20%	98,57%	79,20%	100,00%	83,51%	100,00%	83,51%	80,47%	85,73%	74,16%	85,68%	97,62%	95,72%	94,11%	95,72%	91,78%	66,96%	94,34%	100,00%
100,00%	94,71%	71,45%	94,71%									55,12%	52,81%	57,37%	52,81%								
72,84%	89,00%	76,88%	90,16%	76,28%	88,58%	79,61%	88,58%	0,00%				84,13%	83,80%	82,30%	83,73%	99,97%	80,00%	99,94%	80,03%	69,68%	100,00%	100,00%	100,00%
89,50%	91,12%	89,08%	91,27%	97,25%	94,93%	99,47%	94,93%	78,86%	100,00%	93,77%	100,00%	79,56%	90,93%	79,04%	91,95%	75,19%	90,02%	75,47%	85,80%	100,00%	100,00%	100,00%	100,00%
				100,00%	100,00%	100,00%	100,00%					79,24%	96,68%	81,32%	96,57%								
94,82%	93,45%	95,91%	93,04%	99,71%	83,77%	99,71%	83,77%	100,00%	100,00%	100,00%	100,00%	97,65%	93,98%	97,91%	94,72%	93,29%	96,92%	93,21%	96,91%	95,51%	100,00%	95,51%	100,00%
96,36%	97,95%	98,15%	97,70%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	95,19%	95,12%	95,49%	95,08%	100,00%	98,39%	99,89%	98,39%				
78,19%	93,65%	80,34%	96,74%	100,00%	100,00%	100,00%	100,00%					95,54%	87,81%	96,05%	90,32%	69,07%	97,61%	69,39%	97,70%				
84,79%	48,65%	84,35%	49,78%	89,40%	100,00%	89,40%	100,00%	100,00%	75,45%	33,73%	75,45%	85,07%	67,77%	85,62%	69,23%	98,41%	95,19%	100,00%	99,97%	100,00%	100,00%	100,00%	100,00%
97,82%	83,21%	98,73%	83,21%	93,53%	90,00%	93,53%	90,00%	18,58%	100,00%	5,45%	25,24%	81,05%	80,62%	82,07%	81,40%								
35,86%	85,71%	35,86%	85,71%	100,00%	67,98%	100,00%	67,98%	100,00%	100,00%	56,45%	100,00%	85,64%	87,79%	89,16%	88,77%								
88,58%	85,12%	82,93%	85,69%	43,93%	87,35%	43,93%	87,35%	70,15%	100,00%	70,15%	100,00%	78,45%	91,80%	78,39%	91,74%	96,45%	92,58%	96,60%	94,23%	100,00%	56,96%	100,00%	56,96%
12,67%	90,09%	12,67%	90,09%	20,16%	100,00%	20,16%	100,00%	100,00%	66,67%	100,00%	66,67%	97,24%	55,47%	97,24%	55,47%					100,00%	100,00%	100,00%	100,00%
65,49%	46,68%	65,49%	46,68%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%	99,48%	85,47%	99,48%	85,67%								
100,00%	24,96%	100,00%	30,54%	100,00%	88,89%	100,00%	88,89%	99,12%	87,34%	98,70%	58,93%	92,39%	72,95%	94,78%	73,15%	100,00%	100,00%	100,00%	100,00%				
												100,00%	48,58%	91,77%	53,37%								
45,92%	86,52%	46,15%	87,19%	100,00%	100,00%	100,00%	100,00%					40,06%	95,34%	40,70%	93,21%	98,91%	95,20%	98,91%	94,98%	100,00%	100,00%	100,00%	100,00%
99,57%	77,04%	99,59%	81,35%					100,00%	100,00%	66,67%	100,00%	0,00%				100,00%	80,00%	100,00%	80,00%				
85,21%	94,62%	85,21%	94,62%	100,00%	100,00%	100,00%	100,00%					95,47%	97,63%	91,71%	97,63%								
100,00%	34,86%	100,00%	34,86%									93,86%	90,68%	93,86%	90,68%								
33,52%	100,00%	100,00%	100,00%	0,00%		0,00%						100,00%	47,27%	100,00%	47,27%								
99,89%	92,19%	99,89%	92,19%	77,58%	90,00%	77,58%	90,00%					96,55%	83,93%	96,55%	83,93%	100,00%	99,52%	100,00%	99,52%	100,00%	100,00%	100,00%	100,00%
97,28%	98,57%	97,46%	98,67%	100,00%	100,00%	100,00%	100,00%					98,11%	98,95%	97,86%	98,64%	99,69%	99,88%	99,56%	100,00%	100,00%	100,00%	100,00%	100,00%
95,90%	91,88%	95,90%	91,88%	100,00%	100,00%	100,00%	100,00%					97,96%	98,09%	97,96%	98,09%								
100,00%	100,00%	100,00%	100,00%	100,00%	90,12%	100,00%	90,12%					100,00%	94,30%	100,00%	94,30%								
100,00%	93,62%	100,00%	100,00%	88,91%	99,96%	88,91%	99,96%	100,00%	100,00%	100,00%	100,00%	98,24%	56,81%	98,24%	56,81%								
0,932		0,936		0,922		0,923		0,940		0,968		0,895		0,896		0,967		0,968		0,965		0,991	

Annex 5. Confusion matrix CLC2018 per Bio-geographical region

Bio-geographical region ALP - Alpine - confusion matrix CLC2018 - Blind and plausibility analysis

Table 67: Confusion matrix CLC 2018 for Alpine region –BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	231	242	243	311	312	313	321	322	323	324	331	332	333	334	335	411	412	423	511	512	523	Total	User_accuracy	
111	0,43327																																				0,43327	1	
112	0,034655	40,34768	0,005322													0,924659	0,924659				1,288562																43,52554	0,926988638	
121		0,081697	2,748302																																		2,829999	0,971131782	
122	0,013678	0,013678	0,009259	0,660793		0,064204																															0,761612	0,867624223	
123					0,002969																																0,002969	1	
124						0,236109																															0,236109	1	
131							0,77784												0,046244	0,008939	0,077412																0,910435	0,854360458	
132							0,014101	0,262014											0,041777																		0,317892	0,82422415	
133									0,041708																												0,041708	1	
141										0,109639																											0,109639	1	
142		0,012072				0,027937					3,117614									0,012072		0,004693				0,005304											3,179692	0,980476854	
211		2,674756					0,046244					50,1516				1,787102	1,665568			0,012587						0,116684					0,009272						56,46382	0,888207837	
212													1,042596																								1,042596	1	
221												0,080858		2,902808	0,237325																						3,220991	0,901215744	
222												0,00754			4,126743				0,768476																		4,90276	0,84171837	
231		0,904037						0,012032								111,5382	3,147565	4,347852	7,175181	0,61275	2,196896					1,631834											131,5664	0,847771535	
242																0,582715	32,22153	0,077412	0,003508	5,065119																	37,95028	0,849045819	
243												0,887893				2,602248	1,072941	86,30905	2,697186	2,164436	4,003951					0,024571											99,76227	0,865147157	
311																499,2432			3,42694	8,376724		2,322866			0,029727	0,218673		3,691336				1,767275					519,0767	0,961790731	
312										0,004621						2,430662			391,3119	5,835852						4,142991										403,726	0,969251111		
313																0,005189			12,28259	214,7358	1,245406				0,020499								0,091662				228,3812	0,940251871	
321																0,212267	3,615073	0,009543	156,0936	5,033456				1,969379		0,058661	0,08996			0,050213						167,1322	0,933953167		
322																5,252902	1,474046	0,602389	0,068054	0,402675	328,9065				0,204509			3,968367	6,298783							347,1783	0,947370769		
323																0,824071	1,127963		0,136454	0,136454	2,525776	2,789892														7,540611	0,334956358		
324		0,878657									0,007087					1,209428	2,382074	4,580832	4,291924						125,5396												138,8896	0,903880464	
331																0,017564				0,017564		0,01712				0,498521		0,080922								0,63169	0,789185493		
332																0,819075	0,017117	0,026647	1,493722	1,192787				0,216255		166,3179	2,225779			0,132819						172,4421	0,96448547		
333																0,293256			11,87734	14,02341						2,30995		332,0465								364,5445	0,910853131		
334																					0,084921				0,129136				0,180619								0,394676	0,457639042	
335																											0,058661			20,64111							20,69977	0,997166117	
411																																0,67322						0,67322	1
412																			0,014485							0,025657						0,187225	101,3601				101,5874	0,997761863	
423																																		0,014576				0,014576	1
511																		0,064501																			1,873494	0,965571988	
512											0,004621							0,004621		0,004621																	90,69585	0,999808461	
523																																					0,368182	0,368182	1
Total	0,481603	44,91258	2,762884	0,660793	0,002969	0,32825	0,838184	0,274046	0,041708	0,109639	3,126857	51,13498	1,042596	2,902808	4,364069	119,8708	35,8847	96,90711	513,4534	425,2327	244,4099	173,5357	351,6326	2,525776	139,156	0,717194	174,3976	344,4333	0,180619	20,77393	0,923437	103,1274	0,014576	1,900655	90,67848	0,368182	2953,108		
Prod_accuracy	0,899641	0,89836	0,994722	1	1	0,719297	0,928005	0,956095	1	1	0,997044	0,980769	1	1	0,945618	0,930487	0,897918	0,890637	0,972324	0,92023	0,878589	0,89949	0,93537	1	0,90215	0,695099	0,953671	0,964037	1	0,993606	0,729037	0,982863	1	0,951774	1	1	0,937986379		

Overall accuracy = 93.8%

Table 68: Confusion matrix CLC 2018 for Alpine region –PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	231	242	243	311	312	313	321	322	323	324	331	332	333	334	335	411	412	423	511	512	523	Total	User_accuracy			
111	0,43327																																				0,43327	1			
112		41,63624	0,005322	0,034655												0,924659	0,924659																				43,52554	0,956593379			
121		0,081697	2,739043	0,009259																																	2,829999	0,96785988			
122	0,013678	0,013678	0,009259	0,660793		0,064204																															0,761612	0,867624223			
123					0,002969																																0,002969	1			
124						0,236109																															0,236109	1			
131							0,77784												0,046244	0,008939	0,077412																0,910435	0,854360458			
132								0,317892																													0,317892	1			
133									0,041708																												0,041708	1			
141										0,109639																											0,109639	1			
142		0,012072				0,027937					3,129686																										3,179692	0,984273461			
211							0,046244					50,16419				1,787102	1,665568	2,674756					0,004693									0,009272						56,46382	0,888430761		
212													1,042596																								1,042596	1			
221												0,080858		2,902808	0,237325																							3,220991	0,901215744		
222												0,00754			4,126743				0,768476																			4,90276	0,84171837		
231		0,904037						0,012032								113,5379		1,833083	4,347852	7,096824		2,196896				1,637758												131,5664	0,862970444		
242																0,582715	32,22153	0,077412		0,003508	5,065119																	37,95028	0,849045819		
243																3,490141	1,072941	89,96365	0,08367	2,164436	2,951599					0,035839												99,76227	0,901780243		
311																			494,1706	3,42694	10,13449		2,322866			1,551352	0,218673		3,691336				3,560453					519,0767	0,952018447		
312											0,009243					2,430662			389,202	7,10356						4,980551												403,726	0,96402507		
313																0,005189				12,40046	213,4131	1,226244				0,035035	1,209428							0,091662				228,3812	0,934460362		
321																			0,212267	2,648156	0,009543	156,0112	5,883363			2,16879		0,058661	0,08996		0,050213						167,1322	0,933460137			
322																			5,252902	1,474046	0,602389	0,068054	0,381569	331,2581		0,204509		3,968367	3,968367									347,1783	0,954144011		
323																			0,76044	1,127963		0,136454	0,136454	2,525776	2,853523													7,540611	0,334956358		
324		0,878657									0,007087					1,209428	2,382074	5,040758	3,788041							125,5835													138,8896	0,904196955	
331																0,017564				0,017564			0,01712			0,498521		0,080922										0,63169	0,789185493		
332																			0,819075	0,017117	0,026647	1,467075	1,143999		0,14082		167,3776	1,316923		0,132819								172,4421	0,970630873		
333																			0,293256			10,96952	14,93122				2,30995	3,236798	332,8037									364,5445	0,912930386		
334																									0,214057					0,180619									0,394676	0,457639042	
335																															20,64111								20,69977	0,997166117	
411																																0,67322							0,67322	1	
412																				0,014485						0,025657						0,187225	101,3601						101,5874	0,997761863	
423																																		0,014576					0,014576	1	
511																																			1,873494				1,873494	1	
512											0,004621							0,004621		0,004621																			90,69585	0,999808461	
523																																							0,368182	0,368182	1
Total	0,446948	43,52638	2,753625	0,704708	0,002969	0,32825	0,824084	0,329924	0,041708	0,109639	3,143551	50,25968	1,042596	2,902808	4,364069	122,7584	35,8847	101,7939	505,7255	422,6306	242,6552	172,3937	355,6931	2,525776	141,8634	1,926622	174,7001	341,9512	0,180619	20,77393	0,923437	104,9205	0,014576	1,965155	90,67848	0,368182	2953,108				
Prod_accuracy	0,969398	0,956575	0,994704	0,937683	1	0,719297	0,943884	0,963531	1	1	0,99559	0,9981	1	1	0,945618	0,924889	0,897918	0,883783	0,977152	0,920903	0,879491	0,904971	0,931303	1	0,885243	0,258754	0,958085	0,973249	1	0,993606	0,729037	0,966065	1	0,953356	1	1	0,938901868				

Overall accuracy = 93.9%

Bio-geographical region ANA - Anatolian- confusion matrix CLC2018 - Blind and plausibility analysis

Table 69: Confusion matrix CLC 2018 for Anatolian region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	335	411	421	422	511	512	521	Total	User_accuracy																			
111	0,578694	0,257551																																							0,836245	0,69201463																		
112		18,62013	0,145233																																						18,76536	0,992260585																		
121		0,126343	3,000716																																						3,190865	0,940408484																		
122			0,049991	0,158443																																					0,366877	0,431869081																		
123					0,002018																																				0,002018	1																		
124						1,035022																																			1,035022	1																		
131							2,292299																																		2,292299	1																		
132								0,079279																																	0,155073	0,511233595																		
133		0,076656	0,076656						1,853593									0,075795																							2,297371	0,806831944																		
141										0,086887								0,145233			0,145233																					0,086887	1																	
142											0,045032																															0,045032	1																	
211												166,0687	13,80209					0,233367																								180,1041	0,922070308																	
212		0,454743										0,189116	180,1101					0,189116			0,145236							4,701601														185,7899	0,969428843																	
213														1,199806																												1,199806	1																	
221												0,113616																														0,113616	0,387597																	
222												1,630094									0,842061		0,113616																				18,89923	0,913748162																
223																		0,030039																								0,030039	1																	
231													4,558836					51,99657																								61,61684	0,843869502																	
241																			1,630094																							1,630094	1																	
242		0,009834										0,387597	6,44414	0,042681				3,292436		52,93752	3,161931																					66,27614	0,798741767																	
243												14,56616						4,866755		0,089487	132,8491																						2,086095	158,7482	0,836854367															
311																																												36,41362	6,376669															
312																																												22,91689	2,086095															
313																																												15,03952	1															
321		0,387597																4,992967																										236,2371	16,4795	0,912620118														
323																																												9,078195	5,470003	16,35229	0,555163397													
324			0,369152																																									5,470003	128,0324	0,91036697														
331																																												116,5564	0,91036697															
332																																												0,095349	0,762793	48,61393	6,324244	1,148849	0,663962262											
333							0,084731																																					64,94037	0,748593366															
335																																												60,7269	348,0257	0,739654767														
411																																												20,02146	257,4189	0,024884	7,44747	0,962923197												
421																																												48,61393	6,324244	0,392884	3,928836	0,410196	1,849845	45,48517	0,011604	0,011604	1658,247	0,864928805						
422																																												1,964418	0,392884	3,928836	0,410196	1,849845	45,48517	0,011604	0,011604	1658,247	0,864928805							
511																																															0,06357	0,0266	0,374906	1,849845	45,48517	0,011604	0,011604	1658,247	0,864928805					
512																																																				0,06357	0,0266	0,374906	1,849845	45,48517	0,011604	0,011604	1658,247	0,864928805
521																																																				0,06357	0,0266	0,374906	1,849845	45,48517	0,011604	0,011604	1658,247	0,864928805
Total	0,578694	19,93285	3,641749	0,158443	0,002018	1,035022	2,37703	0,079279	1,853593	0,108614	0,045032	182,9552	204,9663	1,242487	0,227232	17,77035	0,030039	66,079	1,630094	54,01431	153,9218	1,917712	37,8536	22,91689	15,03952	328,771	4,941475	9,191811	144,7202	1,233048	48,61393	270,7739	0,024884	7,854473	3,928836	0,410196	1,849845	45,54498	0,011604	0,011604	1658,247	0,864928805																		
Prod_accuracy	1	0,934143	0,823977	1	1	1	0,964354	1	1	0,799964	1	0,907701	0,87873	0,965649	1	0,971795	1	0,786885	1	0,980065	0,863094	0,961959	1	1	1	0,718546	0,987639	0,805392	0,618624	1	0,950678	1	0,948182	1	1	1	0,998687	1	1	1	0,864928805																			

Overall accuracy = 86.5%

Table 70: Confusion matrix CLC 2018 for Anatolian region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	335	411	421	422	511	512	521	Total	User_accuracy						
111	0,578694	0,257551																																						0,836245	0,69201463						
112		18,62013	0,145233																																					18,76536	0,992260585						
121		0,126343	3,000716																								0,049686								0,014119						3,190865	0,940408484					
122			0,049991	0,158443																						0,081786		0,076656													0,366877	0,431869081					
123					0,002018																																				0,002018	1					
124						1,035022																																			1,035022	1					
131							2,292299																																		2,292299	1					
132								0,079279										0,075795																							0,155073	0,511233595					
133		0,076656	0,076656						1,853593									0,145233			0,145233																					2,297371	0,806831944				
141										0,086887																																0,086887	1				
142											0,045032																															0,045032	1				
211			0,065062									159,6757	13,80209					6,561305																								180,1041	0,886574178				
212		0,454743										0,189116	180,1386					0,116684		0,116684							4,701601		0,072433													185,7899	0,969582524				
213														1,199806																												1,199806	1				
221												0,113616			0,227232	0,728445				0,728445	0,387597						0,113616									0,026665						2,325616	0,09770836				
222																17,26914				1,630094																						18,89923	0,913748162				
223																	0,030039																									0,030039	1				
231													4,558836					51,92624																								61,61684	0,842728049				
241																			1,630094																							1,630094	1				
242		0,009834											3,282209	0,042681			0,971648	3,680033		55,1278	3,161931																					66,27614	0,831789593				
243													14,5491					4,866755		3,251418	129,5875	0,116684																					158,7482	0,816308868			
311																																											40,58581	0,897200775			
312																																											22,91689	1			
313																																											15,03952	16,4795	0,912620118		
321		0,387597																0,051492																									251,7209	0,951398965			
323																																												16,35229	0,555163397		
324			0,369152																																									128,0324	0,91036697		
331										0,021727																																		1,148849	0,663962262		
332																																												64,94037	0,748593366		
333							0,084731																																					348,0257	0,753672417		
335																																												0,024884	1		
411																		0,286761																										7,734231	0,962923197		
421																																												6,286137	0,6875		
422																																												0,410196	1		
511													0,051175																															1,849845	0,03315		
512																																														45,48517	1
521																																														0,011604	1
Total	0,578694	19,93285	3,706811	0,158443	0,002018	1,035022	2,37703	0,079279	1,853593	0,108614	0,045032	174,5275	201,833	1,242487	0,227232	17,99758	1,001688	67,7103	1,630094	60,85444	151,0479	1,92078	37,8536	22,91689	15,03952	327,2127	4,941475	9,191811	144,5925	1,045595	50,30551	275,2596	0,024884	7,854473	4,321719	0,410196	1,849845	45,54498	0,011604	1658,247							
Prod_accuracy	1	0,934143	0,809514	1	1	1	0,964354	1	1	0,799964	1	0,914903	0,892514	0,965649	1	0,959525	0,029989	0,766888	1	0,905896	0,857924	0,961959	1	1	0,7319	0,987639	0,806103	0,72953	0,966374	0,952909	1	0,948182	1	1	1	0,998687	1					0,865541034					

Overall accuracy = 86.5%

Bio-geographical region ARC - Arctic - confusion matrix CLC2018 - Blind and plausibility analysis

Table 71: Confusion matrix CLC 2018 for Arctic region – BLIND ANALYSIS-In column the validation and line the production

CLC Classes	112	121	122	123	124	131	133	141	142	211	231	242	243	311	312	313	321	322	324	331	332	333	335	411	412	421	423	511	512	521	522	523	Total	User_accuracy		
112	0,434877																																0,434877	1		
121		0,246524																0,01837				0,01837											0,283265	0,870295642		
122			0,012019																														0,012019	1		
123				0,029903																													0,029903	1		
124					0,118513																												0,118513	1		
131						0,049736																											0,049736	1		
133	0,025971						0,02608																										0,052051	0,501044546		
141								0,070456																									0,070456	1		
142									0,655897																								0,655897	1		
211										0,098413																							0,098413	1		
231											10,74798	0,002925																					10,7509	0,999727923		
242												0,188863	0,010938																				0,199801	0,945257773		
311														1,010877				0,067392							0,067392								1,14566	0,882352941		
312															0,116659																		0,116659	1		
313																0,169149		0,021144															0,190292	0,888888889		
321																	13,51783																13,51783	1		
322																		149,5998										0,034721					149,6345	0,999767962		
324											0,01304							0,206876	0,1313	1,135995													1,487211	0,763842388		
331																				14,96519	0,075577							0,095987					15,13676	0,988665713		
332																					107,4618							0,045036					107,5068	0,999581089		
333																				1,894309	1,969886	54,37007											58,23427	0,933643963		
335																						44,76702											44,76702	1		
411																								1,838871									1,838871	1		
412																									28,49133								28,49133	1		
421																										0,109774								0,109774	1	
423																											0,363332							0,363332	1	
511																				0,0266									3,348642					3,375241	0,99211918	
512																														6,637691					6,637691	1
521																															0,622728				0,622728	1
522																																0,119445			0,119445	1
523																																	0,132759		0,132759	1
Total	0,460849	0,246524	0,012019	0,029903	0,118513	0,049736	0,02608	0,070456	0,655897	0,098413	10,76102	0,191788	0,010938	1,010877	0,116659	0,169149	13,72471	149,838	1,135995	16,8861	109,5073	54,38844	44,76702	1,838871	28,55872	0,109774	0,363332	3,47935	6,682727	0,622728	0,119445	0,132759	446,184			
Prod_accuracy	0,943645	1	1	1	1	1	1	1	1	1	0,998788	0,984748		1	1	1	0,984927	0,99841	1	0,886243	0,981321	0,999662	1	1	0,99764	1	1	0,962433	0,993261	1	1	1	0,989408331			

Overall accuracy = 98.94%

Table 72: Confusion matrix CLC 2018 for Arctic region – PLAUSIBILITY ANALYSIS-In column the validation and line the production

CLC Classes	112	121	122	123	124	131	133	141	142	211	231	242	311	312	313	321	322	324	331	332	333	335	411	412	421	423	511	512	521	522	523	Total	User_accuracy
112	0,434877																				0,01837										0,434877	1	
121		0,264894																													0,283265	0,935147821	
122			0,012019																												0,012019	1	
123				0,029903																											0,029903	1	
124					0,118513																										0,118513	1	
131						0,049736																									0,049736	1	
133	0,025971						0,02608																								0,052051	0,501044546	
141								0,070456																							0,070456	1	
142									0,655897																						0,655897	1	
211										0,098413																					0,098413	1	
231											10,74798	0,002925																			10,7509	0,999727923	
242												0,199801																			0,199801	1	
311													1,010877				0,067392							0,067392							1,14566	0,882352941	
312														0,116659																	0,116659	1	
313															0,169149																0,190292	0,888888889	
321																13,51783															13,51783	1	
322																	149,5998										0,034721				149,6345	0,999767962	
324											0,01304						0,206876	0,1313	1,135995												1,487211	0,763842388	
331																			14,9332	0,075577							0,127983				15,13676	0,98655193	
332																				107,4748								0,031996			107,5068	0,999702383	
333																			1,894309	1,969886	54,37007										58,23427	0,933643963	
335																						44,76702									44,76702	1	
411																							1,838871								1,838871	1	
412																								28,49133							28,49133	1	
421																									0,109774						0,109774	1	
423																										0,363332					0,363332	1	
511																			0,0266								3,348642				3,375241	0,99211918	
512																												6,637691				6,637691	1
521																													0,622728			0,622728	1
522																													0,119445			0,119445	1
523																														0,132759	0,132759	1	
Total	0,460849	0,264894	0,012019	0,029903	0,118513	0,049736	0,02608	0,070456	0,655897	0,098413	10,76102	0,202726	1,010877	0,116659	0,169149	13,72471	149,8196	1,135995	16,8541	109,5203	54,38844	44,76702	1,838871	28,55872	0,109774	0,363332	3,511346	6,669687	0,622728	0,119445	0,132759	446,184	
Prod_accuracy	0,943645	1	1	1	1	1	1	1	1	1	0,998788	0,985571	1	1	1	0,984927	0,998533	1	0,886027	0,981323	0,999662	1	1	0,99764	1	1	0,953663	0,995203	1	1	1	0,989431532	

Overall accuracy = 98.94%

Bio-geographical region ATL - Atlantic - confusion matrix CLC2018 - Blind and plausibility analysis

Table 73: Confusion matrix CLC 2018 for Atlantic region – BLIND ANALYSIS- In column the validation and line the production

CL Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User_accuracy						
111	3,180092	0,078513								0,074695																														3,333299	0,954037154					
112	0,641663	193,6754	0,564279	2,256526							0,072887	2,268052				0,005989		1,99625			0,228801																			203,7964	0,950337706					
121		0,103345	31,92115		0,155306	0,323006	0,076656			0,145233	0,059242	0,081786				0,169561	0,021559	0,145233					0,562875	0,117457						0,243022										34,12543	0,935406481					
122		0,256449		4,008522						0,145233						0,145233																									4,555437	0,879942304				
123					1,915313																																				1,915313	1				
124						4,038951																																			4,038951	1				
131							5,097328									0,007689															0,025742				0,036353					5,167112	0,986494682					
132			0,021706					0,86922																																0,890926	0,975637051					
133		0,145233	0,010044				0,025742		2,109705	0,145233									0,102398		0,018975																				2,55733	0,824964129				
141			0,14599							5,309671	0,209638																														5,6653	0,937226837				
142											26,30145						0,343534	0,248707																	0,036353						26,93004	0,976658236				
211		13,0474	0,063039				0,003007			3,565778		968,1922	0,072433			6,428277	4,779048	1,378453	0,072433	0,025587	0,012794	0,010352		0,012794						0,016246											997,6798	0,970443789				
212													1,543349					0,188733																							1,732081	0,89103708				
221														21,54757																												21,54757	1			
222		0,053307										0,122249			2,71278	0,37487	0,41053			0,026654																						3,700389	0,733106611			
231		0,013954	0,146122	0,341255			0,029569			0,038733	3,086813	10,38766				621,4094	3,086813	19,9215	5,49686	0,064362		0,059138	0,021706	10,6442					0,029569					0,315513	0,470348						675,5635	0,9198386				
241												0,69357					2,080711				0,69357																					3,467851				
242		1,020937	0,029569									10,17997	208,0201	0,07142		10,17997	208,0201	0,07142	1,818258		0,194081									0,194081											230,7435	0,90152093				
243		1,575054										3,955312				3,955312		79,52181	1,293925		0,007361	0,517228	0,60482					1,572047					0,020732	0,010495							102,5719	0,775278801				
311							0,006382											314,0091	0,15775	8,005816	0,004777																				322,5319	0,973575251				
312		0,822427																0,337065	252,8736	3,703297		1,471516	4,24922																			263,4571	0,959828281			
313		0,437439								0,031701						0,228048	0,293931	0,587862	0,488012	1,406277	65,88682			3,710593																		73,07069	0,901686123			
321			0,024657				0,031393				0,207224					10,73349	0,051492	2,685922			62,98999		0,103463	0,155732		3,113921		3,333086		0,100618				0,051492								83,58247	0,753626752			
322			0,018275	0,010044												0,228801	0,83663	0,653451	1,209428	5,359681		11,95445	167,0688	4,335367								0,060261										191,9201	0,870512577			
324		0,031393		0,031393												0,134376	0,051492	0,188043	2,677193	5,542614	2,539811	0,010352	1,119819	110,8434					0,027542													123,1974	0,899721732			
331		0,027321																						2,06812										0,018994									2,114435	0,978095854		
332																			0,074638							6,467139	0,431636																6,973413	0,927399341		
333																				0,184845			8,195078	0,65658		0,28923	85,17257																94,4983	0,901313234		
334																												0,174996		0,132956													0,307952	0,568257806		
411		0,143608														0,138192		2,79926		0,653451		2,79926		0,021706					15,94028					0,165034	0,031779								16,4406	0,969568091		
412																																151,6251											161,3851	0,939524026		
421																																4,372941											4,38906	0,996327446		
422																													0,053398					0,213591										0,266988	0,8	
423					0,004859						0,004859					0,060261									0,060261		0,004859			0,053398		0,074367		1,726197	0,014106			0,0445						1,994268	0,865579228	
511	0,018477																																											6,836219	0,932492983	
512																																												40,63005	0,966389168	
521																																												0,097869	0,868524832	
522																																												2,320738	0,932144381	
523																																												2,307532	2,430259	0,949500352
Total	3,840231	211,4317	32,94483	6,647739	2,075478	4,361957	5,270078	0,86922	4,486719	9,193352	29,67564	1004,693	1,615781	21,54757	2,71278	654,1994	219,9759	110,8464	325,424	268,1131	81,07827	78,60425	178,3942	135,6638	2,284113	6,756368	91,70709	0,174996	20,92838	152,2647	4,668448	0,213591	1,823928	7,582155	40,60995	0,085002	3,337213	2,326008	3728,427							
Prod_accuracy	0,828099	0,916018	0,968927	0,60299	0,92283	0,925949	0,967221	1	0,470211	0,577555	0,886297	0,96367	0,955172	1	1	0,949878	0,94565	0,717405	0,964923	0,94316	0,812632	0,801356	0,936515	0,817045	0,905437	0,957192	0,928746	1	0,761659	0,995799	0,936701	1	0,946417	0,840754	0,966868	1	0,648224	0,992056		0,931492451						

Overall accuracy = 93.15%

Table 74: Confusion matrix CLC 2018 for Atlantic region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	221	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User_accuracy					
111	3,205658	0,052946								0,074695																														3,333299	0,961707281				
112		194,3723	0,564279	2,256526				0,005415	2,086548		0,192227	0,011526					2,311276		1,99625																					203,7964	0,953757527				
121		0,021559	32,11238		0,07352	0,323006			0,145233	0,059242	0,019846	0,103345					0,198714		0,145233				0,562875		0,117457					0,243022										34,12543	0,941010213				
122		0,217717	0,038733	4,153755													0,145233																							4,555437	0,911823534				
123					1,915313																																			1,915313	1				
124						4,038951																																		4,038951	1				
131								5,097328									0,007689															0,025742				0,036353				5,167112	0,986494682				
132									0,890926																															0,890926	1				
133			0,145233	0,010044				0,025742		2,109705	0,145233									0,102398			0,018975																		2,55733	0,824964129			
141				0,14599							5,309671	0,209638																													5,6653	0,937226837			
142											26,30145						0,343534	0,248707																	0,036353					26,93004	0,976658236				
211		13,0474	0,100387				0,003007			3,565778	0,198635	967,8821	0,072433			6,502401	4,779048	1,378453	0,072433	0,025587	0,012794	0,010352		0,012794					0,016246											997,6798	0,970132961				
212													1,543349				0,188733																							1,732081	0,89103708				
221														21,54757																										21,54757	1				
222		0,053307										0,122249			2,71278	0,37487	0,41053				0,026654																				3,700389	0,733106611			
231		0,013954	0,146122	0,341255			0,029569	0,021706		0,038733	3,086813	10,30617				626,6417	3,218966	19,9215	0,052397	0,064362		0,059138	0,021706	10,78393					0,029569					0,315513	0,470348						675,5635	0,927583798			
241												0,69357				0,69357	0,69357	0,69357						0,69357																	3,467851				
242		1,020937	0,029569									3,276218				10,17997	213,959	0,07142		1,818258		0,194081								0,194081											230,7435	0,927258936			
243		1,575054										13,4931				3,955312		79,51444	1,293925	0,007361	0,007361		0,517228	0,60482				1,572047					0,020732	0,010495							102,5719	0,775207036			
311							0,006382												313,8999	0,15775	8,005816	0,004777																			322,5319	0,973236609			
312		0,822427																0,337065		257,3748	0,400937		1,471516	3,050349																	263,4571	0,976913533			
313		0,437439	0,012233								0,031701					0,228048	0,293931	0,781943	0,293931	0,646092	66,62715			3,718216																	73,07069	0,911817804			
321			0,024657				0,031393									10,75422	0,051492	2,628995				63,28603							3,220237		0,244226				0,051492						83,58247	0,757168613			
322			0,018275	0,010044				0,007689								0,228801	0,83663	0,653451	1,209428	5,359681		9,870744	169,2018	4,27844			3,113921						0,060261								191,9201	0,88162631			
324		0,031393		0,031393												0,134376	0,051492	0,136551	2,728684	3,48221	1,436087	0,010352	1,119819	114,0076					0,027542												123,1974	0,925405123			
331		0,027321																							2,087114																	2,114435	0,987078715		
332																								0,074638	0,100618	6,441159	0,356998															6,973413	0,923673811		
333																																										94,4983	0,901111017		
334																																										0,307952	0,568257806		
411		0,143608														0,138192								0,021706					0,174996						0,165034	0,031779						16,4406	0,969568091		
412																				0,653451				0,008548					15,94028													161,3851	0,991506733		
421																																4,38906										4,38906	1		
422																																										0,266988	1		
423					0,004859						0,004859						0,060261									0,060261						0,074367			1,731055	0,014106			0,0445			1,994268	0,868015511		
511	0,018477																																										6,836219	0,932492983	
512																																	0,018477					39,93801		0,673567			40,63005	0,982967207	
521																																												0,097869	1
522										0,018477																						0,120522			0,018477				2,163263		2,320738	0,932144381			
523																		0,122727																									2,307532	2,430259	0,949500352
Total	3,224135	211,9826	33,20267	6,792972	1,993691	4,361957	5,193421	0,925735	4,341486	9,211828	30,01347	995,92	1,615781	21,54757	2,71278	662,5547	224,6382	108,819	319,6531	269,8011	76,49015	74,01732	180,5271	138,6617	2,247993	6,749498	88,80922	0,174996	20,76214	160,654	4,828175	0,266988	1,809793	6,908589	41,28351	0,097869	3,324345	2,307532	3728,427						
Prod_accuracy	0,994269	0,916926	0,967163	0,611478	0,960687	0,925949	0,981497	0,962398	0,485941	0,576397	0,876322	0,971847	0,955172	1	1	0,945796	0,95246	0,730703	0,982002	0,953943	0,871055	0,855016	0,937265	0,822199	0,928434	0,954317	0,958836	1	0,767757	0,996019	0,909052	1	0,956494	0,922725	0,967408	1	0,650734	1			0,940017189				

Overall accuracy = 94%

Bio-geographical region BLS – Black sea - confusion matrix CLC2018 - Blind and plausibility analysis

Table 75: Confusion matrix CLC 2018 for Black sea region – BLIND ANALYSIS -In column the validation and line the production

CLC classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	311	312	313	321	322	324	331	332	333	411	422	511	512	521	523	Total	User_accuracy		
111	0,257551																																		0,257551	1		
112		2,590362																																	2,590362	1		
121			1,576848																															1,576848	1			
122			0,012134	0,955949																														0,968083	0,987465663			
123					0,112661																													0,112661	1			
124						0,064693																												0,064693	1			
131							0,581703																		0,077412									0,659115	0,882551829			
132								0,015425																										0,015425	1			
133				0,014758					0,318875																										0,333634	0,95576563		
141										0,02972																									0,02972	1		
142		0,007629								0,012072	0,267488															0,027448									0,314637	0,850149776		
211			0,063381									263,6126					3,245727	6,747776	20,24333		6,747776					6,747776			6,747776						314,1561	0,839113303		
212													4,558836					4,558836																	9,117672	0,5		
213														0,87658																					0,87658	1		
221												0,662866			0,294313																				0,957179	0,307479631		
222															5,297885			3,563985	1,690658	1,630094															12,18262	0,43487235		
231																	4,780192								0,339233										5,119425	0,933736039		
242																3,161931		12,76457	4,849701							3,161931									23,93813	0,533231649		
243																		55,12049								0,072433									55,19292	0,998687647		
244																	0,072433																		0,072433			
311																				52,03793		5,387756													57,42568	0,906178641		
312																					24,60553														24,60553	1		
313																				4,533182	0,466368	27,50943													32,50898	0,846210171		
321																							24,08976												24,08976	1		
324																		0,035525	2,693878		0,005156	4,941475			29,88961							0,292193			37,85783	0,789522394		
331																										0,495304									0,504328	0,98210737		
332																											5,001098								0,009024	5,001098	1	
333																4,878502													9,773741						14,65224	0,667047399		
411																															4,997429					4,997429	1	
421																															0,281568					0,281568		
422																																				0,033373	1	
511																				0,24684											1,23729					1,484129	0,833680399	
512												0,052021						0,03315																		1,914021	0,955501685	
521																																				2,925155	1	
523																																				0,337053	0,337053	1
Total	0,257551	2,59799	1,652364	0,970707	0,112661	0,064693	0,581703	0,015425	0,318875	0,041792	0,267488	264,3275	4,558836	0,87658	0,294313	8,459816	12,97685	27,66832	81,9397	61,14192	31,81968	32,90234	29,03123	0,339233	39,97661	0,495304	5,001098	16,52152	5,278997	0,033373	1,23729	2,121044	2,925155	0,346077	637,154			
Prod_accuracy	1	0,997064	0,954298	0,984797	1	1	1	1	1	0,711142	1	0,997295	1	1	1	0,626241	0,368363	0,461342	0,672696	0,851101	0,77328	0,836093	0,829788		0,747677	1	1	0,591576	0,946663	1	1	0,862241	1	0,973926		0,8456955		

Overall accuracy = 84.5%

Table 76: Confusion matrix CLC 2018 for Black sea region – PLAUSIBILITY ANALYSIS -In column the validation and line the production

CLC classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	311	312	313	321	324	331	332	333	411	422	511	512	521	523	Total	User_accuracy		
111	0,257551																																	0,257551	1		
112		2,590362																																2,590362	1		
121			1,576848																															1,576848	1		
122			0,012134	0,955949																														0,968083	0,987465663		
123					0,112661																													0,112661	1		
124						0,064693																												0,064693	1		
131							0,514871									0,066832									0,077412									0,659115	0,781155141		
132								0,015425																										0,015425	1		
133				0,014758					0,318875																										0,333634	0,95576563	
141										0,02972																								0,02972	1		
142		0,007629								0,012072	0,267488														0,027448									0,314637	0,850149776		
211			0,063381									263,6126					3,245727	6,747776	20,24333		6,747776			6,747776			6,747776							314,1561	0,839113303		
212													4,558836					4,558836																9,117672	0,5		
213														0,87658																				0,87658	1		
221												0,662866			0,294313																			0,957179	0,307479631		
222																5,297885		4,912303	1,972434															12,18262	0,43487235		
231																	5,119425																	5,119425	1		
242																	3,161931	12,76457	4,849701					3,161931										23,93813	0,533231649		
243											6,747776							48,28798	0,084731					0,072433										55,19292	0,874894471		
244																		0,072433																0,072433			
311																				52,03793		5,387756												57,42568	0,906178641		
312																					24,60553													24,60553	1		
313																				1,439977	0,466368	30,60263												32,50898	0,941359438		
321																						24,08976												24,08976	1		
324																		0,035525	2,693878			0,005156	4,941475	29,88961					0,292193				37,85783	0,789522394			
331																									0,495304								0,009024	0,504328	0,98210737		
332																									5,001098									5,001098	1		
333																4,878502											9,773741							14,65224	0,667047399		
411																													4,997429						4,997429	1	
421																													0,281568						0,281568		
422																												0,033373							0,033373	1	
511												0,052021								0,24684											1,20414	0,03315			1,484129	0,811344367	
512																																1,862			1,914021	0,972821012	
521																																	2,925155			2,925155	1
523																																	0,337053			0,337053	1
Total	0,257551	2,59799	1,652364	0,970707	0,112661	0,064693	0,514871	0,015425	0,318875	0,041792	0,267488	271,0752	4,558836	0,87658	0,294313	8,526648	13,31609	28,98348	75,38897	56,50335	31,81968	35,99554	29,03123	39,97661	0,495304	5,001098	16,52152	5,278997	0,033373	1,20414	2,187343	2,925155	0,346077	637,154			
Prod_accuracy	1	0,997064	0,954298	0,984797	1	1	1	1	1	0,711142	1	0,97247	1	1	1	0,621333	0,384454	0,440408	0,640518	0,920971	0,77328	0,850178	0,829788	0,747677	1	1	0,591576	0,946663	1	1	0,851261	1	0,973926	0,840254272			

Overall accuracy = 84%

Bio-geographical region BOR - Boreal - confusion matrix CLC2018 - Blind and plausibility analysis

Table 77: Confusion matrix CLC 2018 for Boreal region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	423	511	512	521	522	523	Total	User_accuracy				
111	0,256558																																			0,256558	1				
112		43,3178	0,840294							0,003803	0,002146				0,012799			0,002473																	44,18179	0,980444663					
121		0,276292	8,663147							0,153313																										9,092752	0,952753031				
122				0,953797															0,004695				0,004695													0,963187	0,990250474				
123					0,347194																																0,347194	1			
124						1,279558																															1,279558	1			
131			0,062603	0,001689				2,366286		0,153313									0,03547	2,860336																	5,479697	0,43182794			
132									1,037099	0,076656										0,018667																	1,132422	0,915823636			
133										0,276522						0,018196														0,0055								0,300218	0,921069741		
141			0,073775	0,004362							2,176656		0,013912																									2,268705	0,959426764		
142								0,004282				4,109416							0,035208																			4,148906	0,990481713		
211			2,52619	0,002473	0,029569			0,072433					332,3499		8,799307		5,06225		2,659784	2,467198									0,099922									354,2863	0,938082776		
222														0,622865	0,031729																							0,684847	0,909494849		
231			0,560061		0,029569								0,779489		71,46122		1,133753		1,685558	0,646125											0,052328								77,02485	0,927768417	
242						0,25202								0,00448	62,95188	1,678506		0,257394																			65,64832	0,958925984			
243		0,008039	0,00964								0,004935	3,095913		0,015937		101,6644		1,153279	6,215352									0,002423	0,002423									112,1844	0,90622589		
311		0,58327															111,0355	0,590347	10,36635																			123,2383	0,900981969		
312		0,158815	0,028335								0,004362	11,93686				1,153279		1879,228	4,410111																			1971,609	0,953144477		
313		1,190651	0,057272								0,004265						0,953548	9,012396	490,2252									0,004265											514,8071	0,952250245	
321				0,00784																5,329638																			5,337478	0,998531175	
322																					9,471974							0,027786											9,49976	0,9970751	
324							0,0055				0,002146	0,063541				0,466368		13,75776	4,102127			458,1587																	476,5562	0,961395025	
331																		0,036405					0,337937																0,374341	0,902749976	
332																	0,004724	0,316534			0,655087	0,392556		2,070477	0,316534														3,755912	0,55125805	
333																				0,042156																			8,64466	0,898935729	
334																										0,023431													0,046862	0,5	
411																												9,055299												9,055299	1
412												0,016119																												190,479	0,999502261
421																																								0,240084	1
423																																								1	
511																0,128402		0,24964	0,128402												0,004859									0,004859	1
512											0,004621							3,214229														8,485341								8,991785	0,943677009
521																																	272,9544							276,1732	0,988344815
522																																			0,08496					0,08496	1
523																																								0,053493	1
Total	0,256558	48,75749	9,607212	1,020775	0,347194	1,531578	2,448501	1,037099	0,659804	2,18046	4,131892	348,2557	0,622865	80,31268	62,98288	111,287	112,068	1912,239	521,5141	5,371794	12,9874	546,7246	0,337937	2,070477	8,087528	0,023431	9,089773	190,492	0,306625	0,004859	8,487814	272,9544	0,08496	0,053493	2,724343	4281,062					
Prod_accuracy	1	0,888434	0,901734	0,934385	1	0,835451	0,966422	1	0,419097	0,998256	0,99456	0,954327	1	0,889788	0,999508	0,913534	0,990786	0,982737	0,940004	0,992152	0,72932	0,838006	1	1	0,960861	1	0,996207	0,999434	0,782989	1	0,999709	1	1	1	1	1	0,984526	0,953368677			

Overall accuracy = 95.3%

Table 78: Confusion matrix CLC 2018 for Boreal region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	222	231	242	243	311	312	313	321	322	324	331	332	333	334	411	412	421	423	511	512	521	522	523	Total	User_accuracy	
111	0,256558																																			0,256558	1	
112		43,33935	0,840294								0,002146																								44,18179	0,980932399		
121		0,276292	8,64448					0,018667	0,153313																										9,092752	0,950700108		
122				0,963187																																0,963187	1	
123				0,347194																																0,347194	1	
124						1,279558																														1,279558	1	
131		0,062603	0,001689				2,476723		0,076656									0,001689	2,860336																	5,479697	0,451981849	
132							0,025953	1,007931	0,076656		0,021881																									1,132422	0,890067043	
133		0,018196							0,276522																				0,0055							0,300218	0,921069741	
141		0,044265	0,019117							2,191411		0,013912																								2,268705	0,965930465	
142							0,004282			0,035208	4,109416																									4,148906	0,990481713	
211		2,52619	0,002473	0,029569			0,072433					329,2105		12,02492		5,06225		2,598622	2,467198									0,074856								354,2863	0,929221616	
222													0,66972					0,015127																		0,684847	0,977912095	
231		0,560061										0,794174		71,51683		1,133753		1,685558	0,646125										0,052328							77,02485	0,928490404	
242						0,25202								0,00448	63,2039	1,678506		0,257394																		65,64832	0,962764924	
243		0,008039	0,00964	0,002383								0,004935	3,095913		0,01836		101,6387	1,153279	6,220852									0,002423	0,012924							112,1844	0,905996425	
311		0,58327																111,0355	0,590347	10,36635																123,2383	0,900981969	
312		0,158815	0,028335				0,008565		0,055125		0,004362	11,93928					1,153279	1884,542	3,949378																	1971,609	0,955839622	
313		1,190651	0,061554	0,002383							0,004265						0,007077	0,473445	9,010706	485,3846		2,860336	15,80015				0,004265	0,007631								514,8071	0,942847592	
321				0,00784																	5,329638															5,337478	0,998531175	
322																					0,027786	9,471974														9,49976	0,9970751	
324							0,0055		0,004282		0,002146	0,012049		0,051492		0,466368	0,013631	13,52156	4,004035																	476,5562	0,962058894	
331																		0,036405						0,337937												0,374341	0,902749976	
332																	0,009448	0,316534	0,013631		0,641456	0,387831		2,070477	0,316534										3,755912	0,55125805		
333																				0,042156							7,770994									8,64466	0,898935729	
334																										0,023431									0,046862	0,5		
411																											9,055299									9,055299	1	
412																						0,048357							190,4164	0,014214						190,479	0,999671509	
421																														0,240084						0,240084	1	
423																															0,004859					0,004859	1	
511																		0,24964	0,128402													8,613743			8,991785	0,957956943		
512										0,004621								3,214229															272,9544			276,1732	0,988344815	
521																																		0,08496		0,08496	1	
522																																			0,053493	1		
523																	0,074301		0,074301																	2,682188	2,830789	0,947505433
Total	0,256558	48,76773	9,607582	1,005362	0,347194	1,531578	2,593456	1,026598	0,642555	2,226619	4,153773	345,0658	0,66972	83,61609	63,2039	111,1399	111,6063	1917,193	516,1152	5,39958	12,97377	547,1214	0,337937	2,070477	8,087528	0,023431	9,059564	190,5068	0,319549	0,004859	8,613743	272,9544	0,08496	0,053493	2,682188	4281,062		
Prod_accuracy	1	0,888689	0,899756	0,95805	1	0,835451	0,95499	0,981817	0,430347	0,984188	0,989321	0,954051	1	0,8553	1	0,914511	0,994885	0,982969	0,940458	0,987047	0,730087	0,837977	1	1	0,960861	1	0,999529	0,999525	0,751321	1	1	1	1	1	1	1	0,952959406	

Overall accuracy = 95.3%

Table 79: Confusion matrix CLC 2018 for Continental region – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy = 93.2%

Table 80: Confusion matrix CLC 2018 for Continental region – PLAUSIBILITY ANALYSIS - In column the validation and line the production

Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	241	242	243	311	312	313	321	322	323	324	331	332	333	411	412	421	422	423	511	512	521	522	523	Total	User_accuracy																																																																																																																																																																																																																																																																																																									
111	4,427662		0,053258																																							4,48092	0,988114457																																																																																																																																																																																																																																																																																																								
112	0,864702	280,2778	0,081979	0,007629						0,035989	0,343534	1,319396							0,203989	0,006405	0,010009	0,006032	0,532022				0,007629																283,6971	0,987947299																																																																																																																																																																																																																																																																																																							
121		0,505018	52,1277	0,008939					0,063371	5,78999		0,091492					0,145233			0,081786	0,161459		0,005614																				58,9806	0,883810911																																																																																																																																																																																																																																																																																																							
122			0,05701	5,303731								0,012134							0,030534	0,006098																							5,409508	0,980446146																																																																																																																																																																																																																																																																																																							
123					0,658564																																						0,658564	1																																																																																																																																																																																																																																																																																																							
124						4,058018						0,030556								0,107596	0,161958																						4,088575	0,992526413																																																																																																																																																																																																																																																																																																							
131		0,076778	0,006098				11,73515	0,171255				0,107596	0,161958						0,016347	0,906171	0,038139			0,076656	0,062472		0,13813																13,83451	0,848251792																																																																																																																																																																																																																																																																																																							
132			0,063371					1,569438													0,019717																						1,677631	0,935508517																																																																																																																																																																																																																																																																																																							
133		0,310546	0,292663	0,118991			0,145233	0,019952	1,62314								0,227019				0,033249																							2,770793	0,585803298																																																																																																																																																																																																																																																																																																						
141			0,012072							5,605151	0,084831	0,037093								0,035989	0,013057																							5,788195	0,968376477																																																																																																																																																																																																																																																																																																						
142	0,012072	0,056723								0,151826	15,58498	0,038148									0,027448																							16,21473	0,961161754																																																																																																																																																																																																																																																																																																						
211		3,060463	0,045499	0,013678			0,035026	0,003986			0,093476	2032,611	0,020335		0,011702	0,004491	3,336606		30,5328	13,21356	0,836009		2,98284	0,855082		0,163359																			16,21473	0,961161754																																																																																																																																																																																																																																																																																																					
212													0,128734																																2093,691	0,970826376																																																																																																																																																																																																																																																																																																					
213												0,622264	0,026083	13,75064																															0,128734	1																																																																																																																																																																																																																																																																																																					
221							0,007629					1,214884	2,077658		22,29142					0,624483	1,612686																								14,39899	0,954972748																																																																																																																																																																																																																																																																																																					
222		0,060174										0,123456				22,85924	0,37487	0,343646	1,465978	0,048378	1,536953		0,035907				0,116684																		27,82876	0,801020976																																																																																																																																																																																																																																																																																																					
231		4,201431	0,028961	0,020094								10,33113			0,904037	0,319517	596,7504		38,90434	1,571971	10,39122	0,257317	9,299572	0,088171			4,533295																		26,96529	0,847728459																																																																																																																																																																																																																																																																																																					
241													0,024036				0,859858																												677,6448	0,880624275																																																																																																																																																																																																																																																																																																					
242		1,987057	0,009948	0,00628												1,916105			290,525	9,732328			6,067413				0,034655																			322,2425	0,901572396																																																																																																																																																																																																																																																																																																				
243			0,072887							0,063648	0,046454				0,033025				5,818197	290,328	1,686188	0,025587	6,776298				4,160091																			318,1046	0,912680954																																																																																																																																																																																																																																																																																																				
311																					5,444464	781,5435	5,38646	31,36471			8,348742																			840,0193	0,930387569																																																																																																																																																																																																																																																																																																				
312						0,040656						0,11453							0,017997	5,00413	564,9783	6,011895					1,584068																			578,2958	0,976971091																																																																																																																																																																																																																																																																																																				
313		0,437439	2,933972									0,084338							2,45681	0,465086	2,701173	277,242					0,1304																			286,5715	0,967444552																																																																																																																																																																																																																																																																																																				
321												1,714316							2,935456	2,103541				33,19004		1,311792	10,05299	9,987669			5,330899																66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
322												1,673261																																	286,5715	0,967444552																																																																																																																																																																																																																																																																																																					
323																																															66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
324																																															286,5715	0,967444552																																																																																																																																																																																																																																																																																																			
331																																															66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
332																																															286,5715	0,967444552																																																																																																																																																																																																																																																																																																			
333																																															66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
411							0,008486					0,040508								0,086786	0,042156																											286,5715	0,967444552																																																																																																																																																																																																																																																																																																		
412																					0,111924	0,070522																									66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
421																				0,012794	0,182598																										286,5715	0,967444552																																																																																																																																																																																																																																																																																																			
422																																															66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
423																																															286,5715	0,967444552																																																																																																																																																																																																																																																																																																			
511																																															66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
512																																															286,5715	0,967444552																																																																																																																																																																																																																																																																																																			
521																																															66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
522																																															286,5715	0,967444552																																																																																																																																																																																																																																																																																																			
523																																															66,67487	0,497789348																																																																																																																																																																																																																																																																																																			
Total	5,304436	290,9734	55,78542	5,479342	0,658564	4,098675	11,97015	1,828003	7,413129	5,856615	16,29142	2052,787	2,276846	13,75064	23,24018	25,10872	628,8126	1,203504	377,9174	325,908	814,8539	575,6103	345,0652	39,34645	8,401569	2,045155	180,1971	11,46634	0,081377	1,999674	21,53163	1,81746	1,148822	0,04168	0,028213	17,75909	49,83284	1,085118	0,026746	0,877274	5929,88																																																																																																																																																																																																																																																																																																										

Overall accuracy = 93.6%

Bio-geographical region - Macaronesia - confusion matrix CLC2018 - Blind and plausibility analysis

Table 81: Confusion matrix CLC 2018 for Macaronesia region – BLIND ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	123	124	131	133	142	211	231	242	243	311	312	321	322	323	324	331	332	333	412	523	Total	User_accuracy
111	0,026473																							0,026473	1
112		0,019449																						0,019449	1
121			0,012497																					0,012497	1
123				0,09044																				0,09044	1
124					0,08288																			0,08288	1
131						0,063283																		0,063283	1
133							0,028736																	0,028736	1
141											0,01225													0,01225	
142								1,129968																1,129968	1
211									2,583306															2,583306	1
231										2,598958		0,513086												3,112044	0,835129091
243												7,720082												7,720082	1
311													2,929121											2,929121	1
312														6,038094										6,038094	1
321															0,792459	0,257828								1,050287	0,754516622
322																0,101016								0,101016	1
323																	6,809594							6,809594	1
324																		1,407237						1,407237	1
331																			0,287237		0,143618			0,430855	0,666666667
332																				1,853499				1,853499	1
333												1,556692				0,184845		0,598724			16,14975			18,49001	0,873431057
412																						0,151048		0,151048	1
523																							0,052655	0,052655	1
Total	0,026473	0,019449	0,012497	0,09044	0,08288	0,063283	0,028736	1,129968	2,583306	2,598958	0,01225	9,78986	2,929121	6,038094	0,792459	0,543689	6,809594	2,005961	0,287237	1,853499	16,29337	0,151048	0,052655	54,19482	
Prod_accuracy	1	1	1	1	1	1	1	1	1	1		0,788579	1	1	1	0,185798	1	0,701528	1	1	0,991185	1	1		0,939716693

Overall accuracy = 94%

Table 82: Confusion matrix CLC 2018 for Macaronesia region – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLC Classes	111	112	121	123	124	131	133	142	212	231	243	311	312	321	322	323	324	331	332	333	412	523	Total	User_accuracy
111	0,026473																						0,026473	1
112		0,019449																					0,019449	1
121			0,012497																				0,012497	1
123				0,09044																			0,09044	1
124					0,08288																		0,08288	1
131						0,063283																	0,063283	1
133							0,028736																0,028736	1
141								0,01225															0,01225	
142								1,129968															1,129968	1
211									2,583306														2,583306	
231										2,598958	0,513086												3,112044	0,835129091
243											7,720082												7,720082	1
311												2,929121											2,929121	1
312													6,038094										6,038094	1
321														0,792459	0,257828								1,050287	0,754516622
322															0,025492						0,075524		0,101016	0,252356676
323																6,809594							6,809594	1
324																	1,407237						1,407237	1
331																		0,287237		0,143618			0,430855	0,666666667
332																			1,853499				1,853499	1
333														1,556692	0,184845	5,252902	0,598724			10,89684			18,49001	0,589336956
412																					0,151048		0,151048	1
523																						0,052655	0,052655	1
Total	0,026473	0,019449	0,012497	0,09044	0,08288	0,063283	0,028736	1,142218	2,583306	2,598958	8,233167	2,929121	6,038094	2,349151	0,468165	12,0625	2,005961	0,287237	1,853499	11,04046	0,226572	0,052655	54,19482	
Prod_accuracy	1	1	1	1	1	1	1	0,989275		1	0,937681	1	1	0,337338	0,054451	0,564526	0,701528	1	1	0,986992	0,666667	1		0,793729826

Overall accuracy = 79.4%

Bio-geographical region MED- Mediterranean - confusion matrix CLC2018 - Blind and plausibility analysis

Table 83: Confusion matrix CLC 2018 for Mediterranean region – BLIND ANALYSIS - In column the validation and line the production

Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	223	231	241	242	243	244	311	312	313	321	322	323	324	331	332	333	334	411	412	421	422	423	511	512	521	522	523	Total	User_accuracy						
111	20,89875	0,888309		4,558836					0,513086												0,145233							0,011771																26,85898	0,778091699						
112		91,26946	0,032214				0,003986		0,074753			0,092366								0,023127							0,168288																	91,62979	0,996067605						
121		0,323711	32,83992					0,006391	0,081786		0,006098										0,090436	0,167					0,081786	0,031948	0,106076		0,040103													33,5235	0,979608741						
122			0,031948	6,329093					0,010009			0,031948									0,090436	0,167					0,081786	0,031948	0,106076															6,880246	0,919893405						
123					0,709934																																							0,709934	1						
124						3,284232	0,082878																																					0,709934	0,96147985						
131			0,198189				9,47969	0,878657						0,045899							0,221674				0,145233		0,003986		0,221674	0,076656		0,107596													11,37925	0,833067761					
132								0,897276			0,006391																																		0,903667	0,992927914					
133		0,197304	0,216653	0,082813			0,076656		3,838282												0,145233																								4,556941	0,842293439					
141		0,161757	0,01225							1,219257																																			1,393264	0,875107959					
142		0,25062				0,015278				6,307031	0,019692																																		0,0863	6,814323	0,925555136				
211		2,118416	2,002193				0,008486					0,019692					0,116684	0,035257	2,925219		12,21193	22,7366	6,882459		0,076656			0,058744																		721,4084	0,922027844				
212												665,1586	4,558836			0,116684	0,035257	2,925219											1,8556	0,055592																247,4209	0,979242627				
213												0,514849	242,2851								3,67234	0,387039							0,072433	0,144865															16,21332	0,994040327					
221												1,545938	0,055939		85,28187	2,300177	1,575348			8,847446	3,489143		0,072433				0,387597	0,007086																		103,563	0,823478371				
222		0,244333										0,85257	0,361017					0,37487	2,416661	4,656626	1,254796																									107,6439	0,889456792				
223		0,13841										0,055939					0,466734	0,816244	211,4082		0,84792	1,251584	5,662608				0,045399		7,342136																	228,0352	0,927085926				
231		0,171001	0,131527	0,029569					0,37894	0,512893	0,072433	0,063445	0,042681						96,58759	0,943079	0,725778	1,32141					4,1362	3,360327	4,118126																112,661	0,857329477					
241		1,387141					0,077204					0,623794						0,429929	9,527996	4,044974	0,015074	0,429929				0,69357	0,120394		0,69357																	18,04358	0,528054778				
242		0,014119	1,159229									12,28169	1,139989			0,030182	7,749702	1,034989	7,519451	1,960393	242,7072	6,947445	3,161931	1,844291				2,080787																		289,6314	0,837986488				
243		0,030398					0,029569					2,548923					2,705185	1,792757	4,185774	1,496855	3,660731	287,8564		1,79855	0,015727	1,580977	1,979256		5,822528	0,923317	0,090397														316,5818	0,909263867					
244											0,74249				0,143855			2,664874			0,718613	124,4694																									128,7392	0,966833479			
311																					6,219006		518,14		1,751243		0,430616	7,797457	15,76319																		551,3443	0,939775662			
312				0,007189																	0,012056	1,494233	1,336888	429,97	1,638619			0,537023	17,01819																		452,0142	0,951231182			
313													0,765979							0,045992	3,224714	2,000761	1,28645	0,088708	149,844																							170,6462	0,87809731		
321		0,904037								0,026184		0,070485								0,045992	3,224714	2,000761	1,28645	0,088708	149,844																								261,0822	0,677734547	
322																				0,045992	3,224714	2,000761	1,28645	0,088708	149,844																							53,05474	0,80621523		
323																				0,257828																											452,8137	0,877883946			
324								0,056058				0,876644	5,252902							0,257828																												460,7928	0,9180131		
331											0,051492									0,878657	1,396516	4,77245	6,856568	3,994758	7,718878																								15,98319	0,55474741	
332																				0,155827																												24,08562	0,854691738		
333							0,031393																																									109,1936	0,729513029		
334																																																	9,746431	0,485843357	
411																																																	4,260951	0,953448156	
421																																																		9,149848	0,976252048
422																																																		2,06185	0,906832377
423																																																		0,168287	

[illegible]

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Bio-geographical region PAN – Pannonian - confusion matrix CLC2018 - Blind and plausibility analysis

Table 85: Confusion matrix CLC 2018 for Pannonian region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	311	312	313	321	324	333	411	412	511	512	Total	User_accuracy
111	0,083362																													0,083362	1
112		28,8919	0,005278																											28,8917	0,999817358
121		0,008833	4,375031																0,025123											4,408988	0,992298381
122				0,829522								0,005194							0,020103											0,854819	0,970406153
123					0,022087																									0,022087	1
124			0,059096			0,465556																								0,524653	0,88736118
131							0,582502	0,038139																						0,620641	0,938548525
132								0,301099																						0,301099	1
133									0,356331			0,00798				0,145233								0,012282						0,521826	0,682853911
141										0,310361																				0,310361	1
142		0,015278									1,405051	0,059079																		1,479408	0,949738282
211			0,011747				0,002903				0,007388	322,9795			0,017429	0,141808	0,145746	0,006805	0,129606	4,129678										327,5726	0,985978345
212													0,012051																	0,012051	1
213														0,348184																0,348184	1
221												0,202839			6,130144								0,11519						6,448174	0,950679176	
222												0,120349				4,139991			0,038326										4,298666	0,963087434	
231												0,954273					34,4867			0,580672				0,007564					36,02921	0,957187245	
242		1,230959								0,678199	0,009587				0,163657		0,006805	23,52812	3,792432										29,40976	0,800010668	
243			0,002903														0,68868		12,52387										13,91267	0,900177336	
311																				87,93083		2,696062		0,096176					90,72307	0,969222407	
312																					3,601252		0,059099						3,66035	0,983854365	
313																				0,140482		6,124336		0,009319					6,274137	0,976124016	
321												0,48205								0,003266			9,611126						10,09644	0,95193197	
324											0,404546					0,051492			0,126816	1,308436		0,046379		24,17497					26,11264	0,925795729	
333																									0,114331					0,114331	1
411																				0,161695						3,62325			0,020897	3,805842	0,952023208
412																							0,039771			0,039771	0,318169			0,397711	0,8
511			0,014061																									2,915865		2,929926	0,995200882
512																								0,008308					6,675829	6,684136	0,998757127
Total	0,083362	30,14697	4,468117	0,829522	0,022087	0,465556	0,585405	0,339239	0,356331	0,310361	2,090637	325,2254	0,012051	0,348184	6,311231	4,478524	35,32793	23,53493	16,65628	94,25506	3,601252	8,866777	9,766087	25,06493	0,114331	3,663021	0,318169	2,915865	6,696726	606,8544	
Prod_accuracy	1	0,958368	0,979167	1	1	1	0,995041	0,887574	1	1	0,672068	0,993094	1	1	0,971307	0,92441	0,976188	0,999711	0,751901	0,932903	1	0,690706	0,984133	0,964494	1	0,989143	1	1	0,99688		0,967054685

Overall accuracy = 96.7%

Table 86: Confusion matrix CLC 2018 for Pannonian region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	133	141	142	211	212	213	221	222	231	242	243	311	312	313	321	324	333	411	412	511	512	Total	User_accuracy	
111	0,083362																													0,083362	1	
112		28,8919	0,005278																											28,89717	0,999817358	
121			4,375031	0,008833															0,025123											4,408988	0,992298381	
122				0,849625								0,005194																		0,854819	0,993923684	
123					0,022087																									0,022087	1	
124			0,059096			0,465556																								0,524653	0,88736118	
131							0,582502	0,038139																						0,620641	0,938548525	
132								0,301099																						0,301099	1	
133									0,513847			0,00798																		0,521826	0,984708015	
141										0,310361																				0,310361	1	
142											1,420329	0,059079																		1,479408	0,960065492	
211			0,00344				0,002903				0,007388	322,4994			0,017429	0,116684	0,726418	0,006805	0,062483	4,129678										327,5726	0,984512666	
212													0,012051																	0,012051	1	
213														0,348184																0,348184	1	
221															6,332984		0,11519													6,448174	0,982136057	
222												0,120349				4,178317														4,298666	0,972003244	
231												0,954273					34,4867			0,580672				0,007564						36,02921	0,957187245	
242		1,230959									0,678199						0,006805	23,53771	3,792432											29,40976	0,800336645	
243			0,002903														1,269351	0,006805	11,9364					0,697217						13,91267	0,857951332	
311																				87,91648		2,703863		0,10273						90,72307	0,969064174	
312																					3,601252			0,059099						3,66035	0,983854365	
313																				0,140482	0,003901	6,129755								6,274137	0,976987657	
321																				0,003266			10,09318							10,09644	0,999676553	
324				0,003266							0,067123					0,051492				1,385089				24,60567						26,11264	0,942289669	
333																									0,114331					0,114331	1	
411																				0,161695						3,614942		0,029205		3,805842	0,949840372	
412																	0,039771									0,039771	0,318169			0,397711	0,8	
511			0,014061																										2,915865		2,929926	0,995200882
512																													6,684136	6,684136	1	
Total	0,083362	30,12285	4,459809	0,861724	0,022087	0,465556	0,585405	0,339239	0,513847	0,310361	2,105915	323,7134	0,012051	0,348184	6,51407	4,346493	36,64423	23,55132	15,81643	94,31736	3,605152	8,833618	10,09318	25,47228	0,114331	3,654713	0,318169	2,915865	6,713341	606,8544		
Prod_accuracy	1	0,959135	0,980991	0,985959	1	1	0,995041	0,887574	1	1	0,674447	0,99625	1	1	0,972201	0,961308	0,941122	0,999422	0,754683	0,932135	0,998918	0,693912	1	0,965978	1	0,989118	1	1	0,99565		0,967515867	

Overall accuracy = 96.75%

Bio-geographical region STE - Steppic - confusion matrix CLC2018 - Blind and plausibility analysis

Table 87: Confusion matrix CLC 2018 for steppic region – BLIND ANALYSS- In column the validation and line the production

CLC Classes	112	121	122	123	124	132	142	211	213	221	222	231	242	243	311	321	324	331	411	511	512	Total	User_accuracy
112	4,635239																					4,635239	1
121		1,33606																				1,33606	1
122			0,15974																			0,15974	1
123				0,026802																		0,026802	1
124					0,065638																	0,065638	1
131		0,062472																				0,062472	
132						0,015425																0,015425	1
142							0,012072															0,012072	1
211								112,3692				0,126042										112,4952	0,998879576
213									0,784322													0,784322	1
221										2,651464		0,662866										3,31433	0,8
222											0,116684											0,116684	1
231	2,173926											8,812388	2,173926									13,16024	0,669622122
242													4,996705									4,996705	1
243														1,728357								1,728357	1
311															12,74315							12,74315	1
321																0,803698						0,803698	1
324															0,599517		0,793537					1,393054	0,569638435
331																		0,140418				0,140418	1
411																			4,185088			4,185088	1
511																				1,533761		1,533761	1
512																					3,121402	3,121402	1
Total	6,809165	1,398532	0,15974	0,026802	0,065638	0,015425	0,012072	112,3692	0,784322	2,651464	0,116684	9,601297	7,170631	1,728357	13,34267	0,803698	0,793537	0,140418	4,185088	1,533761	3,121402	166,8299	
Prod_accuracy	0,680735	0,95533	1	1	1	1	1	1	1	1	1	0,917833	0,696829	1	0,955068	1	1	1	1	1	1		0,965241539

Overall accuracy = 96.5%

Table 88: Confusion matrix CLC 2018 for steppic region – PLAUSIBILITY ANALYSIS In column the validation and line the production

CLC Classes	112	121	122	123	124	132	142	211	213	221	222	231	242	243	311	321	324	331	411	511	512	Total	User_accuracy
112	4,635239																					4,635239	1
121		1,33606																				1,33606	1
122			0,15974																			0,15974	1
123				0,026802																		0,026802	1
124					0,065638																	0,065638	1
131		0,062472																				0,062472	
132						0,015425																0,015425	1
142							0,012072															0,012072	1
211								112,3692				0,126042										112,4952	0,998879576
213									0,784322													0,784322	1
221										2,651464		0,662866										3,31433	0,8
222											0,116684											0,116684	1
231												13,16024										13,16024	1
242													4,996705									4,996705	1
243														1,728357								1,728357	1
311															12,74315							12,74315	1
321																0,803698						0,803698	1
324															0,599517		0,793537					1,393054	0,569638435
331																		0,140418				0,140418	1
411																			4,185088			4,185088	1
511																				1,533761		1,533761	1
512																					3,121402	3,121402	1
Total	4,635239	1,398532	0,15974	0,026802	0,065638	0,015425	0,012072	112,3692	0,784322	2,651464	0,116684	13,94915	4,996705	1,728357	13,34267	0,803698	0,793537	0,140418	4,185088	1,533761	3,121402	166,8299	
Prod_accuracy	1	0,95533	1	1	1	1	1	1	1	1	1	0,943444	1	1	0,955068	1	1	1	1	1	1		0,991303134

Overall accuracy = 99.1%

Bio-geographical region DOM - Overseas department (tropical region) - confusion matrix CLC2018 - Blind and plausibility analysis

Table 89: Confusion matrix CLC 2018 for Tropical region – BLIND ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	141	142	211	213	221	222	231	242	243	244	311	312	313	321	322	323	324	331	332	333	411	421	423	511	512	521	522	523	Total	User_accuracy	
111	0,01789																																				0,01789	1	
112		2,916862															0,005092																				2,921954	0,998257283	
121			0,161577												0,005092																						0,166669	0,969447525	
122				0,074629													0,006825																				0,081454	0,916208237	
123					0,023021																																0,023021	1	
124						0,064595																															0,064595	1	
131							0,044961																														0,044961	1	
132								0,003158																													0,003158	1	
141									0,019232																												0,019232	1	
142										0,031181																											0,031181	1	
211											2,38941				0,132553																						2,527055	0,94553126	
213												0,277662																									0,277662	1	
222													0,577819				0,071374	0,071374																			0,720568	0,801894381	
231		0,110781													0,899972							0,004496				0,117606						0,003449					1,136305	0,792016677	
242		0,005092	0,16235													1,298797																					1,466239	0,885801843	
243															0,003449	0,16235	3,107423																			3,273221	0,94934696		
244																		0,092179																			0,092179	1	
311		0,165748																	40,52414			0,114887				1,330481						0,003055				6,234796	48,37311	0,837741095	
312																				0,093109																	0,093109	1	
313																					0,23056																0,23056	1	
321																						1,006519															1,006519	1	
322																							0,679496		0,0755												0,754996	0,9	
323		0,165748																	0,093768					0,750142	0,093768												1,103425	0,679830082	
324															0,004496	0,170244	0,004496		0,004333						0,750142	1,267278										1,450848	0,873474357		
331																										0,011514											0,011514	1	
332																											0,311411										0,311411	1	
333																												0,376002										0,423002	0,888888889
411																													0,376002	4,676151								4,676151	1
421																														0,031668								0,031668	1
511																										0,045584							0,410257					0,455841	0,9
512																																		0,018621				0,018621	1
521																																			0,007951			0,007951	1
522																			0,020023																		1,164754	0,901171947	
523																																					49,9967	50,01673	0,999599669
Total	0,01789	3,364232	0,323926	0,079721	0,023021	0,064595	0,044961	0,003158	0,019232	0,031181	2,38941	0,277662	0,132553	0,577819	0,913009	1,702765	3,19521	0,092179	40,64226	0,093109	0,23056	1,125902	0,726496	0,750142	2,884633	0,057098	0,311411	0,376002	4,676151	0,031668	0,003055	0,528816	0,018621	0,007951	1,049643	56,2315	122,9975		
Prod_accuracy	1	0,867022	0,498807	0,936125	1	1	1	1	1	1	1	1	1	1	0,985721	0,762758	0,972525	1	0,997094	1	1	0,893967	0,935306	1	0,43932	0,201649	1	1	1	1	1	0,775802	1	1	1	0,889123		0,922307277	

Overall accuracy = 92.2%

Table 90: Confusion matrix CLC 2018 for Tropical region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLC Classes	111	112	121	122	123	124	131	132	141	142	211	213	221	222	231	242	243	244	311	312	313	321	322	323	324	331	332	333	411	421	423	511	512	521	522	523	Total	User_accuracy	
111	0,01789																																				0,01789	1	
112		2,916862															0,005092																				2,921954	0,998257283	
121			0,161577												0,005092																						0,166669	0,969447525	
122				0,074629													0,006825																				0,081454	0,916208237	
123					0,023021																																0,023021	1	
124						0,064595																															0,064595	1	
131							0,044961																														0,044961	1	
132								0,003158																													0,003158	1	
141									0,019232																												0,019232	1	
142										0,031181																											0,031181	1	
211											2,38941			0,132553																							2,527055	0,94553126	
213												0,277662																									0,277662	1	
222														0,720568																							0,720568	1	
231			0,110781												0,899972							0,004496				0,117606							0,003449				1,136305	0,792016677	
242			0,005092	0,16235												1,298797																					1,466239	0,885801843	
243																0,003449	0,16235	3,107423																		3,273221	0,94934696		
244																		0,092179																			0,092179	1	
311		0,165748																	40,52414			0,114887				1,330481											48,37311	0,837741095	
312																				0,093109																	0,093109	1	
313																					0,23056																0,23056	1	
321																						1,006519															1,006519	1	
322																							0,679496		0,0755												0,754996	0,9	
323			0,165748																	0,093768					0,750142	0,093768											1,103425	0,679830082	
324																0,004496	0,170244	0,004496		0,004333						1,267278											1,450848	0,873474357	
331																										0,011514											0,011514	1	
332																											0,311411										0,311411	1	
333																																						0,423002	0,888888889
411																																						4,676151	1
421																																						0,031668	1
511																										0,045584											0,455841	0,9	
512																																						0,018621	1
521																																						0,007951	1
522																																						1,164754	0,901171947
523																																						0,11511	1,049643
Total	0,01789	3,364232	0,323926	0,079721	0,023021	0,064595	0,044961	0,003158	0,019232	0,031181	2,38941	0,277662	0,132553	0,720568	0,913009	1,631391	3,123836	0,092179	40,64226	0,093109	0,23056	1,125902	0,726496	0,750142	2,884633	0,057098	0,311411	0,376002	4,676151	0,031668	0,003055	0,528816	0,018621	0,007951	1,049643	49,9967	50,01673	0,999599669	
Prod_accuracy	1	0,867022	0,498807	0,936125	1	1	1	1	1	1	1	1	1	1	0,985721	0,796129	0,994746	1	0,997094	1	1	0,893967	0,935306	1	0,43932	0,201649	1	1	1	1	1	0,775802	1	1	1	0,889123	0,923467858		

Overall accuracy = 92.34%

Figure 91: Confusion matrix of change CLCCH2012-2018 for blind analysis (at level 2 of CLC nomenclature) between the validation and the production

For the blind analysis, the overall accuracy of CLCCH2012-2018 at the third level of nomenclature is **87.12%**. The overall accuracy of CLCCH2012-2018 at the level 2 is **89.56%**.

[illegible]

For the plausibility analysis, the overall accuracy of CLCCH2012-2018 at the third level is **87.8%**. The overall accuracy of CLCCH2012-2018 at the second level is **90.1%**.

Zone 1: Turkey - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

	11-11	12-12	13-11	13-12	13-13	13-32	13-51	14-14	21-11	21-12	21-13	21-21	21-22	21-51	22-11	22-13	22-22	23-13	23-21	23-23	23-24	23-51	24-11	24-13	24-21	24-24	24-51	31-13	31-31	31-32	31-51	32-11	32-12	32-13	32-21	32-22	32-23	32-31	32-32	32-51	33-11	33-12	33-13	33-21	33-24	33-33	33-51	41-21	41-41	42-42	51-42	51-51	52-52	Total	User_accuracy					
11-11	39,39143				0,054592							0,031787																																										39,39143	1					
11-13	0,031787																																																										0,086379	
11-21	0,454743																																																										0,48653	
11-23																																																											0,070333	
11-24																																																											0,009834	
11-51															0,06357									0,009834																															0,06357					
12-11	0,070333																																																										0,070333	
12-12	0,049686	10,707																																																									10,80637	0,990804246
12-13		0,070333			0,059534																																																			0,129867				
12-21		0,070333																																																									0,070333	
12-23																																																											0,049686	
12-24																																																											0,453883	
12-31																																																											0,149839	
12-32		0,369152																																																										

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[illegible]

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Zone 2: France - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 95: Confusion matrix CLCCH2012-2018 for zone 2 (France) –BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-11	13-12	13-13	13-23	13-24	13-32	13-51	14-14	21-21	22-12	22-21	22-22	23-11	23-23	24-13	24-24	31-11	31-13	31-31	31-32	32-12	32-21	32-31	32-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy	
11-11	100,0008																0,038112														100,0389	0,99961903		
12-11	0,082104																														0,082104			
12-12	0,239275	22,95987																0,021559													23,22071	0,98876718		
12-13																0,081786															0,081786			
12-21		0,021559									0,198635							0,009432														0,229627		
12-23	0,013954															0,009432				0,009432												0,032819		
12-24																		0,048471														0,048471		
13-11	0,796039		0,245359																													1,041398	0,235605178	
13-12		0,373949		0,490717			0,081786	0,081786			0,081786																					1,110025	0,442077596	
13-13		0,063371			1,497394																											1,560765	0,959397221	
13-14										0,081786																						0,081786		
13-21											0,257928																					0,257928		
13-22														0,264245																		0,264245		
13-23						0,059138										0,414966																0,474105	0,124737165	
13-24					0,029569													0,47472														0,504289		
13-31																				0,414966				0,031393								0,446359		
13-32							0,031393	0,031393																		0,414966						0,477752	0,0657095	
13-33																										0,414966						0,063181		
13-51									0,04238																						1,334067	0,030789383		
14-12		0,248707																														0,248707		
14-14										5,285678								0,248707														5,534385	0,955061446	
14-21											0,447343																					0,447343		
21-13					0,00978																											0,00978		
21-21											625,4897					0,022921		13,05424														638,5669	0,979521076	
21-24											12,9952						0,101291															13,09649		
22-11	0,96765																															0,96765		
22-13												0,145233																				0,145233		
22-21											0,37487		0,116684																			0,491554	0,237377042	
22-22	0,045399										0,37487			46,77715		1,12461	3,480178			0,045399												51,84761	0,902204641	
22-23																																0,922251		
23-11	0,145233																															0,145233		
23-21																0,116684																0,116684		
23-23																332,832		21,70657						0,019326								354,5579	0,938723991	
23-24																	7,232364															7,232364		
24-11																	0,145233															0,145233		
24-12																	0,290466															0,435699		
24-13																																0,290466	1	
24-21										0,071319																							0,071319	
24-22													4,953601																			4,98926		
24-24										4,917941						9,835883		284,7779			6,211867											305,7436	0,931427217	
24-31																				0,02544												0,02544		
24-32																		0,051492														0,051492		
31-11																			0,076656													0,076656	1	
31-13																				0,076656												0,076656	1	
31-31																					613,7362			0,466254	0,466254							614,6687	0,998482911	
31-32																					1,274712	10,26009			0,341976							11,87678	0,863878288	
31-51																						0,0266										0,0266		
32-12																							0,076656									0,076656	1	
32-13					0,092284																											0,092284		
32-21											0,009459													0,072433								0,081892	0,884494296	
32-22																							0,072433									0,072433		
32-23															1,112619																	1,112619		
32-24																	0,018918															0,018918		
32-31																					16,79582	0,466368		5,130047	1,15662							23,54885	0,217847001	
32-32															3,753634		6,206805			4,963062						144,501						159,4245	0,906391396	
32-33																										0,466368						0,466368		
33-24																	0,13311															0,13311		
33-33																										3,764374	34,3186					38,08297	0,901153355	
41-41																												3,648897				3,648897	1	
42-42																												0,053398	3,64972			3,703117	0,985580357	
51-12		0,010495																														0,010495		
51-24																																0,010495		
51-51																				0,459499						0,249265						13,519	14,22777	0,950184414
52-42																												0,274382				0,274382		
52-51																																0,104126		
52-52																																3,090929	3,090929	1
Total	102,2905	23,67796	0,245359	0,490717	1,629027	0,172318	0,081786	0,031393	0,04238	5,367464	645,2191	0,145233	0,116684	51,995	0,145233	350,2268	0,290466	338,0753	0,076656	0,076656	643,9364	10,75306	0,076656	0,144865</										

Table 96: Confusion matrix CLCCH2012-2018 for zone 2 (France) –PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-11	13-12	13-13	13-23	13-24	13-32	13-51	14-14	21-21	22-12	22-21	22-22	23-11	23-23	23-24	24-13	24-24	31-11	31-13	31-31	31-32	32-12	32-21	32-31	32-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy	
11-11	100,0008																		0,038112														100,0389	0,99961903	
12-11	0,082104																																0,082104		
12-12	0,239275	22,95987									0,021559																						23,22071	0,98876718	
12-13																0,081786																	0,081786		
12-21		0,021559									0,198635								0,009432														0,229627		
12-23	0,013954															0,009432					0,009432												0,032819		
12-24																			0,048471														0,048471		
13-11	0,747277		0,245359													0,048761																	1,041398	0,235605178	
13-12		0,310578		0,490717	0,063371	0,081786	0,081786				0,081786																						1,110025	0,442077596	
13-13		0,063371			1,497394																												1,560765	0,959397221	
13-14										0,081786																							0,081786		
13-21											0,257928																						0,257928		
13-22													0,201064														0,063181						0,264245		
13-23						0,059138										0,414966																	0,474105	0,124737165	
13-24							0,029569												0,47472														0,504289	0,058635494	
13-31																						0,414966				0,031393							0,446359		
13-32						0,031393		0,446359																									0,477752	0,9342905	
13-33																												0,063181					0,063181		
13-51									0,072103																								1,376447	0,052383361	
14-12		0,248707																															0,248707		
14-14										5,285678									0,248707														5,534385	0,955061446	
14-21										0,198635	0,248707																						0,447343		
21-13					0,00978																												0,00978		
21-21											625,4897					0,022921			13,05424														638,5669	0,979521076	
21-24											12,9952								0,101291														13,09649		
22-11	0,96765																																0,96765		
22-13												0,145233																					0,145233		
22-21											0,37487		0,116684																				0,491554	0,237377042	
22-22	0,045399										0,37487			46,77715		1,12461			3,480178			0,045399											51,84761	0,902204641	
22-23																0,922251																	0,922251		
23-11	0,145233																																0,145233		
23-21																0,116684																	0,116684		
23-23																332,832	7,232364		14,4742						0,019326							354,5579	0,938723991		
23-24																		7,232364															7,232364		
24-11																		0,145233															0,145233		
24-12															0,145233				0,290466														0,435699		
24-13																		0,290466															0,290466	1	
24-21										0,071319																							0,071319		
24-22													4,953601						0,035659														4,98926		
24-24																9,835883			289,6959			6,211867										305,7436	0,947512398		
24-31																						0,02544											0,02544		
24-32																			0,051492														0,051492		
31-11																			0,076656														0,076656	1	
31-13																					0,076656												0,076656	1	
31-31																						614,2024					0,466254						614,6687	0,999241455	
31-32																						1,103724	10,26009		0,170988	0,341976							11,87678	0,863878288	
31-51																							0,0266										0,0266		
32-12																								0,076656									0,076656	1	
32-13					0,092284																												0,092284		
32-21											0,009459															0,072433							0,081892	0,884494296	
32-22																									0,072433								0,072433		
32-23																1,112619																	1,112619		
32-24																			0,018918														0,018918		
32-31																																	23,54885	0,262175465	
32-32																3,753634			6,206805			14,429	0,497048			6,173932	1,239445	1,209428				159,4245	0,912916033		
32-33																						3,086245				0,83663	145,5412						0,466368		
33-24																			0,13311														0,13311		
33-33																											3,764374	34,3186					38,08297	0,901153355	
41-41																													3,648897					3,648897	1
42-42																																	3,703117	1	
51-12		0,010495																																0,010495	
51-24																																	0,010495		
51-51																																	13,519	14,22777	0,950184414
52-42																																	0,274382		
52-51																																	0,104126	0,104126	
52-52																																	3,090929	3,090929	1
Total	102,2417	23,61458	0,245359	0																															

[illegible]

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Zone 4: Sweden - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 99: Confusion matrix CLCCH2012-2018 for zone 4 (Sweden) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-11	13-12	13-13	14-14	21-21	22-22	23-23	24-12	24-24	31-11	31-13	31-31	31-32	31-41	32-21	32-31	32-32	33-32	33-33	41-13	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	17,94618	0,025868																									17,97205	0,998560634
11-12		0,009682																									0,009682	
11-14		0,004362				0,00532																					0,009682	
11-23												0,028335															0,028335	
11-31						0,004362																					0,004362	
12-11	0,116012	0,116012																									0,232023	
12-12		4,193442																									4,193442	1
12-21							0,277172																				0,277172	
12-24							0,116012			0,00964																	0,125652	
12-31		0,028335												0,241663													0,269998	
13-11	0,012065		0,081786											0,006032													0,099884	0,818815793
13-12				0,081786																							0,081786	1
13-13	0,062603				1,084068									0,041777													1,188448	0,912171265
13-21				0,029569																							0,029569	
13-24											0,037965																0,037965	
13-31														0,111138													0,111138	
14-12		0,158846																									0,158846	
14-14						3,581866																					3,581866	1
21-11	0,011695																										0,011695	
21-21	2,537884						108,6486		5,062234		7,578569			2,52619													126,3535	0,859878198
21-24											0,00448																0,00448	
21-31														0,007064													0,007064	
22-22								0,045381							0,015127				0,015127								0,075634	0,6
23-21							0,01344		0,00448																		0,017921	
23-23	0,560061								5,619426		1,680184			2,80568					0,565435								11,23079	0,500359114
23-24									0,00448																		0,00448	
23-31														0,010748													0,010748	
24-11	0,789355																										0,789355	
24-24		0,25202					0,789355				23,76619			1,836104					0,50404								27,14771	0,875440003
24-33																			0,789355								0,789355	
31-12													0,076656														0,076656	
31-13													0,153313														0,153313	1
31-21							0,011352							0,072433													0,083785	
31-31														1258,154	18,92246				26,15569				1,767275				1305	0,964103124
31-32														34,04486				0,466368	20,78793								55,29915	0,615648801
31-41															0,016119												0,016119	1
32-13				0,010389																							0,010389	
32-21							0,00987				0,00987																0,01974	
32-24						0,004935					0,00987																0,014805	
32-31	0,466368													30,71968	3,834132		0,004935	24,25113	12,85692								72,13317	0,336199448
32-32														0,862182					292,5772		2,330416						295,7698	0,9892058
32-41																						0,023389					0,023389	
33-33									0,036405					0,036405						0,023431	46,41476						46,511	0,997930801
41-13																					0,066832						0,066832	1
41-32														0,056981													0,056981	
41-41																							122,3144				122,3144	1
41-51																									0,170851		0,170851	
42-32																		0,011099									0,011099	
42-42														0,011099										0,033298			0,044397	0,75
51-31														0,24964													0,24964	
51-51														3,305159											154,2139		157,5191	0,979017409
52-31														0,045495													0,045495	
52-33																			0,026746								0,026746	
52-41																							0,026746				0,026746	
52-52																										1,761697	1,761697	1
Total	22,50223	4,788566	0,081786	0,111355	1,094458	3,596483	109,8658	0,045381	10,72703	0,00964	33,08713	0,028335	0,229969	1300,971	56,94599	0,016119	0,004935	24,7175	354,2895	0,023431	48,74518	0,066832	124,1318	0,033298	154,3848	1,761697	2252,26	
Prod_accuracy	0,797529	0,87572	1	0,734461	0,990507	0,995936	0,988921	1	0,523857		0,718291		0,666667	0,967089	0,597845	1		0,981132	0,825814		0,952192	1	0,985359	1	0,998893	1		0,931975411

Overall accuracy at the second level = 93,2%

Overall accuracy at the third level = 92,6%

Table 100: Confusion matrix CLCCH2012-2018 for zone 4 (Sweden) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lv2	11-11	12-12	13-11	13-12	13-13	14-14	21-21	22-22	23-23	24-12	24-24	31-11	31-13	31-31	31-32	31-41	32-21	32-31	32-32	33-32	33-33	41-13	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	17,94618	0,025868																									17,97205	0,998560634
11-12		0,009682																									0,009682	
11-14		0,004362				0,00532																					0,009682	
11-23												0,028335															0,028335	
11-31						0,004362																					0,004362	
12-11	0,116012	0,116012																									0,232023	
12-12		4,193442																									4,193442	1
12-21							0,277172																				0,277172	
12-24							0,116012			0,00964																	0,125652	
12-31		0,028335												0,241663													0,269998	
13-11	0,012065		0,081786											0,006032													0,099884	0,818815793
13-12				0,081786																							0,081786	1
13-13	0,062603				1,084068									0,041777													1,188448	0,912171265
13-21				0,029569																							0,029569	
13-24											0,037965																0,037965	
13-31														0,111138													0,111138	
14-12		0,158846																									0,158846	
14-14						3,581866																					3,581866	1
21-11	0,011695																										0,011695	
21-21	2,537884						108,6486		5,062234		7,578569			2,52619													126,3535	0,859878198
21-24											0,00448																0,00448	
21-31														0,007064													0,007064	
22-22								0,045381							0,015127				0,015127								0,075634	0,6
23-21							0,01344		0,00448																		0,017921	
23-23	0,560061								5,619426		1,680184			2,80568					0,565435								11,23079	0,500359114
23-24									0,00448																		0,00448	
23-31														0,010748													0,010748	
24-11	0,789355																										0,789355	
24-24		0,25202					0,789355				23,76619			1,836104					0,50404								27,14771	0,875440003
24-33																			0,789355								0,789355	
31-12													0,076656														0,076656	
31-13													0,153313														0,153313	1
31-21							0,011352								0,072433												0,083785	
31-31														1258,154	18,92246				26,15569				1,767275				1305	0,964103124
31-32														34,04486				0,466368	20,78793								55,29915	0,615648801
31-41																0,016119											0,016119	1
32-13					0,010389																						0,010389	
32-21							0,00987				0,00987																0,01974	
32-24						0,004935					0,00987																0,014805	
32-31	0,466368													30,71968	3,834132		0,004935	24,25113	12,85692								72,13317	0,336199448
32-32														0,862182					292,5772		2,330416						295,7698	0,9892058
32-41																						0,023389					0,023389	
33-33									0,036405					0,036405						0,023431	46,41476						46,511	0,997930801
41-13																					0,066832						0,066832	1
41-32														0,056981													0,056981	
41-41																							122,3144				122,3144	1
41-51																									0,170851		0,170851	
42-32																			0,011099								0,011099	
42-42													0,011099											0,033298			0,044397	0,75
51-31														0,24964													0,24964	
51-51														3,305159											154,2139		157,5191	0,979017409
52-31														0,045495													0,045495	
52-33																			0,026746								0,026746	
52-41																							0,026746				0,026746	
52-52																										1,761697	1,761697	1
Total	22,50223	4,788566	0,081786	0,111355	1,094458	3,596483	109,8658	0,045381	10,72703	0,00964	33,08713	0,028335	0,229969	1300,971	56,94599	0,016119	0,004935	24,7175	354,2895	0,023431	48,74518	0,066832	124,1318	0,033298	154,3848	1,761697	2252,26	
Prod_accuracy	0,797529	0,87572	1	0,734461	0,990507	0,995936	0,988921	1	0,523857		0,718291		0,666667	0,967089	0,597845	1		0,981132	0,825814		0,952192	1	0,985359	1	0,998893	1		0,931975411

Overall accuracy at the second level =93.5 %
Overall accuracy at the third level =93 %

Table 101: Confusion matrix CLCCH2012-2018 for zone 5 (Germany) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level =87.5 %
Overall accuracy at the third level =84,4 %

[illegible]

Overall accuracy at the second level =88%
Overall accuracy at the third level =85.5 %

Zone 6: Finland - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 103: Confusion matrix CLCCH2012-2018 for zone 6 (Finland) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-11	13-12	13-13	13-31	13-32	14-14	21-21	23-23	23-32	24-24	31-12	31-13	31-23	31-31	31-32	31-41	32-21	32-23	32-31	32-32	32-41	33-33	41-21	41-23	41-32	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	11,7353	0,724283																													12,45959	0,941869428	
11-24												0,724283																				0,724283	
11-31	0,724283																															0,724283	
12-11	0,002383							0,002146																								0,004529	
12-12	0,276292	2,871173																														3,147465	0,912217584
12-13					0,01268																											0,01268	
12-21									0,138146																							0,138146	
12-24												0,002383																				0,002383	
12-31	0,138146	0,002146														0,009059																0,149351	
12-32								0,002146																								0,002146	
13-11	0,029878		0,081786					0,003803																								0,115468	0,708301611
13-12		0,053058		0,081786																												0,134844	0,60652441
13-13				0,001689	1,159319											0,001689																1,162697	0,997095047
13-14					0,004282																											0,004282	
13-24												0,001689																				0,001689	
13-31						0,004282							0,055125			0,357229						0,001689									0,418325	0,010236684	
13-32							0,031393														0,031393	0,217155									0,279941	0,112141	
14-11	0,014755																															0,014755	
14-12		0,014755																														0,014755	
14-13					0,021881																											0,021881	
14-14	0,073775							0,81801																								0,891785	0,917272754
21-21									62,81112							1,304419												0,002423				64,11796	0,979618153
21-24												0,010565																0,002423				0,012988	
21-31																1,332818						0,002423										1,335241	
21-32											0,051492																					0,051492	
21-41																												0,002423				0,002423	
23-23										0,070577		0,013631				0,013631																0,1251	0,564163879
23-24												0,004724																0,002423				0,007147	
23-31																0,120119																0,120119	
23-32																						0,065122									0,065122		
23-33																0,004724						0,072878									0,077602		
24-21									6,919673							1,155702																8,075375	
24-24									2,306558			35,75407				5,768817																43,82944	0,815754558
24-31												1,153279				2,306558																3,459836	
24-32																						1,153279										1,153279	
31-13														0,153313																		0,153313	1
31-21															0,072433																	0,072433	
31-23																0,072433																0,072433	
31-24																0,018268																0,018268	
31-31	0,58327								11,92184							860,1495						17,61689					0,003632					890,2752	0,966161437
31-32																0,56446	17,72198					0,767908									19,05435	0,930075385	
31-41																	0,048357	0,048357								0,016119	0,58327				0,696103	0,069468431	
32-13					0,029333											2,860336																2,889669	
32-21									0,076698							0,007077			0,072433	0,144865		0,072433					0,072433				0,445938	0,162427578	
32-24												0,022648				0,002836																0,025484	
32-31								0,004265	0,007077							24,60744	0,310865				8,868068	10,87874					0,004265				44,68072	0,198476361	
32-32																13,23473					0,002836	137,5355		0,08996						150,863	0,911658137		
32-41								0,016119															0,209548		0,08996			0,037065				0,262732	0,797572598
33-33																0,633069								5,653343								8,2279	0,68709426
41-21																										0,050132	0,025066		0,025066			0,100264	0,5
41-31																1,812589																1,812589	
41-32																																	

Overall accuracy at the second level =92 %

Overall accuracy at the third level = 91.1%

Table 104: Confusion matrix CLCCH2012-2018 for zone 6 (Finland) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-11	13-12	13-13	13-32	14-14	21-21	21-41	23-23	23-32	24-24	31-12	31-13	31-23	31-31	31-32	31-41	32-13	32-21	32-23	32-31	32-32	32-41	33-33	41-21	41-23	41-32	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	11,7353	0,724283																															12,45959	0,941869428
11-24												0,724283																					0,724283	
11-31	0,724283																																0,724283	
12-11	0,002383						0,002146																										0,004529	
12-12	0,276292	2,871173																															3,147465	0,912217584
12-13					0,01268																												0,01268	
12-21								0,138146																									0,138146	
12-24		0,002383																															0,002383	
12-31	0,138146	0,002383											0,002146			0,006676																	0,149351	
12-32							0,002146																										0,002146	
13-11	0,033682		0,081786																														0,115468	0,708301611
13-12		0,027373		0,107471																													0,134844	0,797000194
13-13				0,001689	1,137438											0,001689			0,021881														1,162697	0,978275755
13-14					0,004282																												0,004282	
13-24												0,001689																					0,001689	
13-31				0,004282	0,06369								0,055125			0,238414						0,056814										0,418325		
13-32					0,004282	0,086518																0,031393	0,157748									0,279941	0,309058311	
14-11	0,014755																																0,014755	
14-12		0,014755																															0,014755	
14-13							0,021881																										0,021881	
14-14	0,044265	0,014755					0,832765																										0,891785	0,933818203
21-21								62,81112								1,304419													0,002423				64,11796	0,979618153
21-24												0,010565																	0,002423				0,012988	
21-31								0,002423								1,327972							0,004846										1,335241	
21-32										0,051492																							0,051492	
21-41																												0,002423					0,002423	
23-23										0,070577		0,013631				0,013631															0,027261		0,1251	0,564163879
23-24										0,002423		0,004724																					0,007147	
23-31																0,115395						0,004724											0,120119	
23-32																0,013631							0,051492										0,065122	
23-33																0,023079							0,054523										0,077602	
24-21								6,919673								0,002423						1,153279											8,075375	
24-24								2,306558				35,75407				4,615538						1,153279											43,82944	0,815754558
24-31												1,153279				2,306558																	3,459836	
24-32																						1,153279											1,153279	
31-13														0,153313																			0,153313	1
31-21															0,072433																		0,072433	
31-23																0,072433																	0,072433	
31-24																0,018268																	0,018268	
31-31	0,58327							11,92184								860,0069						0,142576	17,61689					0,003632					890,2752	0,966001289
31-32																0,466368	17,82007						0,767908										19,05435	0,935223403
31-41																	0,048357	0,048357											0,599389				0,696103	0,069468431
32-13				0,008387												2,860336			0,020946														2,889669	0,007248587
32-21								0,076698								0,007077				0,072433	0,072433	0,072433	0,072433					0,072433				0,445938	0,162427578	
32-24												0,022648				0,002836																	0,025484	
32-31							0,004265	0,007077				0,007077				12,37718	0,310865					21,24676	10,72324					0,004265				44,68072	0,475524049	
32-32																12,92386						0,313701	137,5355		0,08996							150,863	0,911658137	
32-41									0,016119															0,209,										

Overall accuracy at the second level =92.85 %
Overall accuracy at the third level = 91.95%

Zone 7: Norway - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 105: Confusion matrix CLCCH2012-2018 for zone 7 (Norway) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-11	13-12	13-13	14-14	21-13	21-21	23-23	24-24	31-13	31-14	31-31	31-32	32-13	32-31	32-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	7,843167												0,002473								0,002473		7,848113	0,999369719
11-12		0,00902																					0,00902	
11-21		0,002473																					0,002473	
11-24										0,004947													0,004947	
11-31	0,002473																						0,002473	
12-11	0,019846																						0,019846	
12-12		1,34793											0,004695				0,004695						1,357321	0,993081502
12-21		0,019846																					0,019846	
12-31								0,003007					0,045193										0,0482	
13-11	0,022853		0,081786																				0,104639	0,78160092
13-13					0,368249								0,014215						0,0055				0,387965	0,949182505
13-21							0,003007												0,016246				0,019253	
13-24			0,003007																				0,003007	
13-31													0,059539										0,059539	
13-33																		0,019846					0,019846	
13-41																			0,019715				0,019715	
14-14						2,479767		0,013912															2,493679	0,994421055
14-31						0,004621							0,004621										0,009243	
14-51						0,009243							0,004621										0,018486	
21-21							21,84986	0,0055		2,42654			0,0055										24,2874	0,899637663
21-24								0,0055		0,0055													0,011	
21-31													0,011										0,011	
23-23									1,005403	0,052916													1,058319	0,95
24-24										45,37497			2,0353										47,41027	0,957070475
31-11	0,076656																						0,076656	
31-12				0,076656							0,076656												0,153313	
31-13					0,076656						0,229969												0,306626	0,75
31-14						0,076656						0,076656											0,153313	0,5
31-21											0,072433		0,072433	0,072433									0,217298	
31-31													447,8871					3,691336					451,5784	0,991825702
31-32										0,466368			0,466368	8,86099			6,609679						16,4034	0,540192143
32-13															0,076656								0,076656	1
32-21										0,003924													0,003924	
32-24										0,119094													0,119094	
32-31													5,133636	0,11517		1,399104	0,699589						7,347499	0,190419054
32-32													0,850984		0,0055		218,2682	7,936733					227,0614	0,961273828
33-31																	0,004777						0,004777	
33-33																	13,71235	434,4536					448,166	0,969403405
41-41																			89,91241				89,91241	1
42-42		0,004859				0,004859											0,004859		0,082596				0,097172	0,85
51-51																					0,091662		0,091662	
51-51																					58,04599		58,04599	1
52-52										0,122727												1,718181	1,840908	0,933333333
Total	7,964996	1,384129	0,084793	0,076656	0,444906	2,575146	0,003007	21,86678	1,016404	48,58161	0,379058	0,076656	456,5977	9,048593	0,082157	1,399104	239,2992	446,1064	89,95387	0,082596	58,14012	1,718181	1386,882	
Prod_accuracy	0,984704	0,973847	0,96454		0,827702	0,962962		0,999226	0,989177	0,933995	0,606686	1	0,980923	0,979267	0,933053	1	0,912114	0,973879	0,999539	1	0,998381	1		0,967178517

Overall accuracy at the second level = 96.7%

Overall accuracy at the third level =96.5 %

Table 106: Confusion matrix CLCCH2012-2018 for zone 7 (Norway) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-11	13-12	13-13	14-14	21-13	21-21	23-23	24-24	31-13	31-14	31-31	31-32	32-13	32-31	32-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	7,848113																						7,848113	1
11-12		0,00902																					0,00902	
11-21		0,002473																					0,002473	
11-24										0,004947													0,004947	
11-31	0,002473																						0,002473	
12-11	0,019846																						0,019846	
12-12		1,337474				0,019846																	1,357321	0,985378288
12-21		0,019846																					0,019846	
12-31								0,003007					0,045193										0,0482	
13-11	0,019846		0,084793																				0,104639	0,810335709
13-13					0,368249								0,014215						0,0055				0,387965	0,949182505
13-21							0,003007												0,016246				0,019253	
13-24			0,003007																				0,003007	
13-31													0,059539										0,059539	
13-33																		0,019846					0,019846	
13-41																			0,019715				0,019715	
14-14						2,479767		0,013912															2,493679	0,994421055
14-31						0,009243																	0,009243	
14-51						0,009243				0,004621			0,004621										0,018486	
21-21								21,85536	0,0055	2,42654													24,2874	0,899864124
21-24									0,0055														0,011	
21-31													0,011										0,011	
23-23									1,005403	0,052916													1,058319	0,95
24-24										45,37497			2,0353										47,41027	0,957070475
31-11	0,076656																						0,076656	
31-12				0,076656							0,076656												0,153313	
31-13					0,076656						0,229969												0,306626	0,75
31-14						0,076656						0,076656											0,153313	0,5
31-21											0,072433		0,072433	0,072433									0,217298	
31-31													447,8871					3,691336					451,5784	0,991825702
31-32										0,466368			0,466368	14,07157			1,399104						16,4034	0,857844173
32-13															0,076656								0,076656	1
32-21										0,003924													0,003924	
32-24										0,119094													0,119094	
32-31													4,891523	0,003924		1,403028	1,049024						7,347499	0,190953162
32-32					0,0055								0,236103				218,883	7,936733					227,0614	0,963981825
33-31																	0,004777						0,004777	
33-33																	13,71235	434,4536					448,166	0,969403405
41-41																			89,91241				89,91241	1
42-42		0,004859				0,004859														0,087455			0,097172	0,9
51-31																					0,091662		0,091662	
51-51																					58,04599		58,04599	1
52-52										0,122727												1,718181	1,840908	0,933333333
Total	7,966936	1,373673	0,0878	0,076656	0,450406	2,599614	0,003007	21,87228	1,016404	48,57611	0,379058	0,076656	455,7289	14,14792	0,076656	1,403028	235,0483	446,1016	89,95387	0,087455	58,13765	1,718181	1386,882	
Prod_accuracy	0,985086	0,973648	0,965754		0,817594	0,953898		0,999226	0,989177	0,934101	0,606686	1	0,982793	0,994603	1	1	0,931226	0,97389	0,999539	1	0,998423	1		0,971387409

Overall accuracy at the second level = 97.14%

Overall accuracy at the third level = 96.9%

Table 107: Confusion matrix CLCCH2012-2018 for zone 8 (Poland) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 92%
Overall accuracy at the third level = 91.8%

[illegible]

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Table 109: Confusion matrix CLCCH2012-2018 for zone 9 (Italy) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 87.1%
Overall accuracy at the third level = 83.9 %

Table 110: Confusion matrix CLCCH2012-2018 for zone 9 (Italy) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lw2	11-11	12-12	13-12	13-13	13-24	13-32	13-33	14-14	21-12	21-21	21-24	21-31	21-41	21-51	22-21	22-22	23-12	23-21	23-23	24-12	24-24	31-21	31-31	31-32	32-21	32-31	32-32	32-33	32-51	33-32	33-33	33-51	41-32	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	47,91667		0,003883																	0,003883																	47,92443	0,999837944	
11-12	0,003883	0,864702																																				0,868585	
11-14								0,003883																													0,003883		
11-21	0,864702																																					0,882933	
11-22																0,077204																					0,077204		
11-24																																						0,003883	
12-11	0,698918																																					0,698918	
12-12		14,06969						0,006098		0,056306											0,040908					0,080648											14,25365	0,987093808	
12-13		0,006098	0,006157																																		0,012255		
12-21		0,030637							0,121787																												0,152424		
12-22																0,080065				0,080065																0,16013			
12-24		0,012002																	0,002861	0,018231	0,006157															0,03925			
13-11	0,202923			0,003986																																	0,206909		
13-12	0,020266	0,194731	0,01849														0,01849			0,020266																0,272245	0,067918614		
13-13				1,883408		0,111582		0,107596														0,01849								0,107596					0,107596		2,336267	0,806161353	
13-21				0,003986	0,003986			0,020266		0,032224										0,003986																	0,064449		
13-24				0,077204																0,154408							0,077204										0,308817		
13-32							0,006157																			0,01849										0,024647			
13-33				0,031393		0,006157																											0,066832			0,03755			
13-41																																					0,066832		
13-42																																		0,01849			0,01849		
13-51																				0,107596																0,107596			
14-11	0,024036							0,077204																													0,10124		
14-12		0,066531																																			0,066531		
14-14								1,728712									0,024036																				1,752748	0,986286621	
14-21								0,006157																													0,006157		
14-24									0,024036									0,024036																			0,048072		
21-12							0,290466																														0,290466	1	
21-21								321,5858									6,810026			20,43008	0,622264	0,020767				0,026083	0,026083										349,5211	0,920075501	
21-22								0,233367								0,719178																					0,952545		
21-24	0,026083								0,116684											0,04685																	0,189616	0,615368441	
21-31																						0,041534		0,051492	0,072259												0,165285		
21-32								0,051492																		6,836108											6,8876		
21-41												0,066832																									0,066832	1	
21-51														0,03315																							0,03315	1	
22-21									1,827247																												1,827247		
22-22									1,214884							73,89726				7,076416							2,537674										84,72623	0,872188652	
22-24																				6,42396																	6,42396		
23-14							0,904037																														0,904037		
23-21								0,904037																													0,904037		
23-23	0,904037															0,904037		10,84844		1,808073																	14,46458	0,75	
23-32	0,904037																										0,91372										1,817756		
24-11	1,844291																																				1,844291		
24-21									0,429929																												0,429929		
24-22								1,844291								3,688582				3,688582																	9,221454		
24-24	1,844291							4,118511								1,844291		7,66987		159,7423		5,441391			1,844291		1,844291	0,026083									182,531	0,875151493	
24-31																																					1,844291		
24-32																				0,051492							2,157856										2,209348		
31-21								0,067921		0,014028											0,020335		0,08646		0,020335												0,209079	0,097261033	
31-31																						330,9185	0,033558		0,033558		2,322866										333,3085	0,99282952	
31-32																						1,55594	1,399104		0,218673	3,52714										6,700857	0,208794756		
31-33																												0,466368									0,466368		
32-11	0,009683																																				0,009683		
32-13				0,878657																																	0,878657		
32-21																		0,016621																		0,016621			
32-24																				0,895278					0,050543											0,945821			
32-31																				0,016621	1,52934			0,050543	0,101085		0,218673									1,916263	0,052751319		
32-32	0,878657						1,755301													5,551739		2,63597				110,7466										121,5683	0,910982819		
32-33																		0,466368		2,909703						14,54851	0,466368			9,042622						27,43357	0,016999897		
32-52				</																																			

Table 111: Confusion matrix CLCCH2012-2018 for zone 10 (United Kingdom, Ireland) – BLIND ANALYSIS - In column the validation and line the production

Overall accuracy at the second level = 87.7%
Overall accuracy at the third level = 87.2 %

CLCC_Lv2	11-11	12-12	12-13	13-11	13-12	13-13	13-21	13-23	13-32	13-41	13-51	14-14	14-21	14-23	21-11	21-12	21-13	21-21	21-23	21-32	21-51	22-22	23-11	23-12	23-13	23-21	23-23	23-32	24-24	24-32	31-11	31-12	31-31	31-32	32-12	32-31	32-32	33-23	33-32	33-33	41-12	41-32	41-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	59,91439											0,074695																	1,99625						0,016087												61,98534	0,966589759	
11-12		0,074695																																0,016087													0,090782		
11-32																																		0,016087													0,016087		
12-11		0,491392																																0,016087													0,491392		
12-12		12,14										0,059242																0,029153																			12,78185	0,949784053	
12-13			0,081786			0,037348																																								0,119134	0,686506062		
12-21		0,037348														0,037348																															0,074696		
12-23												0,038733																																			0,185134		
12-32																																															0,008626		
13-11	0,247461		0,124624			0,005415																																									0,477253	0,26112733	
13-12		0,112253		0,251348																				0,093764																							0,451376	0,556847897	
13-13					2,877238																																										2,90298	0,99113264	
13-14												0,054605																																			0,054605		
13-21						0,029569																																									0,029569		
13-23		0,055311				0,051275		0,035558	0,029569	0,029569																																						0,029569	
13-24					</																																												

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Table 113: Confusion matrix CLCCH2012-2018 for zone 11 (Romania) – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level =84.4 %
Overall accuracy at the third level = 82%

[illegible]

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Zone 12: Greece - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 115: Confusion matrix CLCCH2012-2018 for zone 12 (Greece) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	14-14	21-13	21-21	22-22	23-23	24-12	24-24	31-31	32-12	32-13	32-22	32-24	32-32	32-51	33-32	33-33	41-21	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	12,35369																									12,35369	1
12-12	0,138024	4,514935									0,058488						0,022621									4,734068	0,953711481
12-13				0,058488																						0,058488	
12-21											0,040392															0,040392	
12-22								0,011164																		0,011164	
12-23																	0,063013									0,063013	
12-24											0,020284															0,020284	
12-32																	0,022621									0,022621	
12-42																						0,011342				0,011342	
13-11	0,085634																									0,085634	
13-12			0,163572														0,040392									0,203964	0,801967406
13-13		0,096214		1,783954																						1,880168	0,948826788
13-14	0,060341																									0,060341	
13-22								0,063013																		0,063013	
13-24								0,060341									0,040392									0,100733	
13-31												0,096214														0,096214	
13-32																	0,136606									0,136606	
14-11	0,022621																									0,022621	
14-13	0,020284																									0,020284	
14-14	0,060634				0,869735																					0,93037	0,934827827
14-21																					0,019623					0,019623	
14-24											0,060634															0,060634	
14-31								0,019623																		0,019623	
21-21		1,829517					81,90519				3,668609						1,829517									89,23283	0,917881754
21-22								2,279979																		2,279979	
21-24							4,799024		0,018203		3,659035															8,476262	
21-32																	0,012593									0,012593	
22-11	0,244333																									0,244333	
22-21							0,244333				0,405403															0,649736	
22-22	0,244333						0,488666	30,22976			3,774853						2,069978									36,80759	0,821291473
22-24											2,27268									1,034989						3,307669	
22-32																	0,202701									0,202701	
23-11	0,171001																									0,171001	
23-21							0,171001																			0,171001	
23-23	0,171001								2,772424		0,171001						0,342002									3,456428	0,802106575
23-32									0,171001								0,342002									0,513003	
24-12		0,096932								0,145233																0,242165	0,599726526
24-21							3,46052				1,580977															5,041497	
24-22								3,88799			1,580977						2,734484									8,203451	
24-23																	1,580977									1,580977	
24-24		1,153507					0,206458	1,153507			66,09556	1,580977					3,161954									73,35196	0,901074186
24-31											1,580977															1,580977	
24-32											0,051492						1,580977									1,632469	
31-24											1,493258															1,493258	
31-31							0,765979	0,765979			0,778035	106,8317					1,493258									110,6349	0,965623406
31-32																	2,179073									2,179073	
32-12		0,076656	0,076656										0,076656													0,229969	0,333333333
32-13						0,153313							0,076656	0,076656												0,306626	0,25
32-22								0,068838			0,053592															0,122429	
32-23									1,4268								1,311792									2,738592	
32-24											0,160114						0,036406									0,196521	
32-31												10,40025					1,343272									11,74352	
32-32								2,630679			3,942471	4,103092					181,3901			1,4268						193,4931	0,937449755
32-33																0,466368	0,466368			1,311792						2,244528	
32-51																		0,0266								0,0266	1
33-12																	0,0863									0,0863	
33-14																	0,058744									0,145044	
33-21							0,0863																		0,0863	0,0863	
33-23									0,090397								0,114501									0,204898	
33-24																				0,090397						0,090397	
33-31												0,218233														0,218233	
33-32											0,466368						3,460386		0,466368							4,393122	0,106158665
33-33												0,489355					2,88572			9,1809						12,55597	0,731197724
33-42																						0,090397				0,090397	
33-51																											

[illegible]

Overall accuracy at the second level = 84.1%
Overall accuracy at the third level = 78.2%

Zone 13: Bulgaria- Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 117: Confusion matrix CLCCH2012-2018 for zone 13 (Bulgaria) – BLIND ANALYSIS In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-13	13-51	14-14	21-12	21-21	22-21	22-22	22-24	23-21	23-23	23-32	24-21	24-24	31-31	31-32	32-31	32-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	16,37283																								16,37283	1
12-11	0,161459	0,007629																							0,169088	
12-12		3,393261					0,019763									0,161459									3,574483	0,949301195
12-21							0,015257																		0,015257	
12-22															0,007629										0,007629	
13-11	0,007629																								0,007629	
13-13			1,098214												0,077412	0,077412			0,077412						1,330449	0,825445823
13-22																			0,007629						0,007629	
13-23												0,154824													0,154824	
13-24							0,005304								0,077412										0,082716	
13-32																0,077412			0,077412						0,154824	
13-51				0,02119																					0,02119	1
14-11	0,027448																								0,027448	
14-14	0,007629				0,682358											0,027448			0,032752						0,750186	0,909585187
21-12						0,145233																			0,145233	1
21-21							157,2744									0,007087		0,007087	0,007593						160,577	0,979433037
21-23							0,007087																		0,007087	
21-24															3,245727										3,245727	
21-32							0,007087																		0,007087	
22-21							0,985201	0,350051		0,116684	0,116684														1,568619	0,223158453
22-22									5,825976						0,565223										6,391199	0,911562289
22-24							0,011702								0,443527										0,455228	
22-31																0,116847									0,116847	
22-32							0,051492																		0,051492	
23-21							0,37432				0,583418	0,233367													1,191105	0,489812046
23-22													0,116684												0,116684	
23-23												14,28289							1,356934						16,31829	0,875268771
24-21							0,035087							0,116684											0,151771	0,768815157
24-24							0,574736					0,574736			50,23694	3,944767									55,33118	0,907931867
31-31																146,7529									146,7529	1
31-32																	1,399104		0,474576						1,87368	0,746714327
32-21							0,007593									0,855082									0,862675	
32-22								1,239305																	1,239305	
32-24															1,259912										1,259912	
32-31															0,364767				0,010312						0,375079	
32-32															1,474734	2,794184		0,029251	42,09158						46,38975	0,90734657
33-33											0,009024				0,086786	0,026647			0,733891	1,425313				0,009024	2,290685	0,62222148
41-21												0,019138													0,019138	
41-41																					0,419136				0,419136	1
42-42																						0,033373			0,033373	1
51-51															0,401921	0,264063							3,434275		4,100259	0,837575126
52-52																								0,074202	0,074202	1
Total	16,57699	3,400889	1,098214	0,02119	0,682358	0,145233	159,369	0,350051	7,06528	0,116684	0,709125	18,54577	0,116684	0,116684	58,55569	155,4701	1,399104	0,036338	44,87009	1,425313	0,419136	0,033373	3,434275	0,083226	474,0408	
Prod_accuracy	0,987684	0,997757	1	1	1	1	0,986857	1	0,824592		0,822729	0,770143		1	0,857934	0,94393	1		0,938077	1	1	1	1	0,891575		0,940875402

Overall accuracy at the second level = 94.1%

Overall accuracy at the third level = 93.1%

Table 118: Confusion matrix CLCCH2012-2018 for zone 13 (Bulgaria) – PLAUSIBILITY ANALYSIS In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-13	13-32	13-51	14-14	21-12	21-21	21-24	22-21	22-22	22-24	23-21	23-23	23-32	24-21	24-24	31-31	31-32	32-22	32-31	32-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	16,37283																											16,37283	1
12-11	0,161459	0,007629																										0,169088	
12-12		3,393261						0,019763										0,161459										3,574483	0,949301195
12-21								0,015257																				0,015257	
12-22																	0,007629											0,007629	
13-11				0,007629																								0,007629	
13-13			1,098214														0,077412	0,077412				0,077412						1,330449	0,825445823
13-22			0,007629																									0,007629	
13-23													0,154824															0,154824	
13-24								0,005304									0,077412											0,082716	
13-32																						0,154824						0,154824	
13-51					0,02119																							0,02119	1
14-11	0,027448																											0,027448	
14-14	0,007629					0,682358												0,027448				0,032752						0,750186	0,909585187
21-12							0,145233																					0,145233	1
21-21								157,2815						3,280814							0,007087	0,007593						160,577	0,979477172
21-23								0,007087																				0,007087	
21-24									3,245727																			3,245727	1
21-32								0,007087																				0,007087	
22-21								0,969009		0,350051	0,016192	0,116684	0,116684															1,568619	0,223158453
22-22											5,825976						0,565223											6,391199	0,911562289
22-24											0,011702						0,443527											0,455228	
22-31																		0,116847										0,116847	
22-32								0,051492																				0,051492	
23-21								0,37432					0,583418	0,233367														1,191105	0,489812046
23-22															0,116684													0,116684	
23-23														14,62212			0,678467					1,0177						16,31829	0,896057309
24-21								0,035087									0,116684											0,151771	0,768815157
24-24														0,574736		0,574736	50,81167	3,370031										55,33118	0,918319061
31-31																		146,7529										146,7529	1
31-32																			1,399104			0,474576						1,87368	0,746714327
32-21								0,007593														0,855082						0,862675	
32-22										0,619652										0,619652								1,239305	0,5
32-24																	1,25232					0,007593						1,259912	
32-31																		0,223682			0,128069	0,023327						0,375079	0,341446441
32-32																	1,474734	1,931243			0,884333	42,09944						46,38975	0,907515983
33-33														0,009024			0,086786	0,026647				0,707244	1,45196				0,009024	2,290685	0,633854063
41-21														0,019138														0,019138	
41-41																								0,419136				0,419136	1
42-42																									0,033373			0,033373	1
51-51																	0,401921	0,264063								3,434275		4,100259	0,837575126
52-52																											0,074202	0,074202	1
Total	16,56936	3,400889	1,105843	0,007629	0,02119	0,682358	0,145233	158,7735	3,245727	0,350051	6,473522	0,116684	0,700101	18,89402	0,116684	0,691419	55,87711	152,9518	1,399104	0,619652	1,01949	45,45754	1,45196	0,419136	0,033373	3,434275	0,083226	474,0408	
Prod_accuracy	0,988139	0,997757	0,993101		1	1	1	0,990603	1	1	0,89997		0,833333	0,773902		0,168759	0,909347	0,959472	1	1	0,125621	0,926127	1	1	1	1	0,891575		0,951315452

Overall accuracy at the second level =95.1 %

Overall accuracy at the third level =94 %

Zone 14: Island- Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 119: Confusion matrix CLCCH2012-2018 for zone 14 (Island) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-13	14-14	21-21	23-23	23-32	24-24	31-31	32-32	33-32	33-33	33-51	41-41	42-42	51-33	51-51	52-52	Total	User_accuracy
11-11	0,434877																		0,434877	1
12-12		0,406959								0,01837		0,01837							0,443699	0,917194748
12-33												0,01837							0,01837	
13-13	0,025971		0,075816																0,101787	0,744847415
13-14				0,025971															0,025971	
14-14				0,700382															0,700382	1
21-21					0,095488														0,095488	1
23-21					0,002925														0,002925	
23-23						10,7261		0,002925											10,72903	0,999727369
23-32							0,051492												0,051492	1
24-23						0,021875													0,021875	
24-24								0,199801											0,199801	1
31-31									1,296685	0,088535				0,067392					1,452612	0,892657394
31-32										0,207215									0,207215	
31-33												0,067392							0,067392	
31-41														0,067392					0,067392	
32-32						0,01304				163,8667									163,8797	0,999920429
32-33												0,075577							0,075577	
33-32											0,466368								0,466368	1
33-33												219,533					0,01304		219,546	0,999940605
33-41														0,58713					0,58713	
33-51												0,0266	0,398995				0,346764		0,772359	0,516592751
33-52																		0,0266	0,0266	
41-33												0,081332							0,081332	
41-41														29,48284					29,48284	1
42-42															0,468248				0,468248	1
51-32																	0,034721		0,034721	
51-33												0,307312				0,7679	0,127983		1,203195	0,638217224
51-41														0,192834					0,192834	
51-51																	9,240573		9,240573	1
52-52																		0,848332	0,848332	1
Total	0,460849	0,406959	0,075816	0,726353	0,098413	10,76102	0,051492	0,202726	1,296685	164,1808	0,466368	220,1279	0,398995	30,39759	0,468248	0,7679	9,763082	0,874931	441,5262	
Prod_accuracy	0,943645	1	1	0,964244	0,970278	0,996755	1	0,985571	1	0,998087	1	0,997297	1	0,969907	1	1	0,946481	0,969598		0,994415648

Overall accuracy at the second level = 99.4%

Overall accuracy at the third level = 98.1%

Table 120: Confusion matrix CLCCH2012-2018 for zone 14 (Island) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lv2	11-11	12-12	13-11	13-13	14-14	21-21	23-23	23-32	24-24	31-31	32-32	33-32	33-33	33-51	41-41	42-42	51-33	51-51	52-52	Total	User_accuracy
11-11	0,434877																			0,434877	1
12-12		0,425329											0,01837							0,443699	0,958597374
12-33													0,01837							0,01837	
13-13			0,025971	0,075816																0,101787	0,744847415
13-14					0,025971															0,025971	
14-14					0,700382															0,700382	1
21-21						0,095488														0,095488	1
23-21						0,002925														0,002925	
23-23							10,7261		0,002925											10,72903	0,999727369
23-32								0,051492												0,051492	1
24-23							0,021875													0,021875	
24-24									0,199801											0,199801	1
31-31										1,296685	0,088535				0,067392					1,452612	0,892657394
31-32											0,207215									0,207215	
31-33													0,067392							0,067392	
31-41															0,067392					0,067392	
32-32							0,01304				163,8667									163,8797	0,999920429
32-33													0,075577							0,075577	
33-32												0,466368								0,466368	1
33-33													219,546							219,546	1
33-41															0,58713					0,58713	
33-51													0,0266	0,398995				0,346764		0,772359	0,516592751
33-52																			0,0266	0,0266	
41-33													0,081332							0,081332	
41-41															29,48284					29,48284	1
42-42																0,468248				0,468248	1
51-32																		0,034721		0,034721	
51-33													0,082482	0,031996			0,960734	0,127983		1,203195	0,798485787
51-41															0,192834					0,192834	
51-51																		9,240573		9,240573	1
52-52																			0,848332	0,848332	1
Total	0,434877	0,425329	0,025971	0,075816	0,726353	0,098413	10,76102	0,051492	0,202726	1,296685	164,1625	0,466368	219,9161	0,430991	30,39759	0,468248	0,960734	9,750042	0,874931	441,5262	
Prod_accuracy	1	1		1	0,964244	0,970278	0,996755	1	0,985571	1	0,998198	1	0,998317	0,925762	0,969907	1	1	0,947747	0,969598		0,994923533

Overall accuracy at the second level =99.5 %

Overall accuracy at the third level = 98.2%

Zone 15: Hungary - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 121: Confusion matrix CLCCH2012-2018 for zone 15 (Hungary) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	13-32	14-14	21-12	21-13	21-21	21-22	22-21	22-22	22-23	23-13	23-21	23-22	23-23	23-32	24-24	31-31	31-32	32-13	32-23	32-31	32-32	33-33	41-23	41-41	51-51	Total	User_accuracy	
11-11	18,73035	0,005278																												18,73563	0,9997183	
11-24																			0,014061											0,014061		
12-11	0,035002																													0,035002		
12-12	0,008833	3,6031																												3,632037	0,992032945	
12-21							0,00344		0,04087								0,00344													0,047749		
12-23																	0,019738								0,003266					0,023004		
12-24	0,020103																		0,020103											0,040206		
12-32																									0,023004					0,023004		
12-51		0,014061																												0,014061		
13-11	0,14787																													0,14787		
13-12		0,072533	0,081786						0,005194										0,025123											0,184636	0,44295822	
13-13				0,642772																										0,655055	0,981249695	
13-21									0,089218																					0,089218		
13-23																	0,046896													0,046896		
13-31																				0,037405										0,037405		
13-32					0,062786															0,003266										0,069317	0,905776045	
13-51																														0,02119	0,02119	
14-11	0,083995																													0,083995		
14-14																														1,484522	1	
14-24									0,072771																					0,072771		
14-51																														0,072771		
21-13								0,290466		0,145233				0,145233																0,580932	0,5	
21-21		0,008308							192,7833			0,036944								4,129678										196,9582	0,978802967	
21-22									0,116684	0,350051		0,213907																		0,680641	0,514295715	
21-23									0,147692								4,137986													4,285677		
21-24												0,163657							4,137986											4,301642		
21-51																									0,008308					0,008308		
22-21									0,115058		0,350051	0,122294	0,116684						0,067123											0,771209	0,453898735	
22-22												6,93054																		6,93054	1	
22-24									0,009587																					0,78344		
22-32																									0,067123					0,067123		
23-12		0,580672																												0,580672		
23-13				0,004299																										0,004299		
23-21									0,96594						0,350051		0,004299													1,320289	0,265131735	
23-22																0,116684														0,116684	1	
23-23									0,806582								26,42123			0,580672					0,004299					27,81278	0,949967163	
23-24																			1,161344											1,161344		
23-31																				0,004299										0,004299		
23-32																0,051492		0,051492							0,810591					0,913574	0,056362956	
23-51																														0,008308	0,008308	
24-11	0,605428																													0,605428		
24-21									0,020414										0,006805											0,027219		
24-22												0,006805																		0,006805		
24-24	1,210856					0,605428											0,612233		15,04091						0,697217					18,16664	0,827941033	
24-31																				0,697217										0,697217		
24-32																				0,163657										0,163657		
31-22																			0,038326											0,038326		
31-31																				71,75931						0,047645				71,80696	0,999336479	
31-32																					2,33184					3,128828				5,460667	0,427024672	
32-12		0,015919																												0,015919		
32-13																						0,076656								0,076656	1	
32-21									0,184666										0,037417											0,222083		
32-22												0,015919																		0,015919		
32-23																							0,072433							0,605148	0,1196941	
32-24																			0,053336											0,053336		
32-31																				3,790592	0,05785			3,264575						7,113017	0,458957912	
32-32									0,886596										0,126816	0,656024					26,78852					28,45795	0,941336731	
32-41																											0,015919			0,015919		
33-33																										0,114331				0,114331	1	
41-23																	0,161695										0,025066			0,186761	0,134214727	
41-32																														0,161695		
41-41																				0,161695										0,201466		
51-13				0,002903																											2,94506	
51-21				0,002903					0,002903																						0,002903	
51-24		0,002903																													0,002903	
51-31																				0,002903										0,002903		
51-41																														0,008308	0,008308	
51-51																														7,517374	7,517374	1
Total	20,84244	4,302774	0,081786	0,652878	0,062786	2,16272	0,00344	0,290466	196,1747	0,495284	0,350051	7,490065	0,116684	0,145233	0,350051	0,168175	31,94023	0,051492	21,5233	81,98672	2,38969	0,076656	0,072433	3,264575	31,95751	0,114331	0,025066	2,969287	7,619643	417,6804		
Prod_accuracy	0,898664	0,83739	1	0,984522	1	0,686414		1	0,982712	0,706768	1	0,925298			1	0,693821	0,827208	1	0,69882	0,875255	0,975792	1	1	1	0,838254	1	1	0,991841	0,986578		0,915018263	

[illegible]

Overall accuracy at the second level = 92.1%
Overall accuracy at the third level = 91.2%

Zone 16: Portugal- Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 123: Confusion matrix CLCCH2012-2018 for zone 16 (Portugal) –BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	13-32	14-14	21-21	21-22	21-32	22-21	22-22	23-21	23-23	23-51	24-21	24-22	24-24	24-32	24-51	31-31	31-32	32-21	32-22	32-31	32-32	32-33	32-51	33-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy	
11-11	11,08755																																	11,08755	1	
11-24																	0,009725																	0,009725		
12-11	0,009725																																	0,009725		
12-12		2,721556															0,021768																	2,743323	0,992065267	
12-23													0,012497																					0,012497		
12-24																	0,018975																	0,018975		
13-11	0,091511																																	0,091511		
13-12		0,060341	0,081786	0,006391			0,009725													0,081786					0,018975									0,259004	0,315772229	
13-13		0,031642		0,812373		0,006391																		0,018975										0,869381	0,934427603	
13-14						0,028736																												0,028736		
13-31																				0,041366														0,041366		
13-32					0,031393																			0,018975				0,031642						0,082009	0,382795509	
13-51				0,012782							0,02119																					0,02119		0,055162		
14-14	0,054953	0,01225				0,727126											0,01225																	0,806579	0,901492873	
14-32																								0,049181									0,049181			
21-21							34,62196										4,387716			0,74249													39,75216	0,870945228		
21-22							0,116684	0,933468			0,143855																							1,194007	0,781794514	
21-23													6,245339																					6,245339		
21-24							0,74249										1,673713																	2,416204		
21-32									0,051492																0,825437									0,876928	0,058718288	
22-21							0,143855			0,233367																								0,377222	0,618645752	
22-22							0,111878	0,116684			23,95791						3,72802																	27,91449	0,858260695	
22-24							0,676724				0,143855		0,055939				0,330779		0,015727															1,223024		
22-32										0,051492								0,051492																0,102983		
22-51																											0,03315							0,03315		
23-21												0,466734	0,116684																					0,583418	0,8	
23-23												5,024926					0,323308								0,323308									5,671542	0,885989467	
23-32																									0,323308									0,323308		
24-21							2,853087								0,116684																			2,969771	0,039290415	
24-22											0,69357					0,116684																		0,810254	0,144008618	
24-23													2,642819				1,32141																	3,964229		
24-24	1,387141						3,402802				1,732416						81,31535		1,387141						2,425986									91,65084	0,887229772	
24-31																	1,388629								1,732416									3,121045		
24-32									0,051492																									0,051492		
24-51														0,03315					0,03315															0,066299	0,5	
31-22																0,072433																		0,072433		
31-24																	7,353017																	7,353017		
31-31													0,5184				7,353017		86,89427						2,645604									97,41129	0,892034887	
31-32																	1,771077	6,529151						0,466368	5,293869								14,52683	0,449454553		
31-33																	0,466368	0,466368																0,932736		
32-21																					0,072433				0,072433					0,257828				0,402693	0,17987041	
32-22										0,007189									0,072433			0,28973			0,072433									0,441785	0,65581799	
32-23																	0,072433					0,072433				0,559534								0,704399		
32-24																	1,382229								0,778452									2,160681		
32-31		0,007189				0,031701											0,045992		2,187168					1,865472	1,873605									6,011126	0,310336457	
32-32																	1,780907		0,077693						93,18942				2,185689					97,23371	0,958406518	
32-33																								0,466368	1,865472									2,33184	0,8	
32-51																			0,0266							1,865472	0,106399							0,132998	0,8	
33-32																					0,466368				2,28914		0,466368	0,184845						3,406721	0,136896435	
33-33	0,027321																						0,184845	0,36969										2,991972	0,796396234	
41-32																								0,075524										0,075524		
41-41																														0,222394					0,222394	1
42-42																																1,095521			1,095521	1
51-21							0,007189																												0,007189	
51-24																																			0,007189	
51-51																								0,129842										3,183554	0,960813013	
51-52																																			0,157571	0,157571
52-42																																			0,068054	0,068054
52-52																																			0,655395	0,655395
Total	12,6582	2,832978	0,081786	0,831546	0,031393	0,762252	42,71809	1,050152	0,102983	0,284859	26,69999	0,466734	14,6166	0,03315	0,116684	0,189116	112,9856	0,051492	0,059749	93,73752	7,461887	0,144865	0,28973	2,516684	113,5525	1,865472	0,139548	0,466368	4,78497	0,480222	1,122842	3,211933	0,88102	447,2289		
Prod_accuracy	0,875918	0,96067	1	0,976944	1	0,953917	0,810475	0,888889	0,5	0,819238	0,8973	1	0,343782		1	0,616994	0,719697		0,554812																	

[illegible]

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Zone 17: Austria / Swiss - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 125: Confusion matrix CLCCH2012-2018 for zone 17 (Austria / Swiss) – BLIND ANALYSIS - In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	13-23	14-14	21-12	21-21	22-22	23-23	24-24	31-12	31-31	31-32	32-31	32-32	33-33	41-41	51-51	Total	User_accuracy
11-11	23,85048							1,288562					1,288562							26,4276	0,902483592
12-11	0,00623	0,004154																		0,010384	
12-12	0,134161	2,479661	0,009259					0,002077												2,625158	0,944575661
12-14		0,027937																		0,027937	
12-21	0,013678	0,013678									0,003508									0,030863	
12-22									0,106806											0,106806	
12-24											0,013678									0,013678	
13-11	0,025061										0,006405									0,031466	
13-12	0,025061	0,031466						0,009259												0,065787	
13-13			0,006405	0,431726	0,044778						0,034321									0,51723	0,834689555
13-14						0,081786														0,081786	
13-21								0,015665		0,006405										0,02207	
13-24											0,009259									0,009259	
13-31													0,004693							0,004693	
14-14						1,489516		0,013057					0,013057			0,004693				1,520323	0,979736183
14-21						0,067053		0,026115												0,093167	
14-31													0,067053							0,067053	
14-32																0,004693				0,004693	
21-12							0,290466													0,290466	1
21-21								63,00221		3,405336	6,810671									73,21822	0,860471801
21-23										1,702668	3,405336									5,108003	
21-24										1,702668	5,108003									6,810671	
22-11	0,024128																			0,024128	
22-22								0,024128	3,550265		0,024128									3,598522	0,986589799
23-21																0,003508				0,003508	
23-23										33,68907			4,498423							38,18749	0,882201652
23-24											4,491408		0,003508							4,494915	
23-32																2,245704				2,245704	
24-11	0,847595																			0,847595	
24-21								0,61275												0,61275	
24-23													1,2255							1,2255	
24-24											25,67346		0,847595							26,52106	0,968040674
31-12												0,076656								0,076656	1
31-23														0,072433						0,072433	
31-31										2,4115			194,5359			0,007976				196,9553	0,987715612
31-32													1,267708			7,719923				8,987631	
31-33													0,485424			2,4115				2,896924	
32-21																0,053005				0,053005	
32-23													0,082145							0,082145	
32-24											0,053005									0,053005	
32-31									0,019162				2,195491			0,110492				2,325145	
32-32													0,239954		0,008875	60,37116				60,61999	0,995895265
32-33																0,053005				0,053005	
33-32																1,167134				1,167134	
33-33																2,519034	51,99445			54,51349	0,953790632
41-21										0,050213										0,050213	
41-23										0,014485										0,014485	
41-32																		0,050213		0,050213	
41-41													0,064698					0,897077		0,961775	0,932730608
51-51																		0,003508	8,981882	8,98539	0,999609624
Total	24,92639	2,556895	0,015665	0,431726	0,044778	1,638355	0,290466	64,99383	3,657071	43,0015	45,63319	0,076656	206,8197	0,072433	0,008875	76,67182	51,99445	0,950798	8,981882	532,7665	
Prod_accuracy	0,956836	0,969794		1		0,909153	1	0,969357	0,970795	0,783439	0,562605	1	0,940606			0,787397	1	0,943499	1		0,884653933

Overall accuracy at the second level = 88.5%

Overall accuracy at the third level =83 %

Table 126: Confusion matrix CLCCH2012-2018 for zone 17 (Austria / Swiss) – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	13-23	14-14	21-12	21-21	22-22	23-23	24-24	31-12	31-31	32-31	32-32	33-33	41-41	51-51	Total	User_accuracy
11-11	25,13904							1,288562											26,4276	0,951241796
12-11	0,00623	0,004154																	0,010384	
12-12	0,134161	2,479661	0,009259					0,002077											2,625158	0,944575661
12-14		0,027937																	0,027937	
12-21	0,013678	0,013678									0,003508								0,030863	
12-22									0,106806										0,106806	
12-24											0,013678								0,013678	
13-11	0,025061										0,006405								0,031466	
13-12	0,025061	0,031466					0,009259												0,065787	
13-13			0,006405	0,431726	0,025061					0,019717	0,034321								0,51723	0,834689555
13-14						0,081786													0,081786	
13-21			0,006405					0,009259		0,006405									0,02207	
13-24											0,009259								0,009259	
13-31												0,004693							0,004693	
14-14						1,489516		0,013057				0,013057			0,004693				1,520323	0,979736183
14-21						0,067053		0,026115											0,093167	
14-31												0,067053							0,067053	
14-32															0,004693				0,004693	
21-12							0,290466												0,290466	1
21-21								63,00221		3,405336	6,810671								73,21822	0,860471801
21-23										1,702668	3,405336								5,108003	
21-24										1,702668	5,108003								6,810671	
22-11	0,024128																		0,024128	
22-22									3,574393		0,024128								3,598522	0,9932949
23-21															0,003508				0,003508	
23-23										33,68907			4,498423						38,18749	0,882201652
23-24											4,491408		0,003508						4,494915	
23-32															2,245704				2,245704	
24-11	0,847595																		0,847595	
24-21								0,61275											0,61275	
24-23										0,61275			0,61275						1,2255	
24-24											25,67346		0,847595						26,52106	0,968040674
31-12												0,076656							0,076656	1
31-23											0,072433								0,072433	
31-31										2,4115			194,5359		0,007976				196,9553	0,987715612
31-32													1,267708		7,719923				8,987631	
31-33													0,485424		2,4115				2,896924	
32-21															0,053005				0,053005	
32-23													0,003788		0,078357				0,082145	
32-24											0,053005								0,053005	
32-31									0,019162			2,130355	0,072788	0,10284					2,325145	0,031304804
32-32												0,286316	0,012663	60,32101					60,61999	0,995067971
32-33															0,053005				0,053005	
33-32															1,167134				1,167134	
33-33															2,519034	51,99445			54,51349	0,953790632
41-21									0,050213										0,050213	
41-23									0,014485										0,014485	
41-32																	0,050213		0,050213	
41-41												0,064698					0,897077		0,961775	0,932730608
51-51																	0,003508	8,981882	8,98539	0,999609624
Total	26,21495	2,556895	0,02207	0,431726	0,025061	1,638355	0,299725	64,95404	3,6812	43,63397	45,70562	0,076656	204,8212	0,085451	76,69238	51,99445	0,950798	8,981882	532,7665	
Prod_accuracy	0,958958	0,969794		1		0,909153	0,969107	0,969951	0,970986	0,772083	0,561714	1	0,949784	0,85181	0,786532	1	0,943499	1		0,887160338

Overall accuracy at the second level = 88.7%

Overall accuracy at the third level =83.3 %

Zone 18: Denmark/Netherland/Belgium/Luxembourg - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 127: Confusion matrix CLCCH2012-2018 for zone 18 (Denmark/Netherland/Belgium/Luxembourg) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	11-51	12-12	13-11	13-12	13-13	13-14	13-23	13-32	13-51	14-14	21-13	21-14	21-21	21-51	22-22	23-12	23-13	23-23	23-51	24-24	31-31	31-32	32-31	32-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	40,03314		2,256526											2,256526																44,54619	0,898688272	
11-21														0,015866																0,015866		
11-31																						2,256526								2,256526		
12-11	0,364113																													0,364113		
12-12			10,34633																											10,34633	1	
12-21	0,005845		0,01169											0,058143																0,075678		
12-24			0,072887																		0,005845									0,078732		
12-32			0,006382																											0,006382		
13-11	0,287852		0,072887	0,081786										0,011526																0,454051	0,180125482	
13-12			0,196145		0,081786																									0,277931	0,294267904	
13-13			0,010044			1,465572																0,011526							0,036353	1,523495	0,961980084	
13-14							0,102518				0,036353	0,012233																		0,151104	0,678462261	
13-21														0,071845																0,071845		
13-23								0,029569											0,079254											0,108823	0,271718745	
13-24			0,029569																		0,020088								0,020732	0,070389		
13-31						0,006382																0,012233								0,018615		
13-32	0,031393		0,041437						0,031393										0,031393						0,036353		0,020732				0,1927	0,162910015
13-51										0,02119																			0,047879	0,069069	0,306792988	
14-11	0,165794										0,072887																			0,238681		
14-12	0,290675																													0,290675		
14-14											7,374413																			7,374413	1	
14-24											0,046454										0,063648									0,110102		
21-13												0,145233	0,145233																	0,290466	0,5	
21-14											3,565778																			3,565778		
21-21	7,131555										3,565778			153,4922							3,565778									167,7553	0,914976679	
21-24																					7,131555									7,131555		
21-32														0,051492																0,051492		
21-51															0,132598															0,132598	1	
22-21														0,009041																0,009041		
22-22														0,122249	0,734083						0,04075									0,897082	0,818300988	
22-24																					0,009041									0,009041		
23-13																		0,145233												0,145233	1	
23-21														0,018081																0,018081		
23-23											3,086813								55,57167		3,086813									61,74529	0,900014642	
23-32																				0,051492										0,051492		
23-41																			0,066832											0,066832		
23-51		0,03315																		0,066299										0,099449	0,666666667	
24-21														4,760582																4,760582		
24-24	2,592984													4,083748					4,083748		62,38912						1,572047			74,72164	0,834953759	
24-32																					0,051492									0,051492		
31-13												0,076656																		0,076656		
31-14											0,021426																			0,021426		
31-21														0,455714																0,455714		
31-31	1,697306																					57,64525								59,34256	0,97139817	
31-32																			0,466368				0,466368		0,113854					1,04659	0,445607197	
32-11																						0,228801								0,228801		
32-13																		0,076656				0,076656								0,153313		
32-14											0,272645																			0,272645		
32-21														0,03655								0,038381	0,012794		0,012794					0,100518		
32-23																			0,039701											0,039701		
32-24														0,006595								0,025587								0,187914		
32-31																					0,70929	0,006595	0,006595	0,228801						0,951281	0,006932613	
32-32			0,018275											0,155732			0,018275		0,59509		0,056927		0,006595		0,228801	0,155732	0,054825					

Overall accuracy at the second level =86.55 %

Overall accuracy at the third level = 86%

CC	11-11	11-51	12-12	12-21	13-11	13-12	13-13	13-14	13-21	13-23	13-32	13-51	14-14	21-11	21-12	21-13	21-14	21-21	21-41	21-51	22-22	23-11	23-12	23-13	23-21	23-23	23-51	24-21	24-24	31-31	31-32	32-31	32-32	33-33	41-41	41-51	42-42	51-51	52-52	Total	per accuracy			
11-11	40,03314		2,256526															0,015866								2,256526														44,54619	0,898688			
11-21																																										44,54619	0,898688	
11-31																																										44,54619	0,898688	
12-11	0,364113																																									44,54619	0,898688	
12-12			10,34633																																							44,54619	0,898688	
12-21			0,017535	0,005845											0,005845			0,046454																								44,54619	0,898688	
12-24			0,072887																											0,005845												44,54619	0,898688	
12-32			0,006382																																							44,54619	0,898688	
13-11	0,21619				0,081786	0,072887									0,011526			0,011526				0,060135																					44,54619	0,898688
13-12			0,143799			0,122606																	0,011526																				44,54619	0,898688
13-13					0,010044	1,465572																									0,011526											44,54619	0,898688	
13-14								0,102518																																		44,54619	0,898688	
13-21									0,025557																																	44,54619	0,898688	
13-23										0,029569																0,025557	0,020732																44,54619	0,898688
13-24			0,029569																								0,079254															44,54619	0,898688	
13-31					0,012233	0,006382																							0,020088													44,54619	0,898688	
13-32	0,031393		0,041437								0,031393																																	

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Table 129: Confusion matrix CLCCH2012-2018 for zone 19 (Albania/Serbia/Macedonia) – BLIND ANALYSIS - in column the validation and line the production

Overall accuracy at the second level = 87.1%
Overall accuracy at the third level = 84%

[illegible]

Overall accuracy at the second level = 87.25 %
Overall accuracy at the third level = 84.2%

Zone 20: Bosnia and Herzegovina/Croatia - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 131: Confusion matrix CLCCH2012-2018 for zone 20 (Bosnia and Herzegovina / Croatia) –BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	14-14	21-21	21-22	22-21	22-22	23-23	23-51	24-24	31-13	31-31	31-32	32-31	32-32	32-51	33-32	33-33	41-41	42-42	51-32	51-51	52-52	Total	User_accuracy
11-11	10,17624													0,542031												10,71827	0,949429227
12-11	0,098378	0,007087				0,007087																				0,112553	
12-12	0,113272	1,828442	0,010009	0,010009													0,034655									1,996387	0,915875552
12-21						0,014174																				0,014174	
12-24		0,007087										0,138622														0,145709	
13-12		0,029163	0,081786	0,030534																						0,141484	0,578061853
13-13	0,030534			0,616469								0,007087					0,030534									0,684625	0,900447659
13-14					0,010009																					0,010009	
13-21						0,007087																				0,007087	
13-24														0,030534												0,030534	
13-31														0,039172			0,029163									0,068334	
13-32				0,030534													0,030026									0,060561	
14-11	0,019692																									0,019692	
14-14	0,078769				0,372691	0,019692																				0,471153	0,791020181
21-21						26,17509				0,027815										0,009272						26,21218	0,998585113
21-22								0,289012																		0,289012	
21-23										0,013503																0,013503	
21-24												2,875444														2,875444	
21-41																				0,040508						0,040508	
22-21						0,071814		0,116684																		0,188498	0,619018749
22-22	0,093011								2,083768			0,536889		0,035907			0,042993									2,792567	0,746183549
22-24												0,823991														0,823991	
22-32																	0,007086									0,007086	
23-23										24,97877		0,013503		3,118804			3,118804									31,22988	0,799835626
23-32												0,051492					0,07275									0,124241	
24-12		0,009272																								0,009272	
24-21						16,8701						4,217524														21,08762	
24-24						8,438373						100,0921		5,269876			1,052352									114,8527	0,87148233
24-31														1,052352												1,052352	
24-32												0,051492														0,051492	
31-14														0,076656												0,076656	
31-31												2,45681		220,9694			9,949578									233,3758	0,946839439
31-32														0,957355	2,33184											3,289194	0,708939457
31-51																		0,0266								0,0266	
32-13													0,076656													0,076656	
32-21						0,016793																				0,016793	
32-22							0,072433																			0,072433	
32-23																	0,016793									0,016793	
32-31													3,406951			0,932736	0,447409									4,787096	0,194843771
32-32												0,430616		7,454386	0,033585		68,4842			1,214705						77,6175	0,882329461
32-33																	0,430616			0,430616						0,861233	
32-51											0,0266															0,0266	
33-31														0,059039												0,059039	
33-32												0,466368					0,857807		0,466368							1,790543	0,260461703
33-33												0,017564		0,288813			1,460194			4,712496						6,479067	0,727341791
41-23									0,055962																	0,055962	
41-24												0,055962														0,055962	
41-31														0,055962												0,055962	
41-41														0,111924						0,89591						1,007835	0,888945805
42-42																				0,030706	0,042478					0,073183	0,580425082
51-51																							0,013503	3,982852		3,996355	0,996621251
52-52																									0,182526	0,182526	1
Total	10,6099	1,881051	0,091795	0,687547	0,3827	51,62021	0,072433	0,116684	2,37278	25,07605	0,0266	112,2355	0,076656	243,4692	2,365425	0,932736	86,06496	0,0266	0,466368	6,357817	0,976396	0,042478	0,013503	3,982852	0,182526	550,1307	
Prod_accuracy	0,959127	0,972032	0,890965	0,896621	0,973847	0,507071		1	0,878197	0,996121		0,891805		0,907587	0,985802	1	0,795727		1	0,741213	0,917568	1		1	1		0,853475107

Overall accuracy at the second level = 85.35%

Overall accuracy at the third level =79.4 %

Table 132: Confusion matrix CLCCH2012-2018 for zone 20 (Bosnia and Herzegovina / Croatia) –PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	13-31	13-32	14-14	21-21	21-22	22-21	22-22	23-23	24-12	24-24	31-31	31-32	32-13	32-31	32-32	32-51	33-32	33-33	41-21	41-41	42-42	51-32	51-51	52-52	Total	User_accuracy
11-11	10,17624							0,007087							0,542031														10,71827	0,949429227
12-11	0,063723	0,041743																											0,112553	
12-12	0,113272	1,828442	0,010009	0,010009															0,034655										1,996387	0,915875552
12-21								0,014174																					0,014174	
12-24		0,007087												0,103966					0,034655										0,145709	
13-12		0,029163	0,081786											0,030534															0,141484	0,578061853
13-13	0,030534			0,616469										0,007087					0,030534										0,684625	0,900447659
13-14							0,010009																						0,010009	
13-21								0,007087																					0,007087	
13-24															0,030534														0,030534	
13-31					0,010009										0,029163				0,029163										0,068334	0,146468289
13-32						0,040543													0,020018										0,060561	0,669462692
14-11	0,019692																												0,019692	
14-14	0,078769						0,372691	0,019692																					0,471153	0,791020181
21-21								26,17509				0,027815												0,009272					26,21218	0,998585113
21-22											0,289012																		0,289012	
21-23								0,013503																					0,013503	
21-24														2,875444															2,875444	
21-41								0,013503															0,027005						0,040508	
22-21								0,071814		0,116684																			0,188498	0,619018749
22-22	0,046505										2,268505	0,102324		0,33224	0,035907				0,007086										2,792567	0,812336586
22-24											0,071814			0,752177															0,823991	
22-32																			0,007086										0,007086	
23-23												24,99227			3,118804				3,118804										31,22988	0,80026799
23-32														0,051492					0,07275										0,124241	
24-12													0,009272																0,009272	1
24-21								16,8701						4,217524															21,08762	
24-24								4,220849						105,362	4,217524				1,052352										114,8527	0,917366111
24-31														1,052352															1,052352	
24-32														0,051492															0,051492	
31-14														0,076656															0,076656	
31-31														2,45681	224,54			0,103947	6,275036										233,3758	0,962139203
31-32														0,957355	2,33184														3,289194	0,708939457
31-51																				0,0266									0,0266	
32-13																	0,076656												0,076656	1
32-21								0,016793																					0,016793	
32-22									0,072433																				0,072433	
32-23																			0,016793										0,016793	
32-31														3,322988	0,430616		0,983114	0,050378											4,787096	0,205367462
32-32														0,430616	7,430342	0,033585		0,013503	67,28004		1,214705	1,214705							77,6175	0,866815393
32-33																			0,861233										0,861233	
32-51																				0,0266									0,0266	1
33-31														0,059039															0,059039	
33-32														0,466368					0,75386		0,570315								1,790543	0,318515042
33-33														0,017564	0,288813				1,460194			4,712496							6,479067	0,727341791
41-23											0,055962																		0,055962	
41-24														0,055962															0,055962	
41-31															0,055962														0,055962	
41-41															0,111924									0,89591					1,007835	0,888945805
42-42																							0,030706	0,042478					0,073183	0,580425082
51-51																										0,013503	3,982852		3,996355	0,996621251
52-52																												0,182526	0,182526	1
Total	10,52873	1,906434	0,091795	0,626478	0,010009	0,040543	0,3827	47,42969	0,072433	0,116684	2,629331	25,17838	0,009272	117,2113	245,8694	2,796041	0,076656	1,100563	81,10464	0,053199	1,785019	5,927201	0,027005	0,935888	0,042478	0,013503	3,982852	0,182526	550,1307	
Prod_accuracy	0,966521	0,95909	0,890965	0,984024	1	1	0,973847	0,551871		1	0,862769	0,992609	1	0,898907	0,913249	0,833979	1	0,893282	0,829546	0,5	0,319501	0,795063		0,957284	1		1			0,868293317

Overall accuracy at the second level = 86.8%
Overall accuracy at the third level = 80.9%

Zone 21: Czech Republic / Slovakia - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 133: Confusion matrix CLCCH2012-2018 for zone 21 (Czech Republic / Slovakia) – BLIND ANALYSIS - In column the validation and line the production

MAP-CLCC_Lvl2	11-11	11-13	12-12	13-12	13-13	13-21	14-14	21-21	21-23	22-21	22-22	23-21	23-23	24-24	31-31	31-32	32-31	32-32	33-33	33-51	41-41	51-51	Total	User_accuracy
11-11	26,35383		0,007676					0,007676															26,36919	0,999417805
11-21								0,007676															0,007676	
12-11	0,038148																						0,038148	
12-12	0,008227		4,947055																				4,955282	0,998339687
12-13															0,008939								0,008939	
12-21								0,095739															0,095739	
12-23												0,198735											0,198735	
12-31															0,005189								0,005189	
12-32																	0,005189						0,005189	
13-11	0,095386																						0,095386	
13-12			0,033299																				0,033299	
13-13		0,081786	0,046244	0,025319	1,274473			0,00798							0,046244							0,046244	1,52829	0,833920862
13-14							0,033299																0,033299	
13-21	0,00798				0,046244	0,029569		0,01596															0,099753	0,296425895
13-23				0,029569									0,066962										0,096531	
13-24													0,00798										0,00798	
14-14							1,559013	0,097228															1,656241	0,94129617
14-23														0,016687									0,016687	
14-24															0,059079								0,059079	
14-31													0,005189										0,005189	
21-13				0,007388																			0,007388	
21-21							0,007388	186,9229															186,9303	0,999960479
21-22											0,014775												0,014775	
21-23	0,016431							0,133115	0,116684				3,823224										4,089453	0,028532796
21-31													0,016431										0,016431	
22-21								0,395083		0,116684			0,00754					0,116684					0,635991	0,183467355
22-22	0,060174							0,161411	0,060174	3,172291													3,454051	0,918426256
22-23													0,00754										0,00754	
22-24														0,00754									0,00754	
23-21							1,741235					0,466734											2,207969	0,211386211
23-22											0,237325												0,237325	
23-23								0,077202					48,89238		0,972509								49,94209	0,978981448
23-24								0,887893															0,887893	
23-31													0,077202										0,077202	
23-32																		0,007414					0,007414	
24-21								0,457753															0,457753	
24-24														49,28255									49,28255	1
31-21								1,84987															1,84987	
31-23													0,011268										0,011268	
31-31															186,1805			0,034302					186,2148	0,999815792
31-32															0,466368	5,130047	0,466368	9,83247					15,89525	0,322740836
32-21								0,011268															0,011268	
32-24															0,094938			0,024571					0,119509	
32-31															5,151136		0,466368	0,842682					6,460187	0,072191091
32-32															0,869763			14,5427		0,007414			15,41987	0,943113844
32-51																				0,0266			0,0266	
33-33															0,022006			0,108872	0,353982				0,484859	0,730070765
41-31													0,020897										0,020897	
41-41																		0,040977			0,535965	0,020897	0,597839	0,896504565
51-21								0,022242															0,022242	
51-24															0,007414								0,007414	
51-51															0,064501							3,768521	3,833021	0,983172371
Total	26,58018	0,081786	5,034274	0,054888	1,328104	0,029569	1,5997	192,8923	0,116684	0,176858	3,424391	0,466734	53,13535	49,37869	193,8767	5,130047	0,932736	25,55586	0,353982	0,0266	0,543379	3,835662	564,5544	
Prod_accuracy	0,991484		0,982675		0,959618	1	0,974566	0,969054	1	0,659759	0,926381	1	0,920148	0,998053	0,960304	1	0,5	0,569055	1		0,986356	0,982496		0,946077602

Overall accuracy at the second level = 94.6%

Overall accuracy at the third level =93.9%

Table 134: Confusion matrix CLCCH2012-2018 for zone 21 (Czech Republic / Slovakia) – PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLCC_Lvl2	11-11	11-13	12-12	13-11	13-12	13-13	13-21	13-23	14-14	21-21	21-22	21-23	22-21	22-22	22-32	23-21	23-23	24-21	24-24	31-31	31-32	32-31	32-32	33-33	33-51	41-41	51-51	Total	User_accuracy
11-11	26,35383		0,007676							0,007676																		26,36919	0,999417805
11-21	0,007676																											0,007676	
12-11	0,038148																											0,038148	
12-12			4,955282																									4,955282	1
12-13																				0,008939								0,008939	
12-21	0,008227									0,087512																		0,095739	
12-23																	0,198735											0,198735	
12-31																				0,005189								0,005189	
12-32																						0,005189						0,005189	
13-11	0,087407			0,00798																								0,095386	0,083657288
13-12			0,033299																									0,033299	
13-13	0,046244	0,081786			0,025319	1,274473				0,00798										0,046244							0,046244	1,52829	0,833920862
13-14				0,00798					0,025319																			0,033299	
13-21	0,00798					0,046244	0,029569			0,01596																		0,099753	0,296425895
13-23							0,050287										0,046244											0,096531	0,520942547
13-24																	0,00798											0,00798	
14-14									1,559013	0,097228																		1,656241	0,94129617
14-23																												0,016687	
14-24																				0,059079								0,059079	
14-31																	0,005189											0,005189	
21-13						0,007388																						0,007388	
21-21								0,007388	186,9229																			186,9303	0,999960479
21-22										0,007388			0,007388															0,014775	0,5
21-23										0,016431		0,116684					3,956338											4,089453	0,028532796
21-31																	0,016431											0,016431	
22-21										0,387543			0,124224		0,116684		0,00754											0,635991	0,195323582
22-22	0,060174									0,075255			0,060174	3,258447														3,454051	0,943369694
22-23																	0,00754											0,00754	
22-24																			0,00754									0,00754	
23-21									1,655557							0,552412												2,207969	0,250189928
23-22													0,237325															0,237325	
23-23																0,077202	48,89238			0,972509								49,94209	0,978981448
23-24																	0,887893											0,887893	
23-31																	0,077202											0,077202	
23-32																							0,007414					0,007414	
24-21									0,228876									0,228876										0,457753	0,5
24-24																			49,28255									49,28255	1
31-21									1,84987																			1,84987	
31-23																	0,011268											0,011268	
31-31																				186,1805		0,034302						186,2148	0,999815792
31-32																				0,466368	7,00399	0,466368	7,958527				15,89525	0,440634092	
32-21									0,011268																			0,011268	
32-24																				0,08367			0,035839					0,119509	
32-31																				3,412639		2,204865	0,842682				6,460187	0,341300564	
32-32																				0,128743			15,28372			0,007414	15,41987	0,991169999	
32-51																									0,0266			0,0266	
33-33																				0,022006			0,108872	0,353982				0,484859	0,730070765
41-31																	0,020897											0,020897	
41-41																							0,040977			0,535965	0,020897	0,597839	0,896504565
51-21									0,007414																			0,014828	0,022242
51-24																			0,007414									0,007414	
51-51																												3,833021	1
Total	26,60969	0,081786	4,996257	0,01596	0,025319	1,328104	0,029569	0,050287	1,59172	191,3715	0,007388	0,116684	0,184398	3,50316	0,116684	0,629613	54,15232	0,228876	49,29751	191,3859	7,00399	2,705535	24,28321	0,353982	0,0266	0,543379	3,914991	564,5544	
od_accura	0,990385		0,991799	0,5		0,959618	1	1	0,979452	0,976754	1	1	0,673672	0,930145	0,877382	0,902868	1	0,999697	0,972802	1	0,814946	0,629394	1		0,986356	0,979063			0,954757178

Overall accuracy at the second level = 95.5%
Overall accuracy at the third level = 94.8%

Zone 22: Estonia/Latvia/Lithuania - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 135: Confusion matrix CLCCH2012-2018 for zone 22 (Estonia / Latvia / Lithuania) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	14-14	21-21	21-22	21-23	22-21	22-22	23-12	23-21	23-23	23-32	24-24	31-31	31-32	31-33	32-13	32-21	32-23	32-31	32-32	32-41	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	11,83196														0,012799															11,84476	0,998919424
11-14					0,60564																									0,60564	
11-21						0,014884																								0,014884	
11-24	0,008039														0,008039															0,016079	
11-32																							0,008039							0,008039	
12-12		3,76525																												3,76525	1
12-21						0,052032																								0,052032	
12-23													0,157911																	0,157911	
12-24															0,157911															0,157911	
12-32		0,00784																												0,00784	
13-11	0,070687																													0,070687	
13-12		0,082742																												0,082742	
13-13				0,949406											0,018196	0,052448														1,020049	0,930744779
13-21						0,014884																								0,014884	
13-23			0,029569										0,014884																	0,044453	
13-24															0,014884															0,014884	
13-31	0,018196															0,022724														0,04092	
13-32																							0,048665							0,048665	
14-11	0,035208																													0,035208	
14-14					1,207899											0,035208														1,243108	0,971677253
14-21						0,012799																								0,012799	
14-23						0,014884																								0,014884	
14-24															0,048007															0,048007	
14-31																0,030615														0,030615	
21-12											0,145233																			0,145233	
21-21						144,2938							0,359032			0,133594														144,7864	0,996597568
21-22							0,116684																							0,116684	1
21-23						0,350051		0,350051				0,116684	17,64716				0,116684													18,58062	0,018839549
21-24															8,218719															8,218719	
21-32																							0,185085							0,185085	
22-21									0,116684																					0,116684	1
22-22										0,475928			0,031729																	0,507656	0,9375
22-24															0,031729															0,031729	
23-21						6,53995						3,733873	3,135677																13,4095	0,278449825	
23-23						0,225438																								49,38079	0,995434693
23-24															0,227869															0,227869	
23-31																0,075956							0,051492							0,127448	
23-32												0,051492		0,102983									3,137914							3,292389	0,031279253
24-21						12,45515																								12,45515	
24-24															90,31923															90,31923	1
24-31																5,683754														5,683754	
24-32																							2,410773							2,410773	
24-51																											3,272981			3,272981	
31-21						5,708747																								5,708747	
31-31																262,9499	0,028083													262,9779	0,999893211
31-32																	9,327358						5,080365							14,40772	0,647385995
31-33																		0,466368												0,466368	1
32-13																			0,153313											0,153313	1
32-21						0,04235															0,072433									0,114782	0,631043618
32-23																					0,072433	0,144865								0,217298	0,666666667
32-24												0,015937		0,025785									0,012049							0,053772	
32-31															3,735015	0,208406						0,932736								4,876156	0,191285059
32-32						0,012049									3,931112								74,54448			0,027786				78,51543	0,949424627
32-41																								0,048357						0,048357	1
33-33																0,042156							0,042156		0,396206				0,042156	0,522673	0,758037702
41-32																							0,671816							0,671816	
41-41																										17,89804				17,89804	1
42-41																											0,014214			0,014214	
51-51															0,15871																

Overall accuracy at the second level =88.95 %

Overall accuracy at the third level = 87.35 %

Table 136: Confusion matrix CLCCH2012-2018 for zone 22 (Estonia / Latvia / Lithuania) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-11	13-12	13-13	14-14	21-21	21-22	21-23	22-21	22-22	23-12	23-21	23-23	23-24	23-32	24-24	31-31	31-32	31-33	32-13	32-21	32-23	32-31	32-32	32-41	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	11,84476																														11,84476	1	
11-14						0,60564																									0,60564		
11-21							0,014884																								0,014884		
11-24	0,008039																0,008039														0,016079		
11-32																									0,008039						0,008039		
12-12		3,76525																													3,76525	1	
12-21							0,052032																								0,052032		
12-23														0,157911																	0,157911		
12-24																	0,157911														0,157911		
12-32		0,00784																													0,00784		
13-11	0,070687																														0,070687		
13-12		0,049191		0,014884	0,018667																										0,082742	0,179881789	
13-13			0,018196		1,001853																										1,020049	0,9821615	
13-21							0,014884																								0,014884		
13-23							0,029569							0,014884																	0,044453		
13-24																	0,014884														0,014884		
13-31	0,018196																	0,022724													0,04092		
13-32																									0,048665						0,048665		
14-11	0,035208																														0,035208		
14-14						1,243108																									1,243108	1	
14-21							0,012799																								0,012799		
14-23														0,014884																	0,014884		
14-24																	0,048007														0,048007		
14-31																		0,030615													0,030615		
21-12												0,145233																			0,145233		
21-21							144,4274							0,359032																	144,7864	0,997520263	
21-22								0,116684																							0,116684	1	
21-23							0,350051		0,466734				0,116684	17,64716																	18,58062	0,025119399	
21-24																	8,218719														8,218719		
21-32																									0,185085						0,185085		
22-21										0,116684																					0,116684	1	
22-22											0,507656																				0,507656	1	
22-24																	0,031729														0,031729		
23-21							6,53995						3,733873	3,135677																	13,4095	0,278449825	
23-23													0,225438	49,0794		0,075956															49,38079	0,993896516	
23-24															0,151913		0,075956														0,227869	0,666666667	
23-31																		0,075956								0,051492					0,127448		
23-32														0,051492		0,102983									3,137914						3,292389	0,031279253	
24-21							9,182173							3,272981																	12,45515		
24-24																	90,31923														90,31923	1	
24-31																	5,683754														5,683754		
24-32																			2,410773												2,410773		
24-51																													3,272981		3,272981		
31-21							5,708747																								5,708747		
31-31																		257,2411	5,73683												262,9779	0,978185129	
31-32																			9,327358						5,080365						14,40772	0,647385995	
31-33																				0,466368											0,466368	1	
32-13																					0,153313										0,153313	1	
32-21							0,04235															0,072433									0,114782	0,631043618	
32-23																						0,072433	0,144865								0,217298	0,666666667	
32-24													0,015937			0,025785									0,012049						0,053772		
32-31																	2,804949	0,462219						1,400582	0,208406						4,876156	0,287230734	
32-32							0,012049										3,441106							0,253814	74,80846						78,51543	0,952786736	
32-41																									0,048357						0,048357	1	
33-33																										0,084311		0,396206			0,522673	0,758037702	
41-32																																	

Overall accuracy at the second level = 88.4%

Overall accuracy at the third level = 87.4%

Zone 23: Oversea departments - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 137: Confusion matrix CLCCH2012-2018 for zone 23 (Guadeloupe /Guyana / La reunion/ Martinique/ Mayotte) – BLIND ANALYSIS- In column the validation and line the production

11-11		2,61046											0,005092									2,615553	0,99805313	
11-21			0,005092																			0,005092		
11-24		0,005092																				0,005092		
12-11		0,01365																				0,01365		
12-12				0,323821						0,005092					0,006825							0,335738	0,964504212	
13-11					0,008707																	0,008707	1	
13-13						0,048119																0,048119	1	
14-14							0,050413															0,050413	1	
21-11								0,132553														0,132553	1	
21-21									2,667072	0,132553												2,799625	0,952653173	
21-23																	0,003449		0,003449			0,006898		
22-22	0,214123									0,49962												0,713743	0,7	
23-22										0,006825												0,006825		
23-23		0,110781								0,896523						0,117606						1,124911	0,796972833	
24-11		0,169381																				0,169381		
24-24											0,16235	4,653922		0,003449								4,819721	0,965599959	
24-32																0,169381						0,169381		
31-31													34,05947									34,05947	1	
31-32													0,004333									0,004333		
31-52													0,020023							0,006674		0,026698		
32-23																0,004496						0,004496		
32-24												0,006825										0,006825		
32-31		0,165748											0,501741			1,445368						2,112857		
32-32		0,165748										0,004496	0,093768	0,004496	0,170244	3,703321						4,142073	0,894074285	
33-31													0,047									0,047		
33-33																0,047	0,698927					0,745927	0,936990834	
41-41																		4,676151				4,676151	1	
42-31																			0,003055			0,003055		
42-42																			0,031668			0,031668	1	
51-51																	0,045584			0,428878		0,474462	0,903924688	
52-31													6,239595									6,234796	12,47439	
52-52																					0,11511	51,04762	51,16273	0,997750113
Total	0,214123	3,240862	0,005092	0,323821	0,008707	0,048119	0,050413	0,132553	2,667072	0,638998	0,901615	0,16235	4,670336	40,96593	0,007945	0,177069	5,487173	0,744511	0,003449	4,676151	0,034723	0,547437	57,28909	122,9975
Prod accuracy		0,805483		1	1	1	1	1	1	0,78188	0,994352		0,996486	0,83141			0,674905	0,938773		1	0,912027	0,783428	0,891053	0,8661738

Overall accuracy at the second level =86.6%

Overall accuracy at the third level =86.4 %

Table 138: Confusion matrix CLCCH2012-2018 for zone 23 (Guadeloupe /Guyana / La reunion/ Martinique/ Mayotte) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	11-12	12-12	13-11	13-13	14-14	21-11	21-21	22-22	23-23	24-12	24-24	31-31	32-23	32-24	32-32	33-33	41-23	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	2,61046											0,005092											2,615553	0,99805313
11-21		0,005092																					0,005092	
11-24	0,005092																						0,005092	
12-11	0,01365																						0,01365	
12-12			0,323821							0,005092					0,006825								0,335738	0,964504212
13-11				0,008707																			0,008707	1
13-13					0,048119																		0,048119	1
14-14						0,050413																	0,050413	1
21-11							0,132553																0,132553	1
21-21								2,667072	0,132553														2,799625	0,952653173
21-23																		0,003449			0,003449		0,006898	
22-22									0,713743														0,713743	1
23-22									0,006825														0,006825	
23-23	0,110781									0,896523						0,117606							1,124911	0,796972833
24-11	0,169381																						0,169381	
24-24											0,16235	4,653922		0,003449									4,819721	0,96559959
24-32																0,169381							0,169381	
31-31													34,05947										34,05947	1
31-32													0,004333										0,004333	
31-52													0,020023								0,006674		0,026698	
32-23																0,004496							0,004496	
32-24												0,006825											0,006825	
32-31	0,165748												0,501741			1,445368							2,112857	
32-32	0,165748											0,004496	0,093768	0,004496	0,170244	3,703321							4,142073	0,894074285
33-31													0,047										0,047	
33-33																0,047	0,698927						0,745927	0,936990834
41-41																			4,676151				4,676151	1
42-31																				0,003055			0,003055	
42-42																				0,031668			0,031668	1
51-51																	0,045584				0,428878		0,474462	0,903924688
52-31												6,239595										6,234796	12,47439	
52-52																				0,11511	51,04762	51,16273	0,997750113	
Total	3,240862	0,005092	0,323821	0,008707	0,048119	0,050413	0,132553	2,667072	0,853121	0,901615	0,16235	4,670336	40,96593	0,007945	0,177069	5,487173	0,744511	0,003449	4,676151	0,034723	0,547437	57,28909	122,9975	
Prod_accuracy	0,805483		1	1	1	1	1	1	0,836625	0,994352		0,996486	0,83141			0,674905	0,938773		1	0,912027	0,783428	0,891053		0,86791467

Overall accuracy at the second level =86.8 %

Overall accuracy at the third level =86.5 %

Annex 8. Confusion matrix of CLCCH2006-12 per bio-geographical region

Bio-geographical region ALP - Alpine - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 139: Confusion matrix CLCCH2012-2018 for Alpine region –BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	14-14	21-21	22-22	23-21	23-23	24-24	31-14	31-31	31-32	32-31	32-32	33-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	39,84709								0,924659	0,924659	1,288562											42,98497	0,927000503
11-24										0,002473												0,002473	
12-11	0,036732	0,005322																				0,042055	
12-12	0,083991	3,563943	0,009259																			3,657193	0,974502145
12-13												0,008939										0,008939	
12-14		0,027937																				0,027937	
12-23								0,198735														0,198735	
12-31											0,005189											0,005189	
13-11	0,05038																					0,05038	
13-12	0,025061	0,06665	0,081786																			0,173497	0,471398332
13-13				1,095662							0,165433											1,261095	0,868818093
13-14					0,107105																	0,107105	
13-21				0,046244		0,00798			0,006405													0,060629	
13-24									0,00798	0,077412												0,085392	
13-31											0,049009											0,049009	
13-32											0,077412				0,117515							0,194927	
14-14	0,012072				2,966835						0,012072				0,009997							3,000976	0,988623398
14-23										0,016687												0,016687	
14-24											0,059079											0,059079	
14-31					0,004621				0,005189													0,081484	
14-32															0,004693							0,004693	
14-51				0,004621					0,004621		0,004621											0,013864	
21-21						50,7046			1,730483		0,012587							0,009272				52,45694	0,966594655
21-23									5,509461													5,509461	
21-24										1,781038												1,781038	
21-31											0,0055											0,0055	
21-32						0,007087																0,007087	
22-11	0,024128																					0,024128	
22-21						0,063631									0,116684							0,180315	
22-22						0,088398	5,422586					0,768476										6,279461	0,86354326
22-32										0,063631												0,063631	
23-21								0,116684														0,116684	1
23-23	0,904037			0,012032					101,0812	3,130879	9,734169				3,756297							118,6186	0,852152933
23-24						0,887893				4,491408	0,003508											5,382809	
23-32											0,212267				2,465056							2,677323	
24-11	0,847595																					0,847595	
24-21	2,674756					0,228876				1,665568												4,5692	
24-22							1,844291															1,844291	
24-23									4,134834			1,2255										5,360334	
24-24									3,176983	110,2774		12,55357										126,0079	0,875162213
24-31												1,464255										1,464255	
24-32												1,127963			0,855189							1,983153	
31-14											0,153313											0,153313	1
31-23												1,09397	0,072433									1,166402	
31-24										0,298417		1,09397										1,392387	
31-31								2,4115				1115,145			3,881829	3,691336	1,767275					1126,897	0,989571411
31-32												5,163241	7,928255		33,90737							46,99886	0,168690355
31-33												0,485424			2,4115							2,896924	
31-41																	0,58327					0,58327	
32-11	0,009683																					0,009683	
32-21						0,072433																0,072433	
32-23									0,599517			0,082145										0,681662	
32-24										2,752792		0,224074			0,024571							3,001437	
32-31									0,019162			14,21049		2,33184	3,879661		0,218673					20,65982	0,112868329
32-32	0,878657								7,222771		11,77441	0,016793	0,008875		574,4196		10,35711					604,6782	0,949959148
32-33															11,69182		6,288975					17,98079	
33-31												0,137249										0,137249	
33-32															1,71986	0,466368	0,058661					2,244888	0,207746612
33-33										0,017564		0,688234			17,24132		519,8879					537,835	0,966630813
41-21									0,050213													0,050213	
41-23									0,014485													0,014485	
41-31												1,793178										1,793178	
41-32															1,837644			0,050213				1,887857	
41-41												0,014485			0,025657			101,6373				101,6774	0,999605205
41-51																				0,170851		0,170851	
42-42																			0,014576			0,014576	1
51-31																				0,091662		0,091662	
51-51																	0,003508			92,31662		92,38463	0,999263857
52-52										0,064501												0,368182	1
Total	45,39418	3,663851	0,091046	1,153938	3,083183	52,0609	7,266877	0,116684	119,8708	132,7918	0,153313	1180,755	8,01748	2,340715	658,3662	0,466368	540,5026	104,0508	0,014576	92,57913	0,368182	2953,108	
Prod_accuracy	0,877802	0,972731	0,898299	0,949498	0,962264	0,973948	0,746206	1	0,843251	0,830453	1	0,944433	0,988871	0,996208	0,872492	1	0,96186	0,976804	1	0,997164	1		0,924390943

Overall accuracy at the second level = 92,44%

Overall accuracy at the third level = 89,86%

Table 140: Confusion matrix CLCCH2012-2018 for Alpine region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	14-14	21-21	22-22	22-32	23-21	23-23	24-21	24-24	31-14	31-31	31-32	32-31	32-32	32-33	33-32	33-33	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	41,13565									0,924659		0,924659													42,98497	0,956977548
11-24												0,002473													0,002473	
12-11	0,002077	0,039978																							0,042055	
12-12	0,083991	3,563943	0,009259																						3,657193	0,974502145
12-13														0,008939											0,008939	
12-14		0,027937																							0,027937	
12-23										0,198735															0,198735	
12-31													0,005189												0,005189	
13-11	0,05038																								0,05038	
13-12	0,025061	0,06665	0,081786																						0,173497	0,471398332
13-13				1,137439										0,123656											1,261095	0,901945637
13-14					0,107105																				0,107105	
13-21						0,00798				0,006405															0,060629	
13-24										0,00798	0,077412														0,085392	
13-31													0,049009												0,049009	
13-32																	0,194927								0,194927	
14-14	0,012072				2,978907												0,009997								3,000976	0,992646102
14-23										0,016687															0,016687	
14-24													0,059079												0,059079	
14-31					0,009243					0,005189			0,067053												0,081484	
14-32																	0,004693								0,004693	
14-51					0,004621						0,004621	0,004621													0,013864	
21-21						50,71718				1,730483											0,009272				52,45694	0,966834606
21-23										5,509461															5,509461	
21-24											1,781038														1,781038	
21-31													0,0055												0,0055	
21-32						0,007087																			0,007087	
22-11	0,024128																								0,024128	
22-21						0,063631		0,116684																	0,180315	
22-22						0,088398	5,422586						0,768476												6,279461	0,86354326
22-32																	0,063631								0,063631	
23-21									0,116684																0,116684	1
23-23	0,904037			0,012032						103,2551		0,956952	9,734169				3,756297								118,6186	0,870479958
23-24										0,887893	4,491408		0,003508												5,382809	
23-32													0,212267				2,465056								2,677323	
24-11	0,847595																								0,847595	
24-21											0,228876	4,340324													4,5692	0,050091099
24-22						1,844291																			1,844291	
24-23										3,943887	0,803698		0,61275												5,360334	
24-24										3,176983	113,932		8,898969												126,0079	0,904165146
24-31													1,464255												1,464255	
24-32													1,127963				0,855189								1,983153	
31-14													0,153313												0,153313	1
31-23											0,072433		1,09397												1,166402	
31-24											0,298417		1,09397												1,392387	
31-31										2,4115			1114,846			0,034302	4,145944			3,691336	1,767275				1126,897	0,989306598
31-32													5,163241	14,9887	1,09397	25,75295									46,99886	0,318916234
31-33													0,485424				2,4115								2,896924	
31-41																					0,58327				0,58327	
32-11	0,009683																								0,009683	
32-21						0,072433																			0,072433	
32-23										0,599517				0,003788			0,078357								0,681662	
32-24												2,752792	0,212805				0,035839								3,001437	
32-31										0,019162			8,661345	0,008781	4,58111	5,961323	0,218673		1,209428						20,65982	0,221740058
32-32	0,878657										7,222771		9,729685	0,016793	0,026166	577,5627		1,214705	8,026693						604,6782	0,955157192
32-33																11,69182			6,288975						17,98079	
33-31													0,137249												0,137249	
33-32																1,71986		0,466368	0,058661						2,244888	0,207746612
33-33										0,017564		0,688234				17,00553		0,084921	520,0388						537,835	0,966911326
41-21										0,050213															0,050213	
41-23										0,014485															0,014485	
41-31																					1,793178				1,793178	
41-32																	1,837644				0,050213				1,887857	
41-41													0,014485				0,025657				101,6373				101,6774	0,999605205
41-51																						0,170851			0,170851	
42-42																						0,014576			0,014576	1
51-31																						0,091662			0,091662	
51-51																					0,003508				92,38112	0,999962032
52-52																								0,368182	0,368182	1
Total	43,97333	3,698507	0,091046	1,195715	3,099876	50,95671	7,266877	0,116684	0,116684	122,7584	0,228876	137,6786	0,153313	1165,276	15,01427	5,735548	655,5789	0,218673	1,765993	539,3138	105,844	0,014576	92,64363	0,368182	2953,108	
Prod_accuracy	0,935468	0,963617	0,898299	0,951263	0,960976	0,995299	0,746206		1	0,841125	1	0,827522	1	0,956723	0,998297	0,798722	0,880996		0,264082	0,96426	0,960256	1	0,997166	1		0,931089721

Overall accuracy at the second level =93,11%
Overall accuracy at the third level = 90,61%

Table 141: Confusion matrix CLCCH2012-2018 for Anatolian region– BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 83,72%
Overall accuracy at the third level = 80,43%

Table 142: Confusion matrix CLCCH2012-2018 for Anatolian region– PLAUSIBILITY ANALYSIS- In column the validation and line the production

CCC_LwI2	11-11	12-12	13-11	13-12	13-13	13-32	13-51	14-14	21-11	21-12	21-13	21-21	21-51	22-22	23-13	23-21	23-23	23-24	23-51	24-11	24-13	24-21	24-24	24-51	31-31	31-51	32-11	32-12	32-13	32-21	32-32	32-51	33-11	33-12	33-13	33-21	33-32	33-33	41-21	41-41	42-42	51-42	51-51	52-52	Total	User_accuracy				
11-11	19,12962																																													19,12962	1			
11-13					0,054592																																								0,054592					
11-21	0,454743											0,031787																																	0,48653					
11-23																	0,070333																												0,070333					
11-24																				0,009834																									0,009834					
11-51													0,06357																																0,06357					
12-11	0,070333																																												0,070333					
12-12	0,049686	3,597522																														0,049686													3,696894	0,973119945				
12-23																	0,049686																												0,049686					
12-24																							0,369152																						0,369152					
12-32		0,369152																					0,369152																						0,369152					
13-11	0,06618		0,081786						0,081786																																				0,229753	0,355974794				
13-12		0,117973		0,163572																												0,081786													0,363332	0,450201127				
13-13				3,214444													0,075795																													3,290239	0,976963803			
13-21		0,065062										0,147382																				0,070333													0,282776					
13-23																																0,070333													0,070333					
13-24																																0,031677													0,031677					
13-32																																0,03068													0,09599					
13-33																																															0,147782			
13-51						0,04238	0,02119					0,02119																				0,02119			0,084731				0,065309						0,10595	0,2				
14-14								0,131919																																					0,131919	1				
14-21												0,003746																																	0,003746					
21-11																																															0,145233			
21-12												0,145233																																	0,145233					
21-13																																															0,145233	1		
21-21											0,145233	335,224						0,145233																												0,290466	0,5			
21-23																																															339,7828	0,986583088		
21-24																4,558836	0,028552																												4,587389					
21-51													0,165748										0,034109																								0,034109			
22-22												0,113616		18,25486									2,746135				0,03315					0,113616		0,03315													0,232047	0,714285714		
22-24														0,971648									1,630094																								21,22822	0,859933289		
22-32	0,387597																0,387597																												2,989339					
23-12																																															0,387597			
23-13																0,145233													0,145233																		0,159352			
23-21																																															0,145233	1		
23-23												6,717087				2,205528	0,350051	0,116684																														9,389349	0,234896782	
23-24																	51,77767																5,061432													56,8391	0,910951583			
23-32																		0,116684																											0,233367	0,5				
23-51																																															4,765696			
24-13					0,145233															0,03315																												0,03315	1	
24-21																					0,290466																										0,580932	0,5		
24-24												9,518421						6,327937				0,116684																											15,96304	0,007309605
24-51												17,87399						8,159192					182,9078										6,344992															215,2859	0,849603856	
31-24													0,03315					0,051492							0,198897								0,03315															0,051492		
31-31																							2,086095																								2,086095			
31-51																							2,086095		75,79444																							0,476804		
32-13																																															0,229969			
32-21																																															0,507028	0,072433		
32-24												</																																						

Overall accuracy at the second level = 84,65%

Overall accuracy at the third level =81,34%

Bio-geographical region ARC - Arctic - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 143: Confusion matrix CLCCH2012-2018 for Arctic region –BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-13	14-14	21-21	23-23	23-32	24-24	31-31	32-32	33-32	33-33	33-51	41-41	42-42	51-33	51-51	52-52	Total	User_accuracy
11-11	0,434877																		0,434877	1
12-12		0,406959								0,01837		0,01837							0,443699	0,917194748
12-33												0,01837							0,01837	
13-13	0,025971		0,075816																0,101787	0,744847415
13-14				0,025971															0,025971	
14-14				0,700382															0,700382	1
21-21					0,095488														0,095488	1
23-21					0,002925														0,002925	
23-23						10,7261		0,002925											10,72903	0,999727369
23-32							0,051492												0,051492	1
24-23						0,021875													0,021875	
24-24								0,199801											0,199801	1
31-31									1,296685	0,088535				0,067392					1,452612	0,892657394
31-32										0,207215									0,207215	
31-33												0,067392							0,067392	
31-41														0,067392					0,067392	
32-32						0,01304				163,8667									163,8797	0,999920429
32-33												0,075577							0,075577	
33-32											0,466368								0,466368	1
33-33												224,186					0,01304		224,199	0,999941837
33-41														0,58713					0,58713	
33-51												0,0266	0,398995				0,346764		0,772359	0,516592751
33-52																		0,0266	0,0266	
41-33												0,081332							0,081332	
41-41														29,48284					29,48284	1
42-42															0,473106				0,473106	1
51-32																	0,034721		0,034721	
51-33												0,307312				0,7679	0,127983		1,203195	0,638217224
51-41														0,192834					0,192834	
51-51																	9,240573		9,240573	1
52-52																		0,848332	0,848332	1
Total	0,460849	0,406959	0,075816	0,726353	0,098413	10,76102	0,051492	0,202726	1,296685	164,1808	0,466368	224,7809	0,398995	30,39759	0,473106	0,7679	9,763082	0,874931	446,184	
Prod_accuracy	0,943645	1	1	0,964244	0,970278	0,996755	1	0,985571	1	0,998087	1	0,997353	1	0,969907	1	1	0,946481	0,969598		0,994473945

Overall accuracy at the second level = 99,45%

Overall accuracy at the third level = 98,14%

Table 144: Confusion matrix CLCCH2012-2018 for Arctic region –PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lv2	11-11	12-12	13-11	13-13	14-14	21-21	23-23	23-32	24-24	31-31	32-32	33-32	33-33	33-51	41-41	42-42	51-33	51-51	52-52	Total	User_accuracy
11-11	0,434877																			0,434877	1
12-12		0,425329											0,01837							0,443699	0,958597374
12-33													0,01837							0,01837	
13-13			0,025971	0,075816																0,101787	0,744847415
13-14					0,025971															0,025971	
14-14					0,700382															0,700382	1
21-21						0,095488														0,095488	1
23-21						0,002925														0,002925	
23-23							10,7261		0,002925											10,72903	0,999727369
23-32								0,051492												0,051492	1
24-23							0,021875													0,021875	
24-24									0,199801											0,199801	1
31-31										1,296685	0,088535				0,067392					1,452612	0,892657394
31-32											0,207215									0,207215	
31-33													0,067392							0,067392	
31-41															0,067392					0,067392	
32-32							0,01304				163,8667									163,8797	0,999920429
32-33													0,075577							0,075577	
33-32												0,466368								0,466368	1
33-33													224,199							224,199	1
33-41															0,58713					0,58713	
33-51													0,0266	0,398995				0,346764		0,772359	0,516592751
33-52																			0,0266	0,0266	
41-33													0,081332							0,081332	
41-41															29,48284					29,48284	1
42-42																0,473106				0,473106	1
51-32																		0,034721		0,034721	
51-33													0,082482	0,031996			0,960734	0,127983		1,203195	0,798485787
51-41															0,192834					0,192834	
51-51																		9,240573		9,240573	1
52-52																			0,848332	0,848332	1
Total	0,434877	0,425329	0,025971	0,075816	0,726353	0,098413	10,76102	0,051492	0,202726	1,296685	164,1625	0,466368	224,5691	0,430991	30,39759	0,473106	0,960734	9,750042	0,874931	446,184	
Prod_accuracy	1	1		1	0,964244	0,970278	0,996755	1	0,985571	1	0,998198	1	0,998352	0,925762	0,969907	1	1	0,947747	0,969598		0,994976528

Overall accuracy at the second level = 99,50%

Overall accuracy at the third level = 98,19%

Table 145: Confusion matrix CLCCH2012-2018 for Atlantic region –BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 90,86%
Overall accuracy at the third level = 89,94%

[illegible]

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Bio-geographical region BLS – Black sea - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 147: Confusion matrix CLCCH2012-2018 for black sea region –BLIND ANALYSIS - In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	14-14	21-21	21-51	22-13	22-21	22-22	23-13	23-21	23-23	23-51	24-24	24-51	31-13	31-31	32-13	32-31	32-32	32-51	33-33	33-51	41-21	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	2,661311																													2,661311	1
12-12		2,722286																												2,722286	1
12-24															0,084731															0,084731	
12-31																		0,08453												0,08453	
13-13		0,014758		0,473969																	0,077412									0,566139	0,837195673
13-21			0,063381																											0,063381	
13-51																												0,058658		0,058658	
14-14	0,007629				0,309281																0,027448									0,344357	0,898139599
21-21						268,1633							3,245727		31,54994			6,747776			6,747776		6,747776							323,2023	0,829707288
21-24															40,48666															40,48666	
21-32																						0,292193								0,292193	
21-51							0,132598							0,03315		0,03315					0,03315									0,232047	0,571428571
22-21						0,282611			0,116684																					0,399295	0,292223855
22-22									0,662866	5,592198					1,972434			1,630094												9,857592	0,567298639
22-24															0,116847															0,116847	
23-13											0,145233																			0,145233	1
23-21						0,035087						0,350051																		0,385138	0,908897351
23-23													4,780192								0,339233									5,119425	0,933736039
24-22															3,282209															3,282209	
24-24										3,161931					32,04653						3,161931									38,37039	0,835189012
24-51															0,03315							0,03315								0,066299	
31-13																	0,076656													0,076656	1
31-31																		113,9893												113,9893	1
31-32																		2,693878												2,693878	
32-13																				0,153313										0,153313	1
32-24																					0,072433									0,072433	
32-31																	0,466368													0,466368	
32-32														0,035525						0,005156	58,92084								58,96152	0,999310049	
32-51														0,053199		0,0266					0,053199			0,0266						0,159598	0,333333333
33-24													0,072433																	0,072433	
33-33													4,878502										15,27014						0,009024	20,15767	0,757535151
33-51																						0,0266		0,0266						0,053199	0,5
41-21																									0,100264					0,100264	1
41-41																										4,997429				4,997429	1
42-42																										0,281568	0,033373			0,314941	0,105966876
51-13								0,066832																						0,066832	
51-51						0,052021												0,24684										2,529489		2,828349	0,894333794
52-11	0,186601																													0,186601	
52-52																													3,262208	3,262208	1
Total	2,855542	2,737044	0,063381	0,473969	0,309281	268,533	0,132598	0,066832	0,779549	8,754129	0,145233	0,350051	12,97685	0,086349	109,608	0,059749	0,076656	125,8588	0,153313	0,005156	69,34707	0,438291	22,01792	0,053199	0,100264	5,278997	0,033373	2,588146	3,271232	637,154	
Prod_accuracy	0,931981	0,994608		1	1	0,998623	1		0,149681	0,638807	1	1	0,368363		0,292374		1	0,905692	1		0,849651	0,121379	0,693533	0,5	1	0,946663	1	0,977336	0,997241		0,811273946

Overall accuracy at the second level = 81,13%

Overall accuracy at the third level = 77,99%

Table 148: Confusion matrix CLCCH2012-2018 for black sea region –PLAUSIBILITY ANALYSIS - In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-12	13-13	13-31	14-14	21-21	21-51	22-21	22-22	23-13	23-21	23-23	23-51	24-24	24-51	31-13	31-31	32-13	32-31	32-32	32-51	33-33	33-51	41-21	41-41	42-42	51-21	51-51	52-52	Total	User_accuracy
11-11	2,661311																													2,661311	1	
12-12		2,722286																												2,722286	1	
12-24					0,084731																									0,084731		
12-31																	0,08453													0,08453		
13-13		0,014758		0,473969																	0,077412									0,566139	0,837195673	
13-21			0,063381																											0,063381		
13-51																												0,058658		0,058658		
14-14	0,007629					0,309281															0,027448									0,344357	0,898139599	
21-21							268,1633						3,245727		31,54994		6,747776				6,747776		6,747776							323,2023	0,829707288	
21-24							6,747776								33,73888															40,48666		
21-32																						0,292193								0,292193		
21-51								0,132598						0,03315		0,03315						0,03315								0,232047	0,571428571	
22-21							0,282611		0,116684																					0,399295	0,292223855	
22-22									0,662866	5,592198						3,602527														9,857592	0,567298639	
22-24																0,116847														0,116847		
23-13											0,145233																			0,145233	1	
23-21							0,035087					0,350051																		0,385138	0,908897351	
23-23													5,119425																	5,119425	1	
24-22															3,282209															3,282209		
24-24										3,161931					32,04653																38,37039	0,835189012
24-51																0,03315						0,03315								0,066299	0,5	
31-13																	0,076656													0,076656	1	
31-31																		113,9893												113,9893	1	
31-32																		2,693878												2,693878		
32-13																			0,153313											0,153313	1	
32-24																					0,072433									0,072433		
32-31																	0,466368													0,466368		
32-32															0,035525					0,005156	58,92084									58,96152	0,999310049	
32-51														0,053199		0,0266						0,053199		0,0266						0,159598	0,333333333	
33-24													0,072433																	0,072433		
33-33													4,878502										15,27014							0,009024	20,15767	0,757535151
33-51																						0,0266		0,0266						0,053199	0,5	
41-21																									0,100264					0,100264	1	
41-41																											4,997429			4,997429	1	
42-42																										0,281568	0,033373			0,314941	0,105966876	
51-13									0,066832																					0,066832		
51-51																	0,24684											0,052021	2,529489	2,828349	0,894333794	
52-11	0,186601																													0,186601		
52-52																														3,262208	3,262208	1
Total	2,855542	2,737044	0,063381	0,473969	0,084731	0,309281	275,2288	0,132598	0,779549	8,820961	0,145233	0,350051	13,31609	0,086349	104,3725	0,092899	0,076656	124,2287	0,153313	0,005156	69,00784	0,438291	22,01792	0,053199	0,100264	5,278997	0,033373	0,052021	2,588146	3,271232	637,154	
Prod_accuracy	0,931981	0,994608		1		1	0,974329	1	0,149681	0,633967	1	1	0,384454		0,30704	0,356835	1	0,917576	1		0,853828	0,121379	0,693533	0,5	1	0,946663	1		0,977336	0,997241	0,811858393	

Overall accuracy at the second level = 81,19%

Overall accuracy at the third level =78,53%

Table 149: Confusion matrix CLCCH2012-2018 for Boreal region – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 91,98%

Overall accuracy at the third level = 91,14%

C_Lv2	11-11	12-12	13-11	13-12	13-13	13-32	14-14	21-21	21-22	21-23	21-41	22-21	22-22	23-12	23-21	23-23	23-24	23-32	24-12	24-24	31-12	31-13	31-23	31-31	31-32	31-33	31-41	32-13	32-21	32-23	32-31	32-32	32-41	33-32	33-33	41-13	41-21	41-23	41-32	41-41	42-42	51-51	52-52	Total	User_accuracy	
11-11	42,24999	0,724283																																									42,97427	0,983146129		
11-12		0,009682																																									0,009682			
11-14		0,004362					0,61096																																					0,615321		
11-21		0,002473						0,014884																																				0,017357		
11-24	0,008039																			0,734795																								0,742835		
11-31	0,726756						0,004362																																					0,731118		
11-32																																											0,008039			
12-11	0,118395	0,116012					0,002146																											0,008039										0,236552		
12-12	0,276292	10,76331																																											11,0396	0,974972633
12-13					0,01268																																							0,01268		
12-21								0,235327																																				0,235327		
12-23																0,157911																												0,157911		
12-24		0,002383																	0,00964	0,157911																								0,169934		
12-31	0,138146	0,030718						0,003007													0,002146				0,268185																				0,442203	
12-32		0,00784					0,002146																																					0,009986		
13-11	0,13628		0,163572																																										0,299853	0,54550913
13-12		0,076565		0,122355	0,018667					</																																				

Overall accuracy at the third level = 91,56%

Table 151: Confusion matrix CLCCH2012-2018 for Continental region – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 89,60%
Overall accuracy at the third level = 87,59%

Table 152: Confusion matrix CLCCH2012-2018 for Continental region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

[illegible]

Overall accuracy at the second level = 90,12%
Overall accuracy at the third level = 88,25%

Bio-geographical region MAC– Macaronesia- Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 153: Confusion matrix CLCCH2012-2018 for Macaronesia region – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-13	14-14	21-21	23-23	24-24	31-31	32-32	33-33	41-41	52-52	Total	User_accuracy
11-11	0,036197												0,036197	1
11-14				1,026852									1,026852	
11-23							0,513086						0,513086	
11-24							0,513086						0,513086	
12-11	0,009725												0,009725	
12-12		0,185816											0,185816	1
12-23						0,012497							0,012497	
13-13			0,092019										0,092019	1
13-24							0,512893						0,512893	
14-14				0,103115			0,01225						0,115366	0,893812924
14-32									0,018975				0,018975	
21-21					2,583306								2,583306	1
23-23						2,586462							2,586462	1
24-24							6,694103						6,694103	1
31-31								8,967216					8,967216	1
32-32									7,716943				7,716943	1
32-33										5,252902			5,252902	
33-32									1,556692				1,556692	
33-33							1,556692		0,783569	13,1812			15,52146	0,849224188
41-32									0,075524				0,075524	
41-41											0,151048		0,151048	1
52-52												0,052655	0,052655	1
Total	0,045922	0,185816	0,092019	1,129968	2,583306	2,598958	9,80211	8,967216	10,1517	18,4341	0,151048	0,052655	54,19482	
Prod_accuracy	0,788238	1	1	0,091255	1	0,995192	0,682925	1	0,760162	0,715044	1	1		0,781441449

Overall accuracy at the second level = 78,14%

Overall accuracy at the third level = 72,64%

Table 154: Confusion matrix CLCCH2012-2018 for Macaronesia region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-13	14-14	21-21	23-23	24-24	31-31	32-32	32-33	33-33	41-41	52-52	Total	User_accuracy
11-11	0,0362													0,0362	1
11-14				1,02685										1,02685	
11-23							0,51309							0,51309	
11-24							0,51309							0,51309	
12-11	0,00972													0,00972	
12-12		0,18582												0,18582	1
12-23						0,0125								0,0125	
13-13			0,09202											0,09202	1
13-24							0,51289							0,51289	
14-14				0,11537										0,11537	1
14-32									0,01898					0,01898	
21-21					2,58331									2,58331	1
23-23						2,58646								2,58646	1
24-24							6,6941							6,6941	1
31-31								8,96722						8,96722	1
32-32									7,71694					7,71694	1
32-33										5,2529				5,2529	1
33-32									1,55669					1,55669	
33-33									0,78357		14,7379			15,5215	0,949517081
41-32									0,07552					0,07552	
41-41												0,15105		0,15105	1
52-52													0,05265	0,05265	1
Total	0,04592	0,18582	0,09202	1,14222	2,58331	2,59896	8,23317	8,96722	10,1517	5,2529	14,7379	0,15105	0,05265	54,1948	
Prod_accuracy	0,78824	1	1	0,101	1	0,99519	0,81307	1	0,76016	1	1	1	1		0,907317766

Overall accuracy at the second level = 90,73%

Overall accuracy at the third level = 89,97%

Table 155: Confusion matrix CLCCH2012-2018 for Mediterranean region – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 86,66%

Overall accuracy at the third level = 81,37%

[illegible]

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Table 157: Confusion matrix CLCCH2012-2018 for Pannonian region – BLIND ANALYSIS- In column the validation and line the production

Overall accuracy at the second level = 92,53%
Overall accuracy at the third level = 91,37%

[illegible]

Overall accuracy at the third level = 91,80%

Bio-geographical region STE – Steppic - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 159: Confusion matrix CLCCH2012-2018 for Steppic region – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-13	14-14	21-21	21-22	22-21	22-22	22-23	23-21	23-23	24-12	24-24	31-31	31-32	32-32	33-33	41-21	41-41	41-51	51-21	51-51	Total	User_accuracy
11-11	4,623297																						4,623297	1
11-24													0,924659										0,924659	
12-12		1,58824																					1,58824	1
13-13			0,015425									0,062472											0,077897	0,19801915
14-11	0,011942																						0,011942	
14-14				0,012072																			0,012072	1
21-21					110,7265																		110,7265	1
21-22						0,116684																	0,116684	1
22-21					0,662866		0,116684																0,779549	0,149680725
22-22								2,651464			0,662866												3,31433	0,8
22-23									0,116684														0,116684	1
23-21										0,350051	0,116684												0,466734	0,75
23-23	2,173926										8,695705		2,173926										13,04356	0,666666667
24-24													5,800402										5,800402	1
24-32																0,803698							0,803698	
31-31														12,44339									12,44339	1
31-32															0,466368								0,466368	1
32-31														0,299758									0,299758	
32-32														0,599517		0,327169							0,926686	0,353052903
33-33																	0,140418						0,140418	1
41-21					0,921417						0,009359							0,175463			0,200529		1,306767	0,134272286
41-41																			4,185088				4,185088	1
41-51																				0,066832			0,066832	1
51-51																						4,588331	4,588331	1
Total	6,809165	1,58824	0,015425	0,012072	112,3108	0,116684	0,116684	2,651464	0,116684	0,350051	9,484613	0,062472	8,898988	13,34267	0,466368	1,130867	0,140418	0,175463	4,185088	0,066832	0,200529	4,588331	166,8299	
Prod_accuracy	0,678981	1	1	1	0,985894	1	1	1	1	1	0,916822		0,651805	0,932602	1	0,289308	1	1	1	1	1	1		0,942314775

Overall accuracy at the second level = 94,23%

Overall accuracy at the third level = 94,23%

Table 160: Confusion matrix CLCCH2012-2018 for Steppic region – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	12-12	13-13	14-14	21-21	21-22	22-21	22-22	22-23	23-21	23-23	24-12	24-24	31-31	31-32	32-31	32-32	33-33	41-21	41-41	41-51	51-21	51-51	Total	User_accuracy
11-11	4,623297																							4,623297	1
11-24												0,924659												0,924659	
12-12		1,58824																						1,58824	1
13-13			0,015425									0,062472												0,077897	0,19801915
14-11	0,011942																							0,011942	
14-14				0,012072																				0,012072	1
21-21					110,7265																			110,7265	1
21-22						0,116684																		0,116684	1
22-21					0,662866		0,116684																	0,779549	0,149680725
22-22								2,651464			0,662866													3,31433	0,8
22-23									0,116684															0,116684	1
23-21										0,350051	0,116684													0,466734	0,75
23-23											13,04356													13,04356	1
24-24													5,800402											5,800402	1
24-32																	0,803698							0,803698	
31-31														12,44339										12,44339	1
31-32															0,466368									0,466368	1
32-31																0,299758								0,299758	1
32-32														0,299758		0,299758	0,327169							0,926686	0,353052903
33-33																		0,140418						0,140418	1
41-21					0,921417						0,009359								0,175463			0,200529		1,306767	0,134272286
41-41																				4,185088				4,185088	1
41-51																					0,066832			0,066832	1
51-51																							4,588331	4,588331	1
Total	4,635239	1,58824	0,015425	0,012072	112,3108	0,116684	0,116684	2,651464	0,116684	0,350051	13,83247	0,062472	6,725062	12,74315	0,466368	0,599517	1,130867	0,140418	0,175463	4,185088	0,066832	0,200529	4,588331	166,8299	
Prod_accuracy	0,997424	1	1	1	0,985894	1	1	1	1	1	0,942967		0,862505	0,976477	1	0,5	0,289308	1	1	1	1	1	1		0,970173161

Overall accuracy at the second level = 97,02%
Overall accuracy at the third level = 97,02%

Bio-geographical region DOM – Oversea departments - Confusion matrix of CLCCH2012-2018 - Blind and plausibility analysis

Table 161: Confusion matrix CLCCH2012-2018 for Oversea departments (Guadeloupe, French Guyana, Martinique, La Reunion, Mayotte) – BLIND ANALYSIS- In column the validation and line the production

CLCC_Lvl2	00	11-11	11-12	12-12	13-11	13-13	14-14	21-11	21-21	22-22	23-23	24-12	24-24	31-31	32-23	32-24	32-32	33-33	41-23	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11		2,61046											0,005092											2,615553	0,99805313
11-21			0,005092																					0,005092	
11-24		0,005092																						0,005092	
12-11		0,01365																						0,01365	
12-12				0,323821							0,005092					0,006825								0,335738	0,964504212
13-11					0,008707																			0,008707	1
13-13						0,048119																		0,048119	1
14-14							0,050413																	0,050413	1
21-11								0,132553																0,132553	1
21-21									2,667072	0,132553														2,799625	0,952653173
21-23																			0,003449			0,003449		0,006898	
22-22	0,214123									0,49962														0,713743	0,7
23-22										0,006825														0,006825	
23-23		0,110781									0,896523						0,117606							1,124911	0,796972833
24-11		0,169381																						0,169381	
24-24												0,16235	4,653922		0,003449									4,819721	0,965599959
24-32																	0,169381							0,169381	
31-31														34,05947										34,05947	1
31-32														0,004333										0,004333	
31-52														0,020023								0,006674		0,026698	
32-23																	0,004496							0,004496	
32-24													0,006825											0,006825	
32-31		0,165748												0,501741			1,445368							2,112857	
32-32		0,165748											0,004496	0,093768	0,004496	0,170244	3,703321							4,142073	0,894074285
33-31														0,047										0,047	
33-33																	0,047	0,698927						0,745927	0,936990834
41-41																			4,676151					4,676151	1
42-31																					0,003055			0,003055	
42-42																					0,031668			0,031668	1
51-51																		0,045584				0,428878		0,474462	0,903924688
52-31													6,239595										6,234796	12,47439	
52-52																					0,11511	51,04762	51,16273	0,997750113	
Total	0,214123	3,240862	0,005092	0,323821	0,008707	0,048119	0,050413	0,132553	2,667072	0,638998	0,901615	0,16235	4,670336	40,96593	0,007945	0,177069	5,487173	0,744511	0,003449	4,676151	0,034723	0,547437	57,28909	122,9975	
Prod_accuracy		0,805483		1	1	1	1	1	1	0,78188	0,994352		0,996486	0,83141			0,674905	0,938773		1	0,912027	0,783428	0,891053		0,8661738

Overall accuracy at the second level = 86,62%

Overall accuracy at the third level = 86,41%

Table 162: Confusion matrix CLCCH2012-2018 for Oversea departments (Guadeloupe, French Guyana, Martinique, La Reunion, Mayotte) – PLAUSIBILITY ANALYSIS- In column the validation and line the production

CLCC_Lvl2	11-11	11-12	12-12	13-11	13-13	14-14	21-11	21-21	22-22	23-23	24-12	24-24	31-31	32-23	32-24	32-32	33-33	41-23	41-41	42-42	51-51	52-52	Total	User_accuracy
11-11	2,61046											0,005092											2,615553	0,99805313
11-21		0,005092																					0,005092	
11-24	0,005092																						0,005092	
12-11	0,01365																						0,01365	
12-12			0,323821							0,005092					0,006825								0,335738	0,964504212
13-11				0,008707																			0,008707	1
13-13					0,048119																		0,048119	1
14-14						0,050413																	0,050413	1
21-11							0,132553																0,132553	1
21-21								2,667072	0,132553														2,799625	0,952653173
21-23																		0,003449			0,003449		0,006898	
22-22									0,713743														0,713743	1
23-22									0,006825														0,006825	
23-23	0,110781									0,896523						0,117606							1,124911	0,796972833
24-11	0,169381																						0,169381	
24-24											0,16235	4,653922		0,003449									4,819721	0,96559959
24-32																0,169381							0,169381	
31-31													34,05947										34,05947	1
31-32													0,004333										0,004333	
31-52													0,020023									0,006674	0,026698	
32-23																0,004496							0,004496	
32-24												0,006825											0,006825	
32-31	0,165748												0,501741			1,445368							2,112857	
32-32	0,165748											0,004496	0,093768	0,004496	0,170244	3,703321							4,142073	0,894074285
33-31													0,047										0,047	
33-33																0,047	0,698927						0,745927	0,936990834
41-41																			4,676151				4,676151	1
42-31																				0,003055			0,003055	
42-42																				0,031668			0,031668	1
51-51																	0,045584				0,428878		0,474462	0,903924688
52-31													6,239595									6,234796	12,47439	
52-52																					0,11511	51,04762	51,16273	0,997750113
Total	3,240862	0,005092	0,323821	0,008707	0,048119	0,050413	0,132553	2,667072	0,853121	0,901615	0,16235	4,670336	40,96593	0,007945	0,177069	5,487173	0,744511	0,003449	4,676151	0,034723	0,547437	57,28909	122,9975	
Prod_accuracy	0,805483		1	1	1	1	1	1	0,836625	0,994352		0,996486	0,83141			0,674905	0,938773		1	0,912027	0,783428	0,891053		0,86791467

Overall accuracy at the second level = 86,79%
Overall accuracy at the third level = 86,52%

